Ophthalmic Artificial Intelligence From Data Integration to Accountable Clinical Impact: Challenges and Opportunities

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Supplementary Material

Tobias Schlosser[a], Michael Friedrich[b], Richard Vogel[c], Benny Platte[d], Matthias Vodel[c], Danny Kowerko[b], Marc Ritter[d], Miriam Goldammer[a], Markus Wolfien[a,e]

[a] *Dresden University of Technology, Faculty of Medicine Carl Gustav Carus, Center for Medical Informatics, Fetscherstraße 74, Dresden, 01307, Saxony, Germany*
[b] *Chemnitz University of Technology, Faculty of Computer Science, Junior Professorship of Media Computing, Straße der Nationen 62, Chemnitz, 09107, Saxony, Germany*
[c] *University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Software Development for Media and Application Systems, Technikumplatz 17, Mittweida, 09648, Saxony, Germany*
[d] *University of Applied Sciences Mittweida, Faculty of Applied Computer and Biosciences, Professorship of Media Informatics, Technikumplatz 17, Mittweida, 09648, Saxony, Germany*
[e] *Dresden University of Technology, Center for Scalable Data Analytics and Artificial Intelligence (ScaDS.AI) Dresden/Leipzig, Strehlener Straße 12, 14, Dresden, 01069, Saxony, Germany*

1. Article-level evidence synthesis and supporting documentation

This supplementary material provides the article-level audit trail for the review. It documents how individual publications informed the main manuscript's thematic, methodological, translational, and ethical synthesis, including dataset overviews, evidence gap mappings, and extended domain-specific analyses. Together, these materials support the transparency, traceability, and reproducibility of the review.

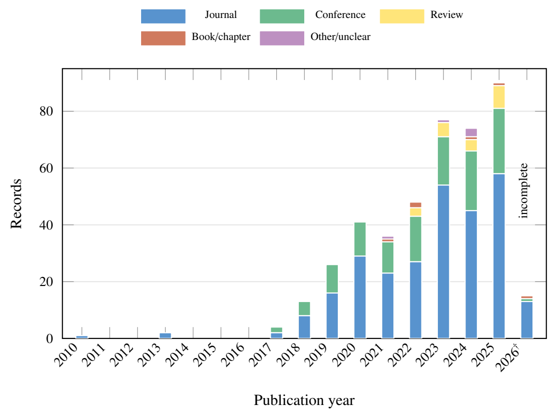
2. Annual publication trend variants

Publication trends were analyzed after deduplication of records across the queried bibliographic sources. Publication years were converted to integer values, and implausible or missing years were excluded from the annual trend analysis but retained in an audit table. Source-specific document-type labels were harmonized into five categories: journal article, conference/proceedings, review, book/chapter, and other/unclear as visualized in Figure 1A. Annual counts were then aggregated by publication year and normalized document type. Because 2026 was ongoing at the time of analysis, records from 2026 were marked as incomplete and interpreted descriptively.

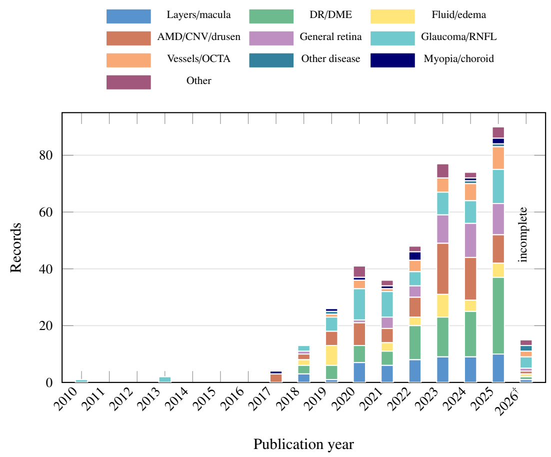
Figure 1D uses the same annual record totals but replaces document type with a single primary task-family assignment per retained record. Primary task labels were assigned from title, abstract, and keyword evidence using the same task-family naming convention used for the task axes in Figures 2A and 2B, while additional task mentions were retained in a separate multi-label audit table.

Figure 1B uses the same annual record totals but stacks records by a single primary clinical-target assignment. Clinical-target labels use the same naming convention as the clinical-target axis in Figure 2A. Disease-specific categories were prioritized over broader anatomical categories when multiple target cues were present, and multi-label target mentions were retained in a separate audit table.

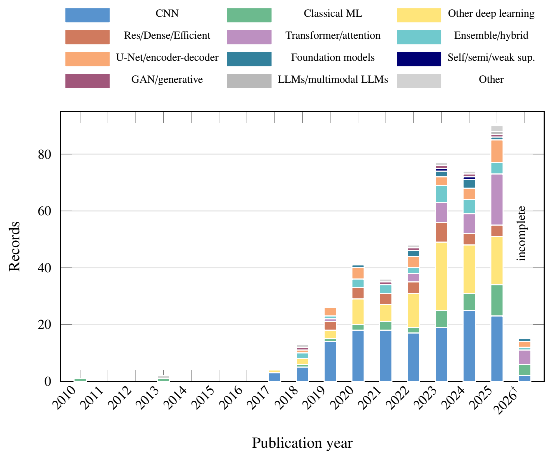
Figure 1C uses the same annual record totals but stacks records by a single primary method-family assignment. Method-family labels use the same naming convention as the method axis in Figure 2B. Specific model families, including U-Net/encoder-decoder segmentation, transformer/attention/ViT, GAN/generative models, foundation models, and large language models/multimodal LLMs, were prioritized over generic deep-learning labels when multiple method cues were present; the generic other-deep-learning category was used only when no more specific method-family cue was available. The LLM audit identified one candidate LLM-related record from title, abstract, and keyword evidence, which was validated as a study that used an LLM.



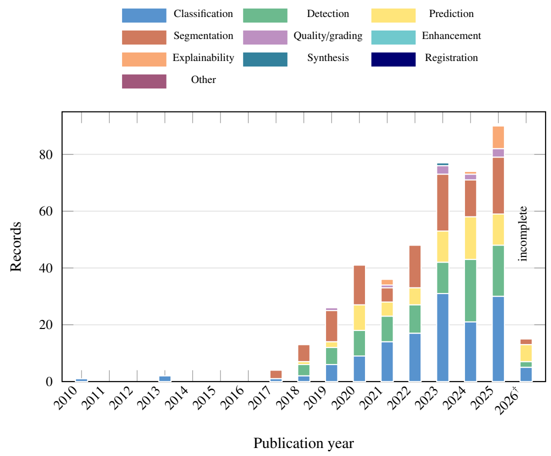
(A) Document type



(B) Clinical target



(C) Method family



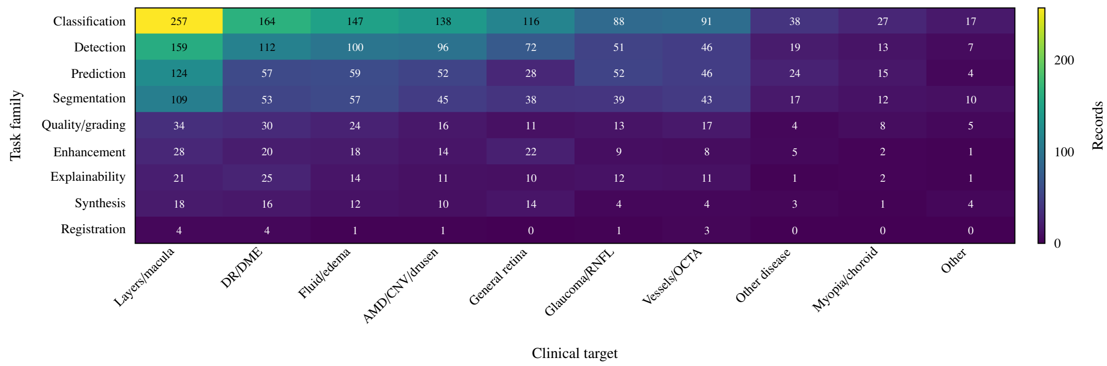
(D) Task family

Figure 1: Annual publication trend variants for retained AI/OCT ophthalmology records. **(A)** Normalized document type. **(B)** Primary clinical target. **(C)** Primary method family. **(D)** Primary task family. Bars show deduplicated literature records per publication year, and stacked colors show the assigned category for each panel. Primary-label encodings preserve annual record totals; multi-label task, clinical-target, and method-family mentions were retained separately for audit purposes. The 2026 count is incomplete because the publication year was ongoing at the time of corpus construction.

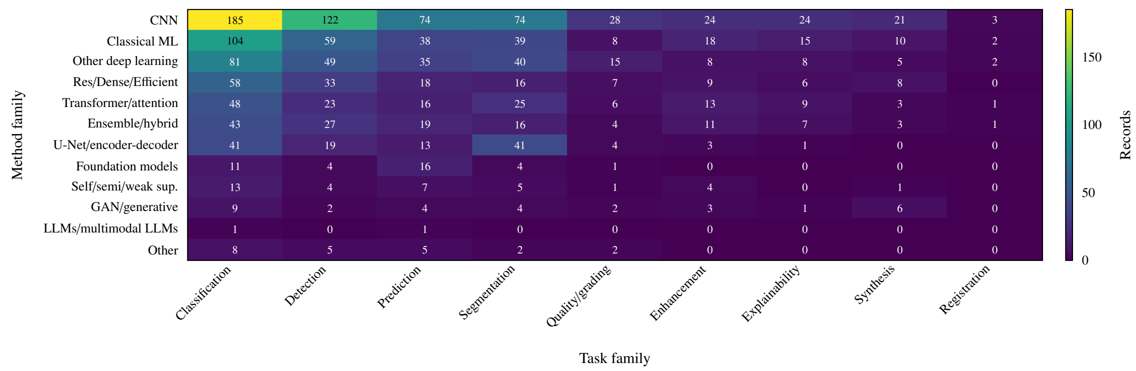
3. Task-family, clinical-target, and method-family heatmaps

Figures 2A and 2B extend the descriptive evidence-base audit by mapping the included records across task families, clinical targets, and method families. Labels were assigned from title, abstract, and keyword text using reproducible multi-label coding, so the figures summarize the thematic scope of the literature rather than mutually exclusive study groups. Consequently, the heatmap counts represent literature-record assignments rather than mutually exclusive study counts, and one publication may contribute to multiple cells when it addresses several tasks, targets, or method families. This structure makes it possible to identify areas where the evidence base is concentrated, such as frequently studied task-target combinations, and areas where comparatively few records link specific clinical needs with specific methodological approaches.

The method-family heatmap also separates studies that explicitly used large language models or multimodal language-model systems. This designation is based only on evidence in the publication metadata fields used for coding and does not use review-screening columns as evidence of study-level LLM use. The separation was included to avoid conflating LLMs used as research objects or study methods with AI tools used during the review workflow itself. It therefore provides a conservative indication of where language-model-based approaches are already visible within ophthalmic AI literature, while preserving the distinction between the analyzed studies and the review process.



(A) Task family by clinical target



(B) Method family by task family

Figure 2: Heatmap summaries of task, clinical-target, and method-family assignments among AI, OCT, and ophthalmology literature records. **(A)** Task-family by clinical-target distribution. **(B)** Method-family by task-family distribution. Cell values are literature-record counts, not experiment counts. Multi-label coding was used, so one record may contribute to multiple cells. Works using large language models or multimodal LLMs were explicitly audited; the LLM row includes only validated papers where LLM use was part of the study, not this review's screening workflow.

4. Identifiable public datasets and benchmark resources

Table 1 provides a consolidated overview of the identifiable public datasets and benchmark resources represented in the included literature. It is restricted to named datasets, challenge resources, and reusable benchmark collections, excluding generic descriptions such as "private dataset," local hospital cohorts, or non-identifiable study-specific data collections. This focus on traceable datasets supports meaningful comparison across studies and separates established resources from datasets that cannot be reliably reused or externally verified.

This overview shows that OCT-based retinal disease classification is dominated by a relatively small number of repeatedly reused public datasets. The Kermany / UCSD / Kaggle / Mendeley / OCT2017 / LOCT retinal OCT dataset [1, 2] is the most prominent example and appears under several names across the literature. Other classification-oriented OCT resources, including OCTDL [3, 4], OCTID [5, 6], OCTMNIST [7, 8], RetinalOCT-C8 / OCT-C8 [9, 10], A2ASDOCT [11, 12], and SERI-CUHK SD-OCT [13, 14], support supervised learning across disease categories such as AMD, CNV, DME, drusen, diabetic retinopathy, macular hole, and normal controls. This concentration facilitates comparability between methods, but it also means that model evaluation may be repeatedly anchored in a narrow set of imaging distributions, disease labels, acquisition protocols, and preprocessing conventions.

Segmentation-oriented OCT and OCTA benchmarks constitute a second major group. The RETOUCH challenge [15, 16], OPTIMA challenge [17, 18], AI Challenger OCT dataset [19, 20], Duke / DUKE SD-OCT / Duke DME OCT dataset [21, 22], and AROI dataset [23, 24] primarily support retinal fluid, retinal layer, and lesion segmentation, whereas OCTA-500 / OCTA-3M / OCTA-6M [25, 26], OCTA-SS [27, 28], ROSE-1 [29, 30], FAROS [31, 32], and DRAC 2022 / OCTA-DR [33, 34] cover vascular, FAZ, capillary, lesion, image-quality, and diabetic-retinopathy grading tasks. Selected fundus benchmarks, including ACRIMA [35, 36], ORIGA [37, 38], and EyePACS [39, 40], are retained because they are frequently used in multimodal OCT or OCTA studies. Overall, Table 1 shows that reusable benchmarks have advanced methodological comparison, but also that the field remains dependent on a limited set of datasets, underscoring the need for more diverse, well-documented, multimodal, and externally validated ophthalmic imaging resources.

Table 1: Consolidated overview of ophthalmic imaging datasets and clinical cohorts used across the included studies. Dataset names were harmonized where multiple entries referred to the same or closely related resource. Non-identifiable public or private datasets were grouped by clinical domain when sufficient contextual information was available.

Dataset	Area	Modality	Scale	Typical tasks / notes
A2ASDOCT [11, 12]	Age-related macular degeneration	SD-OCT	AREDS2 ancillary SD-OCT cohort with 38,400 B-scans from 384 subjects; 269 intermediate AMD and 115 control subjects	Public AMD-focused OCT cohort used for quantitative classification of eyes with and without intermediate AMD.
ACRIMA [35, 36]	Glaucoma	Color fundus photographs	705 fundus images	Optic disc-centered fundus benchmark with 396 glaucomatous and 309 normal images.
AI Challenger OCT dataset [19, 20]	Macular edema / retinal OCT lesion segmentation	Retinal OCT volumes	AI Challenger 2018 macular edema dataset with 83 OCT volumes and 10,880 OCT slices	Used for macular edema-related lesion segmentation, including subretinal fluid and pigment epithelial detachment.
AROI dataset [23, 24]	Retinal layers and fluid in neovascular AMD	Retinal OCT	1,136 annotated B-scans from 25 patients out of 3,200 B-scans	Public annotated retinal OCT image database. Used for joint retinal layer, lesion, and fluid segmentation in healthy and neovascular AMD eyes.
DRAC 2022 / OCTA-DR [33, 34]	Diabetic retinopathy	Ultra-widefield OCTA	1,103 UW-OCTA images	Consolidates DRAC 2022, Diabetic Retinopathy Analysis Challenge 2022, DRAC OCTA, and OCTA-DR where used for the same DR challenge context. Tasks include DR lesion segmentation, image-quality assessment, and DR grading.
Duke / DUKE SD-OCT / Duke DME OCT dataset [21, 22]	Diabetic macular edema; retinal layer and fluid analysis	SD-OCT B-scans	110 annotated B-scans from 10 DME patients	Consolidates Duke dataset, DUKE, DUKE OCT dataset, Duke SD-OCT, Duke spectral domain OCT dataset, and Chiu-style Duke DME references. Used for retinal layer, boundary, and fluid segmentation.
EyePACS [39, 40]	Diabetic retinopathy	Color fundus photographs	35,126 labeled training images in the Kaggle DR release	Fundus benchmark for diabetic retinopathy grading. Included where studies combine fundus and OCT / OCTA resources.

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Dataset	Area	Modality	Scale	Typical tasks / notes
FAROS [31, 32]	Retinal OCTA microstructure segmentation	OCTA	40 OCTA images from 40 subjects	Fully Annotated Retinal OCTA Segmentation dataset. Used for retinal microstructure segmentation and quantitative OCTA analysis.
Kermany / UCSD / Kaggle / Mendeley / OCT2017 / LOCT retinal OCT dataset [1, 2]	Retinal disease; CNV, DME, drusen, normal	Retinal OCT B-scans	108,312 labeled OCT images in the original report; approximately 84,495 images in common Kaggle mirrors	Consolidates UCSD, Kermany, Kaggle OCT, Mendeley OCT, OCT2017, OCT 2017, labeled OCT, publicly available validated OCT, and similar aliases. Used mainly for four-class retinal disease classification.
OCTDL [3, 4]	Multi-disease retinal OCT	Retinal OCT B-scans	2,064 high-resolution OCT images from 821 patients; seven disease / condition groups	Consolidates OCTDL and OCTDL dataset entries. Used for image-level retinal disease classification across AMD, DME, ERM, RAO, RVO, VID, and normal classes.
OCTID [5, 6]	Multi-disease retinal OCT	Retinal OCT B-scans	572 OCT images; five classes	Consolidates OCTID benchmark / public entries. Used for classification of normal, AMD, CSR, macular hole, and diabetic retinopathy.
OCTMNIST [7, 8]	Retinal disease benchmark	Downsampled retinal OCT	109,309 source OCT images standardized to a resolution of 28 $\times$ 28 pixel	MedMNIST benchmark for four-class OCT classification derived from OCT2017 / Kermany-style OCT data. Kept separate from the clinical-resolution Kermany / UCSD dataset because MedMNIST provides standardized low-resolution benchmark images and predefined splits.
OCTA-500 / OCTA-3M / OCTA-6M [25, 26]	Retinal vasculature; multi-disease OCTA	OCT and OCTA volumes, projections, labels	500 subjects with 3 mm $\times$ 3 mm and 6 mm $\times$ 6 mm fields of view	Consolidates OCTA-500 benchmark / public entries and OCTA-3M / OCTA-6M aliases where used as OCTA-500 subsets. Used for vessel, FAZ, artery, vein, capillary, and retinal layer segmentation.
OCTA-SS [27, 28]	Retinal vasculature	OCTA	55 parafoveal ROI slices from 3 mm $\times$ 3 mm OCTA images of 11 participants	Used for OCTA vessel segmentation. Kept separate because it is referenced as a distinct OCTA benchmark and provides detailed superficial-layer vessel annotations.

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Dataset	Area	Modality	Scale	Typical tasks / notes
OPTIMA challenge [17, 18]	Retinal fluid / macular disease	Retinal OCT	Challenge retinal fluid dataset commonly described as 30 OCT volumes	Consolidates Optima and OPTIMA challenge. Used for intraretinal cystoid fluid segmentation in OCT. Kept separate from RETOUCH because the dataset names and challenge lineage differ.
ORIGA [37, 38]	Glaucoma	Color fundus photographs	650 fundus images in ORIGA-light; commonly reported as 168 glaucoma and 482 normal images	Fundus benchmark for optic disc / cup analysis and glaucoma detection. Included where paired or externally compared with OCT measurements.
RetinalOCT-C8 / OCT-C8 [9, 10]	Multi-disease retinal OCT	Retinal OCT B-scans	24,000 OCT images across eight classes, balanced at 3,000 images per class	Consolidates OCT-C8, OCT-C8 dataset, and RetinalOCT-C8. Used for eight-class retinal OCT disease classification across AMD, CNV, CSR, DME, DR, drusen, macular hole, and normal.
RETOUCH challenge [15, 16]	Retinal fluid; AMD and retinal vein occlusion	Multi-vendor retinal OCT volumes	112 OCT volumes and 11,334 B-scans overall	Consolidates RETOUCH, ReTOUCH, MICCAI RETOUCH challenge, RETOUCH challenge dataset, and RETOUCH retinal OCT fluid detection challenge. Used for IRF, SRF, and PED detection and segmentation.
ROSE-1 [29, 30]	Retinal OCTA vessel segmentation	En face OCTA	117 OCTA images from 39 subjects in ROSE-1	Consolidates ROSE-1 and Retinal OCTA vessel Segmentation-1. Used for OCTA vessel segmentation.
SERI-CUHK SD-OCT [13, 14]	Diabetes-related retinal disease	SD-OCT	SERI dataset reported as 32 SD-OCT volumes; CUHK dataset reported as 43 SD-OCT volumes	Used for diabetes-related retinal disease classification, including healthy, DME, and DR-DME categories in source studies. Kept separate from A2A because the names indicate distinct source datasets.

5. Article-level mappings for the evidence gap analysis

Tables 3 and 4 provide the article-level reference assignments underlying the evidence gap mapping. Rather than presenting only aggregated counts, these tables make the classification process explicit by listing the publications assigned to each combination of dimension and stance. They therefore provide a source-level audit trail for the synthesis and allow the distribution of evidence across categories to be inspected directly.

The general reference table documents how publications were assigned to AI-related dimensions relevant to ophthalmic research, clinical translation, and implementation. These dimensions include model performance; technological limitations; real-world impact; data availability, integration, quality, and standardization; infrastructural demands; financial costs; regulatory requirements; cybersecurity risks; and ethical concerns. Within each dimension, studies are grouped according to whether their reported implications were interpreted as predominantly positive, predominantly negative, or balanced. The revised assignments were generated from one article-level decision matrix after a source-level audit of all 427 publications. A category-specific rationale was recorded for all 7,686 article–category decisions, and every retained assignment has a source locator and evidence excerpt. Each retained assignment required explicit, unambiguous evidence connecting both the dimension and stance to AI; ambiguous or merely procedural mentions were left unassigned. The best locally available evidence source was used for each publication, yielding 314 full-text and 113 abstract-only evaluations. All 427 sources were reassessed de novo across all 18 dimensions in a source-only pass blinded to the preceding allocations; no article-level decision was carried forward solely because its source hash was unchanged. The resulting 7,686 decisions were then compared with the preceding matrix, and every status or stance disagreement and every initial decision below high confidence underwent an additional source-based comparison and reconciliation. Before aggregate counts were regenerated, every provisional assignment, every decision changed from the preceding matrix, and every remaining decision below high confidence received a category-specific semantic check to reject evidence that was textually present but did not satisfy the operational category rule. Source-identity validation rejected two historical candidate files: one contained only conference front matter and the other a different article. The corresponding records were evaluated from their correct abstracts. The current canonical evidence set contains no unresolved source-identity mismatch.

The ethical reference table provides a more focused source-level mapping of publications to ethical dimensions, including privacy, transparency, integrity, security, non-maleficence, trust, accountability, equity, and autonomy. It distinguishes whether an issue was described as caused by AI, resolved by an AI-related approach, or both caused and resolved. Routine ethics approval, consent, anonymization, restricted data availability, or author accountability statements were not treated as evidence that AI itself resolved an ethical issue. Likewise, a technical method was not assigned an ethical stance unless the evaluated text made the ethical connection explicit. This conservative rule permits an article to remain unassigned and prevents neutral or mixed categories from becoming substitutes for insufficient evidence. Table 2 states the dimension-specific inclusion and exclusion rules used for every allocation.

Table 2: Operational inclusion, exclusion, and judgment rules for the evidence gap maps. For the general map, positive means that the article frames AI as enabling or improving the named dimension. Negative means that it frames AI as creating or exposing a barrier, risk, failure, or constraint. Balanced requires explicit qualifying evidence in both directions within that dimension. For the ethics map, caused means that the article frames AI as introducing or amplifying the issue. Resolved means that it reports an AI-related mechanism that mitigates the issue. Caused and resolved requires explicit evidence in both directions. Absence, ambiguity, or an excluded basis resulted in no assignment rather than a neutral category.

Dimension	Evidence required for assignment	Evidence not sufficient for assignment
General map		
Model performance	Reported quantitative or qualitative performance, comparison, error, agreement, calibration, or robustness outcome.	Architecture or method description without an evaluated outcome.
Technological limitations	An explicit AI/model/system failure mode, generalization or robustness constraint, image artifact, technical restriction, or a method that explicitly overcomes such a constraint.	A limitation inferred from study design, a sample-size or data limitation without a stated technical consequence, or generic future work.
Real-world impact	A deployed or actually used system, or a measured patient, clinician, workflow, referral, treatment, access, or service consequence.	Retrospective benchmark performance, claimed clinical relevance, prospective utility, or an unmeasured promise of workload reduction.
Data availability, integration, quality, and standardization	An explicit AI-linked effect of data scarcity, missingness, quality, representativeness, annotation, access, harmonization, integration, interoperability, or standardization.	Merely naming or using a dataset, an ordinary train/test split or preprocessing step, or a routine data-availability statement.
Infrastructural demands	An explicit deployment or integration burden or enabler involving hardware, cloud or network capacity, connectivity, staffing, workflow, maintenance, or institutional resources.	A list of devices, software, GPUs, memory, or runtime in the methods section without a stated implementation consequence.
Financial costs	An explicit consequence involving monetary cost, affordability, reimbursement, economic burden, staffing costs, or health-system resources.	Parameter count, runtime, hardware specification, or "computational cost" without a monetary or affordability consequence.
Regulatory requirements	An explicit AI-related approval, certification, liability, legal, compliance, governance, or postmarket obligation or barrier.	Routine research ethics approval, consent, Declaration of Helsinki compliance, or adherence to reporting guidelines.
Cybersecurity risks	An explicit hostile attack, vulnerability, security safeguard, adversarial threat, or resilience claim involving an AI or data system.	Generic privacy, ordinary model robustness, access details, or an inferred security risk.
Ethical concerns	An explicit cross-cutting AI-related ethical harm, benefit, dilemma, or responsible-AI implication.	Routine ethics approval or consent, or an ethical issue supported only by procedural boilerplate.
Ethics map		
Privacy	An explicit AI-linked privacy risk or a concrete privacy-specific AI mechanism that is reported as mitigating it.	Routine anonymization, de-identification, consent, restricted data access, or a non-public dataset without an explicit AI-related privacy effect.

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Dimension	Evidence required for assignment	Evidence not sufficient for assignment
Transparency	An explicit effect on explainability, interpretability, disclosure, traceability, or clinician understanding of an AI decision.	Routine methods reporting, performance metrics, model diagrams, heatmaps, or visualization without a stated interpretive function.
Integrity	An explicit effect on truthfulness, authenticity, manipulation, corruption, or evidentiary integrity of AI/data outputs.	Ordinary accuracy, ground-truth labeling, validation, reproducibility, data cleaning, or data-quality methods.
Security	An explicit ethical threat or protection affecting the confidentiality, integrity, or availability of an AI or data system.	Generic privacy, robustness, login/access details, or a security effect inferred by the reviewer.
Non-maleficence	An explicit AI-related harm, safety risk, consequence of an incorrect output, or reported harm-prevention mechanism.	A generic diagnostic purpose, accuracy improvement, or patient benefit inferred without a stated harm connection.
Trust	Trust, confidence, acceptance, reliance, or credibility of the AI system is explicitly named, investigated, or measured.	Trust inferred solely from accuracy, robustness, expert agreement, validation, or explainability.
Accountability	Explicit responsibility, liability, oversight, contestability, or answerability for an AI-supported decision.	Author contribution or accountability statements, ethics approval, or expert labeling alone.
Equity	An explicit AI-related subgroup fairness, bias, disparity, representativeness, access, or distributional effect.	Ordinary class balancing unless linked to affected patient groups, access, or outcome inequity.
Autonomy	An explicit effect on patient or clinician agency, informed choice about AI use, or meaningful human control.	The technical term "autonomous", generic personalization, or clinician involvement without an agency effect.

Table 3: Article-level reference assignments underlying the re-audited general evidence gap map of AI-related challenges, opportunities, and implementation dimensions in ophthalmology. Each cell is generated from the canonical article-level decision matrix; a slash indicates that no publication met the explicit-evidence rule for that dimension and stance.

General dimension	Positive impact	Negative impact	Balanced
Model performance	[41–379]	/	[380–456]

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General dimension	Positive impact	Negative impact	Balanced
Technological limitations	[41, 49, 53, 54, 64, 69, 73, 74, 84, 88, 102, 105, 111, 116, 133, 137, 140, 153, 180, 195, 206, 221, 228, 232, 239, 246, 267, 275, 279, 296, 311, 312, 315, 317, 318, 332–334, 336, 339, 340, 342, 348, 351, 367, 376, 384, 402, 420, 427, 443]	[55, 65, 75, 91, 94, 98, 100, 115, 119, 124, 146, 147, 157, 174, 177, 182, 185, 194, 199–201, 210, 218, 220, 231, 237, 244, 251, 258, 268, 280, 282, 286, 290, 291, 300, 323, 327, 337, 344, 345, 347, 354, 356, 360, 366, 381, 385, 390–392, 394, 395, 398, 404, 405, 410, 416, 421, 423, 426, 428, 432, 433, 436, 440, 442, 445, 448, 450–453, 456, 457]	[43, 44, 46–48, 51, 52, 58, 60, 62, 76, 78–80, 86, 95–97, 101, 104, 113, 118, 122, 125, 127, 130, 134, 141, 143, 156, 158–160, 165–167, 169, 170, 172, 176, 181, 183, 188, 189, 191, 196, 203, 204, 208, 211, 214, 215, 224, 225, 227, 229, 230, 234, 245, 247, 250, 253, 255, 256, 259, 264, 269, 271, 273, 278, 285, 288, 293, 295, 297, 299, 306, 309, 314, 320, 322, 324, 325, 335, 338, 341, 357, 359, 361, 363, 372, 373, 377, 378, 380, 382, 383, 388, 389, 393, 396, 400, 401, 406–409, 413–415, 418, 419, 422, 429, 430, 435, 437–439, 441, 446, 447, 454, 455, 458]
Real-world impact	[53, 132, 144, 147, 155, 156, 158, 203, 212, 218, 234, 258, 263, 342, 379, 399, 422]	/	/

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General dimension	Positive impact	Negative impact	Balanced
Data availability, integration, quality, and standardization	[41, 45, 60, 64, 67, 73, 78, 79, 86, 99, 108, 127, 133, 134, 142, 153, 157, 181, 186, 199, 226, 238, 239, 243, 252, 264, 272, 279, 283, 309, 310, 312, 326, 339, 345, 362, 369, 373, 389, 409, 410, 423, 434]	[52, 53, 55, 70, 74, 91, 97, 114, 119–122, 124, 147, 149–151, 156, 172, 176, 177, 182, 188, 189, 200–202, 209, 218, 220, 224, 229–231, 237, 244, 245, 256, 258, 268, 290, 291, 305, 314, 323, 328, 343, 346–350, 356, 358, 360, 366, 372, 377, 385, 392, 395–397, 402, 407, 413, 416, 418, 419, 425, 427–429, 435, 436, 439, 444, 446, 448, 449, 451, 452, 455]	[43, 44, 46, 49–51, 54, 58, 61, 62, 65, 68, 69, 76, 81, 82, 88, 93, 94, 98, 101, 102, 105, 110, 139, 141, 144, 146, 160, 165, 183, 185, 191, 194, 203, 208, 211, 214, 215, 217, 221, 225, 227, 228, 233, 247, 250, 255, 259, 260, 269, 271, 273, 277, 278, 280, 292, 293, 297–299, 302, 311, 313, 315, 318, 320, 322, 325, 327, 329, 331, 334, 336, 337, 340, 341, 344, 354, 357, 359, 363, 378–380, 382, 387, 393, 394, 404, 406, 414, 420, 422, 430, 431, 437, 438, 441, 442, 445, 447, 456]
Infrastructural demands	[82, 90, 94, 97, 99, 104, 116, 144, 148, 177, 180, 181, 185, 199, 224, 229, 232, 233, 242, 244, 258, 282, 310, 334, 342, 356, 365, 382, 393, 395, 399, 400, 406, 407, 409, 420, 427, 459]	[54, 124, 146, 201, 230, 247, 250, 297, 349, 350, 358, 388, 413, 441]	[41, 53, 59, 65, 156, 158, 203, 223, 281, 341, 379, 418, 429]
Financial costs	[65, 95, 97, 98, 131, 136, 141, 165, 177, 180, 181, 185, 189, 194, 200, 201, 203, 238, 258, 269, 277, 281, 282, 291, 310, 316, 321, 380, 395, 406, 407, 409, 410, 417, 418, 427, 436, 450, 459]	[43, 76, 349, 350]	[77, 153, 158, 323]
Regulatory requirements	[291, 397]	[124, 156, 194, 224, 323, 334, 361, 399, 422, 433]	/
Cybersecurity risks	[230]	[347]	[334]
Ethical concerns	/	[68, 124, 156, 194, 224, 291, 323, 334, 361]	/

Table 4: Article-level reference assignments underlying the re-audited ethical evidence gap map. Caused denotes an explicit AI-related ethical risk, resolved an explicit AI-related mitigation, and caused and resolved explicit evidence in both directions. Routine ethics approval, consent, anonymization, or restricted data access alone was not treated as an AI mitigation.

Ethical dimension	Caused	Resolved	Caused and resolved
Privacy	[76, 124, 144, 156, 194, 291, 347]	/	[323, 381]
Transparency	[61, 64, 74, 96, 146, 194, 208, 224, 231, 244, 250, 258, 302, 334, 343, 347, 349, 363, 379, 413, 414, 418, 430, 432, 448, 451]	[51, 58, 62, 76, 80, 88, 91, 92, 101, 103, 105, 108, 117, 120, 132, 136, 143, 144, 147, 156, 160, 162, 165, 167, 175, 177, 181, 183, 188, 195, 199–201, 209, 216, 232, 245, 259, 260, 274, 290, 294, 296, 317, 322, 327, 332, 345, 350, 351, 358–360, 368, 372, 380, 382, 394, 407, 411, 416, 425, 431, 438, 440–443, 449, 450, 455]	[68, 124, 171, 173, 178, 179, 193, 203, 214, 223, 238, 291, 293, 297, 316, 323, 329, 335, 381, 397, 427]
Integrity	/	/	/
Security	[347]	[230]	[334]
Non-maleficence	[142, 146, 214, 220, 323, 392, 456]	[62, 165, 174, 189, 263, 271, 282, 296, 310, 354, 356, 380, 439, 450]	[50, 54, 223, 363, 385]
Trust	[96, 144, 146, 194, 237, 244, 291, 347]	[68, 84, 105, 156, 162, 173, 214, 259, 316, 317, 380, 397, 436]	[124, 203, 223, 323, 334, 363, 381]
Accountability	[124]	[53, 68, 277, 341, 394]	/
Equity	[46, 68, 70, 91, 124, 171, 178, 179, 182, 193, 194, 209, 247, 277, 291, 323, 349, 360, 382, 387, 408, 418, 434, 440, 448]	[74, 134, 269, 281, 282, 310, 320, 365, 380, 395]	[62, 156, 441]
Autonomy	/	[53, 277, 341, 350, 394, 399, 456]	/

6. Interpretive synthesis of the included evidence base

Table 5 is more than an inventory of included studies. In aggregate, it shows how artificial intelligence (AI) in ophthalmic optical coherence tomography (OCT) and optical coherence tomography angiography (OCTA) has evolved from relatively narrow proof-of-concept classification studies into a broader research landscape that now includes segmentation, quantitative biomarker extraction, prognosis, treatment-response prediction, multimodal fusion, and workflow-oriented clinical support [123, 164, 194, 197, 224, 348, 350, 399, 408, 441]. Read as a whole, the table suggests that the literature is no longer defined solely by the question of whether AI can detect disease from ophthalmic imaging. Instead, a central question has become how AI can produce clinically meaningful, reproducible, and operationally useful information from OCT-derived data [124, 259, 399, 413].

The table also shows that this evolution has not been uniform. Some application domains, especially retinal disease and glaucoma, already exhibit a relatively dense and methodologically layered evidence base, whereas other topics remain more exploratory or specialized [61, 68, 108, 123, 389]. The resulting picture is therefore one of substantial growth and uneven maturity. This combination of breadth and heterogeneity is one of the most important interpretive results that emerges from Table 5.

6.1. Clinical concentration and expansion across ophthalmic domains

A first major pattern visible in Table 5 is the strong concentration of studies in a limited number of disease areas. A large share of the included evidence addresses age-related macular degeneration (AMD), diabetic macular edema (DME), diabetic retinopathy, retinal fluid abnormalities, and glaucoma [123, 152, 225, 337, 389]. Although this concentration is expected, it remains analytically important. These are precisely the disease areas in which OCT and OCTA already play a central clinical role, either for diagnosis, staging, treatment monitoring, or longitudinal surveillance. The table therefore suggests that AI development has been most intense where the underlying imaging modality already has high clinical relevance and where automated analysis can plausibly improve efficiency, standardization, or sensitivity to change.

This concentration also indicates that the current literature is not evenly distributed across ophthalmology, but instead follows the practical structure of contemporary eye care. In retinal disease, OCT-based fluid detection, atrophy assessment, lesion characterization, and neovascular activity monitoring are already integral to routine management, which makes these domains particularly attractive for AI-based automation and quantification [123, 152, 261]. In glaucoma, structural imaging has long been linked to disease detection and follow-up, which helps explain the large number of studies addressing classification, staging, progression, and structure-function mapping [61, 389, 408].

Table 5 also demonstrates that the field is expanding beyond these dominant indications. Included studies address cataract, keratoconus, corneal edema, dry eye disease, inherited retinal disorders, optic neuropathies, amblyopia, and extra-ophthalmic prediction tasks derived from retinal imaging [108, 199, 346, 392, 451, 452]. This broadening indicates that ophthalmic AI is not merely refining a few high-frequency classification problems, but increasingly exploring OCT and OCTA as general-purpose phenotyping modalities. Even so, the table makes clear that the depth of evidence remains greatest where OCT is already most deeply embedded in established clinical workflows.

6.2. From diagnostic classification to quantitative imaging biomarkers

A second major insight from Table 5 is that the literature has moved well beyond simple disease classification. Classification studies remain numerous and foundational, including work on differentiating normal from pathological scans and on multi-class categorization of choroidal neovascularization (CNV), DME, drusen, AMD stages, or glaucomatous change [68, 164, 225]. Yet the table simultaneously documents an extensive body of work devoted to segmentation, structural delineation, and quantitative biomarker extraction [123, 234, 261, 267, 311, 348].

This development is particularly important for interpreting the state of the field. Classification compresses image information into a diagnostic label, whereas segmentation and quantification transform images into structured measurements that can be linked directly to anatomy, pathology, and temporal change. The studies summarized in the table increasingly produce outputs such as retinal layer boundaries, fluid volumes, atrophic regions, ellipsoid zone metrics, nonperfusion areas, vessel densities, smoothness indices, and other OCT-derived biomarkers [123, 152, 225, 234, 261]. This is a substantial conceptual shift because it aligns AI more closely with the measurement-oriented logic of ophthalmology, where quantification often matters at least as much as categorical diagnosis.

The prominence of segmentation and quantification in Table 5 also suggests that the field is increasingly attempting to integrate AI into clinically interpretable workflows rather than positioning it only as an opaque prediction engine. Biomarker extraction and quantitative lesion assessment can support disease monitoring, treatment decisions, and longitudinal research in ways that map more naturally onto existing clinical reasoning than isolated class labels [123, 261, 399]. Accordingly, the table reflects not only technical diversification, but also broader maturation in the purposes for which AI is being developed.

6.3. *Prediction, prognosis, and clinically actionable decision support*

A third central pattern in Table 5 is the emergence of prognostic and predictive modeling as a major theme. A notable subset of the included studies no longer focuses primarily on current-state recognition, but instead addresses treatment response, recurrence, retreatment burden, postoperative visual outcome, progression, or functional decline [61, 197, 236, 337, 408]. This is one of the strongest signals that the literature has begun to move from automated diagnosis toward clinically actionable decision support.

In glaucoma, this predictive turn is especially striking. Multiple studies in the table seek to infer visual field loss, mean deviation, or progression risk from OCT-derived structural information [61, 124, 408]. This line of work is important because it addresses a core translational challenge in glaucoma care, namely the relationship between structural change and functional impairment. The repeated attempt to bridge this structure-function gap suggests that the field increasingly aims to provide clinically meaningful surrogates or complements for functional testing, rather than merely reproducing diagnostic status from imaging alone.

A comparable trend is visible in retinal disease. The table includes studies predicting anti-vascular endothelial growth factor response, disease recurrence, postoperative outcome, and individualized treatment requirements in AMD, DME, and macular hole settings [197, 236, 337, 379]. These contributions indicate that AI is increasingly being positioned as a component of precision ophthalmology, where OCT-derived structure is used not only to characterize disease, but also to anticipate how disease may evolve or respond to therapy. This is a more ambitious translational goal than image classification and suggests that the literature is progressively orienting itself toward clinically consequential endpoints.

The importance of this shift is substantial. Prediction requires more than high discrimination on static datasets. It requires meaningful linkage between imaging biomarkers, temporal dynamics, and relevant patient outcomes. The fact that such studies now occupy a visible part of Table 5 indicates that the field is moving beyond image-level automation and is increasingly attempting to support anticipatory clinical reasoning.

6.4. *Methodological diversification, interpretability, and multimodal integration*

Table 5 also reveals a highly diverse methodological landscape. Earlier studies often rely on conventional machine-learning strategies, handcrafted features, support vector machines, random forests, or transformed OCT-derived parameters [116]. Over time, this has expanded into a landscape dominated by convolutional neural networks, three-dimensional models, ensemble learning, transformer architectures, hybrid convolutional neural network-transformer systems, self-supervised approaches, uncertainty-aware systems, and multimodal fusion pipelines [68, 225, 301, 441]. The table therefore documents not simply growth in application volume, but also a sustained search for architectures capable of addressing increasingly complex clinical and technical requirements.

Yet the evidence base summarized in the table also suggests that methodological novelty is not identical with substantive clinical progress. Several comparative studies indicate that more complex architectures do not always yield large or practically decisive gains over simpler baselines [102]. This is an important corrective to a purely innovation-driven reading of the literature. The central question is not only whether newer architectures can improve benchmark metrics, but whether those improvements correspond to better generalizability, more useful quantification, stronger robustness, or more credible deployment potential.

This point connects directly to the large number of studies in Table 5 that incorporate interpretability mechanisms, uncertainty modeling, or clinician-facing explanatory tools. Gradient-weighted Class Activation Mapping, Local Interpretable Model-agnostic Explanations, Shapley Additive Explanations, integrated gradients, uncertainty maps, occlusion testing, and related approaches appear across many domains [259, 436]. Their repeated presence suggests that the field increasingly recognizes that clinical adoption depends not only on performance, but also on whether outputs can be understood, challenged, and related back to known disease structures or lesions.

A similar movement toward richer clinical coherence is visible in the rise of multimodal studies. Several works combine OCT with OCTA, fundus imaging, structured clinical variables, or electronic health record-derived information [68, 197, 199, 441]. This suggests that the field increasingly understands ophthalmic AI as a problem of integrating complementary phenotypic signals rather than optimizing isolated modality-specific classifiers. In glaucoma, for example, combining structural, vascular, photographic, and functional inputs may better capture disease complexity than any single modality alone [61, 389, 441]. In retinal disease, combining structural OCT with angiographic or metadata-derived information can support more nuanced stratification of disease activity or treatment response [68, 197, 379]. The increasing prevalence of such multimodal systems in the table is therefore another sign of conceptual maturation.

6.5. Validation, translational readiness, and clinical accountability

Perhaps the most consequential issue revealed by Table 5 concerns validation and translational readiness. The table includes many studies that explicitly address external testing, multicenter evaluation, cross-device robustness, prospective validation, or real-world deployment contexts [68, 311, 389, 413]. Such studies are especially valuable in ophthalmic imaging because OCT and OCTA performance can be influenced by device-specific acquisition properties, patient population differences, annotation practices, and image-quality variation. Their presence in the table indicates that a meaningful portion of the literature has moved beyond internal proof-of-concept design and is actively engaging with the challenges of real-world generalizability [68, 311, 413].

Table 5 also makes clear that validation rigor remains uneven. Many studies continue to rely on retrospective institutional datasets, internal splits, public benchmarks, or relatively narrow cohort designs [116, 164, 225]. Thus, the table should not be interpreted as evidence of a uniformly mature translational field. Rather, it documents a literature in which some subdomains already approach clinically credible validation standards, whereas others remain closer to exploratory performance optimization.

This heterogeneity becomes even more visible when considering workflow-oriented studies and synthesis literature. The table includes work on graphical interfaces, cloud platforms, real-time systems, point-of-care deployment, image-quality control, anomaly screening, and hybrid human-AI workflows [350, 399, 413, 436]. These studies are important because they imply that part of the field now understands AI not merely as an algorithmic output generator, but as a component within a larger clinical workflow requiring safeguards, quality checks, user interaction, and operational reliability.

In parallel, the growing number of systematic reviews, scoping reviews, and meta-analyses included in the table shows that the field has become large enough to require higher-order synthesis [124, 194, 224, 399]. This is itself a marker of maturity, but also of unresolved heterogeneity. A field that generates repeated reviews is one that has accumulated substantial evidence, yet still requires structured interpretation to make sense of diverse tasks, datasets, endpoints, and validation approaches.

Taken together, these observations suggest that the principal message of Table 5 is not merely that ophthalmic OCT AI performs well. The deeper conclusion is that the field is currently transitioning from technical feasibility toward clinical accountability. Earlier work often asked whether disease could be recognized automatically from ophthalmic imaging. The broader evidence base now increasingly asks whether AI systems are robust across devices and populations, whether they generate clinically interpretable and quantifiable outputs, whether they can anticipate outcomes, and whether they can be embedded safely and efficiently into real-world workflows [68, 124, 259, 399]. In that sense, the included literature reflects both substantial progress and a rising standard of what counts as meaningful progress.

For this review, the evidence base should be regarded as strong but uneven. It demonstrates a broad range of successful applications across clinically important ophthalmic domains, but evidentiary depth, validation quality, and translational readiness vary substantially. Retinal disease and glaucoma currently show the most mature bodies of work, whereas other topics remain promising but less consolidated [61, 108, 123, 389, 452]. Overall, OCT- and OCTA-based ophthalmic AI has moved beyond early experimentation, but future progress will depend less on architectural novelty alone than on robust validation, clinically meaningful endpoints, interpretable quantification, and careful integration into routine care and research workflows.

Table 5: Supplementary table of studies included in the final analysis of this structured review on artificial intelligence in ophthalmology. The table summarizes all 427 included studies and provides the bibliographic reference, year of publication, an integrated rationale for inclusion, and a contribution-focused summary for each work. It complements the PRISMA-style flow diagram by detailing the final body of evidence underlying the review.

Work	Year	Overall reasoning	Contribution and summary
[167]: 3D-CNN for Glaucoma Detection Using Optical Coherence Tomography	2019	The study evaluates an end-to-end 3D convolutional neural network that classifies glaucoma directly from volumetric optical coherence tomography (OCT) without retinal segmentation and reports AUC improvements alongside grad-CAM interpretability tests that occlude highlighted regions.	By doubling the input voxel count relative to a prior downsampled 3D OCT glaucoma model and quantifying the performance drop after occluding grad-CAM regions, the study suggests that higher-resolution volumetric input and the identified anatomical regions drive classification accuracy.
[389]: A 3D Deep Learning System for Detecting Referable Glaucoma Using Full OCT Macular Cube Scans	2020	In ophthalmic practice, this work trains a fully 3D convolutional neural network on SD-OCT macular cube volumes, uses multimodal clinical data to label referable versus nonreferable glaucoma, and tests the model externally on independent real-world datasets using AUROC and cross-site generalizability analyses.	The research demonstrates that standardizing OCT cube geometry via layer-segmentation homogenization improves referable glaucoma discrimination compared with training on raw volumes and supports clinically relevant use across myopia severity, including international validation where glaucoma definitions differed.
[152]: A column-based deep learning method for the detection and quantification of atrophy associated with AMD in OCT scans	2021	The analysis introduces a column-based convolutional neural network that analyzes volumetric spectral-domain OCT (vertical A-scan columns within B-scans) to detect and quantify age-related macular degeneration-associated atrophy, then projects the resulting atrophy segments onto corresponding infrared imaging for lesion-level characterization, with evidence reported as performance on 106 clinical OCT scans using metrics reported to be near observer variability.	This research delivers an end-to-end automated workflow for expert-level identification and measurement of atrophy lesions in OCT, producing clinically relevant outputs including lesion area and distances from the fovea that could support diagnosis, follow-up, and research quantification.
[291]: A Comparative Study Between Clinical Optical Coherence Tomography (OCT) Analysis and Artificial Intelligence-Based Quantitative Evaluation in the Diagnosis of Diabetic Macular Edema	2025	This prospective, non-randomized study in an ophthalmic setting evaluates AI-based quantitative analysis on clinical OCT images for diabetic macular edema, using paraconsistent feature engineering with machine-learning classifiers, and reports diagnostic evidence for both binary DME presence detection and multiclass DME phenotyping.	The analysis benchmarks multiple conventional classifiers against routine clinical assessment on 700 OCT exams and shows that paraconsistent feature engineering can reduce input features while maintaining clinically relevant diagnostic accuracy, supporting AI-assisted DME screening workflows.

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Work	Year	Overall reasoning	Contribution and summary
[102]: A comparison of deep learning U-Net architectures for posterior segment OCT retinal layer segmentation	2022	The research evaluates U-Net family semantic segmentation models on posterior segment OCT B-scans for retinal layer (pixel-wise) delineation, benchmarking eight architectural variants across four OCT datasets spanning multiple populations, pathologies, acquisition settings, instruments, and annotation schemes using Dice-based testing results to support an evidence-based, ophthalmic imaging comparison.	The reported work isolates the practical impact of architectural changes and convolution depth by running matched U-Net variants (two vs three convolutions per block) across heterogeneous datasets, showing that vanilla U-Net performance remains close to more complex variants with only marginal gains, informing simpler model selection for OCT retinal layer segmentation.
[98]: A Comprehensive CNN Model for Age-Related Macular Degeneration Classification Using OCT: Integrating Inception Modules, SE Blocks, and ConvMixer	2024	The reported work evaluates a CNN-based deep learning framework for OCT-driven classification of age-related macular degeneration (dry AMD, wet AMD, and normal), combining modified Inception modules, depthwise squeeze-and-excitation blocks, and ConvMixer in an efficiency-focused architecture and reporting quantitative evidence from experiments on both a private OCT dataset and the public Noor dataset.	The study introduces the proposed CNN design and demonstrates its diagnostic effectiveness with metric-based results on the two OCT datasets, highlighting a parameter-efficient, patch-based feature extraction strategy that produced markedly high performance for AMD and normal categorization.
[224]: A Comprehensive Review of Deep Learning-Based Retinal Layer Integrity Assessment in Optical Coherence Tomography (OCT)	2025	This review focuses on deep learning for retinal layer integrity assessment of the ELM and EZ in OCT, synthesizing evidence from 43 peer-reviewed studies that reported quantitative segmentation or volumetric outcomes, with relevance to clinical ophthalmic imaging and translational evaluation of model performance and limitations.	This investigation consolidates trends in AI model families for ELM/EZ segmentation and emphasizes key barriers for clinical use, including generalization, smoothness quantification, and real-time deployment readiness, while proposing practical directions such as standardized annotations and benchmark development.
[180]: A comprehensive set of novel residual blocks for deep learning architectures for diagnosis of retinal diseases from optical coherence tomography images	2021	This chapter develops a deep learning classification approach for retinal diseases using SD-OCT images, employing residual network blocks designed to address gradient explosion and reporting evidence from performance comparisons on two separate OCT datasets, including real-time prediction versus expert ophthalmologists.	This analysis contributes a set of novel residual blocks integrated into a deep residual architecture to improve SD-OCT pathology differentiation and demonstrate that real-time model outputs can exceed expert ophthalmologists on the reported dataset evaluations.
[267]: A Deep Ensemble Learning-Based CNN Architecture for Multiclass Retinal Fluid Segmentation in OCT Images	2023	The study develops a convolutional neural network deep-ensemble approach for OCT imaging that segments and differentiates three classes of retinal cysts, with evidence based on quantitative evaluation on the publicly available RETOUCH challenge dataset.	This work details a multiclass retinal cyst segmentation pipeline that surpasses prior methods by an overall 1.8% on the RETOUCH benchmark, supporting more efficient, automated cyst identification for OCT interpretation.

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Work	Year	Overall reasoning	Contribution and summary
[337]: A Deep Learning Approach for Predicting the Response to Anti-VEGF Treatment in Diabetic Macular Edema Patients Using Optical Coherence Tomography Images	2025	This work focuses on diabetic macular edema in an ophthalmology treatment setting by applying a Siamese deep learning framework to optical coherence tomography images, aiming to predict whether patients will be good versus poor responders to anti-VEGF therapy using post-injection central macular thickness reduction as the evidence for a binary classification task.	The research proposes ESSDP, a Siamese EfficientNetB2 design coupled with feature-distance comparison for OCT-based responder stratification, and reports overall classification results that suggest the approach may help support more individualized anti-VEGF decision-making in DME, including under limited-data conditions.
[454]: A deep learning approach to hard exudates detection and disorganization of retinal inner layers identification on OCT images	2024	In this retrospective, single-center study, a deep learning pipeline is evaluated on spectral-domain OCT images from eyes with diabetic macular edema to detect hard exudates via object detection and classify disorganization of retinal inner layers via image classification, with evidence drawn from test-set quantitative evaluation using standard detection and classification metrics and corresponding statistical agreement tests.	The authors present a dual-branch architecture that combines YOLO-based hard-exudate localization with an ensemble of CNN classifiers for DRIL presence, offering an integrated OCT biomarker workflow that supports ophthalmic feature identification and decision support in DME imaging.
[408]: A deep learning approach to predict visual field using optical coherence tomography	2020	This investigation evaluates an Inception V3 deep learning model that takes combined spectral-domain OCT macular mGCIPL and peripapillary pRNFL thickness maps to predict Humphrey 24-2 visual field threshold values in glaucoma and normal eyes, using RMSE as the primary error metric on a held-out test set.	The analysis shows that structure-function learning from combined OCT thickness inputs yields accurate global and region-specific visual field predictions and identifies visual field mean deviation alongside pRNFL and mGCIPL thickness as key correlates of prediction error, supporting use when perimetry is difficult to perform.
[156]: A Deep Learning-Based Eye Disease Diagnosis Using OCT Imaging and SE-Enhanced CNNs	2025	This ophthalmic study develops an OCT-imaging deep learning classifier using a custom squeeze-and-excitation enhanced CNN to automatically distinguish cataract, glaucoma, and diabetic retinopathy (plus healthy controls), supported by stratified 5-fold cross-validation and a small independent clinical case validation at a private radiology center with Grad-CAM/saliency-based interpretability.	DEEPSIGHT's key contribution is an end-to-end OCT diagnostic pipeline with clinician-facing visual explanations and prototype deployment, demonstrating high overall classification performance and practical usability for early, interpretable decision support across the three targeted ocular conditions.

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Work	Year	Overall reasoning	Contribution and summary
[350]: A Deep Learning-Based Graphical User Interface for Predicting Corneal Ectasia Scores from Raw Optical Coherence Tomography Data	2026	In an ophthalmic setting, the reported work builds a Tkinter-based GUI around a modified EfficientNet-B0 convolutional model that ingests raw Casia2 anterior-segment OCT 3dv data (cropped and resized 16-image stacks) to predict continuous corneal ectasia scores and derive two- and three-class ectasia categories based on the Ectasia Screening Index, with evidence from reported test-set regression and classification results.	The study provides an end-to-end workflow that outputs a numerical ectasia score plus corresponding diagnostic labels from raw OCT files, addressing consistency across software updates and enabling users to review extracted images alongside the predicted scores and classes.
[348]: A Deep Learning Framework for Segmentation of Retinal Layers from OCT Images	2017	In an ophthalmic OCT setting, the study uses a CNN followed by a bidirectional LSTM to segment multiple retinal layer boundaries by first extracting layer-of-interest regions and edges and then tracing eight continuous boundaries, with performance assessed using pixel-wise MAE on normal and AMD (pathology) data against inter-marker baselines and dataset-provided benchmarks.	This investigation introduces an end-to-end deep framework that jointly replaces preprocessing and boundary tracing with learned CNN feature extraction and recurrent continuity modeling, and shows that training on mixed normal and AMD cases yields lower pixel-level boundary error than inter-marker disagreement on public OCT datasets.
[274]: A Deep Learning Model Using VGG19 and Random Forest for AMD Stage Classification on Retinal OCT Images	2025	An ophthalmic study evaluates a deep learning workflow that uses VGG19-derived CNN features with a Random Forest classifier to stage age-related macular degeneration from retinal OCT images, providing evidence of performance on a public OCT dataset for the classification task across Early, Intermediate, and Advanced stages.	This analysis details a VGG19-RF feature-extraction and classification pipeline, including global average pooling and use of VGG19's penultimate-layer features, demonstrating reported high accuracy for distinguishing AMD stages and positioning the approach as a computationally efficient alternative to a standard CNN classifier.
[447]: A deep learning multi-capture segmentation modality for retinal OCT imaging	2022	The study targets retinal and choroidal layer segmentation on optical coherence tomography using a deep learning approach that combines CycleGAN image-to-image translation to standardize multi-capture focus and denoising conditions with a vision-transformer-based TransU-Net, reporting segmentation evidence via Dice coefficient comparisons against a baseline U-Net.	This work shows that applying CycleGAN-based standardization as an effective denoising and capture-modality adaptation step enables TransU-Net to outperform U-Net for enhanced depth imaging OCT, supporting instrument-agnostic segmentation feasibility.

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Work	Year	Overall reasoning	Contribution and summary
[218]: A deep learning system for the detection of optic disc neovascularization in diabetic retinopathy using optical coherence tomography angiography images	2025	This work uses a deep learning pipeline on optical coherence tomography angiography to detect, quantify, and visualize optic disc neovascularization in diabetic retinopathy, providing evaluation against other deep learning methods and comparison with retina specialists for the reported detection and region-identification tasks.	The research specifies a central system that performs optic disc boundary detection, NVD region identification, and generation of 3D visualizations from OCTA to convey NVD location and blood-flow information, which supports more intuitive automated assessment for clinical interpretation.
[87]: A Deep Learning System Model For Detecting Neurodegenerative Diseases From Retinal Optical Coherence Tomography	2025	The analysis evaluates an ophthalmic deep learning approach that combines enhanced adaptive filtering with a hybrid CNN on OCT-derived retinal images to classify neurodegenerative disease-related retinal abnormalities, reporting accuracy of 92.3% as evidence of its effectiveness.	By integrating enhanced adaptive filtering and data augmentation directly into the preprocessing and training pipeline for OCT feature extraction, it aims to improve detection of subtle, disease-associated structural changes in a noninvasive imaging workflow.
[329]: A Deep Learning System Using Optical Coherence Tomography Angiography to Detect Glaucoma and Anterior Ischemic Optic Neuropathy	2023	This retrospective ophthalmology study trains a ResNet50 CNN with integrated gradients on OCT angiography of the macular superficial capillary plexus and radial peripapillary capillary plexus to classify glaucoma, non-arteritic anterior ischemic optic neuropathy, and normal controls, reporting test-set discrimination using ROC-AUC and agreement metrics.	The analysis demonstrates that combining SCP and RPC OCTA inputs yields better three-class performance than either plexus alone and provides model attribution maps that highlight diagnostically relevant regions, suggesting clinically useful support for differentiating optic neuropathies in difficult cases.
[334]: A fusion of deep neural networks and game theory for retinal disease diagnosis with OCT images	2024	The research uses hybrid deep learning on OCT retinal images, integrating Inception-based feature extraction, GAN-based classification, and game-theoretic evaluation to support retinal disease diagnosis and to examine robustness under adversarial conditions.	The reported work introduces the GIGT hybrid pipeline in Python with OCT preprocessing and a game-theoretic reliability assessment, claiming improved diagnostic accuracy for retinal disease identification based on reported results.
[301]: A GCN-assisted deep learning method for peripapillary retinal layer segmentation in OCT images	2021	The reported work introduces a GCN-assisted U-shaped segmentation approach for peripapillary retinal layer segmentation from peripapillary OCT imaging, targeting nine layers that include the RNFL, and it provides evidence via training and testing on a collected dataset with Dice-based comparisons against prior methods.	The study refines an ophthalmic segmentation pipeline by inserting a graph reasoning block between the encoder and decoder to exploit the retina's stratified organization around the optic disc, enabling more accurate RNFL and multilayer delineation in the peripapillary region.

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Work	Year	Overall reasoning	Contribution and summary
[312]: A Hybrid CBAM-Vision Transformer for Enhanced Classification of Retinal Diseases in Optical Coherence Tomography Imagery	2025	In this ophthalmic computer-vision study, a CBAM-augmented Vision Transformer is evaluated for retinal disease classification using Optical Coherence Tomography images, tackling low resolution and class imbalance, with evidence drawn from reported test-set metrics on OCTMNIST.	This investigation proposes a CBAM-ViT-tiny hybrid that refines OCT feature embeddings and is trained with full fine-tuning plus weighted loss and augmentation, achieving state-of-the-art test performance on OCTMNIST and thereby offering an effective attention-enhanced modeling strategy for imbalanced OCT classification.
[369]: A Hybrid Deep Learning Classification of Perimetric Glaucoma Using Peripapillary Nerve Fiber Layer Reflectance and Other OCT Parameters from Three Anatomy Regions	2024	This glaucoma diagnostic study applies a hybrid deep-learning framework to spectral-domain OCT data from peripapillary and macular regions, using peripapillary NFL reflectance and thickness maps plus optic nerve head parameters to classify normal versus perimetric glaucoma and reports performance comparisons on a held-out test set against logistic regression, including ROC-based evidence.	By coupling a CNN over superpixelized NFL reflectance and thickness with an FCN for OCT-derived disc and macular measures and demographics, it shows that incorporating NFL reflectance improves discrimination beyond logistic regression and remains competitive even when reflectance is omitted, informing how OCT structural signals can be fused for screening-oriented glaucoma classification.
[116]: A Hybrid Machine Learning Approach Using LBP Descriptor and PCA for Age-Related Macular Degeneration Classification in OCTA Images	2020	The study applies a hybrid LBP descriptor and PCA feature pipeline with an SVM classifier to OCTA imaging for automated categorization of age-related macular degeneration, using stratified k-fold cross-validation and ROC-AUC to provide performance evidence for binary discrimination between AMD groups.	This work demonstrates how rotation-invariant texture features from full-field OCTA, followed by PCA decorrelation, can distinguish healthy from neovascular AMD and separate CNV-related categories from non-neovascular disease using AUC-based evaluation.
[97]: A Hybrid Model Composed of Two Convolutional Neural Networks (CNNs) for Automatic Retinal Layer Segmentation of OCT Images in Retinitis Pigmentosa (RP)	2021	For SD-OCT B-scans in retinitis pigmentosa, this hybrid deep learning study uses a CNN-based semantic segmentation network (U-Net) plus a sliding-window refinement CNN to delineate retinal layer boundaries and derive photoreceptor metrics, with performance evaluated against Spectralis manual grader corrections using pixel-wise and Bland-Altman/correlation evidence.	The research demonstrates that combining fast U-Net semantic segmentation with sliding-window error correction improves boundary localization and yields more accurate EZ and photoreceptor outer segment measurements than either component alone, which can reduce manual effort for OCT-based RP assessment.
[365]: A Lightweight CNN Net for AMD Detection Using OCT Volumes	2022	The study describes a lightweight convolutional neural network trained on optical coherence tomography volumes to classify whether a patient has age-related macular degeneration, providing evidence from reported model performance aimed at enabling diagnosis in rural and remote ophthalmic settings.	This research specifies an efficient AMD detection CNN with a small parameter and compute footprint and reports quantization-aware fixed-point results, which is relevant for deploying AI on resource-limited hardware.

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Work	Year	Overall reasoning	Contribution and summary
[43]: A lightweight deep learning model for retinal optical coherence tomography image classification	2023	A study uses a lightweight deep learning approach for OCT image classification to detect and grade diabetic macular edema and drusen macular degeneration, with evaluation reported on publicly available Mendeley OCT and Duke datasets to provide evidence for retinal disease diagnosis in an ophthalmic imaging setting.	The analysis introduces an end-to-end lightweight model built from residual blocks and channel attention modules and reports high classification accuracy on the two external public datasets, supporting parameter-efficient OCT-based screening potential for these conditions.
[92]: A machine learning classification model for primary open-angle glaucoma based on optical coherence tomography optic disc parameters and interocular asymmetry features	2026	This prospective case-control study applies OCT-based machine learning classification to distinguish primary open-angle glaucoma from healthy controls using peripapillary pRNFL and BMO-MRW quadrant measurements together with interocular asymmetry indices, validated with discrimination metrics (AUC, accuracy, F1) and feature attribution via SHAP for interpretability.	Notably, elastic-net selection followed by a best-performing support vector machine using five OCT-derived features shows that interocular BMO-MRW asymmetry, especially in temporal and superotemporal regions, drives the POAG versus control separation, offering clinically interpretable evidence for an imaging-based risk signal.
[166]: A Multi-branch and Attention Based CNN Architecture for the Classification of Retinal Diseases from OCT Images	2023	The reported work evaluates an ophthalmic deep learning approach that applies a multi-branch, attention-augmented CNN with CBAM to OCT image data for multi-class retinal disease classification, supported by reported quantitative results (96% accuracy and 96% F1-score) against competing methods.	The study specifically introduces a CBAM-enhanced multi-branch CNN design that routes deep and shallow feature extraction to emphasize informative regions and suppress irrelevant signals, improving multi-class diagnostic performance from OCT.
[139]: A multi-scale anomaly detection framework for retinal OCT images based on the Bayesian neural network	2022	This analysis applies a Bayesian neural network with a multi-scale Bayesian U-Net to retinal OCT images for unspecific anomaly detection, using epistemic uncertainty to separate healthy from non-healthy cases and reporting quantitative evaluation on a public UCSD dataset plus a private dataset.	This investigation demonstrates an automated pipeline that trains only on healthy OCT data, reduces uncertainty from healthy tissue regions, and uses a threshold rule to flag anomalous inputs, offering a practical option for screening when labeled abnormal examples are limited.
[388]: A Neural Network Approach to Quantify Blood Flow from Retinal OCT Intensity Time-Series Measurements	2020	This Scientific Reports study uses a convolutional neural network trained on forward-modeled retinal OCT intensity time-series to estimate vessel blood-flow rates in an ophthalmic OCT angiography setting, with validation performed against Doppler OCT in blood-flow phantoms and further checked in vivo for flow conservation across retinal vessel branch points and expected power-law scaling with vessel diameter.	This analysis introduces a complete, deployable analysis framework that converts OCT intensity time series into label-free flow-rate maps for individual vessels, including SNR-aware model selection and spatial masking to reduce sensitivity to Doppler-angle effects, thereby enabling largely angle-, hematocrit-, and signal-to-noise-robust flow quantification.

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Work	Year	Overall reasoning	Contribution and summary
[316]: A new computer-aided diagnosis tool based on deep learning methods for automatic detection of retinal disorders from OCT images	2024	The study evaluates a deep learning computer-aided diagnosis tool for OCT-based retinal disorder classification, using ophthalmologist-provided semantic descriptions to improve interpretability and report test-set evidence from a UCSD-derived OCT dataset, relevant to screening and second-opinion workflows.	This work demonstrates that embedding clinician-derived semantic information into the network yields strong diagnostic performance on test OCT images while heat-map outputs highlight abnormal regions, supporting interpretability alongside screening-oriented clinical decision support.
[332]: A new convolutional neural network based on combination of circlets and wavelets for macular OCT classification	2023	This ophthalmic deep-learning study uses a 2D CNN on macular OCT B-scans represented in a non-data-adaptive multi-scale circlet and wavelet time-frequency transform domain, aiming to classify normal, AMD, and DME with evidence from subject-wise 5-fold nested cross-validation and external testing on a second OCT dataset plus Grad-CAM interpretability.	The research proposes the CircWave transform and corresponding CircWaveNet classifier by concatenating circlet and 2D-DWT sub-band channels, then demonstrates improved discrimination over original images and single-transform inputs while producing class-relevant Grad-CAM attention for both circle-like (DME) and line-like (normal) OCT features across datasets.
[221]: A new intelligent system based deep learning to detect DME and AMD in OCT images	2024	The analysis uses a convolutional neural network CAD approach on OCT retinal images to classify eyes into AMD, DME, and normal categories, evaluating multiple CNN architectures (including transfer learning models) on two datasets as evidence for automated ophthalmic image analysis.	This research reports that an Inception_V3-based model using features from a custom CNN achieved the top classification accuracy (99.53%) on the DUKE dataset, supporting the feasibility of an efficient OCT-based system for detecting AMD and DME.
[54]: A novel deep learning approach for diabetic retinopathy classification using optical coherence tomography angiography	2025	In an ophthalmic DR screening setting, the study applies a CNN-based classification model to OCTA-derived retinal images, supported by cGAN-driven augmentation and a dedicated preprocessing pipeline to address limited data and imaging quality, with evidence from reported performance results.	The analysis offers an automated, noninvasive DR versus non-DR pipeline that couples cGAN-enhanced dataset expansion with contrast enhancement, denoising, and vessel segmentation before CNN inference, aiming to improve early detection reliability from OCTA.
[452]: A Novel Deep Learning Method for Nuclear Cataract Classification Based on Anterior Segment Optical Coherence Tomography Images	2020	The research uses a CNN (GraNet) with an AS-OCT imaging modality to perform automated nuclear cataract grading, and reports evidence from experiments showing improved classification over baseline approaches using a focal loss plus cross-entropy cross-training scheme.	The reported work specifically introduces the GraNet architecture with a grading block for learning high-level features from AS-OCT and couples it with focal loss and cross-entropy cross-training to improve nuclear cataract classification accuracy versus existing methods.

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Work	Year	Overall reasoning	Contribution and summary
[441]: A Novel Multimodal Implementation of a Foundation Artificial Intelligence Model Using Optic Nerve Head Fundus Photographs and OCT Imaging for Glaucoma Detection	2026	In an ophthalmic diagnostic evaluation, the study applies a self-supervised RETFound foundation model to glaucoma detection using paired optic nerve head color fundus photographs and OCT images, comparing multimodal versus unimodal (fundus-only or OCT-only) performance with evidence based on AUC, precision, and recall across key clinical subgroups (race, age, and disease severity).	The study demonstrates that multimodal fundus-OCT RETFound improves discrimination over the fundus-only model while remaining statistically comparable to the OCT-only model, with subgroup results showing no significant race or age effects and stronger detection for moderate-to-severe versus mild glaucoma.
[381]: A Review of Machine Learning Algorithms for Retinal Cyst Segmentation on Optical Coherence Tomography	2023	This 2023 review focuses on machine learning for retinal cyst and fluid segmentation from optical coherence tomography (OCT) by covering classical and deep-learning method families across a three-step pipeline (denoising, retinal layer segmentation, and cyst/fluid segmentation) and the publicly available OCT datasets, providing evidence primarily in the form of synthesized algorithmic and dataset overviews relevant to quantitative assessment of OCT structural changes in ophthalmic disease.	This investigation distills the technical design choices for an OCT cyst/fluid assessment system by surveying denoising and layer-specific strategies alongside cyst/fluid segmentation and by compiling dataset resources and future challenges, helping researchers benchmark model development targets and pipeline components for this imaging task.
[413]: A robust deep learning classifier for screening multiple retinal diseases on optical coherence tomography	2025	This research develops FlexiVarViT, a high-resolution transformer-based classifier for OCT B-scans that is tailored to variable slice count and native in-plane resolution to screen multiple retinal disease categories, trained and cross-validated on a multi-disease Spectralis dataset and evaluated on two external OCT datasets from different devices and populations, providing evidence of robustness and generalizability via quantitative out-of-domain testing.	FlexiVarViT's dynamic patching enables processing at native resolution without resizing, and the authors show that Kermany-supervised pretraining plus volume-level attention aggregation yields the strongest cross-dataset multi-disease classification performance, supporting a practical approach for OCT screening under real-world imaging variability.
[194]: A scoping review of the use of artificial intelligence models in automated OCT analysis and prediction of treatment outcomes in diabetic macular oedema	2025	This Cochrane-methodology scoping review maps AI model studies using SD-OCT imaging for diabetic macular edema, covering diagnostic classification, automated OCT biomarker segmentation/detection (e.g., IRF, SRF, HRF), and prediction of visual or anatomical treatment outcomes, based on a synthesis of 40 eligible human primary research reports selected after quality screening.	This work consolidates the current landscape of OCT-AI workflows in DMO by distinguishing evidence for diagnosis and biomarker analysis from that for prognostic and treatment-outcome prediction, highlighting the limited external validation and the need for standardized evaluation and more robust prognostic modeling to support clinical translation.

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Work	Year	Overall reasoning	Contribution and summary
[225]: A Vision Transformer Based Deep Learning Architecture for Automatic Diagnosis of Diabetic Retinopathy in Optical Coherence Tomography Angiography	2023	Using a vision transformer deep learning architecture on optical coherence tomography angiography images, the study targets diabetic retinopathy by performing lesion segmentation, image quality assessment, and DR grading, matching the DRAC 2022 multitask evaluation structure and providing competition-level evidence from an OCTA dataset.	The research reports benchmark results for all three DRAC 2022 subtasks, including segmentation overlap and agreement measures plus grading accuracy, thereby demonstrating an OCTA-based automatic DR pipeline that can support practical end-to-end screening within the contest framework.
[440]: Accuracy of a deep convolutional neural network in the detection of myopic macular diseases using swept-source optical coherence tomography	2020	The analysis evaluates deep convolutional neural networks on swept-source OCT images to automatically classify myopic macular disease categories (including myopic choroidal neovascularization and retinoschisis) versus high myopia and non-high myopia, using k-fold cross-validation and reporting diagnostic performance for binary and ternary imaging tasks.	Using an ensemble of nine CNN architectures with Grad-CAM interpretability, the authors report highly accurate lesion-focused SS-OCT classification and show no statistically significant performance difference from three ophthalmologists on the same comparative datasets, supporting AI-assisted screening for myopia-related macular complications.
[149]: Accuracy of generative deep learning model for macular anatomy prediction from optical coherence tomography images in macular hole surgery	2024	This 2024 study develops a conditional variational autoencoder generative deep learning model to predict postoperative 6-month swept-source OCT macular anatomy from preoperative OCT in patients with successfully closed full-thickness macular holes, evaluating image-to-image synthesis using agreement and accuracy measures for neurosensory retinal areas and foveal structural metrics against ground-truth postoperative OCT.	The analysis quantifies how closely generated AI-OCT preserves foveolar height and mean foveal thickness while also estimating ELM and EZ restoration, providing evidence that the model can generate clinically interpretable postoperative retinal layer reconstructions rather than only scalar surgical outcomes.
[313]: Accurate Detection of Non-Proliferative Diabetic Retinopathy in Optical Coherence Tomography Images Using Convolutional Neural Networks	2020	The research evaluates a CNN-based computer-aided diagnosis approach using OCT B-scans to classify non-proliferative diabetic retinopathy versus normal eyes, combining transfer learning with fovea-guided nasal and temporal retina patch extraction and assessed via five-fold cross-validation to estimate diagnostic performance.	The reported work identifies an optimized deployment strategy in which two independently trained, ImageNet-pretrained AlexNet models, fed by nasal and temporal patches aligned using the ONL, extract pool5 features that are fused in a linear SVM, achieving the reported 94% accuracy and highlighting how preprocessing and feature-layer choices impact OCT-based NPDR detection.

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Work	Year	Overall reasoning	Contribution and summary
[259]: Adaptive and Explainable Deep Learning for Automated Retinal Disease Diagnosis Using OCT Images	2025	The reported work proposes an end-to-end, adaptive and explainable deep-learning approach that classifies retinal disease states from OCT imaging, using an MSRA-CNN with hierarchical Grad-CAM++ plus uncertainty weighting to generate clinician-interpretable heatmaps and reporting quantitative evaluation with inter-observer agreement for evidence of clinical relevance.	The study specifically advances an OCT pipeline that mitigates intensity variation, speckle noise, and class imbalance through adaptive illumination normalization and elastic deformation with adversarial perturbation, enabling reliable automated detection of CNV, DME, drusen, and normal eyes while pairing high-performing predictions with uncertainty-aware visual explanations for decision support.
[159]: ADCFormer: A hybrid model of Adaptive Depth-wise Convolution and Transformer for retinal macular edema segmentation in OCT images	2025	This analysis presents a transformer/CNN hybrid, ADCFormer, for segmenting diabetic macular edema lesion regions in OCT images, using global efficient self-attention with adaptive depth-wise convolution and reporting comparative evidence from multiple external and hospital-collected OCT datasets.	This investigation introduces ADCFormer's ESA, ADWConv, and GLFF components plus a residual multiscale feed-forward design to better fuse global and local multiscale features, and the reported cross-dataset segmentation results indicate practical value for delineating complete DME lesion areas under noisy OCT conditions.
[215]: Advanced Deep Learning Techniques for Evaluating OCT Image Quality and Detecting Retinal Pathologies	2025	In a retinal OCT context, this research uses an image-quality assessment model (pretrained ARNIQA) together with CNN classifiers (ResNet50 and Xception) to address how dataset and image quality affect a three-class task that couples OCT image-quality evaluation with retinal pathology detection across high-, low-, and original image sets, using a Tunisian OCT dataset to provide evidence in a medical imaging classification setting.	This analysis introduces an OCT-specific quality-aware deep learning workflow that leverages ARNIQA to guide autonomous disease-related classification from OCT images, highlighting the practical impact of separating high-quality versus low-quality inputs for more reliable predictions.
[395]: Advancement of cataract classification with artificial intelligence using anterior segment optical coherence tomography images with self-supervised vision transformer	2026	This clinical investigation evaluates a self-supervised Vision Transformer trained on uncropped anterior segment optical coherence tomography (AS-OCT) images to grade cataract severity by tasking the model with nucleus, cortical, and posterior subcapsular classification using LOCS III labels, reporting evidence from multiple supervised training and evaluation classification tasks measured primarily with AUPRC across labeled cohorts.	This work establishes the usefulness of AS-OCT domain self-supervised pretraining for ViT-based cataract grading by comparing SS-ViT against ImageNet-pretrained and non-pretrained baselines across five LOCS III-based tasks, including subregion-specific assessments and input configurations that leverage all angular cross-sections.

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Work	Year	Overall reasoning	Contribution and summary
[349]: Advancing Diabetic Retinopathy Diagnosis: Leveraging Optical Coherence Tomography Imaging with Convolutional Neural Networks	2023	This review evaluates ophthalmic applications of convolutional neural networks using optical coherence tomography retinal imaging to automate diabetic retinopathy lesion detection, microaneurysm segmentation, and severity grading, summarizing evidence across the OCT-to-CNN workflow and its reported diagnostic potential.	The research consolidates the technical considerations for OCT-based CNN pipelines and highlights practical implementation issues, such as data availability and model interpretability, along with future directions like multimodal OCT integration and real-time point-of-care screening.
[323]: Advancing Diabetic Retinopathy Screening: A Systematic Review of Artificial Intelligence and Optical Coherence Tomography Angiography Innovations	2025	In the ophthalmic setting of diabetic retinopathy screening, this systematic review synthesizes evidence on AI methods, predominantly deep learning alongside related ML, applied to optical coherence tomography angiography data, targeting DR detection and severity classification using pooled diagnostic performance outcomes reported across studies with OCTA acquisition and parameter variations.	By consolidating 32 controlled human studies and extracting how different OCTA image types, tasks, and model families perform using metrics such as accuracy, sensitivity, specificity, and AUC, the review clarifies which AI-OCTA approaches have the strongest reported diagnostic evidence and where cross-study variability may affect clinical translation.
[213]: Advancing Ophthalmic Diagnostics: Employing Deep Learning for Precision Detection of Retinal Damage in OCT Images	2024	The study uses VGG16 and InceptionV3 on retinal OCT images to classify diabetic macular edema, age-related macular degeneration, choroidal neovascularization, and healthy eyes, reporting performance on an 85,000-image dataset to support automated diagnostic decision-making.	The analysis compares two CNN model families for OCT interpretation and documents model-specific strengths (VGG16 for CNV and InceptionV3 for normal retinas), showing a reported overall accuracy of 92.76% that supports practical relevance for retinal classification.
[281]: Agreement of a Novel Artificial Intelligence Software With Optical Coherence Tomography and Manual Grading of the Optic Disc in Glaucoma	2023	In this prospective cross-sectional study, offline AI running on a smartphone fundus camera estimates vertical cup-to-disc ratio from optic disc photographs and is benchmarked against SD-OCT vCDR and expert manual grading on stereoscopic fundus images in glaucoma and normal eyes, providing evidence of agreement in an ophthalmic clinical measurement task.	The study reports quantitative concordance between the new AI vCDR outputs and both SD-OCT and specialist manual segmentation across 473 eyes, highlighting that on-device inference can support glaucoma-relevant disc assessment without cloud-based processing.
[422]: An artificial intelligence cloud platform for OCT-based retinal anomalies screening system in real clinical environments	2025	The reported work evaluates AI-PORAS, an ophthalmology-oriented cloud platform using a domain-adaptive object detection and classification model on OCT B-scans to localize and screen 15 retinal anomalies, with evidence from large-scale validation on 165,384 eyes and real-world deployment across 207 medical institutions in China.	The study details the end-to-end clinical workflow and cross-scanner adaptation strategy for OCT-AI and reports external real-world screening outcomes for 116,717 diagnosed patients, providing practical support for scalable remote retinal anomaly assessment in routine care settings.

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Work	Year	Overall reasoning	Contribution and summary
[319]: AN EARLY RETINAL DISEASE DIAGNOSIS SYSTEM USING OCT IMAGES VIA CNN-BASED STACKING ENSEMBLE LEARNING	2023	This analysis uses OCT retinal imaging and CNN-based feature extraction with stacking ensemble learning to detect diabetic macular edema, drusen, and choroidal neovascularization, providing evidence in an ophthalmic diagnostic setting via two publicly available OCT datasets (Duke and UCSD).	This investigation reports that fine-tuned AlexNet features combined with homogeneous and heterogeneous stacking ensembles yield near-unity classification metrics across both datasets and claims improved performance over prior approaches for these OCT-based retinal disease triage tasks.
[460]: An Efficient Retinal Layer Segmentation Based on Deep Learning Regression Technique for Early Diagnosis of Retinal Diseases in OCT and FUNDUS Images	2023	This research evaluates a deep learning regression approach for retinal-layer segmentation from OCT and fundus imaging to support early retinal disease detection, using layer-thickness changes and subsequent segmentation, feature extraction, clustering, and classification for conditions including diabetic retinopathy.	This analysis focuses on estimating retinal layer attributes and clustering/segmenting image content from both modalities with regression-based learning, aiming to translate thickness variations into disease-specific classification signals that are clinically relevant for early diabetic retinopathy screening.
[430]: An Open-Source Deep Learning Framework for Automated Corneal Segmentation in Anterior Segment Optical Coherence Tomography With Cross-Device External Validation	2026	Within an ophthalmology workflow using anterior segment optical coherence tomography, the study introduces a deep learning nnU-Net-based full-thickness corneal segmentation model (CUNEX) and validates it across four externally acquired OCT devices without fine-tuning, then tests how the resulting segmentations affect downstream deep learning classification of age, sex, and disease staging, providing internal and cross-device evidence with segmentation quality and downstream utility outcomes.	By releasing CUNEX with pretrained weights and demonstrating cross-device segmentation robustness that maintains anatomically meaningful full-thickness corneal boundaries where prior models fail, the study supplies a reproducible segmentation tool for enabling multimodal AI research and clinical integration on AS-OCT data.
[442]: An Optical Coherence Tomography-Based Deep Learning Algorithm for Visual Acuity Prediction of Highly Myopic Eyes After Cataract Surgery	2021	This work develops an OCT-based deep learning ensemble that ingests preoperative macular OCT images to predict 4-week postoperative best-corrected visual acuity in highly myopic eyes after cataract surgery, with evaluation of prediction accuracy and error patterns using internal and external test datasets.	The research compares multiple CNN backbones with an ensemble strategy trained on internal data and then externally tested, yielding clinically actionable BCVA prediction performance that may assist preoperative counseling and surgical planning for highly myopic cataract patients.

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Work	Year	Overall reasoning	Contribution and summary
[155]: Analysis of early screening results for high-risk populations of retinal diseases based on artificial intelligence optical coherence tomography technology	2026	This diagnostic trial evaluates an ophthalmic deep learning AI-OCT system for early screening in a high-risk "Five-High" population, using wide-field macular and peripheral OCT imaging and measuring agreement with ophthalmologist review plus downstream referral and treatment outcomes.	This research quantifies the AI-OCT screening positive rate and lesion distribution in the "Five-High" cohort and shows high concordance with physician diagnosis while reporting referral and treatment follow-through after positive screening.
[297]: Anomaly Detection in Retinal OCT Images With Deep Learning-Based Knowledge Distillation	2025	This investigation develops an unsupervised deep learning anomaly-detection system for ophthalmic fovea-centered retinal OCT screening, using a teacher-student knowledge distillation framework trained only on normal scans to assign volume- and B-scan-level anomaly scores and produce localized anomaly maps, with evidence from ROC/PR evaluation on an internal OCT test set spanning multiple macular conditions and external public OCT datasets for B-scan detection.	The analysis provides a general-purpose, annotation-free alternative to supervised OCT pathology recognition by showing consistent anomaly detection and clinically interpretable localization across diverse retinal disease patterns, supporting the use of feature-space deviations plus anomaly mapping as a scalable screening workflow.
[236]: Anti-VEGF treatment outcome prediction based on optical coherence tomography images in neovascular age-related macular degeneration using a deep neural network	2024	The research evaluates a DenseNet201-based deep learning model that uses spectral-domain OCT images collected before and during anti-VEGF loading to predict anatomical nAMD treatment response, with results benchmarked against ophthalmologists for clinical relevance.	The reported work shows that incorporating additional OCT timepoints after the first and second injections and using concatenation-based sequence fusion improves treatment-outcome prediction accuracy compared with human graders, supporting OCT-driven individualized guidance.
[263]: Application of artificial intelligence-based dual-modality analysis combining fundus photography and optical coherence tomography in diabetic retinopathy screening in a community hospital	2022	This community-hospital study evaluates deep-learning AI for task-specific diabetic retinopathy screening by combining fundus photography and OCT, with ophthalmologists' ICDR-graded findings as the reference standard and with evidence presented as diagnostic accuracy and referral-rate comparisons in a real-world workflow.	The study extends AI-based screening beyond fundus-only grading by integrating OCT-derived macular edema detection alongside AI DR classification, showing that dual-modality screening increases the number of referable cases flagged for ophthalmologist confirmation.

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Work	Year	Overall reasoning	Contribution and summary
[364]: Application program interface for automatic segmentation of retinal layers and fluids in Optical Coherence Tomography - Neovascular Age related Macular degeneration retinal images using deep learning models	2024	Within an OCT-based ophthalmic imaging context for neovascular age-related macular degeneration, the study applies a deep learning segmentation approach (AR U-Net++) to delineate retinal layer boundaries and fluid compartments from retinal B-scans and reports standard overlap-metric comparisons against prior U-Net variants.	This investigation introduces an AR U-Net++ pipeline that outputs spatially resolved retinal fluid locations and depths across multiple layers while automatically generating clinician-facing reports, offering a practical mechanism for supporting AMD diagnosis beyond pixel-wise segmentation.
[202]: Applications of artificial intelligence in diagnosis of uncommon cystoid macular edema using optical coherence tomography imaging: A systematic review	2024	This systematic review evaluates AI approaches applied to ophthalmic OCT retinal imaging for diagnosing cystoid macular edema in less common clinical contexts, covering the diagnostic tasks of detection, classification, and segmentation and summarizing evidence from 23 included studies published between 2018 and November 2023, with CNN-based models reported as most consistently effective.	This analysis synthesizes cross-database findings on how CNN-driven AI can automate identification and delineation of uncommon OCT-based CME phenotypes such as postoperative (Irvine Gass) CME and retinitis pigmentosa without evident vasculopathy, while also noting constraints related to dataset size and ethical considerations that affect clinical translation.
[85]: Applications of Deep Learning algorithms for retinal diseases diagnosis based on Optical Coherence Tomography imaging	2023	The study evaluates deep learning classifiers for OCT-based automated detection and diagnosis of retinal diseases, comparing five convolutional neural network architectures on an imaging-driven diagnostic task using reported quantitative metrics (accuracy, AUC, F1-score, and loss).	This work benchmarks DenseNet121, DenseNet169, DenseNet201, InceptionResNet, and a 12-layer CNN on OCT images and reports that DenseNet169 achieved the best overall metric values, supporting OCT image-based deep learning as a clinically relevant decision-support direction.
[174]: Artificial Intelligence Aided Analysis of Anterior Segment Optical Coherence Tomography Imaging to Monitor the Device-Cornea Joint After Synthetic Cornea Implantation	2025	In a rabbit keratoprosthesis translational study, the authors train a convolutional neural network to segment anterior lamellar tissue on serial anterior segment optical coherence tomography, then use quantified device-cornea joint tissue volume changes to flag subclinical peri-prosthetic thinning and retraction ahead of slit-lamp findings during 6 to 12 months of longitudinal imaging.	They introduce an AI-assisted, aligned AS-OCT analysis workflow that converts optic-cornea joint segmentation into absolute and percentage tissue-volume change estimates, identifying a practical change threshold that can detect tissue loss up to about 3 months before clinical detection, supporting objective postoperative surveillance after synthetic cornea implantation.

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Work	Year	Overall reasoning	Contribution and summary
[161]: Artificial Intelligence applied to the classification of retinal diseases in Optical Coherence Tomography images	2022	This ophthalmology study uses machine learning (KNN) and deep learning (two CNN architectures, including ResNet50) on grayscale OCT image data to perform retinal disease presence classification for CNV, DME, and drusen versus normal, with reported accuracy on a dataset of 83,484 images.	This research evaluates a hybrid workflow that feeds KNN handcrafted histogram statistics and a CNN end-to-end approach, providing quantitative performance comparisons that may support AI-assisted diagnostic decision-making in OCT-based screening.
[386]: Artificial intelligence-based prediction of neurocardiovascular risk score from retinal swept-source optical coherence tomography-angiography	2024	This investigation leverages machine-learning and deep-learning models applied to swept-source retinal OCT-A microvasculature images to predict the CHA2DS2-VASc neurocardiovascular risk score from the publicly available RASTA dataset, providing quantitative evidence of model performance for this ophthalmic imaging-to-cardiovascular risk stratification task.	The analysis compares conventional ML classifiers using clinical plus OCT-A-derived quantitative features against an image-only EfficientNetV2-B3 deep network trained on SS OCT-A, highlighting how retinal microvascular imaging alone can estimate CHA2DS2-VASc risk categories in a clinically defined two-group scheme.
[189]: Artificial Intelligence-Based Quantification of Central Macular Fluid Volume and VA Prediction for Diabetic Macular Edema Using OCT Images	2023	In a retrospective OCT study of treatment-naive diabetic macular edema receiving anti-VEGF therapy, two deep learning models automatically quantified central macular fluid volume and central subfield thickness from OCT volume scans to relate these metrics to one-month posttreatment BCVA, with multivariable regression and ROC analyses providing prognostic evidence.	The work demonstrates that automatically measured CMFV, rather than CST, was the only significant OCT biomarker for predicting one-month logMAR BCVA and showed slightly better discrimination for achieving BCVA of at least 20/40, supporting fluid-volume quantification as a more informative early outcome predictor.
[240]: Artificial intelligence-based retinal disease classification using optical coherence tomography images	2022	The study describes a deep learning, CNN-based retinal disease classification method for ophthalmic optical coherence tomography images, targeting drusen, diabetic macular edema, and choroidal neovascularization and reporting test-set performance to support faster automated diagnostic screening and user/clinician alerting.	By combining CNNs with pretrained image classification features and evaluating accuracy and F1 on a test set, the study shows that OCT image analysis can generate high-performing suspicion alerts that reduce reliance on manual review of large OCT datasets.
[345]: Artificial Intelligence-Driven Detection of LASIK Using Corneal Optical Coherence Tomography Maps	2025	In this ophthalmology study, a lightweight convolutional neural network was trained on corneal optical coherence tomography parameter maps (pachymetry, epithelial thickness, posterior mean curvature, anterior axial power, and anterior stroma reflectance) to classify prior LASIK status, with repeated fivefold patient-stratified cross-validation and separate out-of-sample testing for evidence of generalization.	The model delivered accurate discrimination of myopic versus hyperopic LASIK using down-sampled OCT maps and maintained correctness in an external post-LASIK test set, offering a practical imaging-based approach for identifying LASIK history when records may be incomplete.

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Work	Year	Overall reasoning	Contribution and summary
[401]: ARTIFICIAL INTELLIGENCE FOR OPTICAL COHERENCE TOMOGRAPHY ANGIOGRAPHY-BASED DISEASE ACTIVITY PREDICTION IN AGE-RELATED MACULAR DEGENERATION	2024	In ophthalmology, this retrospective cross-instrument deep learning study evaluates whether a deep neural network trained on OCTA en-face vascular 2D projections can classify age-related macular degeneration disease activity into four clinically defined categories across Heidelberg Spectralis and Optovue Solix, using clinician-labeled fluid status from same-day OCT as the labeling evidence and reporting performance on held-out test sets.	By expanding a Heidelberg Spectralis dataset and adding a second manufacturer's OCTA images, the authors show that combining cross-instrument training can improve AMD activity classification accuracy and quantify how model generalization varies with instrument and imaging field of view, supporting OCTA-based AI for manufacturer-robust disease activity prediction.
[231]: Artificial intelligence for the detection of glaucoma with SD-OCT images: a systematic review and Meta-analysis	2024	This systematic review and meta-analysis evaluates ophthalmic AI for glaucoma detection using spectral-domain optical coherence tomography (SD-OCT) imaging, quantitatively synthesizing diagnostic accuracy evidence across multiple included studies using pooled sensitivity, specificity, likelihood ratios, diagnostic odds ratio, and AUC.	By aggregating results from 20 studies and 51 AI models, it provides pooled performance estimates that help benchmark SD-OCT-based AI as an adjunct to clinical decision-making for glaucoma screening or diagnosis.
[163]: Artificial intelligence hybrid system for enhancing retinal diseases classification using automated deep features extracted from oct images	2021	This work examines an ophthalmic CAD system that uses a hybrid AI workflow for OCT-based retinal disease classification by extracting automated deep features from spectral-domain OCT with AOCTNet and feeding them into conventional supervised classifiers to distinguish normal eyes from four abnormal classes (AMD, CNV, DME, and drusen), with evidence based on quantitative performance reported on an SD-OCT test set.	The research provides a multiclass SD-OCT classification pipeline that couples AOCTNet feature extraction with eight different machine-learning classifiers, reporting strong test-stage results for the best-performing models and illustrating the utility of deep features for efficient discrimination among retinal disease categories.
[456]: Artificial intelligence in predicting macular hole surgery outcomes: a focus on optical coherence tomography parameters	2025	This retrospective ophthalmic study uses preoperative Spectralis OCT-derived macular hole indices (MHI, THI, HFF, BHD, MHD) to train and test a GPT-based generative AI classifier that predicts anatomical macular hole closure after idiopathic vitrectomy, with performance compared against a logistic regression benchmark using standard classification metrics including AUC, predictive values, and Kappa agreement.	This research provides a direct head-to-head evaluation showing how OCT index-conditioned GPT predictions perform relative to a data-driven logistic regression model for predicting closure, highlighting that MHI, THI, and HFF carry more prognostic value than simple diameter measures and that GPT may under-identify non-closure cases.

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Work	Year	Overall reasoning	Contribution and summary
[458]: Artificial intelligence in retinal screening using OCT images: A review of the last decade (2013-2023)	2024	This PRISMA-structured systematic review covers machine learning and deep learning methods applied to OCT-derived retinal and choroidal images for screening and decision-support tasks in glaucoma, AMD, and related macular and vascular conditions, summarizing the available evidence mainly from peer-reviewed journal studies.	The analysis consolidates what OCT-based ML versus DL approaches imply for feature handling and usability, highlighting how pretrained deep networks tend to support plug-and-play classification and pointing to continuous learning as a future direction for detecting subtle disease progression from OCT.
[360]: Artificial intelligence method based on multi-feature fusion for automatic macular edema (ME) classification on spectral-domain optical coherence tomography (SD-OCT) images	2023	This ophthalmology study develops a multiclass computer-aided diagnosis system that fuses handcrafted radiomics-style features with deep features extracted from SD-OCT (Heidelberg Spectralis) to classify macular edema subtypes (DME, AMD, RVO, and CSC), using standard test-set evaluation with accuracy and ROC/AUC plus model interpretability via Grad-CAM.	The reported work demonstrates that early multi-feature fusion followed by dimensionality reduction and LASSO-driven feature selection yields the best SVM performance on the held-out SD-OCT test set, while Grad-CAM heatmaps help indicate the retinal regions driving the subtype decisions.
[383]: Artificial Intelligence Versus Rules-Based Approach for Segmenting NonPerfusion Area in a DRCR Retina Network Optical Coherence Tomography Angiography Dataset	2025	In a multicenter diabetic retinopathy OCT angiography dataset, the study compares convolutional neural network-based versus rules-based nonperfusion area segmentation across four macular slabs, using manual grading and ETDRS severity-linked diagnostics to quantify differences in segmentation agreement and clinical staging performance.	The authors show that the AI-derived nonperfusion outputs match manual segmentation more closely and yield higher DR-severity discrimination for referable status in most slabs, supporting more scalable OCTA quantification than the rules-based approach for screening and triage.
[335]: Assessment of a Segmentation-Free Deep Learning Algorithm for Diagnosing Glaucoma From Optical Coherence Tomography Scans	2020	In a single-institution cross-sectional ophthalmic dataset, the study evaluates a segmentation-free convolutional neural network that classifies glaucoma versus normal eyes from full peripapillary SD-OCT circular B-scans and reports ROC AUC and sensitivity at fixed specificities.	This investigation demonstrates that probability-based deep learning on raw circle B-scans can outperform conventional RNFL thickness metrics for glaucoma discrimination, with benefits particularly evident in preperimetric and early perimetric disease.
[71]: Attention Assisted Patch-Wise CNN for the Segmentation of Fluids from the Retinal Optical Coherence Tomography Images	2024	This research evaluates an attention-assisted patch-wise CNN for retinal OCT imaging, using encoder-decoder and multi-scale modules to segment and quantify three retinal fluid types, with evidence derived from qualitative and quantitative results on the publicly available RETOUCH OCT fluid detection challenge dataset.	This analysis reports an attention-enhanced, computationally efficient segmentation framework that predicts cyst subtype and estimates volume, and it claims improved precision, recall, and Dice coefficient over prior approaches on the RETOUCH benchmark, supporting its potential value for treatment-planning workflows.

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Work	Year	Overall reasoning	Contribution and summary
[294]: Attention-Guided Neural Networks for Explainable Retinal Disease Detection in OCT Images	2025	The study describes an attention-guided convolutional neural network for automated retinal disease classification from OCT images, using Grad-CAM to generate explanation maps tied to clinically relevant regions and reporting category-level accuracy as evidence in an ophthalmic imaging task.	This work further evaluates multiple attention-enhanced architectures, highlighting a CBAM-augmented model whose Grad-CAM outputs and strong category performance (98.34%) support the feasibility of accurate, interpretable OCT screening in settings needing decision support.
[434]: Augmenting Kalman Filter Machine Learning Models with Data from OCT to Predict Future Visual Field Loss: An Analysis Using Data from the African Descent and Glaucoma Evaluation Study and the Diagnostic Innovation in Glaucoma Study	2022	This retrospective longitudinal cohort study uses Kalman filter machine learning models in an ophthalmic glaucoma setting to forecast 36-month visual field loss from perimetry (and tonometry), evaluating whether adding global OCT retinal nerve fiber layer data improves prediction and whether performance varies by race-matched versus nonmatched training and testing composition in participants from ADAGES and DIGS.	The research shows that incorporating global RNFL from OCT into Kalman filter forecasting produces minimal gains in mean deviation prediction accuracy over tonometry-plus-perimetry models, while race alignment between training and testing yields only modest improvement in mean absolute prediction error.
[83]: Automated Age-Related Macular Degeneration and Diabetic Macular Edema Detection on OCT Images using Deep Learning	2018	The analysis evaluates deep learning for ophthalmic OCT imaging, performing automated multi-class classification of retinal status (healthy, dry and wet AMD, and DME) and reporting comparison to a transfer learning baseline.	This research proposes an end-to-end OCT-based deep learning pipeline that improves over an earlier transfer learning approach for distinguishing healthy eyes, AMD subtypes, and diabetic macular edema, supporting more accurate automated screening at the image level.
[426]: Automated assessment of the smoothness of retinal layers in optical coherence tomography images using a machine learning algorithm	2023	In this ophthalmology study, a support vector regression model with a wavelet kernel was used on spectral-domain OCT B-scans to automatically segment the inner plexiform layer (IPL) and outer plexiform layer (OPL) and compute a smoothness index, with agreement assessed against manual expert delineations on 50 slabs from patients with diabetic retinopathy using Bland-Altman and error statistics.	By replacing time-consuming manual layer tracing with an end-to-end OCT pipeline for IPL/OPL smoothness quantification, the authors show reproducible segmentation and smoothness-index agreement sufficient to support rapid, biomarker-style measurement in images spanning regular to disorganized layers.

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Work	Year	Overall reasoning	Contribution and summary
[96]: Automated classification of age-related macular degeneration from optical coherence tomography images using deep learning approach	2023	The research applies deep learning to ophthalmology by segmenting and classifying OCT macular findings related to AMD, using an attention-based nested U-Net with flower pollination hyperparameter optimization for a drusen-versus-CNV-versus-normal workflow and reporting UCSD dataset-based evaluation with overall diagnostic performance metrics.	The reported work develops an OCT pre-processing and ANU-Net plus SqueezeNet classification pipeline, where retinal flattening and cropping precede hyperparameter-tuned learning to enable automated preliminary screening in hospital or eye-clinic settings.
[183]: Automated deep learning-based AMD detection and staging in real-world OCT datasets (PINNACLE study report 5)	2023	This clinical ophthalmology study develops a two-stage deep learning CNN for OCT-based age-related macular degeneration staging, using macula-centered 3D Topcon volumes processed via a B-scan ResNet50 stage followed by volume-level ResNets and Monte-Carlo dropout uncertainty to produce biomarker-based classifications of Normal, iAMD, GA, and nAMD, with evaluation on real-world retinal OCT test data using ROC-AUC.	The study provides a practical framework for retrospective AMD volume grading from routine 3D OCT by combining B-scan localization, biomarker presence prediction at the volume level, and uncertainty-aware inference, enabling clinically interpretable staging across normal and early/intermediate and late AMD categories with external assessment on independent OCT datasets.
[270]: Automated Deep Learning Framework for Retinal Disease Classification from OCT Images: A Large-Scale Study Across Four Categories	2024	The study describes an ophthalmology-focused deep learning framework using convolutional neural networks to classify high-resolution retinal OCT cross-sectional images into four retinal disease categories, supported by a large-scale dataset and reported classification accuracy evidence.	This investigation contributes an automated CNN-based retinal OCT image classification pipeline trained on 84,495 images, offering a practical route to reduce manual reading burden and speed diagnostic decision-making from OCT.
[147]: Automated detection of a nonperfusion area caused by retinal vein occlusion in optical coherence tomography angiography images using deep learning	2019	In this ophthalmic deep-learning study, OCTA images of the superficial and deep capillary plexus are used to classify retinal vein occlusion eyes with nonperfusion area versus normal eyes, using a VGG-16-based DNN compared with an SVM and evaluated with ROC-derived AUC plus sensitivity, specificity, and reading time against seven ophthalmologists.	The authors provide an OCTA-based automated pipeline for detecting nonperfusion area due to retinal vein occlusion and show DNN-based performance and Grad-CAM heat maps that emphasize the foveal avascular zone and nonperfusion regions, supported by head-to-head benchmarking of model outputs against clinician interpretation.

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Work	Year	Overall reasoning	Contribution and summary
[201]: Automated Detection of Central Retinal Artery Occlusion Using OCT Imaging via Explainable Deep Learning	2025	In this retrospective external validation study from two German institutions, a transfer-learned explainable deep learning classifier analyzes Spectralis OCT volumes and slides to triage an acute emergency 3-class task (central retinal artery occlusion versus control and differential diagnoses), with performance estimated using nested five-fold cross-validation and reported primarily via one-vs-all AUC.	The model attained high multiclass discrimination for CRAO and differentiating non-CRAO etiologies on OCT, while Grad-CAM feature maps highlighted inner-to-middle retinal layer regions relevant to the decision process, supporting practical use as image-based decision support in urgent ocular presentations.
[287]: Automated detection of exudative age-related macular degeneration in spectral domain optical coherence tomography using deep learning	2018	The study reports a TensorFlow-based deep convolutional neural network that analyzes spectral-domain optical coherence tomography cross-sectional images to detect exudative age-related macular degeneration, using held-out testing-stage evaluation on untrained scans as evidence of an image-based AI detection task in ophthalmology.	The research provides experimental results from training a pretrained DCNN on SD-OCT and then testing on a separate set of cross-sections, yielding an AMD testing score that supports automated identification of exudative AMD and suggests utility as decision support in the clinic.
[222]: Automated Detection of Glaucoma using OCT RNFLT Maps and Supervised Deep Learning Models: A Comparative Study of CNN, MobileNet, ResNet50, and VGG16 Architectures	2025	The analysis tests supervised deep learning classification of glaucoma using OCT RNFLT maps with several CNN families (CNN, MobileNet, ResNet50, and VGG16), reporting validation results with multiple diagnostic metrics (accuracy, sensitivity, specificity, and F1-score) to determine model effectiveness for automated early detection.	This research directly compares the diagnostic accuracy, error balance, and computational efficiency implications of four architectures on a dataset of 1000 balanced OCT RNFLT samples, highlighting VGG16's top precision and validation accuracy alongside MobileNet's best F1-score.
[445]: Automated Detection of Macular Diseases by Optical Coherence Tomography and Artificial Intelligence Machine Learning of Optical Coherence Tomography Images	2019	Using Cirrus HD-OCT macular B-scans with masked ophthalmologist labels, this 2019 study trains a CNN with image augmentation and evaluates automated classification of multiple macular disorders in an ophthalmic diagnostic setting, reporting precision and recall on a held-out set of 100 OCT images.	By expanding limited training data through horizontal flipping, rotation, and translation, it demonstrates that the augmented CNN can generate clinically plausible top candidate disease predictions and quantifies detection accuracy across major classes including normal eyes, wet AMD, DR, and ERM.

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Work	Year	Overall reasoning	Contribution and summary
[145]: Automated Detection of Unhealthy Retinal Optical Coherence Tomography Images Using Geometrical Features for Machine Learning Modeling	2026	The study describes an ophthalmic OCT imaging study using a supervised machine-learning classifier that inputs ILM and RPE geometry (symmetry and smoothness) to distinguish unhealthy from healthy retinal layer patterns, providing evidence from reported comparative results against a profile-feature baseline with emphasis on reduced false negatives.	The reported work specifies a feature-driven modeling strategy for OCT layer quality assessment that improves classification over a vertical profile baseline and highlights lower missed-unhealthy cases, supporting more efficient ophthalmologist prescreening workflows.
[41]: Automated Diagnosis of Eight Retinal Diseases from OCT Images Using Deep Learning with Attention Mechanisms	2025	The reported work applies an attention-augmented ResNet deep learning model to classify eight retinal disease categories and normal retina from optical coherence tomography (OCT) images, using the RetinalOCT-C8 dataset and reporting validation results with per-class precision-recall and macro F1 to support diagnostic utility.	The study contributes an Attention Residual Classifiers design with a squeeze-and-excitation attention block and residual connections, achieving high validation accuracy and macro F1 while indicating strong class-level performance for several conditions relevant to OCT-based interpretation.
[185]: Automated Identification and Segmentation of Ellipsoid Zone At-Risk Using Deep Learning on SD-OCT for Predicting Progression in Dry AMD	2023	In patients with nonexudative dry age-related macular degeneration, the study uses a deep learning model on SD-OCT B-scans to automatically detect and pixel-wise segment ellipsoid zone at-risk regions, with performance benchmarked against expert ground truth and evaluated for prognostic relevance to geographic atrophy progression using an independent longitudinal test dataset.	This investigation provides an automated, high-accuracy EZ at-risk quantification pipeline and shows that baseline EZ at-risk area is positively associated with later geographic atrophy growth, supporting its use as a quantitative OCT biomarker for progression risk stratification.
[278]: Automated inter-device 3D OCT image registration using deep learning and retinal layer segmentation	2023	This ophthalmic study develops a two-stage deep-learning registration pipeline for aligning 3D OCT volumes from different devices, using unsupervised multimodal en-face feature matching between PS-OCT and Spectralis (SLO) followed by retinal-layer segmentation-guided axial (Z-axis) alignment, with performance evaluated on paired clinical and experimental OCT data using landmark-based error and RPE-layer-derived axial metrics.	This analysis provides an automated way to transfer spatial correspondence across Heidelberg Spectralis and a custom PS-OCT system by coupling R2D2-style keypoint detection and RANSAC-based 2D alignment with ILM-referenced B-scan adjustment, enabling consistent fusion and cross-device mapping of signals such as PS-OCT DOPU onto conventional OCT.

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Work	Year	Overall reasoning	Contribution and summary
[241]: Automated machine learning-based classification of proliferative and non-proliferative diabetic retinopathy using optical coherence tomography angiography vascular density maps	2023	The study evaluates a supervised SVM classifier, tuned with a genetic evolutionary algorithm, to distinguish proliferative from non-proliferative diabetic retinopathy using OCTA vascular density maps across retinal layers, reporting classification accuracy on 148 eyes from 78 patients (PDR vs NPDR) as evidence of diagnostic feasibility.	By comparing vascular density maps at superficial, deep, and total retina levels, it identifies the deep retinal layer map as the most informative input, with the best reported accuracy reaching 90% for PDR versus NPDR discrimination and supporting layer-specific OCTA feature use for retinal disease classification.
[90]: AUTOMATED MACHINE LEARNING CLASSIFICATION OF OPTICAL COHERENCE TOMOGRAPHY IMAGES OF RETINAL CONDITIONS USING GOOGLE CLOUD VERTEX ARTIFICIAL INTELLIGENCE	2025	Using Google Cloud Vertex AI automated machine learning, the study evaluates single-label classification of optical coherence tomography images for five retinal categories, including a neovascular AMD subanalysis, with evidence from a validated publicly available dataset and reported diagnostic performance metrics.	The research shows that an off-the-shelf automated ML pipeline can deliver very high AUPRC and strong sensitivity and specificity for differentiating retinal conditions on OCT, supporting practical scalability for AI-assisted retinal diagnosis in an ophthalmic imaging workflow.
[254]: Automated Macular Disease Detection using Retinal Optical Coherence Tomography images by Fusion of Deep Learning Networks	2021	A study applies deep learning to retinal OCT for automated macular disease image classification, using a two-network fusion strategy that concatenates learned feature vectors from pretrained DCNNs to support screening and clinician diagnosis-recommendation workflows, with comparative evidence reported as improved classification accuracy over single models on the same dataset.	This research introduces a feature-fusion DCNN framework for macular OCT classification in which concatenated representations from two pretrained networks yield better accuracy than each constituent model, offering a practical approach for OCT-based decision support.
[127]: Automated Macular OCT Retinal Surface Segmentation in Cases of Severe Glaucoma Using Deep Learning	2022	In an ophthalmic glaucoma setting, the study describes an independently evaluated deep learning approach for automated macular OCT retinal surface segmentation that targets known failure modes in severe disease, using three-dimensional OCT imaging through data augmentation, contextual multi-slice input from adjacent B-scans, and topology-preserving fusion of predictions to produce more reliable surface delineations.	The analysis introduces and tests three concrete design elements: vertical and jittered B-scan augmentation, contrast-enhanced six-channel contextual stacking from adjacent slices, and combined horizontal/vertical surface merging with topological order to improve segmentation quality (mean absolute distance, Dice, and Hausdorff) relative to state-of-the-art methods on a dataset containing severe glaucoma cases.

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Work	Year	Overall reasoning	Contribution and summary
[455]: Automated Quantification of Photoreceptor alteration in macular disease using Optical Coherence Tomography and Deep Learning	2020	This 2020 study evaluates a deep learning ensemble of U-shaped convolutional neural networks to automatically segment the photoreceptor layer in Spectralis OCT and generate en-face thickness and uncertainty maps for macular photoreceptor alteration in diabetic macular edema and retinal vein occlusion, with quantitative benchmarking against consensus expert annotations and a second retina specialist using segmentation and thickness agreement metrics.	The reported work delivers an OCT-based, consensus-inspired segmentation pipeline that averages multiple model score maps to produce binarized photoreceptor boundaries plus disagreement maps, enabling reliable layer thickness quantification while reducing expert-dependent variability and highlighting ambiguous regions needing human review.
[186]: Automated Retinal Disease Detection: VGG16-Based Deep Learning Approach for OCT Image Classification	2025	The reported work uses a VGG16 deep learning approach on retinal OCT images to perform four-class disease classification (CNV, DME, DRUSEN, NORMAL), reporting diagnostic evidence via standard evaluation metrics and confusion-matrix analysis on a train/validation/test split.	The study details an OCT preprocessing and augmentation pipeline with VGG16 and reports accuracy plus precision, recall, and F1 across all classes, supporting the feasibility of automated retinal diagnostic support for image-based screening.
[262]: Automated retinal disorders classification: leveraging digital image enhancement techniques and deep learning on OCT Images	2024	This ophthalmology study uses deep learning with transfer learning and data augmentation on retinal OCT images to classify CNV, DME, DRUSEN, and normal, reporting quantitative evidence via test-set classification accuracy.	This investigation evaluates VGG16, ResNet18, and DenseNet with image preprocessing and enhancement, highlighting DenseNet's top test accuracy (92.41%) as support for automated OCT-based retinal disorder recognition.
[47]: Automated segmentation of choroidal neovascularization on optical coherence tomography angiography images of neovascular age-related macular degeneration patients based on deep learning	2023	This research develops a deep learning segmentation model to delineate choroidal neovascularization regions from en face OCT angiography in neovascular age-related macular degeneration, using a modified U-net architecture with ResNeSt and spatial pyramid pooling to address CNV scale variability, and it provides clinical imaging evidence via testing on manually annotated OCTA images.	This analysis introduces and evaluates a ResNeSt-based U-net with multi-scale spatial pyramid pooling to improve automated CNV boundary delineation on OCTA, offering quantitative segmentation results that outperform a standard saliency approach and a baseline U-net.

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Work	Year	Overall reasoning	Contribution and summary
[377]: Automated Segmentation of Retinal Fluid Volumes From Structural and Angiographic Optical Coherence Tomography Using Deep Learning	2020	In a diabetic macular edema setting, the study evaluates a U-Net-like deep CNN (ReF-Net) for volumetric retinal fluid segmentation using structural OCT together with angiographic OCTA on simultaneously acquired 3D OCT/OCTA volumes, reporting agreement-based evidence from manual ground-truth labeling and quantitative segmentation metrics.	This work introduces an OCT/OCTA fusion approach within ReF-Net to improve cystic retinal fluid volume delineation versus structural OCT alone and demonstrates robustness to typical OCT/OCTA shadow artifacts, highlighting volumetric quantification as more informative than 2D projections for macular edema assessment.
[86]: Automated segmentation of the choroid in EDI-OCT images with retinal pathology using convolution neural networks	2017	This 2017 study uses a convolutional neural network with SegNet-style pixel-wise boundary probability maps plus seam carving to segment the choroid on retinal enhanced depth imaging OCT (EDI-OCT) B-scans, evaluated on 62 manually delineated choroid segmentations from patients with age-related macular degeneration to show effectiveness in pathological scans without requiring retinal layer boundary preprocessing.	The research specifically demonstrates a pathology-robust pipeline that directly estimates the Bruch's membrane and choroid-sclera interface as interior and exterior choroid boundaries, enabling accurate choroid segmentation in disease-affected OCT where approaches dependent on retinal boundary detection can degrade.
[340]: Automatic classification of retinal OCT images based on convolutional neural network	2020	The analysis uses a convolutional neural network to automatically classify retinal OCT images, incorporating OCT-focused preprocessing and augmentation, and reports diagnostic-style evidence with accuracy, sensitivity, and specificity from train/validation splits.	This research details an end-to-end CNN pipeline that couples pathological-area centering, noise-reducing cropping, and augmentation via random translation and horizontal flips, achieving reported performance of 93.43% accuracy, 91.38% sensitivity, and 95.88% specificity.
[230]: Automatic Detection and Classification of Diabetic Retinopathy from Optical Coherence Tomography Angiography Images using Deep Learning-A Review	2025	This review focuses on ophthalmic diabetic retinopathy assessment using optical coherence tomography angiography (OCTA) microvascular imaging, surveying deep learning and machine learning approaches for DR detection and classification while summarizing the reported OCTA datasets and evaluation approaches.	The analysis synthesizes recent CNN-centered OCTA DR classification studies and contrasts a proposed deep learning workflow with traditional machine learning pipelines that rely on manually engineered features, underscoring clinical challenges such as data variability, interpretability, and workflow integration.

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Work	Year	Overall reasoning	Contribution and summary
[298]: Automatic detection of microaneurysms in optical coherence tomography images of retina using convolutional neural networks and transfer learning	2022	This ophthalmic deep learning study targets microaneurysm detection from diabetic retinopathy OCT B-scans by converting them into labeled OCT strips using OCT-fluorescein angiography registration, then classifying strips with a stacked ensemble of transfer-learned CNNs for evidence from model evaluation on a hold-out test split.	The reported work provides a practical labeled OCT strip creation workflow for four classes (microaneurysm, normal, abnormal, and vessel) and uses stacking across fine-tuned ImageNet-pretrained networks to improve strip-level discrimination, supporting OCT-based microaneurysm localization without needing fluorescein angiography at inference.
[137]: Automatic diagnosis of macular diseases from OCT volume based on its two-dimensional feature map and convolutional neural network with attention mechanism	2020	This review synthesizes evidence on AI methods for age-related macular degeneration using retinal imaging modalities, particularly OCT and OCT angiography, with comparisons to fundus photography, covering tasks such as classification, segmentation, and prediction, based on findings from published studies and summarized performance reported across datasets.	The study organizes the literature by AI approach and ophthalmic objective, highlighting practical considerations for OCT-based AMD analysis (including interpretability and data quality limits) and delineating gaps in generalizability that affect real-world deployment.
[347]: AUTOMATIC METHOD OF MACULAR DISEASES DETECTION USING DEEP CNN-GRU NETWORK IN OCT IMAGES	2024	Using OCT B-scan images from the labeled OCT Retina dataset, this analysis develops a Deep CNN-GRU architecture that performs four-class macular disease recognition (drusen, choroidal neovascularization, diabetic macular edema, and normal), supported by reported accuracy, precision, recall, and F1 score with repeat experiments, LOOCV, and comparison to prior deep learning baselines.	The paper's main advance is the use of GRU units connected to a CNN feature extractor to improve OCT-based classification performance for early and late macular disease states, yielding strong results on the established dataset and positioning the method among the higher-performing approaches in the study's state-of-the-art comparison.
[268]: Automatic Retinal and Choroidal Boundary Segmentation in OCT Images Using Patch-Based Supervised Machine Learning Methods	2019	In SD-OCT, this research evaluates patch-based supervised deep learning, using convolutional neural networks and recurrent neural networks to predict per-layer probability maps for retinal and choroidal boundary locations, then derives boundary positions through graph-search tracing and compares results with manual and prior image-analysis methods.	This analysis provides a practical framework for automated retinal and choroidal boundary localization that reports how CNN versus RNN choice, patch size, and intensity compensation affect probability maps and the resulting thickness-related boundary tracing.
[352]: Automatic segmentation of multitype retinal fluid from optical coherence tomography images using semisupervised deep learning network	2023	The study addresses an ophthalmology imaging task by using a semisupervised deep learning network to segment subretinal and intraretinal fluid on OCT, with performance assessed using quantitative and qualitative analyses, component ablations, and cross-dataset testing on an unseen Kermany dataset.	This work reports a coarse-to-refine Ref-Net that can delineate multitype retinal fluid with limited labeled OCT images, providing evidence that expert-level segmentation and reasonable generalization are achievable when training data are scarce.

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Work	Year	Overall reasoning	Contribution and summary
[432]: Automatic segmentation of nine retinal layer boundaries in OCT images of non-exudative AMD patients using deep learning and graph search	2017	This work evaluates a CNN-guided graph search (CNN-GS) for automated segmentation of nine retinal layer boundaries in spectral-domain OCT B-scans from non-exudative AMD eyes, validated against expert/manual corrected annotations in a held-out test set.	The research details an end-to-end pipeline that converts CNN-derived probability maps into refined layer boundaries via modified GTDP graph search, supporting practical AI-assisted OCT layer segmentation without relying on many ad hoc rules.
[295]: Automatic segmentation of OCT retinal boundaries using recurrent neural networks and graph search	2018	In an ophthalmology OCT setting, the analysis proposes a recurrent neural network patch classifier integrated with graph search (RNN-GS) to segment retinal layer boundaries in cross-sectional B-scan images and evaluates boundary-position errors against expert-corrected labels in healthy children and patients with AMD, with comparisons to CNN-based graph-search methods.	The authors replace a CNN with an RNN within a graph-search segmentation pipeline and show competitive, sometimes marginally improved boundary localization accuracy and consistency, providing an evidence-backed approach for delineating multiple OCT retinal boundaries in both normal and AMD data.
[119]: Automatic Segmentation of Retinal Fluid and Photoreceptor Layer from Optical Coherence Tomography Images of Diabetic Macular Edema Patients Using Deep Learning and Associations with Visual Acuity	2022	This deep learning study in diabetic macular edema uses a modified U-Net with an EfficientNet-B5 encoder to automatically segment OCT biomarkers: intraretinal cystoid fluid, subretinal fluid, and ellipsoid zone structure. It evaluates these outputs against manually labeled ground truth and uses ROC-based detection of ellipsoid zone disruption to support clinical interpretation.	The analysis further links AI-derived biomarker quantification to visual function by modeling associations with logMAR best-corrected visual acuity, showing that ellipsoid zone disruption has the strongest relationship and thereby supporting OCT-derived severity assessment beyond central subfield thickness.
[358]: Breaking the Mold: ViT-CNN Fusion for Enhanced Glaucoma Prediction in OCT Images	2025	The research targets an ophthalmic glaucoma prediction task using OCT imaging, and evaluates a ViT-CNN fusion model that combines localized CNN feature extraction with global Vision Transformer context followed by ML-based classification, with evidence reported from experiments across four datasets showing improved results versus single-model baselines.	The reported work advances an automated glaucoma detection pipeline by demonstrating that fusing CNN and ViT representations improves OCT image feature richness and supports more reliable classification, further supported by GRAD-CAM visual evidence of the model's attention patterns.

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Work	Year	Overall reasoning	Contribution and summary
[375]: CATALYZE: a deep learning approach for cataract assessment and grading on SS-OCT images	2025	This prospective single-center study evaluates a deep learning cataract grading pipeline from Anterion swept-source OCT (SS-OCT) that outputs layerwise metrics and a clinical significance index, then compares its discrimination with an established optical ocular scatter index using correlation and ROC analyses across development and validation cohorts.	CATALYZE yields an SS-OCT-derived cataract clinical significance index that closely tracks the ocular scatter index and shows comparable validation AUROC, supporting an objective, image-based approach to cataract assessment tied to optical scattering measures.
[443]: Circumpapillary OCT-focused hybrid learning for glaucoma grading using tailored prototypical neural networks	2021	This analysis develops a deep learning-driven glaucoma grading model from raw circumpapillary spectral-domain OCT B-scans, using a hybrid learning strategy that fuses an RNFL-focused OCT descriptor with a residual-attention CNN and supervised prototypical (few-shot/k-shot) classification, with evidence from training and evaluation on Heidelberg Spectralis circumpapillary B-scan datasets that report categorical glaucoma grading results and clinically consistent localization via class activation maps.	This investigation further refines glaucoma severity modeling by formulating static and dynamic tailored prototypical networks on top of an OCT-specific hybrid backbone that incorporates handcrafted RNFL thickness histogram features, enabling discrimination among healthy, early, and advanced glaucoma while avoiding reliance on a conventional projection-head classification stage.
[284]: Classification and Prediction Retinal Oct Images by CNN Algorithm	2022	The study describes a MATLAB-based convolutional neural network trained on well-lit ophthalmic fundus images to classify and predict diabetic retinopathy status for early detection and stage-related characterization, reporting classification metrics from a single training/testing split.	This analysis provides an end-to-end CNN pipeline for automated diabetic retinopathy screening using extracted fundus image features and reports high diagnostic performance, offering quantitative support for image-driven triage in an ophthalmic monitoring context.
[353]: Classification and Quantification of Retinal Cysts in OCT B-Scans: Efficacy of Machine Learning Methods	2019	The study addresses an ophthalmic OCT task by using an end-to-end machine learning approach that combines a random forest classifier with a DeepLab-style segmentation algorithm to detect, label, and quantify retinal fluid compartments in B-scans of highly distorted florid disease, with evidence reported as agreement versus expert manual delineations using an average Dice score.	This work reports an automated segmentation and quantification pipeline for retinal fluid in OCT B-scans, highlighting clinically relevant measurement capability in challenging cases through Dice-based performance relative to trained expert labeling.

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Work	Year	Overall reasoning	Contribution and summary
[260]: Classification of diabetes-related retinal diseases using a deep learning approach in optical coherence tomography	2019	This 2019 ophthalmology study uses an OCT-NET convolutional neural network to classify diabetes-related retinal disease categories from spectral-domain optical coherence tomography B-scan volumes, with evidence from evaluation on public SD-OCT datasets that report quantitative performance and include clinician-oriented class activation map localization for interpretability.	The research extends end-to-end SD-OCT volume classification by adding a CAM-based feedback stage that highlights diagnostically relevant regions within each scan, helping specialists understand and review the basis of model predictions beyond the disease label alone.
[111]: Classification of Fluid Filled Abnormalities In Retinal Optical Coherence Tomography Images Using Chicken Swarm Optimized Deep Learning Classifier	2023	In an ophthalmic OCT context, the study evaluates a deep learning pipeline for classifying fluid-filled retinal abnormalities using CLAHE-preprocessed OCT images, retinal-layer segmentation, Gabor-based feature extraction, and an RNN classifier optimized with chicken swarm optimization, reporting accuracy and processing time from dataset-based MATLAB validation.	This research improves OCT-based identification of fluid-related retinal findings by coupling segmentation-guided feature selection with CSO-trained RNN classification, and reports accuracy and reduced processing time to support clinical feasibility.
[304]: Classification of range of OCT-angiography capillary density using multichannel deep learning models in diabetic retinopathy, aging macular degeneration, and radiation retinopathy	2025	The study describes a nnU-Net-based deep learning approach for segmenting OCT-angiography capillary density into four severity categories in retinal vascular disease, using OCTA (alone and with FAZ and/or large vessel tree) and evaluating agreement with expert annotations via Dice coefficient comparisons.	The analysis compares four multi-input configurations to determine how adding FAZ and large vessel tree information affects consistency of the categorized capillary-density maps, highlighting the three-channel OCTA plus FAZ plus large vessel tree model as a more objective option than OCTA-only labeling.
[315]: Classification of Retinal Diseases in Optical Coherence Tomography Images Using Artificial Intelligence and Firefly Algorithm	2023	This 2023 study addresses retinal disease classification from optical coherence tomography images by combining retinal layer extraction with a hybrid feature pipeline that merges classic texture/shape descriptors and transfer-learned deep features, then applies Firefly algorithm feature selection and hierarchy (multi-binary) classification using SVM, logistic regression, and random forest on two public OCT datasets, with mean accuracy reported for the resulting classifiers.	The reported work adds an efficiency-oriented workflow that reduces the high-dimensional fused feature set via Firefly-based selection and improves training feasibility while maintaining diagnostic performance across the abnormal versus normal and AMD-related binary splits, highlighting how preprocessing, hybrid features, and hierarchical multi-binary decisions work together to support automated OCT-based screening.

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Work	Year	Overall reasoning	Contribution and summary
[235]: Classification of Retinal OCT Images Using Deep Learning	2022	The reported work applies a convolutional neural network to classify retinal disorder categories from optical coherence tomography (OCT) cross-sectional images, using the ophthalmic imaging modality to address automated diagnostic image classification with reported performance on a retinal OCT dataset.	The study reports an OCT-based multi-class classification approach that uses a lightweight deep learning model built on VGG16 to assign images to eight disease states plus Normal Eye, supporting its use for early identification of retinal pathology from OCT.
[191]: CNN-Hyperparameter Optimization for Diabetic Maculopathy Diagnosis in Optical Coherence Tomography and Fundus Retinography	2022	This analysis evaluates a CNN-based accept-or-reject classifier for diabetic maculopathy using retinal imaging modalities (OCT and fundus retinography) and focuses on improving performance through hyperparameter optimization with Bayesian search, providing evidence from an end-to-end deep learning diagnostic pipeline rather than a purely algorithmic benchmark.	This investigation specifically details how Bayesian optimization is used to tune CNN hyperparameters and design choices for diabetic maculopathy detection across OCT and fundus retinography, offering a practical optimization strategy that can improve automated retinal screening performance.
[58]: CNN Layer-wise explanations for the prediction of Diabetic Retinopathy from OCTA images	2025	This research evaluates an explainable CNN for diabetic retinopathy severity classification using OCTA imaging in an ophthalmic diagnostic context, leveraging Signature Activation to produce holistic, class-agnostic explanations of the model's predictions.	This analysis applies a layer-wise explanation method to clarify what the CNN attends to when grading OCTA-derived DR severity, supporting clinician interpretability needs for AI-driven retinal screening.
[429]: Comparative Analysis of Deep Learning Architectures for Macular Hole Segmentation in OCT Images: A Performance Evaluation of U-Net Variants	2025	This OCT macular hole segmentation study evaluates multiple U-Net variants that swap in different CNN and Transformer-style encoders, using the OIMHS dataset and reporting overlap and boundary evidence via Dice coefficient and HD95 to compare architectural families on the MH task.	This work provides a structured, metric-based architectural comparison that identifies InceptionNetV4 plus U-Net as the top-performing option on the reported Dice scores while highlighting unreliable HD95 behavior for small MH regions and contrasting weaker Transformer-based results for MH and intraretinal cyst segmentation.
[177]: Comparing code-free deep learning models to expert-designed models for detecting retinal diseases from optical coherence tomography	2024	This ophthalmology study compares code-free deep learning and clinician-built AutoML models with expert-designed bespoke deep learning models for multiclass retinal pathology detection from OCT macular videos and fovea-centered images, using an internal dataset and reporting point-estimate discriminative performance metrics.	The research shows that trainee-developed code-free models can match expert performance on the same OCT-based classification task, supporting practical democratization of deep learning for retinal disease screening without coding expertise.

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Work	Year	Overall reasoning	Contribution and summary
[392]: Comparison of Autonomous AS-OCT Deep Learning Algorithm and Clinical Dry Eye Tests in Diagnosis of Dry Eye Disease	2021	In this ophthalmology study, a deep learning classification model autonomously detects dry eye disease from anterior segment optical coherence tomography images, with diagnostic agreement assessed against masked cornea specialist diagnoses and compared to standard clinical dry eye tests (OSDI, TBUT, corneal and conjunctival staining, and Schirmer's test).	The authors provide a head-to-head evaluation showing that AS-OCT deep learning performs significantly better than several conventional tests and is comparable to others, supporting the feasibility of autonomous imaging-based DED diagnosis for clinic-relevant screening.
[193]: Comparison of Different Machine Learning Classifiers for Glaucoma Diagnosis Based on Spectralis OCT	2021	This glaucoma diagnostic study evaluates several classical machine learning classifier families on Heidelberg Spectralis spectral-domain OCT measurements (cRNFL thickness, BMO-MRW, ETDRS macular thickness, and PPAA), targeting discrimination of normal versus glaucomatous eyes and stratified early, moderate, and severe stages, with evidence provided through cross-validated/bootstrapped model testing and comparative performance against logistic regression.	The analysis identifies random forest as the top-performing approach across glaucoma subsets and uses model-derived variable-importance rankings to pinpoint which OCT-derived ganglion cell layer, cRNFL, and BMO-MRW features and locations matter most as disease severity increases.
[331]: Context-aware diagnosis of age-related macular degeneration (AMD) stages: a unified approach using self-supervised U-Net and graph neural networks on OCT imaging	2025	The research targets context-aware staging of age-related macular degeneration from optical coherence tomography using a self-supervised U-Net to segment superpixels and graph neural networks to model spatial and structural relationships, providing quantitative diagnostic evidence via reported accuracy, F1-score, and AUC.	The reported work introduces a unified, self-supervised pretext inpainting training scheme with a context-aware loss that links superpixel interdependencies, and reports improved AMD stage classification over baselines, supporting the clinical relevance of OCT-based, relation-informed modeling.
[118]: Convolution neural network model for predicting various lesion-based diseases in diabetic macula edema in optical coherence tomography images	2023	The study describes a lesion-based convolutional neural network that uses OCT retinal images to detect and distinguish DME-related lesions (including hemorrhages, microaneurysms, and exudate), evaluated against established CNN architectures as evidence from an ophthalmic imaging diagnosis task.	The study introduces an OCT lesion-focused CNN pipeline aimed at improving DME lesion prediction and reports higher accuracy than standard CNN baselines such as ResNet, VGG16, Inception, and AlexNet, supporting more effective automated OCT interpretation.

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Work	Year	Overall reasoning	Contribution and summary
[122]: CorneaNet: fast segmentation of cornea OCT scans of healthy and keratoconic eyes using deep learning	2019	This 2019 study develops CorneaNet, a supervised deep fully-convolutional neural network that segments corneal OCT B-scans into epithelium, Bowman's layer, stroma, and background for both healthy and keratoconic (keratoconus) eyes, with validation performed using six-fold cross-validation and layer-wise segmentation with thickness-map outputs.	CorneaNet delivers fast, automated layer segmentation and corresponding corneal thickness mapping while substantially reducing runtime versus the authors' prior iterative robust-fitting algorithm, supporting practical, OCT-based workflows for keratoconus assessment and early detection.
[414]: Cross-instrument optical coherence tomography-angiography (OCTA)-based prediction of age-related macular degeneration (AMD) disease activity using artificial intelligence	2024	This retrospective ophthalmic study evaluates DNN-based multiclass prediction of AMD disease activity from 2D vascular en-face OCTA images using cross-instrument training data from Heidelberg Spectralis and Optovue Solix, with labels derived from clinical diagnoses and contemporaneous B-scan OCT fluid to test generalizability across devices via separate test sets.	By training EfficientNet-b5-based classifiers on combined Heidelberg and Optovue OCTA vascular morphology, the authors demonstrate cross-manufacturer feasibility for activity classification and quantify how performance varies by training source, supporting practical deployment considerations for OCTA-based AMD activity grading.
[80]: CSANet: a lightweight channel and spatial attention neural network for grading diabetic retinopathy with optical coherence tomography angiography	2024	This ophthalmology study proposes CSANet, an attention-based CNN for fine-grained diabetic retinopathy grading using OCTA en-face images, tackling small-lesion visibility and adjacent-grade confusion, and evaluates the approach with benchmark experiments on OCTA-DR and DRAC 2022 classification evidence.	This work introduces a lightweight hybrid attention module combining channel and spatial attention to better capture lesion-relevant OCTA features and reduce missed small findings, supported by strong grading results across the reported datasets and module ablations.
[457]: Cynomolgus monkey's retina volume reference database based on hybrid deep learning optical coherence tomography segmentation	2023	Using spectral-domain OCT from healthy cynomolgus monkeys, the study applies deep learning semantic retinal segmentation plus classical computer-vision foveolar landmark detection to compute retinal volumes across predefined zones, quadrants, and slices, providing evidence for natural variation and analyzing effects of sex, geographic origin, and eye laterality using a large retrospective dataset of 374 eyes from 203 animals.	The research establishes an OCT-derived macaque retinal volume reference database and quantifies how origin and sex systematically shift foveolar subfield volume estimates, which is critical for interpreting future OCT-based preclinical measurements in light of biologic baseline differences.

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Work	Year	Overall reasoning	Contribution and summary
[115]: DAISY Descriptors Combined with Deep Learning to Diagnose Retinal Disease from High Resolution 2D OCT Images	2020	In this ophthalmic study, deep learning is used to classify high-resolution 2D spectral-domain OCT B-scans for four retinal conditions using DAISY-descriptor-driven intelligent downsampling followed by a GRU, with evaluation on a publicly available dataset that includes independent testing with bootstrapped confidence intervals.	The authors show that replacing random downsampling with DAISY-based feature-preserving reduction enables effective OCT disease classification from descriptors, improving test accuracy and AUC over an InceptionV3 baseline.
[318]: Deep CNN framework for retinal disease diagnosis using optical coherence tomography images	2021	This investigation uses a deep convolutional neural network to perform automated retinal disease detection and multi-class classification (Normal, DMD, DME) from optical coherence tomography images, incorporating speckle denoising and hyperparameter tuning, and evaluates the approach on labeled OCT data with K-fold validation using reported accuracy as evidence of performance.	The analysis provides a fully automated end-to-end OCT workflow that combines Kuan-filter despeckling with a tuned CNN and a comparison with pre-trained models, and reports 95.7% classification accuracy, which supports the practicality of CNN-based AI for early identification of DMD and DME from OCT.
[292]: DEEP CONVOLUTIONAL NEURAL NETWORK PREDICTION FOR GLAUCOMA DETECTION USING OCT AND OCT-ANGIOGRAPHY DISC- AND MACULA-CENTERED IMAGES AND THEIR COMBINED POWER	2024	This cross-sectional ophthalmic deep learning study used a deep CNN intermediate fusion framework on en-face OCT and OCTA disc- and macula-centered images to classify eyes as healthy versus glaucomatous, evaluating discriminative evidence using standard accuracy and AUC metrics (with additional sensitivity and specificity).	The study reports that combining disc and macula regions improves binary glaucoma detection across modality pairings, while OCTA alone shows consistently superior sensitivity on disc-and-macula inputs, offering practical guidance on which OCT/OCTA image sets to prioritize for early diagnosis.
[210]: Deep learning algorithms for detection of diabetic macular edema in OCT images: A systematic review and meta-analysis	2023	This systematic review and diagnostic test accuracy meta-analysis evaluates deep learning neural networks, predominantly convolutional architectures, using OCT and fundus retinal imaging to detect diabetic macular edema, synthesizing pooled evidence on diagnostic performance and heterogeneity across included studies.	The study aggregates results from 53 studies to quantify overall sensitivity and report stronger detection performance for OCT versus fundus imaging, clarifying the evidence base for OCT-guided AI screening and clinical decision support for DME.

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Work	Year	Overall reasoning	Contribution and summary
[228]: Deep Learning and OCT Imaging: A Novel Ensemble Approach for Eye Disease Diagnosis	2026	Using retinal OCT imaging, the study describes a deep learning ensemble of three CNNs (VGG16, InceptionV3, and InceptionResNetV2) for automated ocular disease classification and reports high classification accuracy.	This investigation proposes an OCT-focused inference approach that combines heterogeneous CNNs and reuses preexisting architectures to improve disease identification performance on an OCT dataset.
[418]: Deep learning and optical coherence tomography in glaucoma: Bridging the diagnostic gap on structural imaging	2022	This 2022 systematic review covers ophthalmic deep learning approaches for glaucoma diagnosis and progression, focusing on models trained on structural OCT (including optic nerve and macular scans and derived OCT maps) and extending OCT-trained learning to fundus photography to support detection in broader clinical screening contexts.	This analysis consolidates a decade of OCT- and fundus-based deep learning strategies, detailing how OCT-derived ground truth supports automated glaucoma assessment and highlighting practical evidence-based gaps and future directions for translation and research use.
[343]: Deep Learning Approaches for Retinal Disease Diagnosis: Insights from Fundus and OCT Analysis	2025	The study benchmarks deep learning for glaucoma and diabetic retinopathy on clinically relevant retinal imaging, using fundus photographs alongside 2D and 3D OCT inputs and reporting task-based diagnostic classification evidence with standard metrics (accuracy, precision, recall, specificity, and F1-score).	This work provides a modality-specific comparison of prominent architectures, highlighting transformer family models (and a ViT-large plus GRU hybrid for 3D OCT) together with EfficientNet results for 2D OCT, to inform which model families may be better suited for retinal disease screening across imaging types.
[200]: Deep Learning Approaches Predict Glaucomatous Visual Field Damage from OCT Optic Nerve Head En Face Images and Retinal Nerve Fiber Layer Thickness Maps	2020	This evaluation tests deep learning on spectral-domain OCT optic nerve head en face images and RNFL thickness maps to classify eyes with and without glaucomatous visual field damage and to predict quantitative functional outcomes from visual field testing, providing independent-test diagnostic and regression evidence (AUC, sensitivity/specificity, R2, MAE) in an ophthalmic glaucoma setting.	The research reports that models using RNFL en face representations outperform mean RNFL thickness measurements for detecting glaucomatous visual field damage and for estimating VF mean deviation, pattern standard deviation, and sectoral pattern deviation severity from SD OCT data, supporting improved interpretation of structural imaging for functional risk stratification.
[179]: Deep learning approaches to predict 10-2 visual field from wide-field swept-source optical coherence tomography en face images in glaucoma	2022	This glaucoma study uses a convolutional deep learning approach based on Inception-ResNet-V2 to predict 10-2 visual field threshold values from wide-field swept-source OCT en face-derived thickness-map inputs, with performance evaluated against ground-truth 10-2 perimetry in a held-out test set.	By contrasting an en face-based input pipeline with an RNFL thickness-based alternative (both combined with macular mGC/IPLT maps), the authors show that the en face model yields lower sector-specific prediction errors and supports clinically relevant central VF estimation from SS-OCT imaging.

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Work	Year	Overall reasoning	Contribution and summary
[204]: Deep learning architectures analysis for age-related macular degeneration segmentation on optical coherence tomography scans	2020	This investigation uses deep learning CNN-based semantic segmentation on OCT scans to delineate three AMD-associated retinal fluid types, applying patch-based training on the RETOUCH challenge data and testing generalization via OPTIMA, with evidence reported using Dice similarity metrics versus human performance.	The analysis compares four widely used segmentation architectures and shows that patching plus fine-tuning improves dense fluid delineation, yielding high mean Dice scores (up to 0.85) that support more accurate OCT-based AMD fluid segmentation.
[361]: Deep Learning Architectures for OCT Images Retinal Disease Classification	2025	The research evaluates deep learning classifier architectures using optical coherence tomography (OCT) images to predict retinal disease classes, testing architectural variations (e.g., layers, kernel sizes, pooling strategies) and training settings, with quantitative metric comparisons that provide evidence for OCT-based diagnostic decision support.	The reported work identifies OCTDeepNet2 as the best-performing configuration under the tested layer and training conditions, offering practical guidance on which architecture and optimization settings may improve retinal disease classification accuracy from OCT scans.
[171]: Deep Learning-Assisted Detection of Glaucoma Progression in Spectral-Domain OCT	2023	This retrospective cohort ophthalmic study trains a convolutional neural network to estimate the probability of glaucoma structural progression from baseline and follow-up spectral-domain OCT RNFL thickness, with performance reported using AUC, sensitivity, and specificity and benchmarked against conventional trend-based change methods.	The study demonstrates that incorporating time between OCT visits into the DL framework yields better diagnostic discrimination than trend-based analyses and provides likelihood ratios that can substantially shift post-test progression probability.
[433]: Deep learning-based algorithm for the detection of idiopathic full thickness macular holes in spectral domain optical coherence tomography	2024	This 2024 cross-sectional ophthalmology study evaluates a Zeiss deep learning "B-scans of interest" model on spectral-domain OCT B-scans to detect idiopathic full-thickness macular hole features (and relate outputs to IFTMH severity), using labeled B-scan-level ground truth from two graders and a severity stage adjudicated per International Vitreomacular Traction Study Group, with evidence provided by internal test-set detection performance and a probability-stage correlation.	The authors validate automated abnormal B-scan detection for IFTMH across two test sets derived from different grader-agreement labeling rules, showing high accuracy for identifying abnormal scans while demonstrating only a weak relationship between the model's probability scores and cube-level severity stages, highlighting the need for task-specific training to support severity stratification.

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Work	Year	Overall reasoning	Contribution and summary
[293]: Deep learning-based automated detection of retinal diseases using optical coherence tomography images	2019	This 2019 study develops a computer-aided diagnosis deep learning framework for multiclass retinal disease recognition on optical coherence tomography B-scans, using an ensemble of four improved ResNet50 models, and reports patient-level 10-fold cross-validation with evaluation against ophthalmologists plus qualitative interpretability via occlusion testing and analysis of misclassifications.	This analysis delivers four-class classification of CNV, DME, drusen, and normal from OCT with validation on an independent testing set and additional external dataset checks, while also examining how integrating OCT with patient medical history affects sensitivity and specificity and showing occlusion-derived regions consistent with clinically relevant pathology.
[77]: Deep Learning-Based Automatic Detection of Ellipsoid Zone Loss in Spectral-Domain OCT for Hydroxychloroquine Retinal Toxicity Screening	2021	In a retrospective, single-center case-control ophthalmic study for hydroxychloroquine retinal toxicity screening, the authors used a dual mask region-based convolutional neural network on spectral-domain OCT to detect and quantify ellipsoid zone loss from B-scans and reconstruct an en face EZ-loss map, with 10-fold cross-validation performance evaluated against human ground-truth annotations and AAO guideline-based toxicity determinations.	They introduce a combined projection architecture that aggregates parallel horizontal and vertical scan detections to produce more robust en face EZ-loss quantification, enabling fast objective metrics that closely match human grading and support AAO-recommended toxicity screening.
[95]: Deep learning-based classification and segmentation of retinal cavitations on optical coherence tomography images of macular telangiectasia type 2	2022	This work in ophthalmology builds a two-stage convolutional neural network to classify and segment retinal cavitations on spectral-domain OCT B-scans from MacTel2 eyes, using manual segmentations as the reference standard and evaluating the method with reported detection, overlap, and volume-change agreement across longitudinal time points in a clinical trial dataset.	The research advances automated cavitation quantification for MacTel2 by adding a B-scan prescreening classifier before U-Net-based segmentation, improving segmentation accuracy and enabling highly correlated retinal cavitation volume and longitudinal change estimates relative to reader measurements for clinical research use.
[374]: Deep learning-based classification of eye diseases using Convolutional Neural Network for OCT images	2023	This original study applies a CNN with transfer learning and fine-tuning to publicly available OCT retinal images for multi-class ophthalmic disease classification, targeting early detection of diabetic macular edema, choroidal neovascular membranes, and age-related macular degeneration versus normal retina using diagnostic evidence reported from model testing.	This research delivers a four-class OCT grading pipeline that couples OCT-specific preprocessing (brightness/contrast adjustment, multiscale retinex enhancement, and Gaussian blur) with a VGG16-based CNN to improve discrimination among the three disease categories and normal, supported by reported classification performance before and after fine-tuning.

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Work	Year	Overall reasoning	Contribution and summary
[121]: Deep learning-based detection of diabetic macular edema using optical coherence tomography and fundus images: A meta-analysis	2023	This systematic review and meta-analysis evaluates deep learning-based detection of diabetic macular edema in ophthalmology using optical coherence tomography and fundus imaging, synthesizing sensitivity evidence from 53 studies and summarizing model performance reported with established diagnostic metrics (e.g., AUROC and sensitivity).	Across large volumes of OCT B-scans and fundus images, it pools results to quantify overall detection sensitivity for OCT versus fundus and discusses risk-of-bias patterns, helping readers interpret evidence strength and relative modality utility for AI-enabled DME screening and referral.
[175]: Deep Learning-Based Detection of Reticular Pseudodrusen in Age-Related Macular Degeneration on Optical Coherence Tomography	2024	The research develops a deep learning model for OCT B-scan-based segmentation of reticular pseudodrusen in age-related macular degeneration, with evaluation that includes expert comparisons and external testing on multiple held-out datasets to substantiate segmentation accuracy and clinical usability.	The reported work delivers an automatically quantifiable RPD segmentation pipeline with higher agreement than human specialists and provides publicly available code, enabling more reliable identification of RPD on OCT for downstream clinical and research workflows.
[108]: Deep Learning-Based Diagnosis of Corneal Condition by Using Raw Optical Coherence Tomography Data	2025	This ophthalmic study applies modified convolutional neural networks to raw anterior-segment OCT corneal images from the Casia2 device to classify each eye into normal, ectasia (keratoconus), or other corneal disease, with diagnostic performance reported on a labeled test set and supportive evidence from Grad-CAM visualizations.	The study demonstrates that stacking sixteen unprocessed meridional Casia2 OCT slices as a multi-channel input enables accurate three-class corneal condition classification and supports clinical usability via a GUI that operates directly on raw.3dv files without relying on preprocessing.
[367]: Deep Learning-based Diagnosis of Glaucoma Using Wide-field Optical Coherence Tomography Images	2021	This retrospective ophthalmic deep-learning study evaluates convolutional neural network-based detection and five-stage classification of glaucoma from wide-field swept-source OCT imaging, comparing DL fusion strategies against conventional retinal thickness parameters using reported diagnostic performance measures.	This investigation shows that two DL feature-fusion architectures (FCN and FFC) applied to multiple wide-field SS-OCT image types can improve glaucoma discrimination, including early disease, beyond thickness-based conventional metrics, while also providing stage-specific confusion-matrix results.
[404]: Deep learning based eye disease classification using Optical Coherence Tomography (OCT) images	2026	Using OCT imaging, the study applies and compares multiple CNN model families (VGG16, VGG19, ResNet50, InceptionV3) for four-class retinal disease classification (CNV, DME, drusen, normal) and evaluates them with internal testing plus an independent external validation set, providing evidence on generalizability for an ophthalmic diagnostic task.	This analysis identifies VGG19 as the most stable performer across preprocessing and latent-feature SMOTE for class imbalance, and it quantifies how performance drops on external data, which is useful for judging readiness and dataset-diversity needs before clinical deployment.

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Work	Year	Overall reasoning	Contribution and summary
[279]: Deep Learning Based Fluid Segmentation in Retinal Optical Coherence Tomography Images	2019	The study uses deep learning with an attention mechanism and dense skip connections to segment retinal fluid in OCT imaging for macular edema, targeting an ophthalmic imaging task that supports quantitative disease monitoring using results reported on a public dataset and described as adaptable across different scanning devices.	This work introduces a single-stage, parameter-efficient attention-based segmentation framework with joint losses to automatically delineate macular edema fluid regions on OCT, emphasizing improved segmentation accuracy and cross-device adaptability for reducing the burden of manual labeling.
[208]: Deep learning-based image quality assessment for optical coherence tomography macular scans: a multicentre study	2024	This multicenter, externally tested deep learning study evaluates imaging-quality assessment for 3D OCT macular scans acquired on Cirrus and Spectralis platforms, using ResNet-18-based architectures to classify scans as gradable versus ungradable across internal and multiple unseen sites, providing evidence to support inclusion of an image-quality gating step prior to downstream automated retinal disease detection.	The research delivers a robust, device-specific gradability classifier for filtering ungradable 3D OCT macular volumes and demonstrates how this quality screen can be integrated into an end-to-end automated workflow for more dependable ophthalmic analysis.
[212]: Deep learning based joint segmentation and characterization of multi-class retinal fluid lesions on OCT scans for clinical use in anti-VEGF therapy	2021	This ophthalmic OCT study develops and evaluates an end-to-end deep learning model (RFS-Net) that jointly segments and recognizes three retinal fluid lesion types (IRF, SRF, PED) from multi-vendor OCT images for anti-VEGF therapy dosing and activity assessment, with evidence coming from training and validation using publicly available datasets and performance reported with quantitative metrics.	This research specifies an automated, multi-class retinal fluid segmentation approach trained on OCT from Topcon, Cirrus, and Spectralis to reduce reliance on labor-intensive manual delineation and thereby support more consistent lesion characterization relevant to treatment planning.
[289]: Deep-learning based multiclass retinal fluid segmentation and detection in optical coherence tomography images using a fully convolutional neural network	2019	In retinal OCT, the study applies a fully convolutional neural network to multiclass fluid segmentation and detection by using OCT intensity together with retinal layer segmentations from a graph-cut step, with evidence from MICCAI RETOUCH challenge performance reported for both segmentation and detection tasks.	The analysis introduces an end-to-end OCT fluid labeling pipeline that adds random-forest classification to identify and reject falsely labeled fluid regions, and this design is supported by challenge results that quantify both segmentation accuracy and detection discrimination.

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Work	Year	Overall reasoning	Contribution and summary
[330]: Deep Learning-Based Neuro Computing to Classify and Diagnosis of Ophthalmology by OCT	2025	The research evaluates deep learning for ophthalmic analysis of OCT images, aiming to screen for retinal disease, classify lesion types, and localize lesions using labeled OCT datasets and reported testing on an independent patient group.	The study reports model performance for binary lesion presence detection and multiclass multilabel identification of age-related macular degeneration and vitreomacular traction syndrome, with additional lesion-localization results that support automated screening and diagnostic assistance from OCT.
[243]: Deep Learning Based on Ensemble to Diagnose of Retinal Disease using Optical Coherence Tomography	2021	Using retinal cross-sectional optical coherence tomography images, the study describes an ensemble transfer-learning approach that fuses class-probability outputs from MobileNetV3Large and ResNet50 (with CLAHE pre-processing) to classify five retinal diseases, supported by 5-fold cross-validation and early stopping.	The study details an end-to-end OCT diagnostic pipeline that enhances image contrast and improves prediction stability through probability-level ensemble fusion, offering a practical deep learning strategy for automatic retinal disease classification across five conditions.
[258]: Deep learning-based optical coherence tomography and retinal images for detection of diabetic retinopathy: a systematic and meta analysis	2025	This systematic review and meta-analysis evaluates ophthalmic deep learning models for diabetic retinopathy detection using OCT and color retinal imaging, pooling diagnostic accuracy evidence (from 47 studies, with 10 contributing to meta-analysis) to quantify performance relevant to clinical screening decisions.	This investigation consolidates results across large numbers of retinal images and OCT scans to produce pooled diagnostic odds ratios and summary sensitivity and specificity estimates, providing a quantitative synthesis of how AI-assisted interpretation of OCT/retinal data performs versus standard care.
[425]: Deep Learning-Based Prediction of Visual Field Mean Deviation from Numeric OCT Data in Glaucoma	2026	This retrospective glaucoma study trained deep learning models on numeric macular GCC thickness exported from spectral-domain OCT to predict HFA 30-2 mean deviation in a regression setting and to classify visual field deterioration at predefined MD thresholds using fivefold cross-validation.	This analysis demonstrates a clinically oriented approach that avoids image rendering by feeding raw 512x128 OCT thickness arrays into multiple CNN and vision-transformer architectures, yielding robust mean deviation estimation and threshold-based discrimination with Grad-CAM-based qualitative explanations.
[393]: Deep learning based retinal hard exudates quantification of optical coherence tomography	2025	This 2025 study develops a modified U-Net (DUCK-Net) deep learning model to segment retinal hard exudates on optical coherence tomography B-scans from diabetic retinopathy and branch retinal vein occlusion, and evaluates it with test-set segmentation accuracy plus OCT-derived hard exudate area and volume compared against independent manual measurements, including a 2D projection for ophthalmic visualization.	This work delivers device-agnostic hard exudate quantification from OCT by combining pixel-wise segmentation with volumetric area and volume readouts, reducing reliance on time-consuming manual grading and improving structural depiction compared with fundus photograph-based assessment.

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Work	Year	Overall reasoning	Contribution and summary
[378]: Deep Learning-Based Retinal Layer Segmentation in Optical Coherence Tomography Scans of Patients with Inherited Retinal Diseases	2025	This work uses a U-Net-based deep learning model with adaptive batch normalization domain adaptation to segment the outer nuclear layer on spectral-domain OCT B-scans from patients with inherited retinal diseases, producing retinal and ONL thickness maps with performance evaluated on a test dataset using Dice score metrics.	The research demonstrates reliable ONL segmentation in pathologically altered IRD OCT where existing tools fail and supplies automatically generated full-retina and ONL thickness maps that can support structural assessment for structure-function and treatment-outcome studies.
[286]: Deep learning based retinal OCT segmentation	2019	The analysis evaluates an FCN-based deep learning approach with Gaussian-process post-processing for segmenting spectral-domain retinal OCT imagery, using mild non-proliferative diabetic retinopathy data, and reports segmentation accuracy against clinician results and other machine-learning baselines.	This research presents a tailored deep learning plus Gaussian-process pipeline for OCT retinal segmentation on a University of Miami dataset and highlights near-human performance with improved mean unsigned error relative to graph-based and other ML methods.
[307]: Deep Learning-Based Retinal OCT Segmentation: A Comparative Study	2025	This investigation addresses ophthalmic OCT image analysis by performing retinal layer segmentation using CNN, hybrid CNN-graph cut, and Transformer U-Net variants on normal and AMD scans, with evidence based on reported segmentation metrics (DSC, IoU, precision, and recall) on the DUKE OCT dataset.	The analysis quantifies how Transformer U-Net compares with U-Net and CNN-graph cut approaches for segmentation accuracy and boundary stability across normal versus AMD subsets, providing practical guidance on architecture trade-offs for retinal OCT layer measurement.
[151]: Deep Learning-Based SD-OCT Layer Segmentation Quantifies Outer Retina Changes in Patients With Biallelic RPE65 Mutations Undergoing Gene Therapy	2025	The research uses deep-learning-based automated segmentation of SD-OCT retinal layers in patients with biallelic RPE65-IRD to quantify outer-retina ellipsoid zone structural biomarkers, including baseline group differences and longitudinal treatment-related changes around voretigene neparvovec, providing evidence from retrospective in-hospital imaging analysis with control comparisons.	The reported work develops and applies a modified U-Net approach to derive five ellipsoid zone biomarkers (including thickness, granularity, reflectivity/intensity, and EZ area-related measures) with outer-layer distance metrics, enabling objective, repeatable OCT-derived quantification that supports monitoring of disease and treatment response beyond manual segmentation.
[283]: Deep learning-based segmentation and quantitative analysis of retinal microstructures in optical coherence tomography angiography images using RS_Unet3+	2026	The reported work develops a deep learning segmentation framework (RS_Unet3+) for OCTA en face images to quantify retinal vascular network and foveal avascular zone metrics, testing the approach on public OCTA vessel segmentation datasets and examining clinically consistent parameter differences in diabetic retinopathy using additional clinical cohorts.	The study reports multi-class RVN/FAZ segmentation with automated extraction of FAZ and vessel parameters (e.g., area, perimeter, circularity, VPD, VLD, FD, tortuosity) and demonstrates that FAZ morphology and vascular metrics vary in a DR cohort in a way that can support diagnosis-related trend analyses.

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Work	Year	Overall reasoning	Contribution and summary
[354]: Deep learning-based signal-independent assessment of macular avascular area on 6×6 mm optical coherence tomography angiogram in diabetic retinopathy: a comparison to instrument-embedded software	2023	This clinical study evaluates a deep-learning method for signal-independent quantification of macular extrafoveal avascular area from 6x6 mm OCT angiography in diabetic retinopathy, comparing it with instrument-embedded vessel density software to predict DR severity using evidence from a cross-sectional dataset with diagnostic accuracy and correlation analyses.	This investigation reports that MEDnet-based extrafoveal avascular area measured on the superficial vascular complex yields higher sensitivity for distinguishing diabetes and DR stages from controls at fixed specificity and shows minimal dependence on signal strength and shadow artifacts compared with commercial vessel density, supporting more robust OCTA-based DR assessment.
[48]: Deep Learning Based Sub-Retinal Fluid Segmentation in Central Serous Chorioretinopathy Optical Coherence Tomography Scans	2019	The study describes a fully convolutional neural network that performs automated sub-retinal fluid segmentation on OCT volumes from patients with central serous chorioretinopathy, tackling noise and motion artifacts and spatial variability typical of this imaging-based task, with evidence provided as reported test-set segmentation accuracy via Dice, precision, and recall.	This analysis proposes an end-to-end learning approach tailored to differentiate subtle sub-retinal fluid patterns in CSC OCT and reports quantitative test-set results on 15 OCT volumes, indicating practical effectiveness for pixel-level fluid delineation despite imaging variability.
[182]: Deep Learning Classification of Drusen, Choroidal Neovascularization, and Diabetic Macular Edema in Optical Coherence Tomography (OCT) Images	2023	This 2023 study uses deep learning with OCT imaging to classify drusen, choroidal neovascularization, and diabetic macular edema, reporting diagnostic classification performance on a labeled dataset of diseased versus healthy scans.	This work trains a CNN-based supervised model on a multiclass OCT dataset and reports test-set precision, recall, sensitivity, specificity, F1 score, and accuracy to support automated retinal pathology detection, while noting the need for external validation and clinical workflow evaluation.
[344]: Deep Learning Classification on Optical Coherence Tomography Retina Images	2020	This work uses a VGG16 transfer-learning deep neural network on grayscale OCT retina scans (converted to RGB via colormaps) for a clinically relevant four-way classification of three retinal diseases versus normal and reports testing accuracy.	The research details an end-to-end OCT retinal image classification pipeline with data augmentation to mitigate limited training data and evaluates a VGG16-based model to achieve 99.48% testing accuracy across all four classes, highlighting feasibility for automated diagnostic categorization.

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Work	Year	Overall reasoning	Contribution and summary
[416]: Deep learning detection of retinal detachment: Optical coherence tomography staging and estimation of duration of macular detachment	2025	This retrospective deep learning study uses transfer-learned 2D and 3D OCT imaging to classify rhegmatogenous retinal detachment, including macula status, six-stage OCT morphology, and a surrogate estimate of macula-off duration, with evaluation based on ROC and precision-recall performance on a single-center dataset of RRD eyes versus controls.	This research shows that a 3D OCT model can stratify macula-on versus macula-off RRD and discriminate short versus longer macula-off duration, while a 2D model supports automated stage classification and provides Grad-CAM feature maps to visualize model attention linked to clinically relevant staging associations.
[76]: Deep learning-driven automated quality assessment of ultra-widefield optical coherence tomography angiography images for diabetic retinopathy	2025	A ViT-based transfer learning framework was developed for automated image quality assessment of ultra-widefield OCTA in diabetic retinopathy, using UW-OCTA images (12 mm x 12 mm) fine-tuned from pre-training on 6 mm x 6 mm OCTA and evaluated with AUC and quadratic weighted Kappa plus ablations on a labeled DRAC2022 test set.	By leveraging scarce UW-OCTA data through pre-training and transformer attention, the study provides an end-to-end quality-grading pipeline that outperforms ResNet baselines in discrimination and rater agreement (AUC and Kappa) and supports clinical usability via explainability heatmaps and component-level ablation results.
[362]: Deep learning for objective OCTA detection of diabetic retinopathy	2020	A study applies a transfer-learning convolutional neural network using VGG16 to perform OCTA-based diabetic retinopathy detection and multi-class classification (control, no DR, and nonproliferative DR), with augmentation and 5-fold cross-validation as evidence from preliminary dataset evaluation.	The study reports how fine-tuning nine layers of the VGG16 model improves OCTA image classification for DR staging, providing early evidence for objective, OCTA-driven deep learning in ophthalmic screening.
[336]: Deep learning for quality assessment of retinal OCT images	2019	The reported work develops a deep-learning OCT image quality assessment (OCT-IQA) pipeline for ophthalmic retinal B-scan images, using transfer-learned CNNs (VGG-16, Inception-V3, ResNet-18, ResNet-50) to classify scan quality based on signal completeness, location, and effectiveness, with evidence from five-fold cross-validation plus downstream retinopathy detection testing showing utility of the quality pre-filtering step.	The study demonstrates that filtering OCT scans with the best-performing ResNet-50 OCT-IQA model improves retinopathy classification outcomes by comparing a "pure" high-quality set against a "mixed" set containing poor-quality scans, yielding higher reported detection accuracy and AUC for the filtered data.

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Work	Year	Overall reasoning	Contribution and summary
[112]: Deep Learning for Retina Structural Biomarker Classification Using OCT Images	2024	This analysis evaluates a multimodal, multi-task deep learning method for ophthalmic retinal structural biomarker classification, using 3D OCT images together with one-dimensional clinical data and combining supervised and semi-supervised training to predict six biomarkers, with evidence reported as an evaluation using macro F1-score and a baseline comparison.	This investigation introduces a custom post-processing step that improves biomarker detection precision and demonstrates a 14.48% macro F1-score gain over baseline for six retinal structural biomarkers, supporting the utility of post-processing-enhanced OCT-based deep learning for diagnostic decision support.
[272]: Deep Learning for Retinal OCT Disease Classification using EfficientNetB0: A Transfer Learning Approach	2025	This research uses transfer learning with EfficientNet (EfficientNetB7) to classify retinal OCT images into CNV, DME, drusen, and normal, targeting early and more consistent ocular diagnosis despite observer variability, with evidence reported from experimental evaluation on the OCT2017 dataset.	This analysis outlines an end-to-end automated OCT disease classifier that incorporates preprocessing to handle scan-quality and demographic variation and reports balanced cross-class performance using confusion-matrix analysis, which is valuable for dependable clinical triage across visually similar findings.
[253]: Deep Learning for Segmentation of Optic Disc and Retinal Layers in Peripapillary Optical Coherence Tomography Images	2023	Using peripapillary spectral-domain OCT imaging, the study applies a deep-learning approach with transfer learning to automatically segment nine retinal layers and the optic disc centered in the peripapillary region, then derives layer thickness measurements along B-scans, with evidence reported as Dice overlap in both a public dataset and a depth-enhanced dataset.	This work reports an end-to-end optic-disc-centered OCT segmentation and thickness quantification pipeline with visualization outputs, achieving a Dice score of 83.6% for nine-layer and optic-disc delineation across both image sets, which supports practical automated analysis of retinal layer structure.
[196]: Deep learning for the segmentation of preserved photoreceptors on en face optical coherence tomography in two inherited retinal diseases	2018	A patch-based convolutional neural network segments preserved ellipsoid zone regions in en face OCT images from patients with choroideremia or retinitis pigmentosa, producing pixelwise masks that are compared with expert manual grading.	The study provides a versatile CNN segmentation workflow that converts B-scan patch classifications into en face preserved photoreceptor maps and removes vessel-shadow artifacts, demonstrating automated quantification of preserved photoreceptors with agreement to manual labels in both included IRDs.
[249]: Deep Learning Image Analysis of Macular Optical Coherence Tomography Angiography Images for Detection of Progression in Glaucoma	2024	The analysis applies a CNN-based longitudinal OCTA image analysis in glaucoma, using macular microvascular density derived from thick-vessel segmentation, thin-vessel masking, Otsu thresholding, and superpixel measurement to distinguish progressive versus stable eyes on baseline and follow-up scans, with evaluation framed by patient-level progression detection evidence.	This research delivers a five-stage processing pipeline that converts serial macular OCTA into superpixelized thin-vessel density measurements and uses t-value patterning to separate progressive from stable glaucoma, offering a practical framework for tracking disease-related vascular dropout over time.

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Work	Year	Overall reasoning	Contribution and summary
[220]: Deep Learning Image Analysis of Optical Coherence Tomography Angiography Measured Vessel Density Improves Classification of Healthy and Glaucoma Eyes	2022	Using a VGG16 convolutional neural network on en face radial peripapillary capillary OCTA vessel density images, and five-fold cross-validation against feature-based gradient-boosting classifiers derived from instrument OCTA/OCT vessel density and OCT RNFL thickness, this ophthalmic diagnostic study evaluates healthy versus glaucomatous eyes with adjusted precision-recall evidence to compare diagnostic accuracy across method families.	The analysis shows that end-to-end deep learning on 4.5 mm by 4.5 mm OCTA en face vessel density maps improves glaucoma classification over gradient-boosting models built from standard OCTA and OCT measurements, including comparisons that remain consistent in an early glaucoma subgroup.
[94]: Deep learning is effective for the classification of OCT images of normal versus Age-related Macular Degeneration	2017	This EMR-linked deep learning study in ophthalmology classifies central macular spectral-domain OCT images into normal versus age-related macular degeneration, evaluating performance with ROC analysis at image, macular scan, and patient levels to provide evidence for automated OCT screening and computer-aided diagnosis workflows.	The reported work trains a modified VGG16 convolutional neural network on a large OCT dataset extracted from EPIC, then reports how aggregating predictions at the scan and patient levels improves discrimination and uses an occlusion test to identify regions driving the AMD calls.
[53]: Deep learning method for semi-automated segmentation of optic nerve head tissues in optical coherence tomography images	2025	For OCT imaging of optic nerve head (ONH) tissues, the study evaluates a CNN-based semantic boundary segmentation pipeline for anterior lamina cribrosa surface, Bruch's membrane, and choroid-scleral interface using spectral-domain OCT radial scans, with a user-in-the-loop correction and iterative model updating, and it reports evidence from quantitative comparison against manual markings on a held-out test set using boundary-wise RMSE (and Euclidean distance for BMO).	The study provides a semi-automated, interactive workflow that lets users accept or adjust predicted boundary markers and then uses the corrected masks to continue training the model, yielding improved BM and CS predictions in the reported test set and enabling boundary-level error quantification with time savings versus fully manual segmentation.
[173]: Deep Learning Model for the Detection of Corneal Edema Before Descemet Membrane Endothelial Keratoplasty on Optical Coherence Tomography Images	2022	This retrospective ophthalmology study evaluates an OCT-based deep learning pipeline that segments corneal scans at the pixel level to classify normal versus edema, using edema fraction derived from 1992 preoperative corneal OCT images to detect minimal pre-DMEK corneal swelling across FECD and non-FECD patients with AUC-based evidence against objective differential central corneal thickness thresholds.	This investigation provides an automated, quantitative decision-support measure for pre-DMEK assessment by mapping epithelial and stromal edema regions and showing strong discrimination of clinically meaningful edema (DCCT 20 m) and consistent thresholds across patient groups, supporting objective detection of subclinical corneal edema.

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Work	Year	Overall reasoning	Contribution and summary
[246]: Deep learning model to predict visual field in central 10° from optical coherence tomography measurement in glaucoma	2021	In open-angle glaucoma care, this research uses a convolutional neural network to predict HFA 10-2 central 10 visual field threshold sensitivities from spectral-domain OCT macular layer thickness, evaluating accuracy on an external test set against support vector machine and multiple linear regression baselines.	This analysis provides an externally tested CNN pipeline that estimates whole-field and point-wise 10-2 threshold sensitivities from SD-OCT-derived layer thickness, demonstrating improved error performance over conventional machine learning and regression models.
[129]: Deep learning network (DL-Net) based classification and segmentation of multi-class retinal diseases using OCT scans	2024	The study applies ophthalmology-oriented deep learning to volumetric retinal OCT scans to automate multi-class recognition and lesion delineation across common macular disease categories (normal, DME, CNV, AMD, and drusen), providing evidence of effectiveness via reported classification and segmentation performance compared with other OCT frameworks using publicly available datasets.	This work proposes a multi-scale DL-Net that modifies ResNet-50 and SqueezeNet for OCT-based multi-class classification and couples convolutional feature aggregation with a Bi-LSTM-based recurrent convolutional segmentation approach, offering an end-to-end diagnostic pipeline intended to improve OCT interpretation for clinical decision support.
[78]: Deep learning network for parallel self-denoising and segmentation in visible light optical coherence tomography of the human retina	2023	For visible-light optical coherence tomography of the retina, the study evaluates a co-learning deep network that denoises speckle-laden B-scans and segments 10 retinal layers from the same VIS-OCT input, using paired noisy and clean data with boundary annotations and reporting evidence from repeated experiments plus cross-protocol generalization.	The research introduces a VIS-OCT dataset with 105 paired noisy and clean B-scans and 10 manually delineated layer boundaries, and shows that parallel self-denoising improves layer segmentation, particularly when only 25% of segmentation labels are available, while remaining robust when applied to a different scanning protocol.
[227]: Deep learning network with differentiable dynamic programming for retina OCT surface segmentation	2023	This 2023 study evaluates an end-to-end ophthalmic deep learning framework for retina OCT multiple-surface segmentation that couples a U-Net feature learner with a constrained differentiable dynamic programming module to enforce smooth surface transitions across B-scan columns, using OCT-derived B-scan images and reporting external test evidence on Duke AMD and JHU MS for subvoxel segmentation accuracy.	This research introduces differentiable dynamic programming integrated into the segmentation network training to explicitly regularize global surface smoothness, and shows that the DDP-constrained model improves segmentation accuracy over a U-Net baseline and ablation without DDP on both AMD and MS retinal layer surfaces.

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Work	Year	Overall reasoning	Contribution and summary
[59]: Deep learning of fundus images and optical coherence tomography images for ocular disease detection – a review	2024	This review is relevant to ophthalmic practice by synthesizing deep learning methods for ocular disease detection and region delineation using fundus and optical coherence tomography imaging, addressing tasks such as diabetic retinopathy classification/grading and glaucoma detection as well as vessel, optic disc, and optic cup segmentation, with evidence drawn from reported literature rather than new experimental results.	The analysis consolidates convolutional, recurrent, generative adversarial, U-Net and Y-Net, and transformer-based approaches across multiple imaging types and task settings, highlighting architectural variants that improved on baselines and offering a unified reference that spans beyond single-disease or single-task reviews.
[461]: Deep Learning Techniques for Diabetic Retinopathy Diagnosis using Optical Coherence Tomography: A Review	2022	This review synthesizes deep-learning techniques for diabetic retinopathy from optical coherence tomography imaging, covering OCT-based detection and classification as well as retinal-layer segmentation, with evidence drawn from method reports published over the prior four years.	The reported work gathers and summarizes OCT-oriented model approaches and the diagnostic features used for diabetic retinopathy, clarifying how segmentation of retinal layers is leveraged within classification-oriented workflows.
[356]: Deep learning to detect macular atrophy in wet age-related macular degeneration using optical coherence tomography	2023	This ophthalmic deep learning study uses a U-Net-based CNN to segment and quantify optical coherence tomography B-scans from patients with wet age-related macular degeneration for automatic detection of six macular atrophy atrophic features defined by CAM criteria, with model performance reported using standard validation metrics against expert manual annotations.	The study delivers a fully automated, one-vs-all segmentation pipeline that produces pixel-level measurements for interrupted outer retina, interrupted RPE, absence of outer retina, absence of RPE, and hypertransmission thresholds, offering a reproducible OCT workflow to support early lesion identification in wet AMD despite the reliance on internal validation only.
[310]: Deep Learning to Detect OCT-derived Diabetic Macular Edema from Color Retinal Photographs: A Multicenter Validation Study	2022	This retrospective multicenter study validates a deep learning system for detecting center-involving diabetic macular edema from 2D color fundus photographs, using OCT-derived retinal thickening and intraretinal fluid as the reference standard and expert grading as the comparator across international DR screening and clinic datasets.	This investigation provides OCT-labeled, device-thresholded evidence that the model achieves higher specificity with noninferior sensitivity than expert graders across multiple populations, supporting potential use to reduce false-positive referrals pending prospective evaluation.

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Work	Year	Overall reasoning	Contribution and summary
[273]: Deep Learning Using Preoperative Optical Coherence Tomography Images to Predict Visual Acuity Following Surgery for Idiopathic Full-Thickness Macular Holes	2024	This research uses fully automated deep learning on preoperative 3D spectral-domain OCT (SD-OCT) imaging in an ophthalmic surgical setting to predict postoperative visual acuity for idiopathic full-thickness macular holes, with quantitative evaluation of multiple model variants after preprocessing, quality assurance, and anomaly detection to support robustness to image-quality variability.	This analysis reports an end-to-end, quality-aware AI workflow that ingests seven central SD-OCT images per patient and benchmarks nine state-of-the-art predictive networks to estimate visual acuity in ETDRS letters, offering a practical route toward automated prognostication after macular hole surgery.
[277]: Deep learning visual field global index prediction with optical coherence tomography parameters in glaucoma patients	2023	This retrospective cross-sectional glaucoma study uses a deep neural network to regress visual field global indices (MD, PSD, and VFI) from spectral-domain OCT-derived Bruch's membrane opening-minimum rim width and retinal nerve fiber layer parameters, evaluated with fivefold cross-validation using prediction error and correlation metrics to support clinical use when visual field testing is delayed.	This work is notable for jointly modeling all three VF global indices from extracted OCT structural parameters, providing a clinic-oriented quantitative surrogate that could help clinicians obtain functional-field estimates without waiting for standard automated perimetry results.
[133]: Deep learning with adaptive convolutions for classification of retinal diseases via optical coherence tomography	2024	This 2024 study evaluates a CNN-based OCT retinal image classification pipeline that performs Zeckendorf-theorem-based indirect noise reduction and applies a texture-sensitive adaptive convolution layer, reporting accuracy gains across seven OCTNet-style CNN architectures on an OCT dataset from the University of California San Diego.	The research introduces a preprocessing-plus-adaptive-convolution substitute for standard convolution in OCT classification and quantifies modest but consistent accuracy improvements (0.44% to 2.44%) to support the practical value of the denoised base/fine representation and feature-content-driven adaptive filtering.
[448]: Deep Learning with Disc Photos or OCT Scans in Glaucoma Detection	2025	Using a retrospective tertiary-glaucoma dataset, the study compares deep learning models trained on OCT RNFLT maps versus disc photos to detect glaucoma defined by functional visual-field impairment, evaluated with AUC (including performance across demographic groups), providing diagnostic evidence relevant to ophthalmic screening workflows.	This research demonstrates that OCT RNFLT-based deep learning achieves higher glaucoma detection accuracy than disc photo-based modeling while maintaining broadly consistent AUC across demographic groups, informing modality choice and highlighting where dataset equity may be necessary.

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Work	Year	Overall reasoning	Contribution and summary
[366]: Deep Neural Network for Scleral Spur Detection in Anterior Segment OCT Images: The Chinese American Eye Study	2020	This population-based anterior segment OCT study uses a ResNet-18 CNN to localize scleral spurs by predicting X- and Y-coordinate positions in AS-OCT images, and it validates performance against a human expert reference using prediction error distributions compared with intragrader variability.	By showing that CNN localization errors were statistically similar to the reference grader's repeatability (and lower than intergrader variability), the study provides automated scleral spur detection that can remove manual, grader-dependent steps for AS-OCT analysis in angle-closure assessment.
[73]: Deep Neural Network Regression for Automated Retinal Layer Segmentation in Optical Coherence Tomography Images	2020	The research proposes an OCT-specific deep neural network regression approach that uses intensity, gradient, and adaptive normalized intensity score features to predict retinal boundary pixels for layer segmentation, with evidence from quantitative comparison to manual references on 114 images using Dice and pixel-distance metrics.	The reported work introduces a boundary-wise regression formulation that avoids classification-style labeling, reports robust performance under low contrast, speckle noise, intensity variance, and vessels, and emphasizes practical efficiency by detailing per-image processing and rapid training per boundary.
[100]: Deep Neural Networks for Automated Outer Plexiform Layer Subsidence Detection on Retinal OCT of Patients With Intermediate AMD	2024	The reported work evaluates a deep learning detection and classification system on Spectralis retinal OCT volumes to automatically localize and score outer plexiform layer subsidence, an early intermediate AMD biomarker linked to progression to geographic atrophy, using evidence from development and an independent external dataset with cross-validation and ROC/PR and FROC-based performance reporting.	The study advances an object-detection plus B-scan presence scoring pipeline that merges bounding boxes across neighboring B-scans into volume-level predictions, enabling rapid, screening-oriented assessment of OPL subsidence with validated external generalization and quantified low false-positive burden.
[130]: DeepRetina: Layer Segmentation of Retina in OCT Images Using Deep Learning	2020	In an ophthalmic optical coherence tomography setting, this paper introduces an encoder-decoder deep learning segmentation method with an improved Xception65 backbone and atrous spatial pyramid pooling to automatically delineate a 10-layer retinal structure, with quantitative evidence from evaluation on a retinal OCT database including both healthy and disease images.	DeepRetina delivers an end-to-end retinal layer auto-segmentation pipeline and reports strong quantitative segmentation accuracy across individual layers, plus efficiency suitable for practical OCT image processing, supporting its use for retinal layer assessment in clinical workflows.
[223]: Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI	2021	This research addresses retinal OCT disease classification in four diagnostic categories using a compact 6-layer CNN with LIME- and Grad-CAM-based explainability, providing clinician-facing interpretability alongside an evaluation protocol and reported classification outcomes.	This analysis introduces a resource-efficient CNN designed for smaller model size and real-time deployment, then demonstrates how LIME feature attributions (with Grad-CAM visualizations) are used to make the model's decisions more transparent for OCT-based clinical use.

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Work	Year	Overall reasoning	Contribution and summary
[178]: Detecting glaucoma based on spectral domain optical coherence tomography imaging of peripapillary retinal nerve fiber layer: a comparison study between hand-crafted features and deep learning model	2020	The study evaluates an ophthalmology-oriented, pretrained CNN deep learning approach that uses spectral-domain OCT peripapillary RNFL images to perform glaucoma detection, with evidence drawn from comparison against handcrafted pRNFL sector and average thickness measures using ROC-derived AROC in a held-out test set.	This work directly contrasts model-based glaucoma discrimination with multiple handcrafted pRNFL features, showing that the deep learning method attains higher ROC performance than conventional thickness parameters, which supports its practical value for automated glaucoma screening from SD-OCT.
[165]: Detecting Retinal Nerve Fibre Layer Segmentation Errors on Spectral Domain-Optical Coherence Tomography with a Deep Learning Algorithm	2019	This ophthalmic study develops a supervised deep learning classifier that analyzes spectral-domain OCT peripapillary circle B-scans to identify retinal nerve fiber layer segmentation errors, using human grader labels as the reference standard and reporting diagnostic evidence via ROC analysis and accuracy against the test set.	The research provides a probability-based RNFL artifact detector trained on 25,250 SDOCT B-scans that can flag both mild and severe segmentation mistakes relative to human grading, offering a practical quality-assurance tool to reduce unreliable RNFL measurements.
[72]: Detection of Choroidal Neovascularization (CNV) in Retina OCT Images Using VGG16 and DenseNet CNN	2022	The analysis evaluates CNN-based detection of choroidal neovascularization from retinal OCT/OCTA imaging using VGG16 and DenseNet, with accuracy reported after a segmentation step that isolates regions of interest for clinical interpretation.	This research refines an end-to-end OCT/OCTA CNV workflow by comparing VGG16 versus DenseNet and then applying region segmentation via OpenCV to distinguish diseased images from background, reporting higher performance for VGG16.
[104]: Detection of diabetic macular edema in optical coherence tomography scans using patch based deep learning	2018	This investigation targets ophthalmic screening of diabetic macular edema by analyzing optical coherence tomography scans using a patch-based deep learning pipeline that integrates image-processing candidate detection with CNN patch labeling and rule-based confidence aggregation to produce scan-level DME predictions.	The analysis specifies a two-stage OCT framework that first localizes potential fluid and hard exudate regions and then uses a deep convolutional neural network to label patches, enabling automated DME detection from scans without relying on manual annotation at inference.

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Work	Year	Overall reasoning	Contribution and summary
[351]: Detection of glaucoma progression on longitudinal series of en-face macular optical coherence tomography angiography images with a deep learning model	2024	In glaucoma monitoring, the research evaluates a custom convolutional neural network that uses longitudinal superficial en-face macular OCTA vessel-density patterns (baseline versus final scans aligned to the foveal avascular zone) to classify statistically defined 24-2 visual field mean deviation progression, reporting test-set discrimination via AUC, sensitivity, specificity, and accuracy with a logistic regression comparator and validation using confusion-matrix-based metrics.	The reported work demonstrates that spatiotemporal learning from sequential macular OCTA images, without relying on explicit vessel-density rate features, can outperform a logistic regression baseline for progression detection and provides gradient-based attention maps to support biologically plausible localization around microvasculature loss and foveal avascular zone changes.
[237]: Detection of Macular Neovascularization in Eyes Presenting with Macular Edema using OCT Angiography and a Deep Learning Model	2025	In this prospective cross-sectional ophthalmic study, a hybrid multitask convolutional neural network was evaluated for macular neovascularization detection and segmentation using OCTA with structural OCT in eyes with macular edema from exudative AMD, diabetic macular edema, or retinal vein occlusion, with diagnostic classification and segmentation quality reported using standard sensitivity/specificity and IoU/F1 metrics.	The authors show that the aiMNV model performs most reliably on denser 3x3-mm OCTA acquisitions, enabling automated MNV membrane area segmentation and maintaining high diagnostic performance in a real-world macular edema cohort, which is clinically relevant for distinguishing exudative AMD MNV despite coexisting IRF and SRF.
[132]: DETECTION OF MORPHOLOGIC PATTERNS OF DIABETIC MACULAR EDEMA USING A DEEP LEARNING APPROACH BASED ON OPTICAL COHERENCE TOMOGRAPHY IMAGES	2021	This analysis develops a deep learning model for ophthalmic OCT image analysis that classifies three morphologic diabetic macular edema patterns (diffused retinal thickening, cystoid macular edema, serous retinal detachment), using internal 5-fold cross-validation and external validation plus occlusion-based testing to provide evidence of model performance and interpretability.	This investigation delivers automated, pattern-level OCT detection with multilabel grader-derived references and uses occlusion maps to identify the image regions most critical for accurate recognition, which may help standardize DME assessment and support treatment planning.
[285]: Detection of Nonexudative Macular Neovascularization on Structural OCT Images Using Vision Transformers	2022	This ophthalmology study uses a vision-transformer framework on structural OCT B-scans, segmenting the double-layer sign and drusen and then producing an eye-level classification of nonexudative macular neovascularization, with performance validated against human-graded labels from a held-out set and reported using diagnostic metrics and intergrader agreement.	By deriving the neMNV decision from structural OCT alone through a segmentation-to-projection-to-classification pipeline, the authors demonstrate clinically actionable detection performance and kappa-based concordance with a senior human grader, supporting transformer-based detection of DLS-associated neMNV in AMD.

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Work	Year	Overall reasoning	Contribution and summary
[303]: Detection of Retinal Disease from Optical Coherence Tomography Images Using CNN Models	2024	The study uses convolutional neural networks on optical coherence tomography to perform four-class classification of retinal disease states (DME, drusen, CNV, and normal), providing evidence from an imaging-based, ophthalmology-relevant diagnostic task with comparative AI model evaluation.	This work introduces a proposed LayerV variant of VGG16 and reports higher accuracy (97.45%) relative to other tested CNN architectures, suggesting an effective OCT-based automated screening tool for differentiating major retinal conditions.
[164]: Detection of Retinal Disorders in Optical Coherence Tomography using Deep Learning	2019	This 2019 study applies DenseNet deep learning to macular optical coherence tomography for screening and detecting normal retina versus CNV, DME, and drusen, with evidence reported via training, validation, and test classification metrics and AUC.	The research demonstrates how DenseNet can be parameter-tuned and evaluated for multi-class disorder recognition on OCT, providing quantitative performance results that support its use as an automated screening aid for treatable retinal conditions.
[382]: Development and validation of a 3-D deep learning system for diabetic macular oedema classification on optical coherence tomography images	2025	This multicenter, platform-based study builds and externally validates a 3D OCT deep learning classifier that grades diabetic macular edema into no DME, non-center-involved DME, and center-involved DME, using retrospective, cross-sectional annotated volumes and evaluating diagnostic performance against retina experts in a screening-relevant ophthalmic setting.	By training on 3D CNN features from multiple OCT device brands and testing on an external set split by center, it provides automated DME localization with human-comparable agreement and quantitative ROC metrics, supporting deployment for population-based DME screening.
[394]: Development and validation of a pixel wise deep learning model to detect cataract on swept-source optical coherence tomography images	2022	This retrospective ophthalmic study uses a U-Net convolutional segmentation model on swept-source OCT to perform pixel-wise cataract detection and localization, outputting per-pixel cataract probability maps summarized by Cataract Fraction and evaluated with ROC AUC, sensitivity, and specificity on a held-out validation set.	Its key contribution is an SS-OCT-based, automated, interpretable probability mapping framework that converts lens imaging into quantitative Cataract Fraction to support objective cataract assessment and help guide clinical diagnosis and surgical planning.
[117]: DEVELOPMENT AND VALIDATION OF AN EXPLAINABLE ARTIFICIAL INTELLIGENCE FRAMEWORK FOR MACULAR DISEASE DIAGNOSIS BASED ON OPTICAL COHERENCE TOMOGRAPHY IMAGES	2022	This ophthalmology study uses a two-stage AI pipeline on optical coherence tomography images to classify nine retinal lesions at the image level and produce an explainable eye-level diagnosis of three macular diseases, validating both levels against human expert readers.	The reported work develops an integrated deep learning plus random-forest framework that links detected central macular lesions to interpretable disease predictions, demonstrating test performance on lesion labeling and eye-level accuracy relative to four ophthalmic experts.

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Work	Year	Overall reasoning	Contribution and summary
[134]: Development of a deep learning system to detect glaucoma using macular vertical optical coherence tomography scans of myopic eyes	2023	This retrospective diagnostic study developed and validated a CNN-based deep learning glaucoma classifier using macular vertical SD-OCT B-scans from myopic eyes, and compared its discrimination against circumpapillary OCT while including an external test set, showing strong performance especially in eyes with large myopic parapapillary atrophy.	By benchmarking macular vertical versus circumpapillary OCT inputs (and refining predictions by adding demographic and ophthalmic features), the authors demonstrate an evidence-based OCT imaging approach that can better support glaucoma detection in challenging high-myopia anatomy.
[44]: Diagnosis of Retinal Diseases Based on Bayesian Optimization Deep Learning Network Using Optical Coherence Tomography Images	2022	This analysis investigates ophthalmic OCT image-based classification of retinal diseases using transfer learning with pretrained CNNs (VGG16, DenseNet201, InceptionV3, Xception), paired with image augmentation and Bayesian hyperparameter optimization, and reports diagnostic performance across an eight-category setup (seven diseases plus normal) using validation and test results derived from the model evaluation metrics.	This investigation compares feature-extractor and fine-tuning strategies for multiple CNN backbones while tuning classifier hyperparameters via Bayesian optimization, providing test accuracies and class-wise precision, recall, and F1-scores to show that DenseNet201 fine-tuning achieves the highest reported performance for the seven-disease versus normal OCT diagnosis setting.
[138]: Diagnosis of retinal diseases using the vision transformer model based on optical coherence tomography images	2023	This ophthalmic deep learning study uses a vision transformer to classify retinal conditions from optical coherence tomography (OCT) images into normal and three abnormal classes (CNV, drusen, and diabetic macular edema) and reports diagnostic evidence via accuracy, sensitivity, and specificity.	This analysis demonstrates that the proposed transformer-based multi-class OCT classifier achieves high reported diagnostic metric values for CNV, DME, and drusen and compares favorably against traditional CNN approaches, supporting transformer use for retinal disease diagnosis.
[148]: Diagnosis of Retinal Disorders from Optical Coherence Tomography Images using CNN	2025	A study assesses a lightweight CNN for automated retinal disorder diagnosis from optical coherence tomography imaging, performing multiclass classification of choroidal neovascularization, diabetic macular edema, drusen, and normal eyes with reported diagnostic performance metrics.	This work proposes OpticNet-71, using depthwise separable convolutions, residual blocks, and an encoder-decoder design to reduce computational cost and support real-time clinical deployment while delivering high accuracy on the specified retinal categories.
[449]: Diagnostic ability of macular microvasculature with swept-source OCT angiography for highly myopic glaucoma using deep learning	2023	This ophthalmic deep-learning diagnostic study evaluates swept-source OCT angiography and swept-source OCT macular imaging to distinguish highly myopic glaucoma from healthy high myopia, using ViT-based classification with ROC analysis across superficial capillary plexus, deep capillary plexus, and ganglion cell layer thickness parameter sets, with both internal testing and external validation reported via AUC.	Notably, the ViT model trained on macular OCTA superficial capillary plexus images achieved glaucoma discrimination comparable to macular ganglion cell layer thickness measures and better than deep capillary plexus, supporting macular OCTA microvasculature as a potentially useful biomarker for glaucoma detection in high myopia.

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Work	Year	Overall reasoning	Contribution and summary
[68]: Diagnostic performance and generalizability of deep learning for multiple retinal diseases using bimodal imaging of fundus photography and optical coherence tomography	2025	This multicenter ophthalmology study evaluates deep learning for detecting and grading seven retinal conditions using bimodal color fundus photography and optical coherence tomography, employing multimodal multi-instance learning fusion and reporting AUC-based diagnostic performance with external testing across different devices and scanning protocols in separate hospitals and dataset splits.	This research introduces and selects a bimodal Fusion-MIL model that improves multi-disease discrimination over single-modality baselines and extends to detailed pathologic myopia ATN component grading, supporting robust performance and clinically usable heatmap-based localization across heterogeneous imaging settings.
[52]: DRUNET: a dilated-residual U-Net deep learning network to segment optic nerve head tissues in optical coherence tomography images	2018	In this glaucoma-focused OCT study, DRUNET-a dilated-residual U-Net-segments multiple optic nerve head tissue layers from Spectralis B-scan images, enabling automatic extraction of clinically relevant structural measures, with performance evaluated against expert manual segmentations using overlap and classification metrics.	The analysis introduces an end-to-end segmentation framework that isolates six neural and connective ONH tissues simultaneously and converts the segmentations into six structural parameters, reducing parameter extraction error versus a prior patch-based approach and supporting more reliable quantitative ONH assessment.
[368]: Dry age-related macular degeneration classification from optical coherence tomography images based on ensemble deep learning architecture	2024	In this ophthalmic study of dry age-related macular degeneration, the authors develop an ensemble deep learning classifier that combines multiple pretrained CNNs to categorize four OCT-derived stages (normal, drusen, nascent geographic atrophy, and geographic atrophy), using three-fold cross-validation to evaluate early-stage recognition, particularly nascent geographic atrophy, in retinal B-scan OCT images from a hospital dataset.	Their specific contribution is a weighted ensemble of ResNet50, EfficientNetB4, MobileNetV3, and Xception applied to non-segmented OCT for computer-aided staging, showing improved discrimination of nascent geographic atrophy relative to the constituent models and supporting clinical feasibility through performance comparisons and Grad-CAM-based visualization of highlighted features.
[405]: Dry and Wet Age-Related Macular Degeneration Classification Using OCT Images and Deep Learning	2019	The reported work evaluates deep convolutional neural networks, including ResNet, on OCT imaging to classify dry versus wet age-related macular degeneration and to support evidence in the form of reported stage-wise ROC-based performance from model evaluation.	The study specifically reports that an 18-layer ResNet improves classification relative to AlexNet for dry and wet AMD and provides stage-specific performance values, which matters for automated, time-sensitive triage toward appropriate treatment selection.
[371]: DTG: Dual transformers-based generative adversarial networks for retinal 2D/3D OCT image classification	2026	This analysis evaluates a dual-transformer, GAN-based deep learning approach for classifying retinal disorders from 2D and 3D optical coherence tomography (OCT) images, using patient instance-based data augmentation and reporting results on real-world datasets with multiple classification metrics.	DTG's integration of Vision Transformer and multi-scale Vision Transformer encoders with a GAN-driven semantic representation step, followed by weighted classification, targets data scarcity in OCT and is shown to outperform prior 2D/3D CNN and transformer baselines on retinal disease classification.

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Work	Year	Overall reasoning	Contribution and summary
[168]: Dual-input convolutional neural network for glaucoma diagnosis using spectral-domain optical coherence tomography	2021	This research evaluates a VGG16-based dual-input convolutional neural network that ingests SD-OCT RNFL and GCIPL inputs to perform glaucoma diagnosis, reporting test-set evidence for separating normal versus glaucoma and early-stage versus normal using accuracy and ROC-AUC.	This analysis shows that jointly using RNFL and GCIPL thickness information within a dual-input architecture improves classification of early-stage primary open-angle glaucoma versus normal compared with single-input approaches, supporting clinically relevant automated SD-OCT discrimination.
[216]: Dual-Stream CNN-ViT Framework for Interpretable Glaucoma Detection from OCT and Fundus Images	2025	The study develops a multimodal, interpretable deep learning approach for glaucoma stage classification by fusing OCT and fundus images in a dual-stream CNN-ViT architecture, and it evaluates performance on public datasets while providing SHAP-based regional explanations that are reported to correlate with established retinal markers.	This work offers an OCT-fundus fusion model that outputs Healthy, Mild, and Severe glaucoma stage predictions together with SHAP attributions linked to clinically meaningful features such as RNFL thinning and optic disc cupping, supporting transparent use for screening-oriented workflows.
[74]: Early Detection of Cystoid Macular Edema in Retinitis Pigmentosa Using Longitudinal Deep Learning Analysis of OCT Scans	2025	This retrospective ophthalmic study uses deep learning on longitudinal spectral-domain OCT macular B-scan images to predict early cystoid macular edema onset in retinitis pigmentosa, using a patient-wise split for evidence from internal validation/testing rather than external cohorts.	ResNet-based models trained on early-visit OCTs provide an OCT-only, noninvasive screening approach for CME risk in RP, with ResNet-34 identified as the top performer to support earlier detection of a vision-threatening complication.
[412]: Early Detection of Optic Nerve Changes on Optical Coherence Tomography Using Deep Learning for Risk-Stratification of Papilledema and Glaucoma	2024	This ophthalmic proof-of-concept study uses deep learning by retraining a VGG-19 model on OCT-derived retinal measurements to forecast progression to papilledema or glaucoma from prediagnostic scans, providing retrospective evidence of AI detecting early optic nerve changes when clinician-identified OCT findings are nonsignificant.	This research shows that model performance depends on the OCT representation, with retinal nerve fiber layer thickness mapping for papilledema and extracted vertical tomograms for glaucoma, suggesting a data-driven route to earlier AI-based risk stratification in these optic nerve diseases.
[402]: Effect of patch size and network architecture on a convolutional neural network approach for automatic segmentation of OCT retinal layers	2018	The study investigates CNN-based retinal boundary segmentation in ophthalmic OCT B-scans by predicting per-pixel boundary probabilities from image patches and converting them to boundary paths with graph search, while systematically varying patch size and CNN architecture to assess segmentation quality under different labeling granularities.	The analysis shows how architectural and input-window design choices, together with the number of boundary classes used for training, measurably influence segmentation agreement and robustness, providing practical design guidance for more reliable OCT layer boundary detection via CNN plus graph search.

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Work	Year	Overall reasoning	Contribution and summary
[265]: Effective blood vessels reconstruction methodology for early detection and classification of diabetic retinopathy using OCTA images by artificial neural network	2020	An ophthalmology study uses OCTA imaging with custom blood-vessel reconstruction and feature extraction, followed by a supervised artificial neural network to classify normal, diabetic without DR, and mild-to-moderate NPDR using reported diagnostic results from 100 processed images.	The reported work provides an OCTA-to-ANN pipeline that reconstructs and quantifies vascular features for automated early DR categorization, reporting high overall accuracy (97%) and low misclassification error to support screening-oriented decision support.
[462]: Efficacy and prognostic factors of anti-VEGF treatment for neovascular age-related macular degeneration: An OCTA imaging-based deep learning analysis	2025	In a retrospective ophthalmic cohort of neovascular age-related macular degeneration, an improved LUNet deep learning model analyzes OCTA images to extract FAZ and vessel/lesion vascular parameters and then uses statistical testing to evaluate how these imaging biomarkers relate to anti-VEGF treatment response at 6 months.	The study identifies specific OCTA-derived predictors of inadequate response, such as Vdisp in macular neovascularization, neovascular surface area, and pigment epithelial detachment, supporting risk stratification and more personalized anti-VEGF treatment decisions.
[427]: End-to-End Deep Learning Model for Predicting Treatment Requirements in Neovascular AMD From Longitudinal Retinal OCT Imaging	2020	This analysis targets an ophthalmic prediction need in neovascular AMD by forecasting individualized anti-VEGF retreatment requirements from longitudinal, treatment-initiation OCT volumes using an end-to-end DenseNet plus recurrent neural network model that integrates temporal information across multiple time points, with evaluation on a held-out test set using regression and classification evidence derived from real-world pro-re-nata scheduling labels.	This investigation introduces a dimension-reduction pre-processing pipeline based on star-shaped sampling planes to enable end-to-end learning from raw volumetric OCT and demonstrates that the approach can recover prognostic spatiotemporal patterns beyond feature-engineered machine learning baselines, supported by occlusion-sensitivity visualizations tied to treatment requirement strata.
[415]: Enhanced AMD detection in OCT images using GLCM texture features with Machine Learning and CNN methods	2025	This research targets age-related macular degeneration (AMD) detection using supervised machine learning and convolutional neural networks on preprocessed retinal OCT image datasets, focusing on gray level co-occurrence matrix (GLCM) texture features and assessing how feature selection affects diagnostic outcomes.	This analysis reports that feature sets combining selected GLCM measures, particularly the sfGLCM group, outperform using all GLCM features or grayscale pixel features alone, which matters for designing more accurate and less overfitting OCT-based computational AMD detection approaches.

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Work	Year	Overall reasoning	Contribution and summary
[135]: Enhanced Deep Learning Model for Classification of Retinal Optical Coherence Tomography Images	2023	The study targets a retinal OCT imaging classification problem by building an enhanced deep learning pipeline that combines a modified ResNet-50 with a random forest and optimizes training using Adam (and dual SGD/Adam), demonstrating multiclass performance across four OCT categories (CNV, DME, drusen, and normal).	The authors report a tailored ResNet-50 plus random forest approach trained with Adam to improve automated OCT image categorization, and they provide quantitative evidence on a multiclass dataset with published diagnostic metrics to support clinical relevance for OCT-based decision support.
[154]: Enhancing pathological myopia diagnosis: a bimodal artificial intelligence approach integrating fundus and optical coherence tomography imaging for precise atrophy, traction and neovascularisation grading	2025	Using a bimodal deep learning approach that fuses paired fundus photographs and optical coherence tomography (OCT) images, this single-center retrospective cross-sectional study performs ATN (atrophy, traction, neovascularization) staging classification for pathological myopia and reports comparative evidence, including multiclass and binary evaluations, supporting improved discrimination for severe disease.	The research introduces a ResNet-50-based multimodal multi-instance learning model with cross-modality interaction to grade ATN components from matched fundus-OCT inputs, providing accuracy improvements over single-modality models that may benefit screening and timely referral workflows.
[103]: Ensemble Convolutional Neural Networks for the Classification and Visualization of Retinal Diseases in Optical Coherence Tomography Images	2023	A clinical study applies a deep-learning CNN ensemble to OCT imaging to classify four retinal disease categories and generate positive/negative heatmaps for activity visualization and region-of-interest estimation, reporting diagnostic performance metrics as evidence of feasibility.	This research introduces an ensemble built from VGG19, ResNet152, and DenseNet121 with soft and hard voting plus a new positive/negative heatmap algorithm to support second-reader interpretation of retinal OCT by highlighting relevant areas.
[372]: Ensemble Deep Learning for Diabetic Retinopathy Detection Using Optical Coherence Tomography Angiography	2020	This ophthalmology study evaluates an ensemble deep learning approach based on fine-tuned VGG19, ResNet50, and DenseNet CNN backbones (using majority soft voting and stacking) to classify referable versus nonreferable diabetic retinopathy from en face OCTA and co-registered structural OCT images, with Grad-CAM used for localization and evidence from five-fold cross-validation performance comparisons to a handcrafted feature classifier.	The analysis shows that combining multiple OCT/OCTA-trained CNNs yields higher referable DR discrimination than single-data-type networks and a conventional pipeline using handcrafted features, while Grad-CAM provides model interpretability by highlighting image regions linked to microvascular and structural disease signals.

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Work	Year	Overall reasoning	Contribution and summary
[69]: Ensemble Learning Based on Convolutional Neural Networks for the Classification of Retinal Diseases from Optical Coherence Tomography Images	2020	The research uses an ensemble of CNN-based architectures to classify spectral-domain OCT images into four retinal categories (CNV, DME, DRUSEN, and NORMAL), applying supervised training with image preprocessing and augmentation and evaluating performance against individual CNNs using metrics reported for the same dataset split.	The reported work develops and compares multiple ensemble configurations, then identifies the best-performing CNN plus DenseNet121 plus InceptionV3 averaged-prediction setup, demonstrating high multi-class diagnostic accuracy for OCT while emphasizing feasibility on a limited dataset.
[327]: Epiretinal Membrane Detection in Optical Coherence Tomography Retinal Images Using Deep Learning	2021	The reported work uses deep learning with transfer learning to adapt four convolutional neural network architectures for automatic epiretinal membrane detection in optical coherence tomography retinal images, reporting quantitative evidence from an exhaustive evaluation with high reported classification performance.	The study refines CNN transfer learning for ERM detection by tuning network hyperparameters and comparing cross-entropy versus focal loss, demonstrating that the approach can work effectively with relatively small training datasets in OCT-based clinical imaging.
[453]: Evaluation of 2D and 3D Deep Learning Approaches for Automatic Segmentation of the Retinal External Limiting Membrane in Spectral Domain Optical Coherence Tomography Images	2022	This analysis evaluates 2D versus 3D deep learning networks for automated segmentation of the retinal external limiting membrane in spectral-domain optical coherence tomography, reporting quantitative evidence that 3D architectures better capture 3D structure, with performance differences across overlap and surface-distance measures.	This investigation directly compares three state-of-the-art model designs in both 2D and 3D forms on a public spectral-domain OCT dataset, showing that 3D variants typically improve Dice, mean surface distance, false positive rate, and surface smoothness while trading off Hausdorff distance.
[65]: Evaluation of Convolutional Neural Networks (CNNs) in Identifying Retinal Conditions Through Classification of Optical Coherence Tomography (OCT) Images	2025	In an ophthalmic OCT imaging setting for diabetic macular disease classification, the study trains supervised convolutional neural networks (VGG16 and InceptionV3) to categorize grayscale OCT images into four diagnostic groups (normal, choroidal neovascularization, diabetic macular edema, and drusen) and reports evidence from model evaluation using accuracy, sensitivity, specificity, and confusion-matrix-based analyses across varying dataset sizes.	This analysis provides a direct, experimentally tested comparison of VGG16 versus InceptionV3 on a four-class OCT dataset, showing how performance (including sensitivity and specificity) changes with training set size and clarifying practical limitations such as missing image augmentation and grayscale-only inputs.

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Work	Year	Overall reasoning	Contribution and summary
[317]: Explainable Deep Learning Approach for Automated Classification of Retinal Diseases Using OCT Images and Grad-CAM	2025	An ophthalmology-focused CNN is trained for automated four-class retinal OCT image classification (CNV, DME, drusen, normal), uses Grad-CAM to generate region-level visual explanations, and reports classification accuracy and related measures.	This work details an explainable OCT-based classification pipeline that leverages extensive labeled training data, preprocessing and augmentation, and Grad-CAM highlighting to support interpretable decision-making for retinal disease triage.
[321]: Eye Disease Prediction Using Deep Learning and Attention on Oct Scans	2024	The study describes an OCT-imaging AI pipeline for ophthalmic eye disease prediction, using a custom UNet for image segmentation followed by an attention-enhanced InceptionV3 classifier to categorize normal eyes versus conditions including DME, CNV, and drusen, with reported classification performance from extensive experimentation.	The research introduces an end-to-end web-based workflow that lets users upload raw OCT scans for automated segmentation and condition categorization, supporting timely decision-making for screening and early detection based on the model's reported precision, recall, accuracy, and F1 score.
[62]: Feasibility study to improve deep learning in OCT diagnosis of rare retinal diseases with few-shot classification	2021	Using few-shot learning for OCT diagnosis of rare retinal diseases, the study generates synthetic OCT images with CycleGAN from normal OCTs and trains an Inception-v3 classifier, with validation on an independent test dataset to demonstrate improved classification accuracy versus several deep-learning and few-shot baselines.	This research develops disease-specific GAN augmentation for five rare retinal classes and shows that the resulting feature learning improves rare-class separability and diagnosis performance, addressing the data-scarcity challenge that limits conventional multiclass OCT deep learning.
[363]: Few-shot out-of-distribution detection for automated screening in retinal OCT images using deep learning	2023	This 2023 study evaluates uncertainty-based out-of-distribution detection for automated age-related macular degeneration screening and staging from retinal OCT B-scans using a CNN and several OOD scoring/uncertainty method families, testing performance on near-OOD OCT outliers (DME, RVO, Stargardt) and far-OOD eye fundus images, with evidence from comparative AUC-based experiments including a few-shot outlier exposure strategy.	The analysis shows that few-shot outlier exposure combined with cosine-distance feature-space scoring yields improved near-OOD rejection (e.g., with eight exposed outliers per class) without reducing AMD classification performance, and argues for cosine-distance as a practical alternative to reject-bucket probability for OCT-based OE setups.

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Work	Year	Overall reasoning	Contribution and summary
[328]: Forme fruste keratoconus detection with OCT corneal topography using artificial intelligence algorithms	2024	This retrospective monocentric case-control study evaluated machine learning classification (random forest and logistic regression) using swept-source OCT corneal topography from CASIA 2 to separate normal corneas from forme fruste keratoconus for refractive surgery risk screening, benchmarking algorithm outputs against the ectasia screening index (ESI).	The study reports a multiclass/automated approach that identifies FFKC with markedly higher sensitivity than the built-in ESI while maintaining high specificity, highlighting discriminative topographic parameters, particularly posterior surface features, that may improve clinical screening workflows.
[181]: From Machine to Machine: An OCT-Trained Deep Learning Algorithm for Objective Quantification of Glaucomatous Damage in Fundus Photographs	2019	In a glaucoma-focused cross-sectional ophthalmic dataset, the authors train a convolutional neural network on optic disc fundus photographs using spectral-domain OCT retinal nerve fiber layer thickness as the quantitative target, then evaluate prediction fidelity and discrimination of glaucomatous versus healthy eyes with correlation/agreement and ROC analyses in a held-out test set.	Rather than learning from subjective optic disc labels, the study demonstrates OCT-trained image-to-measurement transfer by mapping disc photographs to continuous RNFL thickness estimates, and validates that these predictions replicate SDOCT-based structure-function diagnostic performance in the test cohort.
[314]: Fully Automated Detection and Quantification of Macular Fluid in OCT Using Deep Learning	2018	This 2018 study develops and validates a fully automated deep-learning semantic segmentation approach for conventional spectral-domain OCT to detect and quantify intraretinal cystoid fluid and subretinal fluid across neovascular AMD, DME, and RVO on both Cirrus and Spectralis platforms, with evaluation against masked expert consensus using volume-level detection metrics and pixel-level localization and extent comparisons.	This investigation demonstrates reproducible, pixel-wise fluid typing and volume quantification in routine macular OCT by using an encoder-decoder pipeline with device-specific training and correlation to manual expert measurements, enabling consistent IRC versus SRF measurement for research and clinical decision support.
[63]: Fully automated detection and quantification of multiple retinal lesions in OCT volumes based on deep learning and improved DRLSE	2019	Within an OCT-based ophthalmic workflow, the study describes a fully automated deep learning approach that integrates improved DRLSE to detect and segment multiple retinal lesion types and evaluates it with quantitative overlap metrics on annotated 3D OCT-derived B-scans.	This analysis specifies five lesion classes (PED, SRF, drusen, CNV, and MH) and reports performance on 500 B-scans from 15 3D OCT volumes, demonstrating an end-to-end method for automated lesion quantification in retinal imaging.

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Work	Year	Overall reasoning	Contribution and summary
[88]: Fused-AETNet: A Variational Transformer-Based Framework for Diabetic Retinopathy Classification Using OCT Biomarkers	2025	The study targets diabetic retinopathy classification from central OCT B-scans by segmenting 11 retinal layers with atlas-guided probabilistic methods, extracting layer-wise handcrafted morphological biomarkers, and using a variational autoencoder with transformer-inspired attention to learn and refine a latent representation for diagnostic decision-making with reported quantitative validation on 481 subjects.	This work introduces Fused-AETNet, which operationalizes clinical interpretability by fusing percentile-based cumulative distribution functions of reflectivity, Laplacian-derived thickness, and boundary tortuosity into a variational latent space that is then selectively emphasized through attention, enabling a compact biomarker-driven OCT DR classifier with experimentally reported performance.
[172]: Fusing Results of Several Deep Learning Architectures for Automatic Classification of Normal and Diabetic Macular Edema in Optical Coherence Tomography	2018	This work evaluates a convolutional neural network ensemble approach on OCT imaging to classify diabetic macular edema versus normal, using principal component analysis with voting and patient-independent Leave-Two-Patients-Out cross-validation as the evidence for an ophthalmic diagnostic task.	The research offers two fusion-based DME classification frameworks that leverage pretrained AlexNet, VggNet, and GoogleNet feature integration and majority/plurality voting, reporting volume-level performance to support automated screening potential on multi-institution OCT datasets.
[384]: GCC Aware Glaucoma Detection Using Macula OCT Image Analysis Based on Deep CNN	2024	The study evaluates a deep CNN pipeline for glaucoma screening using macular SD-OCT B-scans, where deep segmentation first extracts the GCC (including the retinal nerve fiber layer and ganglion cell complex with the inner plexiform layer) and subsequent CNN classification distinguishes glaucomatous from non-glaucomatous eyes while explicitly modeling multiple macular GCC subregions, with performance reported from training and validation on a single locally acquired OCT dataset.	This research demonstrates that GCC-aware, region-specific OCT analysis can improve glaucomatous discrimination beyond using OCT without targeted GCC focus, highlighting a previously unreported approach that accounts for how different macular GCC regions affect screening performance.
[387]: Generative adversarial network-based deep learning approach in classification of retinal conditions with optical coherence tomography images	2023	This investigation evaluates a GAN-based data synthesis strategy for classifying retinal conditions from optical coherence tomography (OCT) images, using pretrained deep neural networks and testing on both a publicly available benchmark dataset and a hospital-collected clinical OCT set, providing comparative performance evidence in an ophthalmic workflow.	The analysis demonstrates that training with GAN-generated class-balanced OCT data can improve classification accuracy relative to training on the original imbalanced dataset across multiple evaluation settings, including a clinical test set, which supports the practical value of synthesis-based balancing for retinal OCT classification.

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Work	Year	Overall reasoning	Contribution and summary
[252]: GENERATIVE DEEP LEARNING APPROACH TO PREDICT POSTTREATMENT OPTICAL COHERENCE TOMOGRAPHY IMAGES OF AGE-RELATED MACULAR DEGENERATION AFTER 12 MONTHS	2025	For neovascular age-related macular degeneration, the study evaluates a conditional GAN-based generative deep learning model that uses pretreatment OCT plus fluorescein and indocyanine green angiography, along with treatment-related staging and clinical variables, to forecast 12-month posttreatment OCT anatomy, with evidence from validation on 533 eyes.	The reported work shows that adding OCT acquired after loading doses and key regimen and treatment descriptors markedly improves compartment-level OCT prediction and wet versus dry macular status, supporting a clinically usable way to anticipate long-term anatomical response from baseline imaging and management context.
[438]: Glaucoma Detection From Raw Circumpapillary OCT Images Using Fully Convolutional Neural Networks	2020	The reported work uses fully convolutional neural networks in ophthalmic imaging to classify glaucoma versus normal from raw circumpapillary OCT B-scans around the optic nerve head, with evidence from an independent test set reported using ROC AUC and accuracy.	The study evaluates two CNN training strategies, training from scratch versus fine-tuning, and shows that fine-tuned VGG-family models perform best on the independent test set, which is particularly informative for settings with limited OCT data.
[431]: Glaucoma Diagnosis Through the Integration of Optical Coherence Tomography/Angiography and Machine Learning Diagnostic Models	2022	In this retrospective cross-sectional glaucoma clinic study, investigators derived OCTA-based structural and vascular parameters and built machine-learning classifiers to distinguish healthy from glaucomatous eyes (and among glaucoma severities), using a rigorous nested stratified group cross-validation framework with held-out test evaluation and interpretable decision-tree outputs.	They also generated North Texas normative ranges across multiple OCT/OCTA metrics and showed which OCTA features most influenced the final model, providing clinician-facing, data-driven support for glaucoma assessment.
[209]: Glaucoma Diagnosis with Machine Learning Based on Optical Coherence Tomography and Color Fundus Images	2019	This research focuses on glaucoma diagnosis in an ophthalmic setting by combining OCT-derived thickness and deviation maps with color fundus images, using transfer-learned CNNs with an ensemble random forest to classify healthy versus glaucomatous eyes, evaluated by 10-fold cross-validated ROC AUC.	This analysis details a multimodal pipeline that builds separate VGG19-based classifiers for disc fundus and multiple OCT feature maps, then fuses their learned representations with a random forest to reach improved cross-validated discrimination of glaucoma from healthy controls.

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Work	Year	Overall reasoning	Contribution and summary
[444]: Glaucoma Diagnostic Accuracy of Machine Learning Classifiers Using Retinal Nerve Fiber Layer and Optic Nerve Data from SD-OCT	2013	This ophthalmic cross-sectional study assessed diagnostic accuracy of 10 machine learning classifiers for distinguishing early-to-moderate primary open-angle glaucoma from healthy eyes using SD-OCT-derived retinal nerve fiber layer and optic nerve parameters, with ROC/AUC and sensitivity/specificity computed under 10-fold cross-validation.	This work systematically compared classifier performance against individual SD-OCT parameters and found that the best random forest model achieved good discrimination without producing statistically significant improvement over the top single structural metric, supporting the clinical value and limits of OCT-based AI for glaucoma screening in this setting.
[299]: GlauNet: Glaucoma Diagnosis for OCTA Imaging Using a New CNN Architecture	2022	GlauNet is a CNN for glaucoma detection in an ophthalmic setting, designed to extract features from OCTA of the optic nerve head and retinal nerve fiber layer for a glaucoma versus non-glaucoma classification task, with evidence from training and separate testing and reported diagnostic metrics including robustness to artifact and poor-quality images.	By replacing handcrafted OCTA parameter pipelines with an end-to-end two-stage convolutional feature-extraction plus fully connected classification architecture, GlauNet highlights vascularly relevant regions and supports clinically actionable performance with maintained sensitivity and specificity under artifact degradation.
[302]: Harnessing deep learning for detection of diabetic retinopathy in geriatric group using optical coherence tomography angiography-OCTA: A promising approach	2024	This 2024 study develops an ophthalmic deep-learning system for diabetic retinopathy severity classification from OCTA imaging in a geriatric case-control cohort, using CNN-based feature extraction with multiple pretrained architectures and reporting model evaluation on a held-out test split with standard classification metrics.	This research operationalizes OCTA layer-focused preprocessing and compares several CNN backbone plus classical classifier combinations (including KNN and feed-forward neural networks) to grade DR into no DR signs, mild, or moderate categories, providing quantitative performance results that support OCTA-enabled automated screening for older adults.
[338]: HTC-retina: A hybrid retinal diseases classification model using transformer-Convolutional Neural Network from optical coherence tomography images	2024	This investigation evaluates a hybrid CNN and visual transformer framework for automated multi-label retinal disease classification using optical coherence tomography images, leveraging CPTE and an RDP-ConvNet module to emphasize lesion-relevant local features and reporting comparative performance across three OCT datasets.	The analysis introduces HTC-Retina, a transformer-convolution design that replaces raw tokenization with convolutional patch and token embedding and adds a residual depthwise-pointwise conv branch, demonstrating high accuracy and sensitivity with computational efficiency for seven retinal diseases.

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Work	Year	Overall reasoning	Contribution and summary
[417]: Human versus Artificial Intelligence: Validation of a Deep Learning Model for Retinal Layer and Fluid Segmentation in Optical Coherence Tomography Images from Patients with Age-Related Macular Degeneration	2024	This external validation study assesses a deep learning model for OCT retinal layer and total fluid segmentation in an ophthalmic setting of age-related macular degeneration by comparing automated outputs to a clinically derived gold standard based on manually adjusted Heidelberg Spectralis segmentation (and MATLAB fluid refinement), evaluating quantitative agreement across healthy, intermediate, and exudative AMD groups.	Results demonstrated near-complete segmentation overlap for retinal boundaries overall and close concordance of key OCT-derived thickness and fluid area measurements, while also pinpointing reduced agreement for specific layers (notably ISE) and increasing method discrepancies with later disease stage.
[198]: Hybrid 3d-2d Deep Learning For Detection Of Neovascularage-Related Macular Degeneration Using Optical Coherence Tomography B-Scans And Angiography Volumes	2021	The reported work uses a CNN-based hybrid 3D-2D deep-learning approach on OCTA angiography volumes and OCT B-scans to classify retinal findings into non-AMD, non-neovascular AMD, and neovascular AMD, targeting automated detection of late-stage disease by capturing choroidal neovascularization and sequelae evidence in an ophthalmic imaging setting.	The study introduces a multi-class automated staging pipeline for AMD based on hybrid 3D-2D CNNs using retinal angiography and structural OCT inputs, which matters for reducing clinician workload in interpreting OCTA-rich datasets for late-stage differentiation.
[176]: Hybrid deep learning and optimal graph search method for optical coherence tomography layer segmentation in diseases affecting the optic nerve	2024	In the ophthalmic OCT setting, the study tests Deep LOGISMOS, a hybrid deep learning plus optimal 3D graph search method, for segmenting retinal layers and quantifying GCIPL thickness in optic nerve diseases with irregular layer topology, using reference-traced and longitudinal OCT cohorts across Cirrus and Spectralis and reporting segmentation accuracy and clinically oriented thickness robustness.	This investigation demonstrates a two-stage true 3D workflow where learned probability maps drive in-region costs while built-in topology constraints avoid extensive dataset-specific post-processing, yielding more reliable GCIPL thickness measurement over time than baseline reference algorithms and deep-learning-only segmentation in the provided NAION, glaucoma, and multiple sclerosis test sets.
[207]: Hybrid Deep Learning/Machine Learning Model for Retinal Diseases Classifications Using OCT Images	2024	In an ophthalmic diagnostic context, the study uses a hybrid deep learning and machine learning pipeline for OCT imaging, extracting features with a DenseNet and then performing multi-condition retinal disease classification using an SVM tuned by Bayesian optimization, with testing on publicly accessible OCT data across eight retinal conditions.	This analysis quantifies the benefit of DenseNet feature extraction combined with Bayesian-optimized SVM hyperparameters for OCT-based classification across eight conditions, offering an end-to-end automated workflow that could support earlier, more consistent clinical screening and decision-making.

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Work	Year	Overall reasoning	Contribution and summary
[146]: Hybrid deep learning models for the screening of Diabetic Macular Edema in optical coherence tomography volumes	2024	This retrospective evaluation in a real-world teleophthalmology diabetic retinopathy screening program uses a hybrid CNN-RNN model to read complete OCT cubes (with bidirectional recurrence over cube-derived slice sequences) for binary diabetic macular edema detection, with clinically oriented threshold tuning and diagnostic metrics reported against retina-specialist ground truth.	Processing unselected OCT volumes from 4 years of screening, the authors show how a tuned multi-CNN bidirectional RNN ensemble can support first-pass identification of diabetic macular edema, particularly for extrafoveal cases, by optimizing sensitivity and specificity tradeoffs relevant to population screening workflows.
[437]: HyFormer: a hybrid transformer-CNN architecture for retinal OCT image segmentation	2024	HyFormer is an ophthalmic OCT image segmentation study that uses a hybrid Transformer-CNN architecture to perform multi-class delineation of retinal layers and lesion types (including traction maculopathy and age-related degeneration) on myopic traction maculopathy OCT data and the public AROI dataset, reporting evidence from comparative evaluations across these imaging tasks and datasets.	The authors develop an efficient HyFormer design with parallel Transformer and convolutional encoders plus multi-scale gated attention, group positional embedding, and a three-path fusion decoder, and show that this combination improves segmentation quality while maintaining computational efficiency through their network-level design and loss formulation with CAM-guided supervision.
[376]: Hyper-reflective foci segmentation in SD-OCT retinal images with diabetic retinopathy using deep convolutional neural networks	2019	The analysis uses deep convolutional classification-network adaptations to perform pixel-wise segmentation of hyper-reflective foci in diabetic retinopathy spectral-domain OCT, evaluated with Dice-based evidence on a small-lesion task using retinal imaging rather than nonimaging signals.	This research details a fused deep convolutional pipeline that segments HRF from small patch-based pixel predictions and reports Dice similarity coefficients for foci area and count, providing quantitative support for lesion-focused delineation in SD-OCT B-scans.
[70]: Identification of diabetic retinopathy classification using machine learning algorithms on clinical data and optical coherence tomography angiography	2024	This two-center cross-sectional ophthalmology study used machine-learning classifiers (random forest, GBM, deep learning, and logistic regression) to assign multiclass diabetic retinopathy categories (DR, referable DR, and vision-threatening DR) from OCTA-derived quantitative vascular and retinal metrics combined with systemic clinical data, and it provides evidence from independent external validation using ROC/AUC.	By comparing feature patterns across clinical-only, OCTA-only, and combined OCTA plus clinical/non-laboratory variables, the study identifies OCTA-informed models, particularly those with non-laboratory factors and vascular density measures, as practical tools for screening and referral decision support for DR severity stratification.

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Work	Year	Overall reasoning	Contribution and summary
[435]: Implementing Predictive Models in Artificial Intelligence through OCT Biomarkers for Age-Related Macular Degeneration	2023	This narrative review focuses on ophthalmic prediction of age-related macular degeneration progression by summarizing AI methods that leverage multimodal retinal imaging, with particular emphasis on OCT-derived biomarkers used to forecast high-risk phenotypes and evolution toward late disease.	The reported work consolidates reported OCT feature predictors, such as retinal layer morphometrics, drusen metrics, reticular pseudodrusen, and hyperreflective foci, within the context of AI-based progression modeling to clarify what evidence currently supports and where future work may refine predictive efforts.
[269]: Improving OCT Image Segmentation of Retinal Layers by Utilizing a Machine Learning Based Multistage System of Stacked Multiscale Encoders and Decoders	2023	The reported work evaluates a deep learning multi-stage stacked multiscale encoder-decoder segmentation pipeline for delineating retinal layers on optical coherence tomography (OCT) B-scans, using peripapillary OCT data plus additional publicly available datasets (Duke SD-OCT, UMN, and Heidelberg) to test robustness on noisier imagery and provide Dice-score-based evidence against prior single-stage approaches.	By stacking a first local-context encoder-decoder with a second refinement/denoising stage and paired attention-based decoders, the authors report improved retinal-layer segmentation quality on the peripapillary dataset (highest Srensen-Dice result) and demonstrate qualitative generalization to datasets with limited or noisier annotations, supporting the value of multi-stage multiscale processing for OCT segmentation.
[399]: Integrating Human Expertise With Artificial Intelligence (AI) Models for Optical Coherence Tomography (OCT) Retinal Fluid and Pathology Quantification: A Systematic Review	2025	This systematic review synthesizes ophthalmology-focused evidence on clinician-in-the-loop hybrid workflows that pair human review with deep-learning AI architectures for OCT-based retinal fluid and biomarker quantification, assessing diagnostic and operational outcomes reported across included studies from 2021 to 2025.	This investigation consolidates results from nine hybrid human-AI OCT studies to clarify where expert-level biomarker reliability holds and where complex features remain challenging, while highlighting clinically relevant efficiency gains of more than 50% that support practical adoption for retinal pathology assessment.
[195]: Integrating lightweight convolutional neural network with entropy-informed channel attention and adaptive spatial attention for OCT-based retinal disease classification	2025	LOCT-Net is a lightweight convolutional neural network for retinal disease classification using OCT B-scan imaging, combining nested residual multi-scale feature extraction with entropy-informed channel attention and adaptive spatial attention, and it provides post-hoc explainable AI with evidence from evaluation on six benchmark OCT datasets including reported classification metrics.	The study delivers a parameter-efficient OCT classifier with attention mechanisms tailored to channel- and region-level feature refinement, showing that strong diagnostic performance can be maintained at low computational cost while adding interpretability via explainable AI.

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Work	Year	Overall reasoning	Contribution and summary
[233]: Integrating Retinal Segmentation Metrics with Machine Learning for Predictions from Mouse SD-OCT Scans	2025	The study describes an ophthalmic SD-OCT study in mice where retinal layer thickness segmentation metrics are used to train and optimize neural networks for categorical prediction of mouse age and species, with validation reported against multiple baseline machine learning methods using accuracy and AUC-like evidence.	This work demonstrates that feature-level OCT segmentation metrics, rather than raw images, can yield high validation performance for mouse group differentiation, supporting automated retinal morphometry as a practical input for experimental predictive modeling.
[341]: Interactive Deep Learning-Based Retinal OCT Layer Segmentation Refinement by Regressing Translation Maps	2024	This work evaluates an interactive deep learning method for retinal OCT layer boundary segmentation that refines an initial model output using clinician-guided point clicks or line scribbles, regressing pixel-wise translation maps to update boundaries and providing coordinate-wise confidence as evidence of failure-prone regions across multiple private and public datasets.	The research introduces a translation-map-based interaction mechanism that turns subjective specialist corrections into localized boundary refinements and couples them with confidence estimates, enabling more reliable, user-supervised segmentation refinement for clinical OCT workflows.
[162]: Interpretable Deep Learning For Automated Retinal Damage Detection in Optical Coherence Tomography	2024	Using transfer-learning CNNs (VGG16, EfficientNetB3, and ResNet50) on optical coherence tomography images, the study targets automatic four-class retinal injury classification and generates Grad-CAM visual explanations.	This research compares three pretrained architectures for Drusen, Normal, DME, and CNV detection and pairs the best-performing model with Grad-CAM region highlighting to support clinically interpretable decision-making.
[120]: Interpretable Retinal Disease Classification from OCT Images Using Deep Neural Network and Explainable AI	2021	This investigation uses a deep neural network with pretrained CNN backbones on macular OCT images to classify CNV, DME, and drusen, and it provides evidence of interpretability by applying LIME to explain misclassifications.	The analysis offers an OCT classification pipeline that couples CNN-based predictions with LIME visual explanations to identify why errors occur, supporting future refinement of model performance.
[420]: IoT based automatic detection of glaucoma disease in OCT and fundus images using deep learning techniques	2020	Within ophthalmic practice for glaucoma detection using retinal imaging, the study applies a two-phase RCNN-style deep learning workflow to first localize the optic nerve head/optic disc and then classify it as glaucoma or stable, with rule-based semi-automatic ground-truth support and evaluation on the ORIGA fundus dataset.	The reported work introduces a semi-automatic, rule-assisted labeling strategy for optic disc localization and couples this with glaucoma-versus-stable classification to show that AUC alone may be insufficient for assessing performance under dataset splits and class imbalance conditions.

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Work	Year	Overall reasoning	Contribution and summary
[170]: JOINT MOTION CORRECTION AND 3D SEGMENTATION WITH GRAPH-ASSISTED NEURAL NETWORKS FOR RETINAL OCT	2022	The reported work uses graph-assisted neural networks on 3D OCT volumes to jointly handle motion artifacts and produce consistent retinal layer segmentation across neighboring B-scans, with evidence reported as experimental visual and quantitative improvements versus conventional and 2D deep-learning approaches.	The study advances a single framework that couples eye motion correction with 3D, cross-B-scan retinal layer segmentation, addressing how misalignment from involuntary eye movements undermines downstream OCTAngiography projection and disease analysis.
[64]: Joint Segmentation and Uncertainty Visualization of Retinal Layers in Optical Coherence Tomography Images Using Bayesian Deep Learning	2018	The study describes Bayesian deep learning for OCT-based retinal layer segmentation that couples end-to-end pixel-wise labeling with a corresponding uncertainty map, with evidence from validation on 1487 OCT volumes from 15 subjects against uncertainty-agnostic state-of-the-art segmentation methods.	This investigation introduces an uncertainty visualization mechanism that highlights likely mis-segmented retinal regions and reports comparable or improved accuracy with increased robustness to noise relative to methods that do not model uncertainty.
[114]: Linking Function and Structure with ReSensNet: Predicting Retinal Sensitivity from OCT using Deep Learning	2022	This retrospective cross-sectional study uses a deep learning model to predict microperimetry retinal sensitivity maps from spectral-domain OCT, evaluated with internal development and external validation across multiple retinal diagnoses by point-wise and mean error and correlation measures.	This analysis shows that OCT-based structure can predict both point-wise and mean retinal sensitivity with measured agreement and correlation against microperimetry in development and external testing, supporting an imaging-derived, objective functional surrogate for clinical monitoring and trials.
[357]: Loss-Modified Transformer-Based U-Net for Accurate Segmentation of Fluids in Optical Coherence Tomography Images of Retinal Diseases	2023	The study targets ophthalmic OCT semantic segmentation by using a transformer-enhanced U-Net (Trans-U-Net) with B-scan imaging and a customized hybrid loss to delineate retinal fluid regions associated with macular edema, providing evidence from quantitative evaluation including 5-fold cross-validation and robustness testing with added Gaussian noise.	This work introduces a weighted combination of Dice, focal Tversky, and weighted binary cross-entropy to address class imbalance and noisy OCT inputs, and reports improved segmentation quality and convergence relative to alternative loss functions on initial and noise-perturbed datasets.
[126]: Machine learning and optical coherence tomography-derived radiomics analysis to predict persistent diabetic macular edema in patients undergoing anti-VEGF intravitreal therapy	2024	Using retrospective OCT-omics radiomics derived from manually segmented spectral-domain OCT inner-retina layers, the study applies logistic regression, SVM, and backpropagation neural network classifiers to predict persistent versus non-persistent diabetic macular edema after the first three anti-VEGF intravitreal injections, with evidence from separate training and test sets and decision-curve analysis to support clinical relevance.	The research delivers a non-invasive OCT-based prediction pipeline with z-scored, reproducibility-filtered radiomics features and recursive feature elimination, showing that OCT-omics scores differentiate persistent from non-persistent DME and track central subfield thickness change, thereby offering a quantitative approach for treatment-response stratification in diabetic macular edema.

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Work	Year	Overall reasoning	Contribution and summary
[359]: Machine learning-based prediction of persistent diabetic macular edema using OCT-derived structural biomarkers and clinical features	2026	In a retrospective single-center ophthalmic setting, the authors used machine learning classifiers that combine baseline clinical variables with OCT-derived structural biomarkers to predict persistent diabetic macular edema after an anti-VEGF loading phase, evaluating performance in an internal train/validation split and using SHAP for evidence-based interpretability of key predictors.	By comparing eight model types and showing that multimodal OCT-and-clinical approaches outperform clinical-only or OCT-only baselines, the study provides practical, OCT-anchored risk stratification for persistent DME, with diabetes duration, EZ/ELM disruption, and hyperreflective foci highlighted as influential factors for clinician-facing decision support.
[205]: Machine learning classifiers for glaucoma diagnosis based on classification of retinal nerve fibre layer thickness parameters measured by Stratus OCT	2010	In an ophthalmic glaucoma diagnostic setting, this investigation evaluates two machine learning classifier families, artificial neural networks and support vector machines, using Stratus OCT retinal nerve fiber layer thickness features (including transformed A-scan measurements) to distinguish glaucoma from healthy eyes, with evidence reported as receiver operating characteristic area under the curve (AROC) comparisons across conventional versus novel input parameterizations.	The analysis shows that classifier choice (ANN versus SVM) yielded no meaningful differences, while transformed A-scan-based RNFLT inputs produced the best AROCs, highlighting that imaging feature/input engineering can be at least as important as the learning algorithm for OCT-based glaucoma diagnosis.
[128]: MACULA OCT Disease Images Classification using Deep Learning Techniques	2024	Using convolutional neural networks for macular optical coherence tomography image analysis, the research addresses the task of automated retinal disease classification across multiple macular conditions, supported by reported classification accuracy with a top-performing ResNet model.	The reported work evaluates several CNN architectures, including ManualNet, AlexNet, and ResNet, and reports that ResNet reaches 90% accuracy, offering an actionable deep-learning approach for differentiating specified retinal diseases on macular OCT.
[459]: Macular edema degeneration classification on OCT and fundus images with portable platform based on artificial intelligence methods	2022	A study develops a portable AI computer-aided diagnosis workflow for ophthalmic staging of age-related macular degeneration, using machine learning neuronal networks with combined OCT and fundus images to classify normal retina, dry AMD, and wet AMD on a Jetson TX2 device, with evidence limited to evaluation on a provided three-class dataset.	The study demonstrates how a Jetson TX2-based model can perform multi-class AMD categorization from multimodal retinal imaging in a point-of-care setting, highlighting practical cost and portability advantages for clinician support.

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Work	Year	Overall reasoning	Contribution and summary
[184]: Macular Holes Detection Using Deep Learning on Optical Coherence Tomography Images	2023	The study applies deep learning to optical coherence tomography imaging to detect macular holes and automate their size measurement, addressing the clinically relevant task of staging MHs and reducing human judgment errors while reporting detection accuracy and measurement efficiency.	This investigation introduces an OCT-based deep learning approach that couples macular hole detection with automated sizing, aiming to speed up and improve stage recognition by distinguishing stage 2 from stage 3 or 4 without the time-consuming caliper workflow.
[60]: Macular OCT Classification Using a Multi-Scale Convolutional Neural Network Ensemble	2018	The study describes a convolutional neural network ensemble in a macular OCT CAD setting, using a multi-scale mixture-of-experts approach to classify OCT images into normal retina versus dry age-related macular degeneration and diabetic macular edema, with evidence reported as classification results on two Heidelberg OCT datasets.	This analysis introduces a multi-scale correlated mixture-of-experts model with a new cost function to enable discriminative, fast feature learning from multiple OCT sub-image scales and reports high average precision and AUC for the three-way macular classification task.
[248]: MBT: Model-Based Transformer for retinal optical coherence tomography image and video multi-classification	2023	The study applies a model-based transformer framework to OCT data for multiclass retinal disease classification, extending transformer-based learning from OCT images to OCT videos, and reports evidence from experiments on three real-world retinal datasets using standard classification metrics.	This work introduces MBT, which uses approximate sparse representation to learn optimal features and improves OCT image and video classification relative to transformer baselines, highlighting a pathway to identify retinal pathologies from OCT videos as well as images.
[288]: MDLGO: Integrated Multimodal Deep Learning Framework for Enhanced Precision in Glaucoma Diagnosis Using OCT A-scans	2024	This work develops a multimodal deep learning system for glaucoma diagnosis that fuses OCT A-scans with fundus images and visual field data, using transfer learning, attention-based segmentation, and anomaly detection to generate diagnostic evidence reported as improved metrics and reduced time to diagnosis.	The research specifically details how pretrained feature extractors combined with V-Net and SegNet attention mechanisms for OCT A-scan region delineation, together with One-Class SVM and Isolation Forest anomaly detection, can better capture subtle early glaucoma signals and improve diagnostic precision.
[296]: Multi-Fundus Diseases Classification Using Retinal Optical Coherence Tomography Images with Swin Transformer V2	2023	The analysis evaluates an ophthalmic deep learning approach that uses the Swin Transformer V2 architecture with windowed self-attention and PolyLoss to perform multi-disease classification from retinal OCT images, training and testing on two independent public datasets (OCT2017 and OCT-C8) with quantitative diagnostic performance and Grad-CAM visual evidence.	The authors introduce a modified Swin Transformer V2 pipeline tailored to OCT classification by combining localized attention computation with PolyLoss, then report consistently high multi-class diagnostic results and interpretability heatmaps that support the model's decision basis across both datasets.

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Work	Year	Overall reasoning	Contribution and summary
[219]: Multi-label classification of Retinal Disorders in Optical Coherence Tomography using Deep Learning	2021	A study uses a CNN deep-learning approach on optical coherence tomography (OCT) images to perform multi-label identification of diabetic macular edema, choroidal neovascularization, and drusen, with an additional normal-macula category, providing evidence of high reported classification performance for an ophthalmic screening task.	The analysis proposes an OCT-based CNN screening pipeline that simultaneously flags the presence of each of three retinal disorders and labels the remaining cases as normal macula, emphasizing early detection of potentially treatable conditions.
[255]: Multi-scale CNN-Swin transformer network with boundary supervision for multiclass biomarker segmentation in retinal OCT images	2025	The research applies a CNN-Swin transformer segmentation model to retinal OCT images to localize and segment multiclass retinal biomarkers, using edge-focused boundary supervision and multiscale dual-encoder fusion, with evaluation reported on both local and public datasets as evidence for its effectiveness against noise, class imbalance, and small-marker errors.	MSCS-Net introduces a parallel dual-encoder CNN-Swin architecture with an added edge detection path, a revised Swin window partition, and modules for dimensionality reduction of small biomarkers plus cross-fusion of global and local features, aiming to improve segmentation accuracy in challenging OCT biomarker settings.
[143]: Multi-scale convolutional neural network for automated AMD classification using retinal OCT images	2022	The reported work develops an FPN-style multi-scale CNN with feature fusion for automated classification of normal, dry AMD, and wet AMD (drusen and choroidal neovascularization) from retinal OCT images, supported by evaluations on two OCT datasets and performance comparisons against other OCT classification frameworks, with heatmaps used as evidence for lesion localization across scales.	The study provides a multi-scale OCT diagnostic approach that improves normal versus dry/wet AMD discrimination across tested backbone models and supplements quantitative results with heatmap-based qualitative evidence for detecting AMD-related pathology at different sizes.
[333]: Multi-scale local-global transformer with contrastive learning for biomarkers segmentation in retinal OCT images	2024	This analysis develops a multi-scale local-global transformer using contrastive learning for pixel-wise segmentation of retinal biomarkers in Optical Coherence Tomography (OCT), and it provides evidence from experiments on one public dataset and one local dataset showing improved performance versus existing methods.	This investigation introduces MsLGT-Net with dedicated multi-scale fusion attention, a local-global transformer module, and a contrastive learning enhancement component to better capture small-structure detail and class-specific biomarker features for retinal OCT segmentation.

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Work	Year	Overall reasoning	Contribution and summary
[214]: Non-transfer Deep Learning of Optical Coherence Tomography for Post-hoc Explanation of Macular Disease Classification	2021	This research develops a non-transfer deep convolutional neural network that classifies four macular disease conditions from monochromatic spectral-domain retinal OCT images, and it provides evidence through post-hoc kernel/feature analysis and expert review of misclassified cases to support explanation of model behavior.	The authors demonstrate that only eight of 256 final-layer convolutional kernels remain active and that the resulting selective features maintain strong class separability in reduced 2D embeddings, while clinical interpretation of the specific misclassification patterns highlights where confounding imaging artifacts and textures drive errors.
[250]: Novel Deep Learning Model for Glaucoma Detection Using Fusion of Fundus and Optical Coherence Tomography Images	2025	The study evaluates a deep learning, two-branch multimodal fusion model that ingests paired fundus photographs and optical coherence tomography scans from the same eye to classify glaucoma, benchmarking against fundus-only and OCT-only baselines and reporting robustness via five-fold cross-validation on a private clinical dataset with fixed test-set results.	This work specifies a ResNet-based single-modality pipeline and a feature-level fusion network, demonstrating that combining fundus and OCT representations yields higher glaucoma detection performance than either modality alone with statistical significance tests on validation predictions.
[206]: OCT(2)Former: A retinal OCT-angiography vessel segmentation transformer	2023	This work introduces OCT2Former, an end-to-end transformer encoder-decoder model for semantic segmentation of retinal vessels in OCTA imaging, designed to better capture thin-vessel structure and preserve vessel connectivity, and it provides evidence from Jaccard-based comparisons on three public OCTA datasets against convolutional baselines.	By combining a dynamic token aggregation transformer with an auxiliary convolution branch and a lightweight decoder, the authors report improved vessel-segmentation accuracy over the best convolutional approaches across multiple OCTA datasets, supporting its potential use for automated retinal screening workflows.
[271]: OCT-angiography based artificial intelligence-inferred fluorescein angiography for leakage detection in retina [Invited]	2023	In an ophthalmic setting, the study builds an OCT-angiography-to-fluorescein angiography image-domain deep learning pipeline using a U-Net-type convolutional neural network to infer retinal vascular leakage, with training evidence derived from diabetic retinopathy FA and OCTA pairs and qualitative comparison to real FA in additional retinal vascular conditions.	This research demonstrates time-aware AI-inferred FA generation from same-day OCTA, including reproducible leakage depiction for diabetic retinopathy and evaluation by structural similarity against corresponding FA, supporting OCTA-based leakage assessment while avoiding contrast-dye acquisition steps.

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Work	Year	Overall reasoning	Contribution and summary
[203]: OCT-based diagnosis of glaucoma and glaucoma stages using explainable machine learning	2025	This cross-sectional OCT study develops explainable machine-learning classifiers for glaucoma diagnosis and stage stratification (normal vs early vs moderate vs advanced) using RNFL/GC-IPL/macular thickness features together with frequency-domain TSNIT signals, and it evaluates evidence from prospectively collected clinical data with SHAP- and partial-dependency-based interpretability to support decision trustworthiness.	In addition to model training with feature selection and an OCT-derived clinical feature set, the authors provide an explainable, user-oriented software workflow that translates SHAP global feature ranking and partial dependency cut-off estimates into feature-level decision support for glaucoma staging.
[56]: OCT DEEPNET 1—A Deep Learning Approach for Retinal OCT Image Classification	2023	The research evaluates OCT DEEPNET 1, a deep learning model for retinal OCT imaging, to classify images into four clinically defined categories (normal, choroidal neovascularization, diabetic macular edema, and Drusen) using random-search hyperparameter tuning and reported classification metrics, supporting evidence for automated ophthalmic diagnosis from OCT data.	The reported work provides an assessed, hyperparameter-tuned OCT image classification pipeline that reports an 88% accuracy at a batch size of 32, which helps quantify the feasibility of deep learning for four-way retinal OCT disease categorization.
[446]: OCT Retinal and Choroidal Layer Instance Segmentation Using Mask R-CNN	2022	The reported work evaluates a Mask R-CNN deep learning approach for instance segmentation of retinal and choroidal layer boundaries in posterior-segment OCT images, targeting automatic delineation of seven retinal layers with quantitative evidence reported via boundary error and Dice coefficients against a U-Net baseline.	The study demonstrates that a region proposal-based instance segmentation strategy can achieve segmentation performance comparable to U-Net while enabling faster boundary position extraction, cutting inference time by 2.5x for the OCT layer segmentation task.
[67]: OCT-Trans: A Novel Transformer Backbone with Multimodal Feature Extraction in OCT-Based Retinal Disease Classification	2025	This analysis targets OCT-based retinal disease classification in an ophthalmic setting, using a transformer backbone that extracts multimodal features from OCT images acquired with two devices, with atlas-based eleven-layer segmentation feeding layerwise clinical indicators to classify normal retina versus DR and AMD, validated on 541 patients and benchmarked against established model families.	This investigation proposes OCT-Trans, which integrates layer-specific morphologic and texture features from automated eleven-layer segmentation into transformer-derived contextual embeddings and a multilayer perceptron classifier, showing statistically significant gains over prior baselines and reporting high agreement and classification scores.

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Work	Year	Overall reasoning	Contribution and summary
[242]: OCT Vision: Retinal Disease Detection Using Deep Learning	2024	This research uses deep learning for retinal disease classification from optical coherence tomography (OCT) images, tested with VGG16 and ResNet50 in TensorFlow and reported as an image-based diagnostic decision-support workflow that outputs real-time, confidence-scored predictions via a Streamlit interface.	This analysis develops and evaluates an OCTDL-trained classification system for AMD, DME, and RVO, and it matters by providing comparative model results plus a clinical-facing interface that could support time-sensitive retinal screening and interpretation.
[232]: OctNET: A Lightweight CNN for Retinal Disease Classification from Optical Coherence Tomography Images	2021	The study describes a lightweight ophthalmic deep CNN that uses gradient-based class activation mapping to classify diabetic macular edema, drusen, and choroidal neovascularization from optical coherence tomography images, with evidence reported from training and validation on a public labeled dataset.	This work presents OctNET, a six-block downsampling and weight-sharing architecture that markedly reduces learnable parameters while producing high classification performance and providing model explanations via class activation maps for OCT-based computer-aided diagnosis.
[409]: Optic Cup and Disc Segmentation of Fundus Images Using Artificial Intelligence Externally Validated With Optical Coherence Tomography Measurements	2025	In a glaucoma-focused ophthalmic diagnostic workflow, the study builds a deep learning fundus optic cup and disc segmentation pipeline to derive vertical cup-to-disc ratio from retinal photographs, then externally validates VCDR measurements against OCT VCDR and benchmarks against EHR- and expert-based assessments on retrospective hospital data and public datasets.	By combining an ONH detector with a transformer-based segmentation model and a standardized crop-and-square-resize procedure, the study demonstrates external, cross-dataset generalizability with VCDR agreement to OCT that is comparable to EHR assessments and shows high same-day repeatability, supporting near-expert use for ONH quantification without OCT-dependent training.
[50]: Optical coherence tomography-based diabetic macula edema screening with artificial intelligence	2020	For diabetic macular edema in an ophthalmic screening and diagnostic-aid context, this retrospective study uses convolutional neural networks to classify macular OCT cross-sectional images into DME versus non-DME, and it provides evidence from internal validation/verification with reported classification accuracy, sensitivity, and specificity.	This research trains two CNN architectures (VGG16 and InceptionV3) on labeled OCT images and packages the resulting models into an OCT-driven screening platform intended to reduce manual clinician grading of DME.
[153]: Optical coherence tomography image based eye disease detection using deep convolutional neural network	2022	Using optical coherence tomography, this investigation builds an attention- and transfer-learning-enhanced deep convolutional neural network for multi-class image classification of CNV, DME, drusen, and normal, with the clinical motivation of reducing diabetic retinopathy screening workload and evaluation reported with standard diagnostic metrics on a predefined training/testing split.	The analysis delivers an OCT-based VGG16_Attention pipeline that couples ImageNet transfer learning with attention and reports comparative results showing the model achieves substantially higher classification performance than several other CNN baselines on a 1000-image test set across the labeled disorders.

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Work	Year	Overall reasoning	Contribution and summary
[463]: Optical coherence tomography image classification for retinal disease detection using artificial intelligence	2024	This chapter addresses an ophthalmic deep learning approach for retinal disease classification from optical coherence tomography images, using convolutional neural networks with pretrained and transfer learning components to detect drusen, diabetic macular edema, and choroidal neovascularization, and discusses evidence from the reported diagnostic strategy and its potential clinical impact.	The reported work outlines a specific OCT-based pipeline that leverages convolutional neural networks plus pretrained image classification models and derived features to support automated screening and earlier identification of clinically important retinal conditions, aiming to streamline diagnostic workflows and reduce health care burden.
[326]: Optimizing Deep Learning Based Retinal Diseases Classification On Optical Coherence Tomography Scans	2023	The reported work applies convolutional neural networks with transfer learning and fine-tuning to OCT B-scans for automated three-way retinal disease classification (CNV, drusen, DME) versus healthy eyes, using a publicly available OCT2017 dataset and reporting evaluation using standard multiclass metrics and confusion matrices.	The study compares multiple EfficientNet variants (B0, B3, B5) under an imbalanced multiclass setting and identifies EfficientNet B0 as best performing, highlighting how image enhancement and resolution scaling with ImageNet-pretrained weights can yield highly accurate OCT-based diagnostic predictions for clinical decision support.
[142]: Optimizing glaucoma diagnosis using fundus and optical coherence tomography image fusion based on multi-modal convolutional neural network approach	2024	This 2024 study targets glaucoma diagnosis using an ophthalmic multimodal deep learning setup that fuses segmented fundus images with segmented optical coherence tomography images, employing a multi-modal convolutional neural network (MMCNN) and using GoogLeNet as a unimodal comparator, with evidence from reported classification experiments on paired, segmented FI and OCTI data.	This investigation introduces an MMCNN fusion pipeline that combines features from segmented fundus and OCT streams and provides evaluation results showing improved glaucoma detection and classification performance versus unimodal processing, supported by segmentation-based preprocessing and ROC/accuracy-related reporting.
[51]: Optimizing the OCTA layer fusion option for deep learning classification of diabetic retinopathy	2023	Using OCTA en-face imaging from the superficial capillary plexus, deep capillary plexus, and choriocapillaris, the study trains a CNN to classify diabetic retinopathy stages (control, NoDR, NPDR) and compares single-layer versus early-, intermediate-, and late-fusion architectures, with Grad-CAM used for interpretability, supported by quantitative evaluation in an ophthalmic DR setting.	This analysis identifies the intermediate-fusion strategy as the best OCTA layer-combination approach for DR staging with the CNN and provides activation-map evidence on which spatial features drive classification decisions.

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Work	Year	Overall reasoning	Contribution and summary
[419]: Patch-based CNN for corneal segmentation of AS-OCT images: Effect of the number of classes and image quality upon performance	2023	The study reports an ophthalmic AI segmentation study using a patch-based CNN on AS-OCT images to automatically delineate three corneal boundaries (epithelium, Bowman's layer, and endothelium), evaluating how label-class configuration and training image-quality distribution affect segmentation accuracy and boundary error using Dice/probability and boundary error metrics.	The authors quantify how moving from 4 to 8 boundary classes and varying training data quality impacts corneal segmentation behavior, providing evidence that patch-based learning can perform competitively while showing limited sensitivity to image-quality shifts in this AS-OCT setting.
[110]: Performance of Artificial Intelligence in Detecting Diabetic Macular Edema From Fundus Photography and Optical Coherence Tomography Images: A Systematic Review and Meta-analysis	2024	This systematic review and meta-analysis evaluates AI models for detecting diabetic macular edema using fundus photography and optical coherence tomography, pooling diagnostic performance and assessing how internal versus external validation and training characteristics relate to evidence quality in ophthalmic imaging.	Across 53 included studies, it reports pooled accuracy estimates for both FP- and OCT-based approaches and identifies factors such as deep learning and training-data diversity that improve performance while underscoring the need for more external validation.
[89]: Performance of retinal fluid monitoring in OCT imaging by automated deep learning versus human expert grading in neovascular AMD	2023	This ophthalmology study in neovascular AMD evaluates a deep learning segmentation approach for intraretinal and subretinal fluid on SD-OCT volumes, comparing automated fluid presence/absence and 3D fluid volume quantification against expert reading-center gradings and relating the results to central retinal thickness and volume-derived biomarkers during anti-VEGF monitoring in clinical-trial OCT data.	This research provides head-to-head evidence that automated fluid quantification from OCT can perform reliably relative to certified experts and shows that central subfield thickness is a weak surrogate for fluid activity, supporting objective, volume-based monitoring of anti-VEGF treatment response.
[308]: Performances of Machine Learning in Detecting Glaucoma Using Fundus and Retinal Optical Coherence Tomography Images: A Meta-Analysis	2022	This 2022 meta-analysis evaluates machine learning diagnostic models for glaucoma detection in an ophthalmic setting by pooling evidence from fundus and retinal OCT imaging, estimating pooled sensitivity, specificity, and AUC using a bivariate random-effects approach across classifier categories and dataset types.	By synthesizing 105 studies with modality-specific pooled performance, the study clarifies how ML accuracy varies between fundus and OCT and across clinical versus external testing, informing whether these imaging sources and model classes are suitable for glaucoma screening or diagnosis.

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Work	Year	Overall reasoning	Contribution and summary
[325]: Predicting 10-2 Visual Field From Optical Coherence Tomography in Glaucoma Using Deep Learning Corrected With 24-2/30-2 Visual Field	2021	This multicenter ophthalmic deep learning study uses macular spectral-domain OCT with a convolutional neural network to predict HFA 10-2 threshold sensitivities in glaucoma/OAG, then tests whether a correction informed by the same eye's HFA 24-2/30-2 results improves the pointwise visual field prediction, with evidence from an independent testing cohort.	The reported work quantifies the benefit of applying an HFA 24-2/30-2-based adjustment to CNN-generated 10-2 thresholds, showing lower absolute and pointwise errors that could reduce the need for additional 10-2 perimetry in routine clinical workflows.
[436]: Predicting Clinician Fixations on Glaucoma OCT Reports via CNN-Based Saliency Prediction Methods	2024	This 2024 glaucoma study uses a CNN-based saliency prediction model (TranSalNet) to learn how ophthalmologists fixate on specific regions of glaucoma OCT reports by training on eye-tracking-derived fixation heat maps and evaluating agreement between predicted and observed fixation saliency with quantitative image-similarity metrics via 5-fold cross-validation.	The study provides an initial, OCT-specific mapping from gaze behavior to model-predicted medically salient regions on glaucoma report sub-images, showing that fixation regions can be predicted with reasonable qualitative correspondence while highlighting that improved data quantity and quality are needed for stronger pixel-level accuracy.
[380]: Predicting HFA 30-2 Visual Fields with Deep Learning from Multimodal OCT-Fundus Feature Fusion and Structure-Function Discordance Analysis	2026	This analysis uses a ViT-B/32 feature-fusion approach on multimodal OCT and fundus-derived structure (disc and red-free photographs plus RNFL thickness/plots) to predict Humphrey 30-2 visual-field outcomes (MD, PSD, and point-wise TS) in glaucoma/OHT. Uncertainty-aware regression and sensitivity analyses excluding SFD cases evaluate how structure-function discordance changes prediction reliability.	This investigation quantifies the practical impact of structure-function discordance on forecast accuracy by reporting improved VF metric agreement after excluding discordant eyes, thereby highlighting that low-performing predictions may reflect SFD rather than failure of the OCT-fundus fusion model.
[403]: Predicting macular hole surgery outcomes: Integrating preoperative OCT features with supervised machine learning statistical models	2025	This retrospective ophthalmic study applies supervised machine learning, evaluating multiple classifiers with a random forest model, to preoperative spectral-domain OCT features from idiopathic full-thickness macular hole eyes to predict dichotomous postoperative anatomical closure status, using an 80:20 training/test split with internal validation and external testing.	Using the OCT-derived macular hole area as the highest-importance preoperative feature, the random forest classifier delivered the best internal and external predictive performance for postoperative hole closure, offering a practical, evidence-based decision-support approach grounded in routine OCT measurements.

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Work	Year	Overall reasoning	Contribution and summary
[407]: Predicting optical coherence tomography-derived diabetic macular edema grades from fundus photographs using deep learning	2020	This ophthalmic deep learning study uses 2D color fundus photographs to predict OCT-derived center-involved diabetic macular edema grade and retinal fluid presence, with ROC-AUC and sensitivity/specificity reported against expert OCT grading as reference on retrospective clinical validation sets.	This work improves on retinal specialists by achieving similar sensitivity with substantially higher specificity for ci-DME detection from routine fundus imaging, and extends prediction to intraretinal and subretinal fluid to support screening-style referral decisions.
[282]: Predicting Optical Coherence Tomography-Derived High Myopia Grades From Fundus Photographs Using Deep Learning	2022	This retrospective ophthalmic deep learning study uses fundus photographs labeled from OCT to predict OCT-derived ATN classification and grading for high myopia, and it reports test-set discrimination and comparison against attending ophthalmologists and retinal specialists.	The research provides a fundus-only prediction framework for OCT-referenced atrophy, traction, and neovascularization risk in high myopia, offering evidence that such triage can match retinal specialists while avoiding OCT-dependent workflows.
[410]: Predicting the central 10 degrees visual field in glaucoma by applying a deep learning algorithm to optical coherence tomography images	2021	This retrospective glaucoma study uses convolutional neural networks with macular OCT retinal layer thickness maps (RNFL, GCC, and OS+RPE) to predict Humphrey Field Analyzer central 10-2 total deviation across the central 10 degrees, with outcomes evaluated against measured HFA 10-2 and corrected using overlapping HFA 24-2 inner-point values.	This research demonstrates that incorporating an HFA 24-2-based adjustment substantially reduces prediction error for HFA 10-2 total deviation compared with uncorrected CNN outputs, supporting an OCT-to-central visual-field modeling strategy informed by clinically available 24-2 data.
[370]: Predicting the Glaucomatous Central 10-Degree Visual Field From Optical Coherence Tomography Using Deep Learning and Tensor Regression	2020	This glaucoma study applies deep learning with CNNs plus tensor regression to predict Humphrey 10-2 pointwise visual-field sensitivity from OCT-derived retinal layer thicknesses, evaluated with 5-fold cross-validation using RMSE to benchmark against linear and support vector regression baselines.	The analysis quantifies the central 10 VF prediction error achievable from macular OCT thickness inputs, showing lower RMSE and mean absolute error for CNN-tensor regression than CNN pointwise mapping, SVR, and multiple linear regression, supporting OCT-to-visual-field translation.

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Work	Year	Overall reasoning	Contribution and summary
[385]: Predicting treat-and-extend outcomes and treatment intervals in neovascular age-related macular degeneration from retinal optical coherence tomography using artificial intelligence	2022	This post-hoc ophthalmology study in neovascular age-related macular degeneration uses CNN-based OCT segmentation to derive quantitative biomarkers (including intraretinal and subretinal fluid, PED, hyperreflective foci, and photoreceptor layer thickness) from baseline and a 4-week follow-up scan, then applies machine learning with 10-fold cross-validation to predict treat-and-extend visual response and the likelihood of extending treatment intervals, providing retrospective evidence from a prospective randomized clinical trial cohort.	The reported work demonstrates that features computed after the first injection from OCT fluid and retinal integrity maps can stratify patients into visual responders and an extendable-vs-non-extendable treatment-interval group, offering an imaging-driven decision-support approach for individualizing treat-and-extend management from early follow-up data.
[144]: Predicting visual field global and local parameters from OCT measurements using explainable machine learning	2025	In a glaucoma-focused ophthalmic context where OCT can help bypass impractical standard perimetry, this cross-sectional study uses interpretable supervised regression (XGBoost, SVM, and random forest) on CIRRUS spectral-domain OCT-derived thickness features (RNFL, GC-IPL, and macular thickness) to predict 24-2 visual field global indices and local pointwise sensitivities, with evidence from five-fold cross-validation and SHAP-based explanation of feature contributions.	The study further introduces an OCT-to-VF Predictor software tool and pairs model performance reporting with SHAP analyses to highlight clinically relevant OCT parameters and support transparent, practitioner-facing interpretation of predicted visual field outcomes.
[140]: Predicting Visual Field Worsening with Longitudinal OCT Data Using a Gated Transformer Network	2023	In this retrospective longitudinal glaucoma cohort from the Johns Hopkins Wilmer Eye Institute, a gated transformer network ingests serial OCT scans to predict the probability of future visual field worsening, using AUC-based comparisons across multiple trend- and event-based progression definitions (including a majority-of-6 consensus) and assessing effects of baseline severity.	By training GTNs with at least five longitudinal OCT images and benchmarking them against linear mixed-effects models and naive Bayes classifiers under matched input and reference standards, the study demonstrates that the transformer framework can discriminate visual field decline across definitions, with performance declining as baseline glaucoma severity increases.
[61]: Prediction and Detection of Glaucomatous Visual Field Progression Using Deep Learning on Macular Optical Coherence Tomography	2024	This retrospective ophthalmic study uses a self-supervised vision transformer trained on unlabeled macular OCT B-scan images to both detect concurrent and predict future glaucomatous visual field progression, with evidence based on ROC-based diagnostic performance for predefined mean deviation progression outcomes.	This analysis develops an OCT-to-visual-field deep learning pipeline that leverages large-scale self-supervised pretraining to identify progressing eyes in clinical practice and stratify patients for near-term risk of functional decline.

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Work	Year	Overall reasoning	Contribution and summary
[396]: Prediction of age-related macular degeneration disease using a sequential deep learning approach on longitudinal SD-OCT imaging biomarkers	2020	This ophthalmology study models sequential exudation risk in non-exudative age-related macular degeneration using longitudinal SD-OCT radiomics biomarkers plus demographic and visual variables, with an RNN (stacked LSTM) to output 3-month and up to 21-month horizon predictions that were evaluated with patient-level cross-validation on the retrospective HARBOR trial dataset and tested on an external real-world SD-OCT dataset from Bascom Palmer Eye Institute.	This work introduces a hybrid deep sequence framework that leverages engineered OCT drusen features aggregated across visits to forecast future conversion timing, providing evidence of strong short-term discriminative ability internally and partial generalization externally that could support risk-stratified follow-up intervals.
[423]: Prediction of Axial Length From Macular Optical Coherence Tomography Using Deep Learning Model	2024	This retrospective ophthalmic deep learning study uses a ResNet-152 model to predict axial length from horizontal and vertical macular OCT images, evaluated with internal and external test sets using mean absolute error, R-squared, and error-range analyses.	The research shows that a dual-input configuration combining horizontal and vertical OCT sections yields the most accurate axial-length estimates, supporting the clinical relevance of macular OCT structure for modeling axial elongation.
[451]: Prediction of Causative Genes in Inherited Retinal Disorders from Spectral-Domain Optical Coherence Tomography Utilizing Deep Learning Techniques	2019	This ophthalmic study applies deep learning to central foveal spectral-domain optical coherence tomography for a four-class task that predicts the causative gene category (ABCA4, RP1L1, EYS, or Normal) in inherited retinal disorders, using repeated random train-test splits to report classification performance evidence.	This research demonstrates an OCT-based gene prediction workflow using the Medic Mind platform on molecularly confirmed inherited retinal disorder cohorts, offering practical support for gene-level screening when clinical phenotyping at general clinics is limited.
[450]: Prediction of Central Visual Field Measures From Macular OCT Volume Scans With Deep Learning	2023	In a glaucoma structure-function setting, this investigation uses a 3D DenseNet121 deep learning model to predict central 10-2 visual field mean deviation, threshold sensitivity, and total deviation from macular OCT volume scans, with evidence based on ground-truth comparison using correlations and mean absolute error across 10-fold stratified cross-validation.	The analysis refines structure-function inference by training and evaluating a segmentation-free network on raw macular OCT volumes and showing that its predictions for global and pointwise central field metrics outperform baseline log-log linear models based on GCIPL thickness, supporting OCT-derived supplementation of central perimetry when reliability is limited.

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Work	Year	Overall reasoning	Contribution and summary
[199]: Prediction of coronary artery disease using retinal optical coherence tomography angiography imaging and electronic health records: a multimodal machine learning approach	2026	This retrospective ophthalmology-linked study evaluates a multimodal AI CAD prediction model that combines OCTA-derived retinal microvascular biomarkers from 3x3 mm macular scans with EHR variables, using Transformer-enhanced multi-structure segmentation to quantify vascular metrics and an XGBoost classifier for CAD detection, with performance reported via internal hold-out testing using standard discrimination and calibration metrics.	The reported work introduces a retina-coronary axis-informed feature pipeline that selects 4 OCTA biomarkers and 13 EHR predictors via LASSO and models non-linearity with restricted cubic splines, then benchmarks the integrated OCTA-EHR approach against EHR-only and multiple deep-learning baselines to demonstrate incremental predictive value for noninvasive CAD risk stratification.
[101]: Prediction of neovascular age-related macular degeneration recurrence using optical coherence tomography images with a deep neural network	2024	This retrospective ophthalmic deep learning study uses spectral-domain OCT images with a DenseNet201-based multi-instance architecture to predict neovascular age-related macular degeneration recurrence within 3 months (after 1 month of follow-up post-third loading anti-VEGF injections), and it evaluates evidence by comparing model prediction accuracy against six ophthalmologists on a held-out set of 100 cases.	The study demonstrates that incorporating the pre-injection OCT alone versus sequential OCTs immediately after the first, second, and third injections meaningfully changes model performance, with the model achieving higher accuracy than clinicians and using Grad-CAM to provide visual support for OCT regions driving recurrence decisions.
[290]: Prediction of postoperative visual acuity in patients with age-related cataracts using macular optical coherence tomography-based deep learning method	2023	This ophthalmic deep learning study used macular OCT B-scans plus extracted macular morphology indices and preoperative best-corrected visual acuity to predict 1-month postoperative visual acuity and whether vision improved by at least two lines in patients with age-related cataracts, with evidence reported from regression and classification performance on held-out validation and test sets.	By testing multiple input configurations, it showed that fusing horizontal and vertical macular OCT with quantitative foveal and retinal morphological features together with preoperative BCVA produced the best postoperative VA prediction and improvement classification, supporting OCT morphology-based prognostication for cataract surgery planning.
[136]: Prediction of response to anti-vascular endothelial growth factor treatment in diabetic macular oedema using an optical coherence tomography-based machine learning method	2021	The study reports an OCT-based machine learning study for diabetic macular edema that uses deep learning feature extraction and a random forest classifier to predict anti-VEGF response at baseline, comparing model outputs with ophthalmologists and using five-fold cross-validation to support evidence for classification of good versus poor responders.	This analysis identifies the most informative OCT feature, the sum of hyperreflective dots, via feature importance analysis and evaluates whether the self-explainable RF model outperforms human assessment for anticipating treatment response, which could inform individualized anti-VEGF decision-making.

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Work	Year	Overall reasoning	Contribution and summary
[391]: Prediction of visual field from swept-source optical coherence tomography using deep learning algorithms	2020	The study uses ophthalmic deep learning to predict 24-2 visual field metrics from wide-angle swept-source OCT structure maps in glaucoma, training and testing multiple Inception family networks with RMSE-based evidence comparing model performance across glaucoma severity and Garway-Heath sectors.	This work identifies Inception-ResNet-v2 as the top performer for OCT-to-VF prediction, showing error increases with glaucoma progression and providing a data-driven approach that could support VF estimation when standard perimetry is not feasible.
[211]: Progressive Neural Networks for Continuous Classification of Retinal Optical Coherence Tomography Images	2023	The study reports a deep learning approach for retinal lesion classification from optical coherence tomography that tackles cross-device performance drop by casting multi-device learning as domain-incremental tasks and evaluating evidence from experiments on public medical image datasets for both stability and forgetting.	The research introduces a multi-granularity feature space constraint implemented with progressive neural networks to balance stability and plasticity across incremental domains, improving average accuracy while reducing forgetting versus selected baselines.
[187]: Proof-of-Concept Analysis of a Deep Learning Model to Conduct Automated Segmentation of OCT Images for Macular Hole Volume	2022	This proof-of-concept study applies a fully three-dimensional convolutional neural network to swept-source OCT to segment macular holes and quantify volume, using correlation with surgeon-defined manual volumetry as the main evidence for task-relevant automated retinal imaging in idiopathic full-thickness macular holes.	The study demonstrates high agreement between automated and manual macular hole volume on OCT and suggests that automated volumetry may better reflect clinically meaningful postoperative change than minimum linear diameter in this 1-year follow-up setting.
[439]: Quantification of Nonperfusion Area in Montaged Widefield OCT Angiography Using Deep Learning in Diabetic Retinopathy	2021	In this cross-sectional diabetic retinopathy study, a deep learning segmentation model processes montaged widefield OCT angiography (three 6x6 mm scans) to automatically detect and quantify nonperfusion area in the superficial vascular complex for staging clinically meaningful DR severity thresholds, with evidence from five-fold cross-validation against majority-voted manual ground truth and diagnostic comparisons using sensitivity/AUC with specificity fixed at 95%.	By training the network to distinguish true nonperfusion from widefield-specific low-signal shadow artifacts and then aggregating results across nasal, macular, and temporal regions, the authors provide an automated, artifact-aware workflow for nonperfusion quantification that improves diagnostic sensitivity and correlates with DR severity (including performance superior to macular-only measurements for several clinically relevant comparisons).

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Work	Year	Overall reasoning	Contribution and summary
[320]: Quantification of Retinal Nerve Fibre Layer Thickness on Optical Coherence Tomography with a Deep Learning Segmentation-Free Approach	2020	The study introduces a segmentation-free deep learning model that predicts global retinal nerve fiber layer thickness directly from raw Spectralis spectral-domain OCT B-scans, and evaluates it against conventional segmentation-based RNFL measurements across image-quality tiers including scans with segmentation errors and other artifacts, providing correlation-based agreement evidence.	The reported work demonstrates that an RNFL thickness regression network trained on unsegmented, high-quality B-scans can match conventional outputs in good-quality imaging and yield closer estimates to an intra-eye, same-day best-available reference when conventional segmentation is unreliable.
[105]: Quantifying the Characteristics of Diabetic Retinopathy in Macular Optical Coherence Tomography Angiography Images: A Few-Shot Learning and Explainable Artificial Intelligence Approach	2025	This retrospective ophthalmic study uses a few-shot learning framework with an intrinsic self-attention explainable AI component (MTUNet/SCOUTER) to stage diabetic retinopathy as PDR versus NPDR from macular OCTA images, supported by model-level evidence including standard classification metrics and class-wise matching/attention outputs.	The study demonstrates an OCTA-based data-efficient DR staging approach by coupling prototypical few-shot classification with pattern-matching explanations, using ResNet-50 as the selected backbone and reporting class-specific matching scores that indicate how the model emphasizes nonperfusion-related regions for NPDR and neovascularization for PDR.
[342]: Real-time retinal layer segmentation of adaptive optics optical coherence tomography angiography with deep learning	2020	This analysis targets ophthalmic en face retinal layer analysis from adaptive optics optical coherence tomography angiography, using a U-Net family deep-learning approach to segment 3D OCT volumes and generate real-time projections suitable for image-guided surgical workflows. The implementation prioritizes low-latency operation rather than clinical comparative outcomes.	By deploying a compressed CNN inference pipeline on GPU hardware, the authors enable very low-latency retinal-layer segmentation and en face OCT/OCTA projection generation, which can support near-instant visual feedback during acquisition.
[158]: Real-time retinal layer segmentation of OCT volumes with GPU accelerated inferencing using a compressed, low-latency neural network	2020	This research targets real-time ophthalmic optical coherence tomography by segmenting eight retinal layer boundaries in volumetric OCT with a compressed U-Net family model accelerated on GPUs via TensorRT, enabling millisecond inference and low-latency en face OCT/OCTA visualization during continuous scanning.	This analysis provides an end-to-end real-time OCTViewer pipeline that compresses and optimizes the segmentation network and runs an integrated GPU workflow to update en face layer projections with only 41 ms total pipeline latency, supporting immediate operator feedback during in vivo imaging.

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Work	Year	Overall reasoning	Contribution and summary
[256]: Recent Advanced Deep Learning Architectures for Retinal Fluid Segmentation on Optical Coherence Tomography Images	2022	This ophthalmic OCT survey reviews deep learning approaches for automated retinal fluid segmentation, focusing on semantic/instance segmentation of OCT-derived retinal fluid classes (IRF, SRF, PED) using CNN/FCN/U-Net and related hybrid architectures and summarizing evidence from published studies and public OCT benchmark datasets.	This work organizes the main deep learning model families used for retinal fluid delineation on OCT, outlines commonly employed datasets and labeling settings, and discusses future directions to support reproducible macular change assessment research.
[464]: Repeatability of Microperimetry in Areas of Retinal Pigment Epithelium and Photoreceptor Loss in Geographic Atrophy Supported by Artificial Intelligence-Based Optical Coherence Tomography Biomarker Quantification	2025	In a prospective ophthalmic reliability study in geographic atrophy, the authors co-registered fundus-controlled microperimetry test-retest data from CenterVue MAIA and NIDEK MP3 to AI-based OCT biomarker quantification of RPE and photoreceptor integrity loss, then evaluated pointwise sensitivity repeatability with Bland-Altman coefficients alongside interdevice correlation, scotoma detection repeatability, and fixation stability metrics.	The research delivers device-specific reference estimates of microperimetry pointwise repeatability across biomarker-defined retinal areas, showing that PR integrity loss patterns (with or without RPE loss) and related OCT features are associated with greater test-retest variability, supporting more accurate longitudinal interpretation and trial endpoint planning.
[169]: Research on deep learning-based intelligent diagnosis algorithms for OCT medical images of macular edema	2019	Using deep learning for macular edema on OCT imaging, the study applies Faster R-CNN for lesion area detection and a U-Net with multi-attention modules for lesion segmentation to support ophthalmic diagnosis with results reported on an OCT fundus dataset.	This research refines Faster R-CNN label generation for OCT lesion localization and introduces multi-attention fusion in U-Net decoding to segment edema regions and enable severity quantification, helping clinicians rapidly identify and locate lesions.
[192]: Residual Neural Network Based Classification of Macular Edema in OCT	2019	This investigation applies a deep residual neural network to classify macular edema imaging biomarkers, SRF and PED, directly from OCT scans, using a multi-label formulation with data augmentation, pretrained fine-tuning, and label-correlation analysis, and reports results on a large competition dataset (AI Challenger).	The analysis demonstrates that a residual-network approach, strengthened by augmentation and transfer from pretrained weights while accounting for co-occurring labels, can accurately recognize SRF and PED in macular edema OCT images from the benchmark competition evaluation.

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Work	Year	Overall reasoning	Contribution and summary
[346]: Retinal and choroidal microvasculature and structural analysis in OCTA for refractive amblyopia diagnosis using machine learning	2025	Using supervised machine learning based on random forest with cross-validation, this case-control study evaluates OCTA-derived retinal thickness, choroidal thickness, and capillary plexus perfusion measurements to classify amblyopic versus non-amblyopic eyes in adolescents with refractive amblyopia, reporting diagnostic classification evidence in a labeled dataset from macular 12 by 12 mm scans.	The authors integrate region-wise structural OCTA metrics from retinal and choroid into a diagnostic pipeline and show that their selected thickness features drive amblyopia discrimination even when SCP and DCP perfusion density differences are not significant, supporting a practical framework for automated refractive amblyopia assessment.
[49]: Retinal Boundary Segmentation in Stargardt Disease Optical Coherence Tomography Images Using Automated Deep Learning	2020	The reported work evaluates a fully automated deep learning approach for OCT retinal boundary segmentation in Stargardt disease, combining a fully semantic pixel-wise network with graph-search boundary extraction and measuring boundary accuracy against manual ground truth as well as agreement for retinal thickness and volume quantification.	The study establishes FS-GS as a high-throughput alternative to both manual correction and existing OCT segmentation tools by delivering accurate, ground-truth-validated inner and outer retinal boundary localization and clinically relevant thickness/volume measurements across the central 6 mm macular zone.
[251]: Retinal disease classification based on optical coherence tomography images using convolutional neural networks	2023	This analysis uses supervised convolutional neural networks on optical coherence tomography images to classify retinal diseases, with evidence provided through test-set comparisons of multiple CNN architectures under hyperparameter tuning.	This investigation evaluates AlexNet, VGG, Inception, and residual networks for OCT-based retinal disease classification and reports that Inception1 with tuned optimization settings achieved the highest accuracy, offering an implementable CNN baseline for OCT diagnosis support.
[81]: Retinal Disease Classification from Bimodal OCT and OCTA Using a CNN–ViT Hybrid Architecture	2025	This ophthalmic study applies a multimodal CNN-ViT hybrid deep learning approach to retinal disease classification using OCT B-scans and OCTA projection maps and reports test-set results on an OCTA dataset spanning AMD, DR, and normal cases.	This analysis introduces MultiModalNet, a dual-branch fusion model that combines vascular and structural representations to deliver multiclass AMD/DR/normal discrimination, supporting the value of bimodal OCT and OCTA inputs for computer-aided screening and decision-making.
[424]: Retinal Disease Classification from OCT Images Using Deep Learning Algorithms	2021	The study uses CNN-based deep learning with transfer-learned feature extractors on optical coherence tomography (OCT) cross-sectional retinal images to classify four retinal conditions (CNV, DME, drusen, and normal) via multiple binary classifier schemes and reports test-set performance using quantitative metrics and ROC/confusion analyses.	This work introduces two extensible four-class OCT classification pipelines built from three or four binary CNNs, combined through probability multiplication after FCN-based denoising and retina-layer cropping, and demonstrates performance besting a multi-class CNN approach in its reported comparisons.

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Work	Year	Overall reasoning	Contribution and summary
[355]: Retinal Disease Classification from Retinal-OCT Images Using Deep Learning Methods	2022	This work uses a convolutional neural network on retinal OCT cross-sectional images to perform automated classification and prediction of retinal disorders into CNV, DME, drusen, and normal categories, providing performance evidence via reported training and validation accuracies.	The research introduces a Retina CNN architecture and reports that, versus an existing comparison model, the proposed approach achieved the highest reported validation accuracy, supporting its use for OCT-based retinal disease categorization.
[465]: Retinal Disease Classification in Optical Coherence Tomography Images Using a Convolutional Neural Network Algorithm	2024	This ophthalmology study uses a convolutional neural network with quantitative feature extraction on retinal optical coherence tomography imaging to classify retinal OCT pathology for diagnostic categorization, reporting model efficacy on an independent dataset with comparisons to additional classical ML baselines.	This research details an OCT-to-label workflow that combines OCT-derived features with logistic regression and random forest classifiers and contrasts them against the CNN, highlighting a quantitatively grounded approach to automated retinal disease identification and categorization.
[109]: Retinal Disease Classification Using Custom CNN Model from OCT Images	2024	A study uses OCT retinal imaging to perform multi-class classification of macular disorders using a custom convolutional neural network, with results reported in comparison to conventional machine learning and transfer learning baselines, providing evidence of an ophthalmology-relevant AI method for OCT-based diagnostic categorization.	The authors introduce a custom CNN trained on a public OCT dataset to distinguish normal eyes from CNV, DME, and Drusen and report improved test accuracy over the evaluated alternative models, supporting the practical value of their architecture for OCT-driven retinal disease screening.
[79]: Retinal Disease Detection by Optical Coherence Tomography (OCT)-Based Deep Learning	2024	The research evaluates OCT-derived retinal image classification for four retinal disorder categories using CNN architectures and transfer learning (including VGG16 and ShuffleNet), with the evidence described as reported classification accuracy across model variants after OCT preprocessing for enhanced image quality and edges.	By incorporating preprocessing steps that target noise reduction and edge and gradient contour enhancement, it demonstrates improved deep learning performance for four-class retinal disease detection from OCT images, highlighting how image enhancement can strengthen ophthalmic OCT classification.
[390]: Retinal Disease Detection: Using Deep Learning for Accurate Diagnosis with Optical Coherence Tomography Scans	2024	The reported work evaluates a CNN-based deep learning approach for ophthalmic retinal disease classification from optical coherence tomography-based imaging, targeting CNV, DME, drusen, and normal labels with evidence reported as supervised training and validation and test-set metrics.	The study reports a four-class retinal classifier trained on 968 evenly distributed images with high overall accuracy (96.9%) and class-specific precision/recall, highlighting where confusion between DME and drusen limits performance and motivating data or feature improvements.

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Work	Year	Overall reasoning	Contribution and summary
[99]: Retinal diseases classification based on hybrid ensemble deep learning and optical coherence tomography images	2023	This hospital-based study uses optical coherence tomography imaging to classify five retinal diseases with a hybrid deep-learning and ensemble machine-learning framework, where transfer learning extracts features and k-NN and XGBoost outputs are fused via soft-voting to produce class probabilities, with performance assessed on a held-out test set and reported classification accuracy metrics.	By demonstrating a tuned OCT feature-extraction stage across multiple transfer-learning CNN backbones and selecting an effective k-NN/XGBoost soft-voting pair, the study provides an OCT-specific multiclass diagnostic pipeline that improves upon standalone classifier results and is deployed as a web service for rapid external predictions.
[466]: Retinal Diseases Classification System Using OCT Images Combined with CNN Models	2023	This research evaluates an OCT-based deep learning classification approach that uses convolutional neural networks to distinguish retinal disease categories, specifically diabetic macular edema versus age-related macular degeneration, from retinal OCT images.	This analysis describes a CNN-driven OCT workflow for automated retinal disease category classification, emphasizing early detection of DME and AMD as the intended clinical use case.
[339]: Retinal fluid segmentation in OCT images using adversarial loss based convolutional neural networks	2018	Using voxel-level convolutional neural networks on 3D OCT volumes from the ReTOUCH challenge, the study targets segmentation of intra-retinal fluid, sub-retinal fluid, and PED, combining cross-entropy and Dice with an adversarial loss to support higher-order mask consistency, and evaluates results on a held-out validation set via overlap and volume-difference metrics.	This work refines a U-Net-inspired encoder-decoder with skip connections, multi-scale feature use, and adversarial training to deliver improved segmentation and separate detection of fluid presence across vendors, providing quantitative validation that the loss design and architecture modifications enhance retinal fluid delineation.
[125]: Retinal Fluids Segmentation Using Volumetric Deep Neural Networks on Optical Coherence Tomography Scans	2020	This work uses an encoder-decoder volumetric deep neural network for semantic segmentation on OCT volumetric imaging to delineate retinal fluids, reporting Dice-based evidence of segmentation accuracy using the RETOUCH challenge dataset and external Cirrus scanner data.	The research extends retinal fluid quantification by treating the task in 3D rather than 2D and reports Dice results that include the strongest performance for intraretinal fluid, supporting more reliable measurement of fluid burden on clinically used OCT scans.
[75]: Retinal Nerve Fiber Layer Features Identified by Unsupervised Machine Learning on Optical Coherence Tomography Scans Predict Glaucoma Progression	2018	This retrospective ophthalmic study in primary open-angle glaucoma applies unsupervised PCA to wide-angle swept-source OCT retinal nerve fiber layer thickness maps to extract latent structural RNFL features and evaluates their evidence for detecting glaucoma and predicting 2-year progression using leave-one-out, AUC-based comparison versus standard perimetry and cpRNFL metrics.	This research demonstrates that PCA-derived RNFL components derived from SS-OCT outperform mean cpRNFL thickness and perimetry-based measures for glaucoma discrimination and provide higher baseline AUC for identifying eyes classified as progressing, supporting a clinically relevant imaging-driven risk assessment approach.

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Work	Year	Overall reasoning	Contribution and summary
[467]: Retinal OCT analysis and prediction with deep learning	2020	Using deep learning on retinal OCT data, the study examines imaging-biomarker quantification and disease-progression prediction for longitudinal ophthalmic monitoring.	The analysis outlines a deep learning approach that links OCT-based biomarker quantification to prognostic prediction, highlighting clinical relevance for tracking retinal disease over time.
[239]: Retinal OCT Image Classification Based on Domain Adaptation Convolutional Neural Networks	2021	The research tackles computer-aided diagnosis from retinal OCT, using an end-to-end domain-adaptation convolutional neural network (Inception V3) to classify six OCT categories and provide evidence via quantitative diagnostic metrics (accuracy, precision, sensitivity, specificity) compared with clinical expert performance.	The reported work combines two publicly available retinal OCT datasets and presents a domain-adaptation classification framework that aims to transfer prior knowledge from a related domain to improve reliable categorization across six OCT classes, with reported results near or above expert levels on key measures.
[113]: Retinal OCT Layer Segmentation via Joint Motion Correction and Graph-Assisted 3D Neural Network	2023	The reported work targets retinal layer segmentation in 3D spectral-domain OCT by pairing an axial motion-correction network with a graph-assisted 3D convolutional neural network using a graph pyramid design, and it evaluates segmentation accuracy with pixel- and boundary-based error on multiple datasets that include healthy eyes and retinal diseases with both simulated and real deformations, providing comparative evidence against commercial software and prior conventional and deep learning methods.	The study demonstrates that jointly correcting motion and performing volumetric (rather than slice-wise) segmentation improves retinal layer delineation, especially under severe deformation, while also releasing/using a large dataset with manually corrected OCT annotations to support clinically relevant benchmarking.
[131]: Retinal Optical Coherence Tomography Classification Using Deep Learning	2023	The study describes a deep learning approach using transfer learning to classify retinal findings from optical coherence tomography for screening treatable conditions such as choroidal neovascularization, diabetic macular edema, drusen, and normal retina, with evidence reported as results comparable to human retinal specialists.	This investigation outlines an OCT-based classification pipeline that leverages pretrained network weights to improve pattern learning and enable patient triage among multiple age-related and diabetic retinal categories, offering clinically relevant support for earlier detection of treatable disease.
[45]: Retinal pathologies detection in OCT images based on Bilinear convolutional neural network	2023	The study uses a bilinear convolutional neural network to detect and classify CNV, drusen, and diabetic macular edema in an image dataset of 6000 OCT scans and compares its classification results with alternative models.	The work demonstrates that the proposed B-CNN attains the best classification score among evaluated alternatives on OCT images, and it emphasizes improved early diagnosis performance for CNV, drusen, and DME relative to other deep learning networks.

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Work	Year	Overall reasoning	Contribution and summary
[373]: Retinal Specialist versus Artificial Intelligence Detection of Retinal Fluid from OCT: Age-Related Eye Disease Study 2: 10-Year Follow-On Study	2021	This prospective ophthalmic comparison in age-related macular degeneration evaluated intraretinal and subretinal fluid detection on spectral-domain OCT macular cube scans by benchmarking a machine-learning AI tool (Notal OCT Analyzer) against retinal specialists, using reading-center grading as the ground truth to provide diagnostic performance evidence across two common OCT devices.	Direct head-to-head results showed that the AI achieved higher accuracy with substantially higher sensitivity than retinal specialists for overall retinal fluid detection, including improved intraretinal and subretinal fluid performance, supporting AI-assisted OCT interpretation for retreatment-relevant disease activity assessment.
[280]: RetOCTNet: Deep Learning-Based Segmentation of OCT Images Following Retinal Ganglion Cell Injury	2025	This rat ophthalmic OCT study develops RetOCTNet, a deep learning semantic segmentation model that extracts RNFL and total retinal thickness from radial and volumetric OCT scans after retinal ganglion cell injury, validated by comparison to human manual ground truth on held-out scan sets and by longitudinal thickness changes after optic nerve crush.	RetOCTNet automates RNFL and total retina segmentation for preclinical RGC injury monitoring and quantifies longitudinal thinning across scan types, demonstrating strong agreement with human annotations and enabling consistent structural measurements for experimental glaucoma research.
[266]: RETRACTION:MAC-Leonets-A novel optimized hybrid convolutional neural networks for the segmentation and diagnosis of Edema diseases using retinal OCT images	2023	Using retinal optical coherence tomography imaging, this report describes a hybrid deep-learning framework that combines convolutional layers with Single Feed Forward Training Networks and Lion-based hyperparameter tuning to segment and classify three edema disease types, evaluated on a Kaggle OCT dataset with standard diagnostic performance metrics reported from model testing.	This research specifically details how SLFTN is fused with convolutional components and optimized via Lion search to perform automated edema segmentation and diagnosis on OCT, offering reported evidence of very high classification accuracy for the targeted task.
[322]: Robust and Interpretable Convolutional Neural Networks to Detect Glaucoma in Optical Coherence Tomography Images	2021	This glaucoma-focused ophthalmic study evaluates convolutional neural networks on OCT report imagery using end-to-end, fine-tuned transfer learning and a CNN ensemble, assessing robustness via performance on a separate "Field" dataset and establishing evidence for interpretability through Testing with Concept Activation Vectors together with clinician eye-fixation tracking on the same OCT reports.	The analysis quantifies which OCT report sub-images and visual concepts (including RNFL and RGCP probability maps and RGCP thickness maps) drive CNN classifications using TCAV, then corroborates these concept scores with where glaucoma experts fixate, offering a combined robustness-and-explainability framework for clinic-relevant OCT decision support.

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Work	Year	Overall reasoning	Contribution and summary
[66]: Robust Deep Learning-Based Approach for Retinal Layer Segmentation in Optical Coherence Tomography Images	2022	The research applies a deep learning method for automated retinal layer segmentation on optical coherence tomography (OCT) images, aiming to support ophthalmic assessment by delineating retinal layers as potential computational biomarkers for neurodegenerative disorders.	The reported work introduces a robust OCT-based segmentation pipeline that outputs the main retinal layers to facilitate evaluation of clinically relevant ocular structures and derive computational biomarkers for disease analysis.
[324]: Robust Fovea Detection in Retinal OCT Imaging Using Deep Learning	2022	The reported work uses a deep learning, pixel-wise regression approach (PRE U-net) applied to retinal OCT B-scans with spatial location encoding to automatically localize the fovea centralis, evaluated on OCT volumes from healthy subjects and patients with nAMD, DME, and RVO to provide evidence of robustness for clinically relevant macular landmark detection.	The study details how PRE U-net is trained and validated using sampled 2D B-scans concatenated with spatial location information for end-to-end fovea detection, reporting improved robustness and performance versus state-of-the-art methods in macular OCT settings where automated landmarking can guide treatment.
[306]: SAE-wAMD: A Self-Attention Enhanced convolution neural network for fine-grained classification of wet Age-Related Macular Degeneration using OCT Images	2022	This analysis evaluates a self-attention enhanced CNN (SAE-VGG16) with contrastive self-supervised pretraining to fine-grainedly classify wet age-related macular degeneration subtypes (neovascular AMD vs PCV) from OCT images, addressing small lesion-region localization and high image similarity using an experimental comparison on its OCT dataset.	This investigation introduces SAE-wAMD, integrating self-attention layers into a VGG16 backbone and leveraging contrastive pretraining to better separate OCT instances for subtype discrimination, which is clinically meaningful because the two wet AMD subtypes require different treatment approaches.
[244]: Segmentation of choroidal area in optical coherence tomography images using a transfer learning-based conventional neural network: a focus on diabetic retinopathy and a literature review	2024	This diabetic retinopathy OCT study uses a transfer learning DeepLabv3+SE semantic segmentation model with EfficientNetB0 to delineate choroidal area in EDI B-scans and supports automated luminal area and choroidal vascularity index measurement by reporting segmentation quality and automated-manual agreement via Bland-Altman analysis.	This analysis identifies DeepLabv3+SE paired with EfficientNetB0 as the top-performing variant among several CNN backbones and provides clinically oriented evidence that automated choroid-derived luminal area and CVI can reproduce manual measurements in the validation set.
[264]: Segmentation of retinal fluid based on deep learning: application of three-dimensional fully convolutional neural networks in optical coherence tomography images	2019	The study uses a deep learning fully convolutional approach on retinal optical coherence tomography (OCT) volumes to segment retinal fluid, contrasting modified 2D versus improved 3D U-Net architectures and evaluating segmentation outputs with standard metrics including accuracy, F1 score, and Kappa coefficient.	This work introduces an improved 3D fully convolutional U-Net for end-to-end volume segmentation aimed at leveraging inter-slice spatial correlations and addressing retinal OCT category imbalance, producing fluid delineations that closely match ophthalmologist annotations.

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Work	Year	Overall reasoning	Contribution and summary
[229]: Segmentation of retinal fluids and hyperreflective foci using deep learning approach in optical coherence tomography scans	2020	This work uses a deep-learning ensemble with OCT B-scan preprocessing and CNN-based segmentation to identify and delineate retinal fluids (intraretinal and subretinal) and hyperreflective foci from expert-annotated images, using Dice coefficient evaluation to quantify segmentation agreement in an ophthalmic diagnostic workflow.	The research demonstrates OCT-driven automated masks for both fluids and hyperreflective foci by combining SEUNet, a modified DRIU, and Pix2Pix, showing improved segmentation for clinically small lesions such as hyperreflective foci relative to baseline architectures.
[57]: Segmentation of Retinal Layers in OCT Images Using Deep Learning Algorithms for Medical Diagnostics	2024	The analysis uses optical coherence tomography imaging to perform retinal layer and abnormality segmentation, comparing classical machine learning with deep learning and reporting development and testing of a neural network model to support medical diagnostic decision-making.	This research adds value by describing a side-by-side algorithm comparison that culminates in training and testing a selected U-Net 3+ variant on the OCTDL dataset, with the stated goal of improving automated delineation of retinal abnormalities for diagnostic support.
[406]: Segmentation of retinal low-cost optical coherence tomography images using deep learning	2020	The study describes an AMD-relevant home-monitoring computer-aided diagnosis pipeline that applies CNN-based segmentation to low-cost self-examination full-field OCT images for delineating the retina and pigment epithelial detachments, with CDAE-based refinement to address imaging artifacts.	The analysis demonstrates how pairing CNN segmentation with denoising autoencoder refinement can improve automated retinal and PED delineation on SELF-OCT, supporting more reliable OCT biomarker quantification for continuous patient monitoring.
[275]: Semantic uncertainty Guided Cross-Transformer for enhanced macular edema segmentation in OCT images	2024	The research evaluates a transformer-based, uncertainty-guided semantic segmentation model for multi-class macular edema on OCT, targeting boundary delineation in low-contrast, noisy regions and improving separation of IRF, SRF, and PED, with evidence reported via evaluations on public datasets and OCT device data.	SuGCTNet's semantic uncertainty guided attention and cross-transformer modules aim to focus learning on ambiguous pixels while reducing inter-class feature confusion, supporting more reliable multi-class lesion segmentation where current methods struggle.
[305]: Sensitivity and specificity of machine learning classifiers for glaucoma diagnosis using Spectral Domain OCT and standard automated perimetry	2013	This 2013 observational cross-sectional study evaluates multiple machine learning classifier types for distinguishing glaucomatous from healthy eyes using multimodal ophthalmic imaging and functional data: Spectral Domain OCT retinal nerve fiber layer parameters combined with standard automated perimetry global indices, assessing diagnostic evidence through ROC curve analysis (aROC) and statistical comparisons between combined versus single-modality inputs.	The study reports a head-to-head comparison across nine classifier families trained on OCT-only, perimetry-only, and OCT-plus-perimetry feature sets, showing that adding OCT measures to perimetry improves diagnostic discrimination over OCT parameters alone.

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Work	Year	Overall reasoning	Contribution and summary
[93]: SF Net: A Pyramid-Based Feature Fusion Convolutional Neural Network With Embedded Squeeze-and-Excitation Mechanism for Retinal OCT Image Classification	2025	This analysis targets retinal OCT image classification in ophthalmology, using an end-to-end multi-scale convolutional neural network with pyramid-based intermediate feature fusion and embedded squeeze-and-excitation to classify normal eyes versus early/late AMD and DME, and it provides evidence from evaluations on two datasets, including a Noor Eye Hospital dataset and a public UCSD dataset, compared with established OCT classification frameworks.	This investigation introduces SF Net to leverage fused intermediate feature maps for more reliable OCT-based diagnosis, reporting that multi-scale feature fusion maintains strong accuracy even when training data are substantially reduced relative to comparable network approaches.
[42]: Simplified Convolutional Neural Network Model for Automatic Classification of Retinal Diseases from Optical Coherence Tomography Images	2023	A simplified convolutional neural network was trained to classify OCT B-scan images from a public dataset into normal retina versus diabetic macular edema, drusen, and choroidal neovascularization, providing multi-class image-based evidence from a train/test evaluation with standard classification metrics.	This analysis details a four-layer CNN architecture and OCT preprocessing approach (resizing and normalization) for end-to-end, OCT-to-disease categorization among four macular classes, reporting strong test accuracy to support practical deployment for ophthalmic screening workflows.
[247]: Spatial-Aware Transformer-GRU Framework for Enhanced Glaucoma Diagnosis From 3D OCT Imaging	2025	This ophthalmology-focused study targets glaucoma detection using 3D OCT volumes, combining a pre-trained Vision Transformer for slice-wise feature extraction with a bidirectional GRU to model inter-slice dependencies, and evaluates the resulting classification model with cross-validation using standard diagnostic performance metrics.	This work introduces a ViT-large plus bidirectional GRU framework that leverages full 3D OCT information through sequential processing of 64 B-scans, achieving improved glaucoma versus normal discrimination over baseline approaches that rely on 3D-CNN features or slice-wise RETFound adaptation.
[226]: State-of-the-Art Deep Learning Strategies for Multi-Class Classification of Retinal OCT Images	2024	Using retinal OCT as the ophthalmic imaging modality, this work applies deep learning with CNN, VGG-19, and DenseNet-121 plus ensemble learning to perform multi-class classification of CNV, DME, drusen, and normal, reporting accuracy evidence via comparisons on balanced versus imbalanced training datasets.	The research demonstrates how undersampling for class imbalance, coupled with model ensembling, can improve multi-class OCT categorization across four retinal and neuro-ophthalmic labels, as reflected by the reported balanced accuracy results for each backbone model.
[107]: Statistical Analysis of Deep Learning Models for Diabetic Macular Edema Classification using OCT Images	2022	The analysis evaluates deep learning architectures for classifying diabetic macular edema from OCT images, using statistical analysis of accuracy measures to support automated DME detection in an ophthalmic diagnostic setting.	This research compares OpticNet and DenseNet on a standard OCT dataset and reports statistically evaluated differences that can guide selection of a model for DME classification.

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Work	Year	Overall reasoning	Contribution and summary
[160]: Strategies to Improve Convolutional Neural Network Generalizability and Reference Standards for Glaucoma Detection From OCT Scans	2021	This investigation evaluates CNNs for glaucoma detection on OCT imaging using RNFL probability maps and circumpapillary disc B-scans, testing generalizability on an unseen dataset from a different site and quantifying how different reference-standard schemes affect accuracy and robustness across models.	The analysis reports practical gains in out-of-site performance by combining data augmentation, multimodal OCT inputs, and training on confidently labeled images, while finding that using a consistent, well-specified reference standard for both label creation and evaluation yields the best accuracy.
[141]: Structural-prior guided and feature-enhanced transformer with masked image modeling pretraining for retinal layers and fluid segmentation in macular edema OCT images	2025	The research develops a transformer-based OCT segmentation pipeline for macular edema that uses masked image modeling self-supervised pretraining (SimMIM) alongside a structural prior-guided, feature-enhanced architecture to delineate retinal layers and intraretinal/subretinal fluid in B-scans, with evidence from reported overlap-based and quantitative thickness/volume assessments on both a public and a private macular edema dataset.	By introducing a structural-prior constraint through customized multi-class synergistic segmentation (MCSS) loss plus dual self-attention and axial feature enhancement, the authors target layer order and topological continuity to improve segmentation robustness to low-contrast lesions and artifacts, which is reflected in substantially improved macular layer and fluid delineation outcomes across the evaluated datasets.
[157]: Studies of Plausibility Using Deep Learning Techniques to Automatically Detect Retinal Neovascularization in Enface Optical Coherence Tomography Angiography Images	2025	The reported work evaluates a deep learning binary classification approach using en face OCT angiography to automatically distinguish retinal neovascularization from non-neovascularization in diabetic retinopathy, addressing a clinically relevant prescreening task with evidence from model performance on a small, imbalanced OCTA dataset.	The study reports a feasibility assessment across five deep learning architectures and identifies Inception V3 as the top performer for RNV detection under limited RNV examples, supporting the practical viability of automated OCTA-based screening in real-world data conditions.
[300]: The Classification of Common Macular Diseases Using Deep Learning on Optical Coherence Tomography Images with and without Prior Automated Segmentation	2023	The study evaluates deep learning for macular disease classification from spectral-domain OCT in an ophthalmic imaging setting, contrasting end-to-end classification against segmentation-assisted classification using automated OCT segmentation (RelayNet and a graph-cut Caserel approach) across seven disease categories plus normal/other, with performance reported using internal 5-fold cross-validation.	This investigation demonstrates that introducing prior automated segmentation of OCT, especially with the graph-cut method, improves validation sensitivity while maintaining high overall discrimination (AUC), offering an evidence-based preprocessing strategy for OCT-based multiclass macular screening.

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Work	Year	Overall reasoning	Contribution and summary
[106]: The classification of optical coherence tomography images using machine learning for macular hole and diabetic macular edema diagnoses	2023	This research uses a Support Vector Machine to classify retinal optical coherence tomography images for macular hole versus diabetic macular edema, evaluated on both varied training data and real images with reported classification accuracy.	This analysis provides an image-based OCT classification approach for these two common retinal conditions and reports high accuracy figures, suggesting a practical aid for earlier detection and treatment planning.
[400]: The effect of optical degradation from cataract using a new Deep Learning optical coherence tomography segmentation algorithm	2024	This ophthalmology study evaluates a freely available online deep learning segmentation method for Spectralis SD-OCT retinal images, testing how cataract-related optical degradation affects segmentation validity by comparing agreement with ETDRS-derived measurements and reviewing sensitivity via post-cataract-surgery imaging.	This work reports excellent agreement with ETDRS measures and suggests that the online OCT segmentation tool remains usable after cataract surgery, highlighting practical robustness to media opacities for retinal layer quantification workflows.
[217]: The possibility of the combination of OCT and fundus images for improving the diagnostic accuracy of deep learning for age-related macular degeneration: a preliminary experiment	2019	This ophthalmology study evaluates multimodal deep learning for age-related macular degeneration by fusing OCT and fundus images and extracting features with transfer learning from a pretrained VGG-19 before random-forest classification, providing experimental evidence that combined imaging improves diagnostic discrimination versus single-modality models.	The research demonstrates, on postmortem Project Macula imaging data, that OCT-fundus combination produces higher AUC and accuracy and is statistically superior to OCT-only and fundus-only deep learning pipelines, indicating that multimodal fusion can materially boost AMD diagnostic performance in this setting.
[397]: The Predictive Capabilities of Artificial Intelligence-Based OCT Analysis for Age-Related Macular Degeneration Progression-A Systematic Review	2023	This PRISMA-guided systematic review evaluates AI-based analyses of spectral-domain OCT to predict future AMD outcomes in ophthalmic care, covering both natural-history progression and post-anti-VEGF follow-up (structure and visual function) using machine learning and deep learning evidence from 19 included studies.	This research consolidates 19 OCT prediction studies across clinically relevant tasks (anti-VEGF treatment needs and response, conversion to nAMD or GA, GA growth, and subsequent visual acuity) and critically summarizes their strengths and limitations to inform where OCT-based AI prognostication is most mature and what gaps remain.

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Work	Year	Overall reasoning	Contribution and summary
[124]: The role of artificial intelligence in monitoring glaucoma progression using optical coherence tomography	2025	This review addresses AI-assisted glaucoma progression monitoring in an ophthalmic setting by synthesizing OCT-based workflows that use deep learning for tasks such as segmentation, detection of subtle structural change, and progression prediction from RNFL, GCC, and ONH imaging biomarkers, drawing on reported diagnostic evidence and emphasizing the need for robust clinical validation.	The analysis distills practical considerations for translating AI-enhanced OCT into care, focusing on sources of performance risk such as device/protocol variability and limited data diversity, plus priorities like interpretability, privacy, and workflow integration, so readers can gauge how and when AI outputs may be trusted for longitudinal monitoring.
[411]: The utility of artificial intelligence in characterization and detecting causes of macular edema: A spectral-domain OCT-based algorithm study	2025	This ophthalmology study uses spectral-domain OCT imaging with pretrained CNNs and Grad-CAM to classify DME, AMD, and normal macular status, reporting retrospective evidence from a KA UH dataset and a public benchmark with interpretability checked against annotated data.	The reported work details a CNN-based OCT framework that achieves high classification accuracy for DME versus AMD and normal conditions while providing Grad-CAM visualizations manually validated against annotations to support clinically interpretable decision-making.
[84]: Three-dimensional transformer-enhanced multi-scale global co-attention network for precise diabetic macular edema segmentation in OCT volumes	2025	In ophthalmic OCT volumes of diabetic macular edema, the authors use a 3D transformer-based encoder-decoder segmentation approach augmented with non-local, multi-scale feature aggregation, and global channel-spatial co-attention to delineate DME lesions while leveraging 3D context, and they report evidence from quantitative test-set comparisons plus consistency between predicted and manual 3D lesion measurements.	They introduce a transformer-enhanced 3D segmentation framework that explicitly models long-range dependencies and 3D morphology in retinal OCT, and demonstrate clinically relevant lesion measurement consistency and improved robustness in challenging cases via ablation of key attention modules and evaluation against baseline 3D architectures.
[309]: Towards label-free 3D segmentation of optical coherence tomography images of the optic nerve head using deep learning	2020	This 2020 study targets optic nerve head glaucoma-relevant 3D segmentation in spectral-domain OCT by pairing a label-free enhancer for cross-device image harmonization with ONH-Net, an ensemble-inspired 3D deep learning model that segments six ONH tissue layers and evaluates evidence through device-wise training and testing with manual slice-wise reference labels.	This investigation shows that enhancer-based harmonization enables ONH-Net trained on one OCT device to produce consistent, high-overlap label-free 3D tissue segmentations on two unseen devices, supporting more practical automated OCT quantification without collecting manual segmentations for every platform.

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Work	Year	Overall reasoning	Contribution and summary
[150]: TransDualSegNet: transformer dual segment network for retinal vasculature segmentation in OCT	2025	This research targets retinal vasculature segmentation in OCT, using a Transformer-based encoder-decoder architecture with Pyramid Vision Transformer encoding plus a dual-branch, edge-aware decoder and feature fusion to predict vessel inner and outer boundaries, with evidence from comparative experiments on an ophthalmic OCT vessel dataset including cross-validation and ablation results.	TransDualSegNet is introduced as a dual-train framework that jointly leverages global OCT features and edge-enhancement cues fused via a feature fusion module, and the reported comparisons show improved segmentation accuracy and efficiency versus U-Net, Attention U-Net, and Trans U-Net.
[238]: Transfer Learning and Multi-Feature Fusion-Based Deep Learning Model for Idiopathic Macular Hole Diagnosis and Grading from Optical Coherence Tomography Images	2025	In an ophthalmic context, this retrospective single-center study uses transfer learning with multi-feature fusion on OCT images to perform idiopathic macular hole diagnosis and grading, supplemented by clinical-factor modeling to estimate surgical risk, with evidence reported from ROC/AUC-based diagnostic evaluation and a surgical-risk nomogram.	This work introduces an OCT radiomics and deep-feature fusion framework using a ResNet101 transfer-learning backbone to classify IMH grades and identify key drivers for surgical decision-making through a logistic-regression nomogram.
[91]: Transformer-Based DME Classification Using Retinal OCT Images Without Data Augmentation: An Evaluation of ViT-B16 and ViT-B32 With Optimizer Impact	2025	This work assesses Vision Transformer models (ViT-B16, ViT-B32) for classifying diabetic macular edema from retinal OCT images, using transfer learning without data augmentation and evaluating fine-tuning behavior across Adam, SGD, and RMSProp with evidence based on 5-fold cross-validation and Grad-CAM visual explanations in an ophthalmic diagnostic task.	The research shows that optimizer choice and transformer resolution affect DME OCT classification outcomes, with ViT-B16 outperforming ViT-B32 and transformer models outperforming CNN baselines while providing interpretable Grad-CAM heatmaps that support decision understanding.
[245]: Two-step hierarchical neural network for classification of dry age-related macular degeneration using optical coherence tomography images	2023	This ophthalmology study uses a two-step hierarchical CNN (with OCT image enhancement) to perform multi-class classification of dry age-related macular degeneration phases (normal, drusen, nascent geographic atrophy, and geographic atrophy) on Heidelberg Spectralis OCT macular B-scans, assessed with five-fold cross-validation and supported by ablation experiments.	This research proposes an OCT-based hierarchical pipeline that isolates nascent geographic atrophy from drusen using a second-stage discriminator, and the ablation results indicate that combining image enhancement with the hierarchical design substantially improves nGA performance over a flat baseline.

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Work	Year	Overall reasoning	Contribution and summary
[188]: Unsupervised machine learning analysis of optical coherence tomography radiomics features for predicting treatment outcomes in diabetic macular edema	2025	Using unsupervised machine learning on pretreatment OCT radiomics from 234 treatment-naive diabetic macular edema eyes, the study clusters eyes into imaging-defined subtypes and compares residual/recurrent disease and 6-month visual outcomes across clusters, with evidence supported by clustering robustness checks (bootstrap stability) and downstream multivariate feature/predictor analyses.	The analysis identifies four distinct DME radiomic clusters, including a lower-residual/recurrent and better-visual subgroup, and reports outcome-associated OCT radiomic biomarkers, most notably logarithm_gldm_DependenceVariance for residual/recurrent DME and wavelet-LH_firstorder_Mean for visual prognosis, offering noninvasive, OCT-omics-based stratification for treatment response heterogeneity.
[46]: Using Deep Learning and Transfer Learning to Accurately Diagnose Early-Onset Glaucoma From Macular Optical Coherence Tomography Images	2019	This multi-institution artificial intelligence diagnostic study applies deep learning with transfer learning to spectral-domain macular OCT retinal layer thickness features, aiming to distinguish early open-angle glaucoma from normal eyes with accuracy evaluated in a testing cohort using AROC and compared against random forests and support vector machines.	The reported work provides evidence that a convolutional neural network trained with two-stage pretraining improves discrimination of early glaucoma on OCT, and quantifies the gain via higher AROC than conventional classifiers, supporting clinical feasibility for early detection from macular imaging.
[190]: Using Machine Learning to Detect Different Eye Diseases from OCT Images	2023	This ophthalmology-focused study uses hyperparameter-tuned deep convolutional neural networks to classify eight retinal disease categories plus normal from OCT tomography images (OCT-C8 dataset), evaluating evidence through reported test-set classification metrics and confusion-matrix analysis.	The study systematically tunes six CNN architectures and highlights a fine-tuned DenseNet121 pipeline as the top-performing approach for multi-disease OCT classification, offering an OCT-based machine-learning alternative for early retinal disease detection.
[82]: Utilization of deep learning to quantify fluid volume of neovascular age-related macular degeneration patients based on swept-source OCT imaging: The ONTARIO study	2022	In a phase IV retrospective proof-of-concept study of neovascular age-related macular degeneration, the authors used a U-Net deep learning model to automatically segment intraretinal fluid, subretinal fluid, and serous pigment epithelial detachments from baseline swept-source OCT (with optional OCT-angiography CNVM lesion size) and correlated these imaging-derived measurements with week-52 change in BCVA to test their value as predictive biomarkers.	The analysis shows that baseline total retinal fluid volume had the strongest association with 12-month visual improvement and that combining total fluid with OCT-A mean CNVM size further increased correlation, supporting automated fluid quantification as a more clinically informative prognostic metric than thickness alone.

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Work	Year	Overall reasoning	Contribution and summary
[276]: Utilizing Swin Transformer for the Classification of Ophthalmic Diseases in Optical Coherence Tomography (OCT) Images: A Novel Approach	2024	Using a Swin Transformer for OCT-based classification, the study describes AMD and diabetic retinopathy detection from a Kaggle dataset, evaluated against CNNs and autoencoder approaches with standard diagnostic metrics (accuracy, precision, recall, and F1 score) as evidence from an experimental benchmarking study.	This analysis provides an application-focused demonstration of how transformer-based representations can be used to classify ophthalmic diseases in OCT images, and it matters because the reported metric-based comparison supports the feasibility of improving automated diagnosis workflows beyond CNN and autoencoder baselines.
[197]: Utilizing Transformers on OCT Imagery and Metadata for Treatment Response Prediction in Macular Edema Patients	2023	The study uses a transformer-based machine learning approach in an ophthalmic setting, combining OCT imaging with patient clinical metadata to predict treatment response for diabetic macular edema and to estimate comparative effectiveness among anti-VEGF injections, with evidence reported as model accuracy exceeding clinician skill.	This work reports that fusing clinical-record metadata with OCT improves response prediction by 12% over OCT-only training and that the model outputs comparative effectiveness probabilities for ranibizumab, aflibercept, and bevacizumab to support physician decision-making.
[428]: Validation and Clinical Applicability of Whole-Volume Automated Segmentation of Optical Coherence Tomography in Retinal Disease Using Deep Learning	2021	This diagnostic validation study in ophthalmology uses deep learning whole-volume OCT segmentation to quantify four macular pathological features in patients with wet age-related macular degeneration and diabetic macular edema across real-world device types, comparing automated outputs with adjudicated expert segmentations via voxelwise agreement, volumetric metrics, and expert qualitative clinical applicability ratings.	The research demonstrates that the automated model can deliver clinically acceptable, rapid retinal disease measurements in most scans by showing favorable specialist stack-ranking and Likert satisfaction alongside grader agreement, while also highlighting that qualitative assessment is necessary because quantitative concordance alone does not fully predict clinical usability.
[234]: VALIDATION OF A DEEP LEARNING-BASED ALGORITHM FOR SEGMENTATION OF THE ELLIPSOID ZONE ON OPTICAL COHERENCE TOMOGRAPHY IMAGES OF AN USH2A-RELATED RETINAL DEGENERATION CLINICAL TRIAL	2022	This ophthalmic validation study evaluated DOCTAD, a fully automatic deep learning en face segmentation algorithm on spectral-domain OCT volumes, to delineate ellipsoid zone on USH2A-related retinal degeneration (RUSH2A clinical trial) with performance tested against trained reader manual segmentations using overlap and measurement-agreement evidence.	By testing the EZ segmentation pipeline developed for macular telangiectasia type 2 on 127 baseline USH2A-related OCT volumes and comparing Dice overlap plus EZ area and length differences with reader measurements, the study supports clinically relevant generalizability of quantitative EZ biomarkers in a trial-grade, multicenter imaging setting.

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Work	Year	Overall reasoning	Contribution and summary
[123]: Validation of a deep learning model for automatic detection and quantification of five OCT critical retinal features associated with neovascular age-related macular degeneration	2024	This validation study uses a U-Net deep learning model to automatically segment five OCT-derived retinal biomarkers (intraretinal fluid, subretinal fluid, subretinal hyperreflective material, drusen/PED, and neovascular PED) linked to neovascular age-related macular degeneration, with evidence based on performance measured against manual annotations using detection (ROC/AUC) and segmentation agreement (correlation and Dice) on sequestered data.	The analysis reports quantitative evidence that the model can detect and quantify these multiple biomarkers with grader-level consistency, supporting its use for standardized, less subjective OCT feature reporting and downstream assessment of morphology and treatment response in clinical workflows.
[421]: Validation of a deep learning model for the automated detection and quantification of cystoid macular oedema on optical coherence tomography in patients with retinitis pigmentosa	2025	The research validates a nnU-Net deep learning segmentation model for automated cystoid macular edema assessment on spectral-domain OCT in retinitis pigmentosa, using expert-annotated external test scans and reporting overlap and area-agreement evidence (Dice and intraclass correlation) against human graders.	By externally testing on RP OCT volumes and benchmarking against three graders under a rotated reference framework, the authors show the model can provide near-human CMO detection and quantification to reduce manual variability in ophthalmic monitoring and clinical-trial workflows.
[261]: Validation of an Automated Artificial Intelligence Algorithm for the Quantification of Major OCT Parameters in Diabetic Macular Edema	2023	This multicenter validation study in diabetic macular edema tests a semi-supervised adversarial deep learning algorithm on spectral-domain OCT (Spectralis map and linear scans) to automatically detect and quantify intraretinal and subretinal fluid, external limiting membrane and ellipsoid zone interruption, and hyperreflective retinal foci, using comparisons against blinded expert manual examination and reporting agreement and accuracy-based evidence.	The work provides reproducible, automated quantification of major OCT biomarkers with expert-benchmarked concordance for fluid and retinal integrity measures and excellent reliability for hyperreflective foci counts, supporting more objective grading and monitoring in clinical DME workflows.
[311]: Validation of Deep Learning-Based Automatic Retinal Layer Segmentation Algorithms for Age-Related Macular Degeneration with 2 Spectral-Domain OCT Devices	2025	This prospective study validates UNet and DeepLabv3 retinal layer segmentation on spectral-domain OCT in an ophthalmic AMD cohort, using expert-labeled ground truth to quantify ILM, RPE, and related thickness and volume outputs, with testing across two OCT devices to provide evidence of device-robust performance rather than single-device accuracy.	The work trains on Heidelberg Spectralis B-scans and evaluates on both Spectralis and Zeiss Cirrus, showing clinically useful segmentation metrics on the held-out device and thereby supporting practical device-independence for generating retinal layer biomarkers when labeling is resource intensive.

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Work	Year	Overall reasoning	Contribution and summary
[398]: VCT-NET: An Octa Retinal Vessel Segmentation Network Based on Convolution and Transformer	2022	This research targets ophthalmic OCTA retinal vessel segmentation using a U-shaped deep learning architecture that combines convolution with a Swin-transformer branch to leverage global and local context, with evidence drawn from comparative experiments on an OCTA-6M dataset.	This analysis introduces VCT-Net as a transformer-convolution model for handling weak vessel boundaries and complex vascular patterns in OCTA, reporting improved segmentation performance relative to other deep learning approaches.
[55]: Vessel Density Features of Optical Coherence Tomography Angiography for Classification of Glaucoma Using Machine Learning	2023	In glaucoma detection, the study applies machine learning classifiers to peripapillary OCT angiography vessel density features generated via threshold-based segmentation, using a supervised classification task (glaucoma versus healthy) and reporting evidence from a train/test split with AUC and accuracy (including mild-to-moderate subgroup results for SVM).	This work develops combined vessel density metrics across multiple thresholds and image regions and shows that an SVM using these features achieves perfect discrimination in the test set (AUC and accuracy of 1) while other classifiers also perform highly, suggesting a practical OCTA-based route for glaucoma classification.
[257]: Vision Transformers for Retinal Disease Classification using Optical Coherence Tomography Images	2025	This work evaluates vision transformer architectures for ophthalmic retinal OCT image categorization, treating retinal disease labeling as a seven-class classification task and reporting comparative evidence between ViT and BEiT versus CNN baselines.	The research specifies an OCT-based seven-label retinal disease classification pipeline using ViT and BEiT and provides reported accuracies that quantify the potential benefit of transformer models relative to CNN approaches for this diagnostic task.
[379]: Visual acuity prediction on real-life patient data using a machine learning based multistage system	2024	This Scientific Reports study uses real-world ophthalmic patient data from heterogeneous clinical information systems, OCT biomarkers, and machine-learning and deep-learning components to predict visual-acuity progression in age-related macular degeneration, diabetic macular edema, and retinal vein occlusion during intravitreal operative medication therapy, with evidence from a multistage system evaluated against ophthalmologist annotations.	This research contributes a research-compatible data-integration and prediction workflow that aggregates more than 49,000 patients, completes incomplete OCT biomarker documentation with a deep neural network ensemble, and predicts winners, stabilizers, and losers of visual-acuity progression, supporting clinically oriented longitudinal decision support from routine-care data.

References

[1] D. Kermany, K. Zhang, M. Goldbaum, Large dataset of labeled optical coherence tomography (oct) and chest x-ray images, Mendeley Data (2018). doi:10.17632/rscbjbr9sj.3.
URL https://doi.org/10.17632/rscbjbr9sj.3

[2] D. S. Kermany, M. Goldbaum, W. Cai, C. C. S. Valentim, H. Liang, S. L. Baxter, A. McKeown, G. Yang, X. Wu, F. Yan, J. Dong, M. K. Prasadha, J. Pei, M. Y. L. Ting, J. Zhu, C. Li, S. Hewett, J. Dong, I. Ziyar, A. Shi, R. Zhang, L. Zheng, R. Hou, W. Shi, X. Fu, Y. Duan, V. A. N. Huu, C. Wen, E. D. Zhang, C. L. Zhang, O. Li, X. Wang, M. A. Singer, X. Sun, J. Xu, A. Tafreshi, M. A. Lewis, H. Xia, K. Zhang, Identifying medical diagnoses and treatable diseases by image-based deep learning, Cell 172 (5) (2018) 1122–1131.e9. doi:10.1016/j.cell.2018.02.010.
URL https://doi.org/10.1016/j.cell.2018.02.010

[3] M. Kulyabin, et al., Octdl: Optical coherence tomography dataset for image-based deep learning methods, Mendeley Data (2024). doi:10.17632/sncdhf53xc.4.
URL https://doi.org/10.17632/sncdhf53xc.4

[4] M. Kulyabin, et al., Octdl: Optical coherence tomography dataset for image-based deep learning methods, Scientific Data 11 (1) (2024) 365. doi:10.1038/s41597-024-03182-7.
URL https://doi.org/10.1038/s41597-024-03182-7

[5] P. Gholami, P. Roy, M. K. Parthasarathy, V. Lakshminarayanan, Optical coherence tomography image retinal database, Borealis / Scholars Portal Dataverse (2018). doi:10.5683/SP2/W43PFI.
URL https://doi.org/10.5683/SP2/W43PFI

[6] P. Gholami, P. Roy, M. K. Parthasarathy, V. Lakshminarayanan, Octid: Optical coherence tomography image database, Computers & Electrical Engineering 81 (2020) 106532. doi:10.1016/j.compeleceng.2019.106532.
URL https://doi.org/10.1016/j.compeleceng.2019.106532

[7] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, B. Ni, Medmnist v2, MedMNIST benchmark dataset (2023).
URL https://medmnist.com/

[8] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, B. Ni, Medmnist v2: A large-scale lightweight benchmark for 2d and 3d biomedical image classification, Scientific Data 10 (1) (2023) 41. doi:10.1038/s41597-022-01721-8.
URL https://doi.org/10.1038/s41597-022-01721-8

[9] O. Sai Naren, Retinal oct image classification - c8, Kaggle dataset (2021). doi:10.34740/KAGGLE/DSV/2736749.
URL https://doi.org/10.34740/KAGGLE/DSV/2736749

[10] M. Subramanian, P. Shanmugavadivu, O. Sai Naren, K. Premkumar, K. Rankish, Classification of retinal oct images using deep learning, in: 2022 International Conference on Computer Communication and Informatics (ICCCI), IEEE, 2022, pp. 1–7. doi:10.1109/ICCCI54379.2022.9740985.
URL https://doi.org/10.1109/ICCCI54379.2022.9740985

[11] Areds 2 ancillary spectral domain optical coherence tomography study (a2asdoct), ClinicalTrials.gov, clinicalTrials.gov Identifier: NCT00734487. Related AREDS2 Ancillary SD-OCT publications may also report NCT00794498; verify the preferred identifier for the exact cited data source. (2008).
URL https://clinicaltrials.gov/study/NCT00734487

[12] S. Farsiu, S. J. Chiu, R. V. O'Connell, F. A. Folgar, E. Yuan, J. A. Izatt, C. A. Toth, A.-R. E. D. S. . A. S. D. O. C. T. S. Group, Quantitative classification of eyes with and without intermediate age-related macular degeneration using optical coherence tomography, Ophthalmology 121 (1) (2014) 162–172. doi:10.1016/j.ophtha.2013.07.013.
URL https://doi.org/10.1016/j.ophtha.2013.07.013

[13] K. Alsaih, G. Lemaitre, M. Rastgoo, J. Massich, D. Sidibé, F. Mériaudeau, Machine learning techniques for diabetic macular edema classification on sd-oct images, BioMedical Engineering OnLine 16 (1) (2017) 68. doi:10.1186/s12938-017-0352-9.
URL https://doi.org/10.1186/s12938-017-0352-9

[14] O. Perdomo, S. Otálora, F. A. González, F. Mériaudeau, H. Müller, Classification of diabetes-related retinal diseases using a deep learning approach in optical coherence tomography, Computer Methods and Programs in Biomedicine 178 (2019) 181–189. doi:10.1016/j.cmpb.2019.06.016.
URL https://doi.org/10.1016/j.cmpb.2019.06.016

[15] Retouch: Retinal oct fluid challenge, Grand Challenge (2017).
URL https://retouch.grand-challenge.org/

[16] H. Bogunović, F. Venhuizen, S. Klimscha, S. Apostolopoulos, A. Bab-Hadiashar, U. Bagci, M. F. Beg, L. Bekalo, Q. Chen, C. Ciller, K. Gopinath, J. Hamwood, X. Han, X. Hu, I. Jang, I. Kirchberger, K. Krissian, S. Lee, F. Mériaudeau, et al., Retouch: The retinal oct fluid detection and segmentation benchmark and challenge, IEEE Transactions on Medical Imaging 38 (8) (2019) 1858–1874. doi:10.1109/TMI.2019.2901398.
URL https://doi.org/10.1109/TMI.2019.2901398

[17] Optima cyst segmentation challenge, Medical University of Vienna OPTIMA challenge (2015).
URL https://optima.meduniwien.ac.at/optima-segmentation-challenge-1/

[18] I. Oguz, L. Zhang, M. D. Abràmoff, M. Sonka, Optimal retinal cyst segmentation from oct images, in: Medical Imaging 2016: Image Processing, Vol. 9784, SPIE, 2016, p. 97841E. doi:10.1117/12.2217355.
URL https://doi.org/10.1117/12.2217355

[19] Ai challenger 2018 retinal edema lesion segmentation challenge, AI Challenger dataset, original challenge page formerly available at https://challenger.ai/competition/fl2018; page no longer reliably accessible. (2018).

[20] J. Hu, Y. Chen, Z. Yi, Automated segmentation of macular edema in oct using deep neural networks, Medical Image Analysis 55 (2019) 216–227. doi:10.1016/j.media.2019.05.002.
URL https://doi.org/10.1016/j.media.2019.05.002

[21] S. J. Chiu, S. Farsiu, Duke dme oct dataset, Duke University (2015).
URL https://people.duke.edu/~sf59/Chiu_BOE_2014_dataset.htm

[22] S. J. Chiu, M. J. Allingham, P. S. Mettu, S. W. Cousins, J. A. Izatt, S. Farsiu, Kernel regression based segmentation of optical coherence tomography images with diabetic macular edema, Biomedical Optics Express 6 (4) (2015) 1172–1194. doi:10.1364/BOE.6.001172.
URL https://doi.org/10.1364/BOE.6.001172

[23] M. Melinščak, M. Radmilović, Z. Vatavuk, S. Lončarić, Annotated retinal oct images database – aroi database, University of Zagreb, Faculty of Electrical Engineering and Computing (2021).
URL https://ipg.fer.hr/ipg/resources/oct_image_database

[24] M. Melinščak, M. Radmilović, Z. Vatavuk, S. Lončarić, Annotated retinal optical coherence tomography images (aroi) database for joint retinal layer and fluid segmentation, Automatika 62 (3–4) (2021) 375–385. doi:10.1080/00051144.2021.1973298.
URL https://doi.org/10.1080/00051144.2021.1973298

[25] M. Li, Y. Chen, S. Yuan, Q. Chen, Octa-500, IEEE DataPort (2020).
URL https://ieee-dataport.org/open-access/octa-500

[26] M. Li, K. Huang, Q. Xu, J. Yang, Y. Zhang, Z. Ji, K. Xie, S. Yuan, Q. Liu, Q. Chen, Octa-500: A retinal dataset for optical coherence tomography angiography study, Medical Image Analysis 93 (2024) 103092. doi:10.1016/j.media.2024.103092.
URL https://doi.org/10.1016/j.media.2024.103092

[27] Y. Giarratano, Optical coherence tomography angiography retinal scans and segmentations, University of Edinburgh DataShare (2019). doi:10.7488/ds/2729.
URL https://doi.org/10.7488/ds/2729

[28] T. Pissas, E. Bloch, B. Flores, O. Georgiadis, S. Jalali, C. Ravasio, D. Stoyanov, L. Da Cruz, C. Bergeles, Deep iterative vessel segmentation in oct angiography, Biomedical Optics Express 11 (5) (2020) 2490–2510. doi:10.1364/BOE.384919.
URL https://doi.org/10.1364/BOE.384919

[29] Y. Ma, H. Hao, H. Fu, J. Zhang, J. Yang, J. Liu, Y. Zheng, Y. Zhao, Rose: Retinal oct-angiography vessel segmentation dataset, Zenodo, dataset release includes ROSE subsets; see also the original ROSE publication. (2024). doi:10.5281/zenodo.12775880.
URL https://doi.org/10.5281/zenodo.12775880

[30] Y. Ma, H. Hao, H. Fu, J. Zhang, J. Yang, J. Liu, Y. Zheng, Y. Zhao, Rose: A retinal oct-angiography vessel segmentation dataset and new model, IEEE Transactions on Medical Imaging 40 (3) (2021) 928–939. doi:10.1109/TMI.2020.3042802.
URL https://doi.org/10.1109/TMI.2020.3042802

[31] Faros: Fully annotated retinal octa segmentation dataset, Zenodo, dataset by Zhongshan Ophthalmic Center; access restricted. (2022). doi:10.5281/zenodo.6381668.
URL https://doi.org/10.5281/zenodo.6381668

[32] P. Xiao, et al., Omsn and faros: Octa microstructure segmentation network and fully annotated retinal octa segmentation dataset, arXiv preprint (2022). doi:10.48550/arXiv.2212.13059.
URL https://doi.org/10.48550/arXiv.2212.13059

[33] Drac 2022: Diabetic retinopathy analysis challenge 2022 dataset, Mendeley Data (2023). doi:10.17632/s8kbw25s3x.1.
URL https://doi.org/10.17632/s8kbw25s3x.1

[34] B. Qian, H. Chen, X. Wang, et al., Drac 2022: A public benchmark for diabetic retinopathy analysis on ultra-wide optical coherence tomography angiography images, Patterns 5 (3) (2024) 100929. doi:10.1016/j.patter.2024.100929.
URL https://doi.org/10.1016/j.patter.2024.100929

[35] A. Diaz-Pinto, S. Morales, V. Naranjo, T. Köhler, J. M. Mossi, A. Navea, Cnns for automatic glaucoma assessment using fundus images: An extensive validation, figshare dataset (2019). doi:10.6084/m9.figshare.7613135.
URL https://doi.org/10.6084/m9.figshare.7613135

[36] A. Diaz-Pinto, S. Morales, V. Naranjo, T. Köhler, J. M. Mossi, A. Navea, Cnns for automatic glaucoma assessment using fundus images: An extensive validation, BioMedical Engineering OnLine 18 (1) (2019) 29. doi:10.1186/s12938-019-0649-y.
URL https://doi.org/10.1186/s12938-019-0649-y

[37] Retinal fundus glaucoma image dataset, Figshare, redistributed dataset including ORIGA-light; original dataset from Zhang et al. (2010). (2023).
URL https://figshare.com/articles/dataset/Retinal_Fundus_Glaucoma_Image_dataset/24549217

[38] Z. Zhang, F. S. Yin, J. Liu, D. W. K. Wong, N. M. Tan, B. H. Lee, J. Cheng, T. Y. Wong, Origa(-light): An online retinal fundus image database for glaucoma analysis and research, in: 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology, IEEE, 2010, pp. 3065–3068. doi:10.1109/IEMBS.2010.5626137.
URL https://doi.org/10.1109/IEMBS.2010.5626137

[39] Diabetic retinopathy detection, Kaggle competition dataset (2015).
URL https://www.kaggle.com/competitions/diabetic-retinopathy-detection

[40] V. Gulshan, L. Peng, M. Coram, M. C. Stumpe, D. Wu, A. Narayanaswamy, S. Venugopalan, K. Widner, T. Madams, J. Cuadros, R. Kim, R. Raman, P. C. Nelson, J. L. Mega, D. R. Webster, Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs, JAMA 316 (22) (2016) 2402–2410. doi:10.1001/jama.2016.17216.
URL https://doi.org/10.1001/jama.2016.17216

[41] W. Li, Z. Zhu, W. Li, Z. Zhu, L. W., Z. Z., Automated diagnosis of eight retinal diseases from oct images using deep learning with attention mechanisms, 2025 10th International Symposium on Advances in Electrical, Electronics and Computer Engineering, ISAEECE 2025 (2025) 328–331doi:10.1109/ISAEECE66033.2025.11160177.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11160177

[42] N. B. Khalaf, H. K. Aljobouri, M. S. Najim, I. Çankaya, K. N.B., A. H.K., N. M.S., Çankaya I., Simplified convolutional neural network model for automatic classification of retinal diseases from optical coherence tomography images, Al-Nahrain Journal for Engineering Sciences 26 (4) (2023) 314–319. doi:10.29194/NJES.26040314.
URL https://www.scopus.com/pages/publications/105013643999?origin=resultslist

[43] M. R. Mathews, S. T. M. Anzar, M. M.R., A. S.T.M., A lightweight deep learning model for retinal optical coherence tomography image classification, International Journal of Imaging Systems and Technology 33 (1) (2023) 204–216. doi:10.1002/ima.22808.
URL https://www.scopus.com/pages/publications/85138543591?origin=resultslist

[44] M. Subramanian, M. S. Kumar, V. E. Sathishkumar, J. Prabhu, A. Karthick, S. S. Ganesh, M. A. Meem, S. M, K. MS, S. VE, P. J, K. A, G. SS, M. MA, K. M.S., G. S.S., M. M.A., Diagnosis of retinal diseases based on bayesian optimization deep learning network using optical coherence tomography images, Computational intelligence and neuroscience 2022 (2022) 8014979. doi:10.1155/2022/8014979.
URL https://pubmed.ncbi.nlm.nih.gov/35463234/

[45] Z. Haddad, B. M. Yaya, H. Zgolli, D. Sidibé, H. Tabia, N. Khlifa, Z. Haddad, B. M. Yaya, H. Zgolli, D. Sidibe, H. Tabia, N. Khlifa, H. Z., Y. B.M., Z. H., S. D., T. H., K. N., Retinal pathologies detection in oct images based on bilinear convolutional neural network, 17th International Conference on INnovations in Intelligent SysTems and Applications, INISTA 2023 - Proceedings (2023) 1–7doi:10.1109/INISTA59065.2023.10310341.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10310341

[46] R. Asaoka, H. Murata, K. Hirasawa, Y. Fujino, M. Matsuura, A. Miki, T. Kanamoto, Y. Ikeda, K. Mori, A. Iwase, N. Shoji, K. Inoue, J. Yamagami, M. Araie, A. R, M. H, H. K, F. Y, M. M, M. A, K. T, I. Y, M. K, I. A, S. N, I. K, Y. J, A. M, Using deep learning and transfer learning to accurately diagnose early-onset glaucoma from macular optical coherence tomography images, American journal of ophthalmology 198 (2019) 136–145. doi:10.1016/j.ajo.2018.10.007.
URL https://pubmed.ncbi.nlm.nih.gov/30316669/

[47] W. Feng, M. Duan, B. Wang, Y. Du, Y. Zhao, B. Wang, L. Zhao, Z. Ge, Y. Hu, F. W., D. M., W. B., D. Y., Z. Y., Z. L., G. Z., H. Y., Automated segmentation of choroidal neovascularization on optical coherence tomography angiography images of neovascular age-related macular degeneration patients based on deep learning, Journal of Big Data 10 (1) (2023). doi:10.1186/s40537-023-00757-w.
URL https://www.scopus.com/pages/publications/85163043888?origin=resultslist

[48] T. J. N. Rao, G. N. Girish, A. R. Kothari, J. Rajan, T. J. Narendra Rao, G. N. Girish, A. R. Kothari, J. Rajan, N. R. TJ, G. GN, K. AR, R. J, K. A.R., Deep learning based sub-retinal fluid segmentation in central serous chorioretinopathy optical coherence tomography scans, Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference 2019 (2019) 978–981. doi:10.1109/EMBC.2019.8857105.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8857105

[49] J. Kugelman, D. Alonso-Caneiro, Y. Chen, S. Arunachalam, D. Huang, N. Vallis, M. J. Collins, F. K. Chen, K. J, A.-C. D, C. Y, A. S, H. D, V. N, C. MJ, C. FK, C. M.J., C. F.K., Retinal boundary segmentation in stargardt disease optical coherence tomography images using automated deep learning, Translational vision science & technology 9 (11) (2020) 12. doi:10.1167/tvst.9.11.12.
URL https://pubmed.ncbi.nlm.nih.gov/33133774/

[50] D.-K. Hwang, Y.-B. Chou, T.-C. Lin, H.-Y. Yang, Z.-K. Kao, C.-L. Kao, Y.-P. Yang, S.-J. Chen, C.-C. Hsu, Y.-C. Jheng, H. DK, C. YB, L. TC, Y. HY, K. ZK, K. CL, Y. YP, C. SJ, H. CC, J. YC, H. D.-K., C. Y.-B., L. T.-C., Y. H.-Y., K. Z.-K., K. C.-L., Y. Y.-P., C. S.-J., H. C.-C., J. Y.-C., Optical coherence tomography-based diabetic macula edema screening with artificial intelligence, Journal of the Chinese Medical Association : JCMA 83 (11) (2020) 1034–1038. doi:10.1097/JCMA.0000000000000351.
URL https://pubmed.ncbi.nlm.nih.gov/32452907/

[51] B. Ebrahimi, D. Le, M. Abtahi, A. K. Dadzie, J. I. Lim, R. V. P. Chan, X. Yao, E. B, L. D, A. M, D. AK, L. JI, C. RVP, Y. X, D. A.K., L. J.I., C. R.V.P., Optimizing the octa layer fusion option for deep learning classification of diabetic retinopathy, Biomedical optics express 14 (9) (2023) 4713–4724. doi:10.1364/BOE.495999.
URL https://pubmed.ncbi.nlm.nih.gov/37791267/

[52] S. K. Devalla, P. K. Renukanand, B. K. Sreedhar, G. Subramanian, L. Zhang, S. Perera, J.-M. Mari, K. S. Chin, T. A. Tun, N. G. Strouthidis, T. Aung, A. H. Thiéry, M. J. A. Girard, D. SK, R. PK, S. BK, S. G, Z. L, P. S, M. JM, C. KS, T. TA, S. NG, A. T, T. AH, G. MJA, D. S.K., R. P.K., S. B.K., M. J.-M., C. K.S., T. T.A., S. N.G., T. A.H., G. M.J.A., Drunet: a dilated-residual u-net deep learning network to segment optic nerve head tissues in optical coherence tomography images, Biomedical optics express 9 (7) (2018) 3244–3265. doi:10.1364/BOE.9.003244.
URL https://pubmed.ncbi.nlm.nih.gov/29984096/

[53] K. A. Clingo, C. A. Czerpak, H. A. Quigley, T. D. Nguyen, C. KA, C. CA, Q. HA, N. TD, C. K.A., C. C.A., Q. H.A., N. T.D., Deep learning method for semi-automated segmentation of optic nerve head tissues in optical coherence tomography images, Experimental eye research 261 (2025) 110678. doi:10.1016/j.exer.2025.110678.
URL https://pubmed.ncbi.nlm.nih.gov/41038371/

[54] A. Abini, S. S. S. Priya, P. S.S.S., A novel deep learning approach for diabetic retinopathy classification using optical coherence tomography angiography, Multimedia Tools and Applications 84 (31) (2025) 38613–38651. doi:10.1007/s11042-025-20708-2.
URL https://www.scopus.com/pages/publications/86000732033?origin=resultslist

[55] J. Jalili, M. Nadimi, B. Jafari, A. Esfandiari, R. Sadeghi, P. Ghahari, M. Sajedi, M. Safizade, M. Aghsaei Fard, J. J, N. M, J. B, E. A, S. R, G. P, S. M, S. M, A. F. M, Vessel density features of optical coherence tomography angiography for classification of glaucoma using machine learning, Journal of glaucoma 32 (12) (2023) 1006–1010. doi:10.1097/IJG.0000000000002329.
URL https://pubmed.ncbi.nlm.nih.gov/37974327/

[56] R. Rajan, S. Kumar, R. R., Oct deepnet 1—a deep learning approach for retinal oct image classification, Lecture Notes in Networks and Systems 608 (2023) 689–701. doi:10.1007/978-981-19-9225-4_50.
URL https://www.scopus.com/pages/publications/85150979937?origin=resultslist

[57] A. Pershin, V. Borisov, A. Pershin, V. Borisov, P. A., B. V., Segmentation of retinal layers in oct images using deep learning algorithms for medical diagnostics, Proceedings - 2024 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology, USBEREIT 2024 (2024) 110–113doi:10.1109/USBEREIT61901.2024.10583993.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10583993

[58] S. Azimov, J. Delgado, E. Romero, A. Vellido, J. Zarranz-Ventura, C. König, S. Azimov, J. Delgado, E. Romero, A. Vellido, J. Zarranz-Ventura, C. Konig, A. S., D. J., R. E., V. A., Z.-V. J., K. C., Cnn layer-wise explanations for the prediction of diabetic retinopathy from octa images, 2025 IEEE Symposium on Trustworthy, Explainable and Responsible Computational Intelligence, CITREx Companion 2025 (2025) 1–5doi:10.1109/CITRExCompanion65208.2025.10981490.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10981490

[59] R. M, S. Narayanan, M. R., N. S., Deep learning of fundus images and optical coherence tomography images for ocular disease detection – a review, Multimedia Tools and Applications 83 (41) (2024) 88745–88789. doi:10.1007/s11042-024-18938-x.
URL https://www.scopus.com/pages/publications/85188740562?origin=resultslist

[60] R. Rasti, H. Rabbani, A. Mehridehnavi, F. Hajizadeh, R. R., R. H., M. A., H. F., Macular oct classification using a multi-scale convolutional neural network ensemble, IEEE Transactions on Medical Imaging 37 (4) (2018) 1024–1034. doi:10.1109/TMI.2017.2780115.
URL https://www.scopus.com/pages/publications/85037579752?origin=resultslist

[61] J. Huang, G. Galal, V. Mukhin, M. Etemadi, A. P. Tanna, H. J, G. G, M. V, E. M, T. AP, T. A.P., Prediction and detection of glaucomatous visual field progression using deep learning on macular optical coherence tomography, Journal of glaucoma 33 (4) (2024) 246–253. doi:10.1097/IJG.0000000000002359.
URL https://pubmed.ncbi.nlm.nih.gov/38245813/

[62] T. K. Yoo, J. Y. Choi, H. K. Kim, Y. TK, C. JY, K. HK, Y. T.K., C. J.Y., K. H.K., Feasibility study to improve deep learning in oct diagnosis of rare retinal diseases with few-shot classification, Medical & biological engineering & computing 59 (2) (2021) 401–415. doi:10.1007/s11517-021-02321-1.
URL https://pubmed.ncbi.nlm.nih.gov/33492598/

[63] L. Guan, K. Yu, X. Chen, G. L., Y. K., C. X., Fully automated detection and quantification of multiple retinal lesions in oct volumes based on deep learning and improved drlse, Progress in Biomedical Optics and Imaging - Proceedings of SPIE 10949 (2019). doi:10.1117/12.2512656.
URL https://www.scopus.com/pages/publications/85068330630?origin=resultslist

[64] S. Sedai, B. Antony, D. Mahapatra, R. Garnavi, S. S., A. B., M. D., G. R., Joint segmentation and uncertainty visualization of retinal layers in optical coherence tomography images using bayesian deep learning, Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 11039 LNCS (2018) 219–227. doi:10.1007/978-3-030-00949-6_26.
URL https://www.scopus.com/pages/publications/85053900739?origin=resultslist

[65] R. R. Teegavarapu, H. A. Sanghvi, T. Belani, G. S. Gill, K. V. Chalam, S. Gupta, T. RR, S. HA, B. T, G. GS, C. KV, G. S, Evaluation of convolutional neural networks (cnns) in identifying retinal conditions through classification of optical coherence tomography (oct) images, Cureus 17 (1) (2025) e77109. doi:10.7759/cureus.77109.
URL https://pubmed.ncbi.nlm.nih.gov/39925554/

[66] A. Budiño, L. Ramos, J. de Moura, J. Novo, M. G. Penedo, M. Ortega, B. no A., R. L., de Moura J., N. J., P. M.G., O. M., Robust deep learning-based approach for retinal layer segmentation in optical coherence tomography images, Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 13789 LNCS (2022) 427–434. doi:10.1007/978-3-031-25312-6_50.
URL https://www.scopus.com/pages/publications/85151140756?origin=resultslist

[67] M. Elsharkawy, I. Abdelhalim, M. Ghazal, A. Mahmoud, H. S. Sandhu, A. Thanos, A. El-Baz, M. Elsharkawy, I. Abdelhalim, M. Ghazal, A. Mahmoud, H. S. Sandhu, A. Thanos, A. El-Baz, E. M., A. I., G. M., M. A., S. H.S., T. A., E.-B. A., Oct-trans: A novel transformer backbone with multimodal feature extraction in oct-based retinal disease classification, 2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI) (2025) 1–4doi:10.1109/ISBI60581.2025.10980790.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10980790

[68] X. Gu, Y. Zhou, J. Zhao, H. Zhang, X. Pan, B. Li, B. Zhang, Y. Wang, S. Xia, H. Lin, J. Wang, D. Ding, X. Li, S. Wu, J. Yang, Y. Chen, G. X, Z. Y, Z. J, Z. H, P. X, L. B, Z. B, W. Y, X. S, L. H, W. J, D. D, L. X, W. S, Y. J, C. Y, Diagnostic performance and generalizability of deep learning for multiple retinal diseases using bimodal imaging of fundus photography and optical coherence tomography, Frontiers in cell and developmental biology 13 (2025) 1665173. doi:10.3389/fcell.2025.1665173.
URL https://pubmed.ncbi.nlm.nih.gov/41018263/

[69] J. Kim, L. Tran, J. Kim, L. Tran, K. J., T. L., Ensemble learning based on convolutional neural networks for the classification of retinal diseases from optical coherence tomography images, 2020 IEEE 33rd International Symposium on Computer-Based Medical Systems (CBMS) 2020-July (2020) 532–537. doi:10.1109/CBMS49503.2020.00106.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9182910

[70] X. Li, X. Wen, X. Shang, J. Liu, L. Zhang, Y. Cui, X. Luo, G. Zhang, J. Xie, T. Huang, Z. Chen, Z. Lyu, X. Wu, Y. Lan, Q. Meng, L. X, W. X, S. X, L. J, Z. L, C. Y, L. X, Z. G, X. J, H. T, C. Z, L. Z, W. X, L. Y, M. Q, Identification of diabetic retinopathy classification using machine learning algorithms on clinical data and optical coherence tomography angiography, Eye (London, England) 38 (14) (2024) 2813–2821. doi:10.1038/s41433-024-03173-3.
URL https://pubmed.ncbi.nlm.nih.gov/38871934/

[71] B. Anoop, S. Parida, B. Ajith, G. Girish, A. R. Kothari, M. S. Kavitha, J. Rajan, P. S., K. A.R., K. M.S., R. J., Attention assisted patch-wise cnn for the segmentation of fluids from the retinal optical coherence tomography images, Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 13102 LNCS (2024) 213–223. doi:10.1007/978-3-031-12700-7_22.
URL https://www.scopus.com/pages/publications/85200697734?origin=resultslist

[72] M. Abirami, B. Vennila, K. Suganthi, S. Kawatra, A. Vaishnava, K. S., V. A., Detection of choroidal neovascularization (cnv) in retina oct images using vgg16 and densenet cnn, Wireless Personal Communications 127 (3) (2022) 2569–2583. doi:10.1007/s11277-021-09086-8.
URL https://www.scopus.com/pages/publications/85115036231?origin=resultslist

[73] L. Ngo, J. Cha, J. H. Han, L. Ngo, J. Cha, J.-H. Han, N. L., C. J., H. J.-H., Deep neural network regression for automated retinal layer segmentation in optical coherence tomography images, IEEE Transactions on Image Processing 29 (2020) 303–312. doi:10.1109/TIP.2019.2931461.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8784409

[74] F. Hosseini, F. Asadi, R. Rabiei, A. Roshanpoor, H. Sabbaghi, M. Eslami, R. E. Harari, H. F, A. F, R. R, R. A, S. H, E. M, H. RE, H. R.E., Early detection of cystoid macular edema in retinitis pigmentosa using longitudinal deep learning analysis of oct scans, Diagnostics (Basel, Switzerland) 16 (1) (2025). doi:10.3390/diagnostics16010046.
URL https://pubmed.ncbi.nlm.nih.gov/41515541/

[75] M. Christopher, A. Belghith, R. N. Weinreb, C. Bowd, M. H. Goldbaum, L. J. Saunders, F. A. Medeiros, L. M. Zangwill, C. M, B. A, W. RN, B. C, G. MH, S. LJ, M. FA, Z. LM, W. R.N., G. M.H., S. L.J., M. F.A., Z. L.M., Retinal nerve fiber layer features identified by unsupervised machine learning on optical coherence tomography scans predict glaucoma progression, Investigative ophthalmology & visual science 59 (7) (2018) 2748–2756. doi:10.1167/iovs.17-23387.
URL https://pubmed.ncbi.nlm.nih.gov/29860461/

[76] Y. Jin, F. Gui, M. Chen, X. Chen, H. Li, J. Zhang, J. Y., G. F., C. M., C. X., L. H., Z. J., Deep learning-driven automated quality assessment of ultra-widefield optical coherence tomography angiography images for diabetic retinopathy, Visual Computer 41 (2) (2025) 1049–1059. doi:10.1007/s00371-024-03383-6.
URL https://www.scopus.com/pages/publications/85192741003?origin=resultslist

[77] T. De Silva, G. Jayakar, P. Grisso, N. Hotaling, E. Y. Chew, C. A. Cukras, D. S. T, J. G, G. P, H. N, C. EY, C. CA, C. E.Y., C. C.A., Deep learning-based automatic detection of ellipsoid zone loss in spectral-domain oct for hydroxychloroquine retinal toxicity screening, Ophthalmology science 1 (4) (2021) 100060. doi:10.1016/j.xops.2021.100060.
URL https://pubmed.ncbi.nlm.nih.gov/36246938/

[78] T. Ye, J. Wang, J. Yi, Y. T, W. J, Y. J, Deep learning network for parallel self-denoising and segmentation in visible light optical coherence tomography of the human retina, Biomedical optics express 14 (11) (2023) 6088–6099. doi:10.1364/BOE.501848.
URL https://pubmed.ncbi.nlm.nih.gov/38021135/

[79] F. Noor, F. T. Nusaiba, T. A. Sami, M. A. Rahman, F. Noor, F. T. Nusaiba, T. A. Sami, M. A. Rahman, N. F., N. F.T., S. T.A., R. M.A., Retinal disease detection by optical coherence tomography (oct)-based deep learning, 2024 6th International Conference on Sustainable Technologies for Industry 5.0, STI 2024 (2024) 1–6doi:10.1109/STI64222.2024.10951065.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10951065

[80] F. Ma, X. Liu, S. Wang, S. Li, C. Dai, J. Meng, M. F, L. X, W. S, L. S, D. C, M. J, Csanet: a lightweight channel and spatial attention neural network for grading diabetic retinopathy with optical coherence tomography angiography, Quantitative imaging in medicine and surgery 14 (2) (2024) 1820–1834. doi:10.21037/qims-23-1270.
URL https://pubmed.ncbi.nlm.nih.gov/38415109/

[81] O. F. Aydın, F. B. Tek, Y. Turkan, O. F. Aydin, F. Boray Tek, Y. Turkan, A. O.F., T. Y., Retinal disease classification from bimodal oct and octa using a cnn–vit hybrid architecture, International Conference on Computer Science and Engineering, UBMK (2025) (2025) 260–264. doi:10.1109/UBMK67458.2025.11206835.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11206835

[82] S. K. Sodhi, A. Pereira, J. D. Oakley, J. Golding, C. Trimboli, D. B. Russakoff, N. Choudhry, S. SK, P. A, O. JD, G. J, T. C, R. DB, C. N, S. S.K., O. J.D., R. D.B., Utilization of deep learning to quantify fluid volume of neovascular age-related macular degeneration patients based on swept-source oct imaging: The ontario study, PloS one 17 (2) (2022) e0262111. doi:10.1371/journal.pone.0262111.
URL https://pubmed.ncbi.nlm.nih.gov/35157713/

[83] S. Kaymak, A. Serener, S. Kaymak, A. Serener, K. S., S. A., Automated age-related macular degeneration and diabetic macular edema detection on oct images using deep learning, Proceedings - 2018 IEEE 14th International Conference on Intelligent Computer Communication and Processing, ICCP 2018 (2018) 265–269doi:10.1109/ICCP.2018.8516635.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8516635

[84] J. Hu, C. Li, S. Lin, M. Qin, R. Wu, J. Zhong, Y. Liu, G. Ni, H. J, L. C, L. S, Q. M, W. R, Z. J, L. Y, N. G, Three-dimensional transformer-enhanced multi-scale global co-attention network for precise diabetic macular edema segmentation in oct volumes, Biomedical optics express 16 (9) (2025) 3608–3622. doi:10.1364/BOE.564842.
URL https://pubmed.ncbi.nlm.nih.gov/41368089/

[85] P. Elena-Anca, P. Elena-Anca, E.-A. P., Applications of deep learning algorithms for retinal diseases diagnosis based on optical coherence tomography imaging, Proceedings - 2023 24th International Conference on Control Systems and Computer Science, CSCS 2023 (2023) 594–597doi:10.1109/CSCS59211.2023.00099.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10214776

[86] M. Chen, J. Wang, I. Oguz, B. L. VanderBeek, J. C. Gee, C. M, W. J, O. I, V. BL, G. JC, V. B.L., G. J.C., Automated segmentation of the choroid in edi-oct images with retinal pathology using convolution neural networks, Fetal, Infant and Ophthalmic Medical Image Analysis : International Workshop, FIFI 2017 and 4th International Workshop, OMIA 2017, Held in Conjunction with MICCAI 2017, Quebec City, QC, Canada, September 14, 2017. FIFI (Workshop) (2017 ... 10554 (2017) 177–184. doi:10.1007/978-3-319-67561-9_20.
URL https://pubmed.ncbi.nlm.nih.gov/29757338/

[87] A. V, V. Asmabi, V. Anoop, A deep learning system model for detecting neurodegenerative diseases from retinal optical coherence tomography, Conference Proceedings - 2025 4th International Conference on Advances in Computing, Communication, Embedded and Secure Systems, ACCESS 2025 (2025) 628–632doi:10.1109/ACCESS65134.2025.11135738.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11135738

[88] M. Elsharkawy, I. Abdelhalim, A. Mahmoud, A. Gamal, M. E. S. Abdel-Hady, A. Sewelam, A. El-Baz, M. Elsharkawy, I. Abdelhalim, A. Mahmoud, A. Gamal, M. E.-S. Abdel-Hady, A. Sewelam, A. El-Baz, E. M., A. I., M. A., G. A., A.-H. M.E.-S., S. A., E.-B. A., Fused-aetnet: A variational transformer-based framework for diabetic retinopathy classification using oct biomarkers, IEEE Access 13 (2025) 163120–163133. doi:10.1109/ACCESS.2025.3609175.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11159226

[89] M. Pawloff, B. S. Gerendas, G. Deak, H. Bogunovic, A. Gruber, U. Schmidt-Erfurth, P. M, G. BS, D. G, B. H, G. A, S.-E. U, G. B.S., Performance of retinal fluid monitoring in oct imaging by automated deep learning versus human expert grading in neovascular amd, Eye (London, England) 37 (18) (2023) 3793–3800. doi:10.1038/s41433-023-02615-8.
URL https://pubmed.ncbi.nlm.nih.gov/37311835/

[90] E. M. Sina, J. Pena, S. Zafar, N. K. Bommakanti, A. E. Kuriyan, Y. Yonekawa, S. EM, P. J, Z. S, B. NK, K. AE, Y. Y, S. E.M., B. N.K., K. A.E., Automated machine learning classification of optical coherence tomography images of retinal conditions using google cloud vertex artificial intelligence, Retina (Philadelphia, Pa.) 45 (10) (2025) 1984–1990. doi:10.1097/IAE.0000000000004555.
URL https://pubmed.ncbi.nlm.nih.gov/40523306/

[91] K. C. Pavithra, P. Kumar, M. Geetha, S. V. Bhandary, K. B. A. Shenoy, G. Rao, S. Fernandes, A. Tulsani, K. Pavithra, P. Kumar, M. Geetha, S. V. Bhandary, K. Ajitha Shenoy, G. Rao, S. Fernandes, A. Tulsani, K. P., B. S.V., R. G., F. S., T. A., Transformer-based dme classification using retinal oct images without data augmentation: An evaluation of vit-b16 and vit-b32 with optimizer impact, IEEE Access 13 (2025) 180781–180798. doi:10.1109/ACCESS.2025.3620945.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11202415

[92] K. Zhang, X. Liang, L. Li, Y. Liu, Z. K., L. X., L. L., L. Y., A machine learning classification model for primary open-angle glaucoma based on optical coherence tomography optic disc parameters and interocular asymmetry features, Chinese Journal of Ocular Fundus Diseases 42 (2) (2026) 144–154. doi:10.3760/cma.j.cn511434-20251021-00461.
URL https://www.scopus.com/pages/publications/105031480334?origin=resultslist

[93] S. Zheng, Y. Wang, Z. S., W. Y., Sf net: A pyramid-based feature fusion convolutional neural network with embedded squeeze-and-excitation mechanism for retinal oct image classification, International Journal of Imaging Systems and Technology 35 (5) (2025). doi:10.1002/ima.70197.
URL https://www.scopus.com/pages/publications/105016393271?origin=resultslist

[94] C. S. Lee, D. M. Baughman, A. Y. Lee, L. CS, B. DM, L. AY, Deep learning is effective for the classification of oct images of normal versus age-related macular degeneration, Ophthalmology. Retina 1 (4) (2017) 322–327. doi:10.1016/j.oret.2016.12.009.
URL https://pubmed.ncbi.nlm.nih.gov/30693348/

[95] J. Loo, C. X. Cai, J. Choong, E. Y. Chew, M. Friedlander, G. J. Jaffe, S. Farsiu, L. J, C. CX, C. J, C. EY, F. M, J. GJ, F. S, C. C.X., C. E.Y., J. G.J., Deep learning-based classification and segmentation of retinal cavitations on optical coherence tomography images of macular telangiectasia type 2, British Journal of Ophthalmology 106 (3) (2022) 396–402. doi:10.1136/bjophthalmol-2020-317131.
URL https://pubmed.ncbi.nlm.nih.gov/33229343/

[96] G. M. Nagamani, T. Sudhakar, N. G.M., S. T., Automated classification of age-related macular degeneration from optical coherence tomography images using deep learning approach, IAES International Journal of Artificial Intelligence 12 (4) (2023) 2011–2021. doi:10.11591/ijai.v12.i4.pp2011-2021.
URL https://www.scopus.com/pages/publications/85167328694?origin=resultslist

[97] Y.-Z. Wang, W. Wu, D. G. Birch, W. YZ, W. W, B. DG, W. Y.-Z., B. D.G., A hybrid model composed of two convolutional neural networks (cnns) for automatic retinal layer segmentation of oct images in retinitis pigmentosa (rp), Translational vision science & technology 10 (13) (2021) 9. doi:10.1167/tvst.10.13.9.
URL https://pubmed.ncbi.nlm.nih.gov/34751740/

[98] E. Yusufoğlu, H. Fırat, H. Üzen, S. T. A. Özçelik, I. B. Çiçek, A. Şengür, O. Atila, N. H. Guldemir, Y. E, F. H, Üzen H, Özçelik STA, Çiçek İB, Şengür A, A. O, G. NH, Özçelik S.T.A., Çiçek İ.B., G. N.H., A comprehensive cnn model for age-related macular degeneration classification using oct: Integrating inception modules, se blocks, and convmixer, Diagnostics (Basel, Switzerland) 14 (24) (2024). doi:10.3390/diagnostics14242836.
URL https://pubmed.ncbi.nlm.nih.gov/39767197/

[99] K. Pin, J. W. Han, Y. Nam, P. K., H. J.W., N. Y., Retinal diseases classification based on hybrid ensemble deep learning and optical coherence tomography images, Electronic Research Archive 31 (8) (2023) 4843–4861. doi:10.3934/era.2023248.
URL https://www.scopus.com/pages/publications/85168584469?origin=resultslist

[100] G. Aresta, T. Araujo, G. S. Reiter, J. Mai, S. Riedl, C. Grechenig, R. H. Guymer, Z. Wu, U. Schmidt-Erfurth, H. Bogunovic, A. G, A. T, R. GS, M. J, R. S, G. C, G. RH, W. Z, S.-E. U, B. H, R. G.S., G. R.H., Deep neural networks for automated outer plexiform layer subsidence detection on retinal oct of patients with intermediate amd, Translational vision science & technology 13 (6) (2024) 7. doi:10.1167/tvst.13.6.7.
URL https://pubmed.ncbi.nlm.nih.gov/38874975/

[101] J. Jung, J. Han, J. M. Han, J. Ko, J. Yoon, J. S. Hwang, J. I. Park, G. Hwang, J. H. Jung, D. D.-J. Hwang, J. J, H. J, H. JM, K. J, Y. J, H. JS, P. JI, H. G, J. JH, H. DD, H. J.M., H. J.S., P. J.I., J. J.H., H. D.D.-J., Prediction of neovascular age-related macular degeneration recurrence using optical coherence tomography images with a deep neural network, Scientific reports 14 (1) (2024) 5854. doi:10.1038/s41598-024-56309-6.
URL https://pubmed.ncbi.nlm.nih.gov/38462646/

[102] J. Kugelman, J. Allman, S. A. Read, S. J. Vincent, J. Tong, M. Kalloniatis, F. K. Chen, M. J. Collins, D. Alonso-Caneiro, K. J, A. J, R. SA, V. SJ, T. J, K. M, C. FK, C. MJ, A.-C. D, R. S.A., V. S.J., C. F.K., C. M.J., A comparison of deep learning u-net architectures for posterior segment oct retinal layer segmentation, Scientific reports 12 (1) (2022) 14888. doi:10.1038/s41598-022-18646-2.
URL https://pubmed.ncbi.nlm.nih.gov/36050364/

[103] J. Kim, L. Tran, J. Kim, L. Tran, K. J., T. L., Ensemble convolutional neural networks for the classification and visualization of retinal diseases in optical coherence tomography images, 2023 IEEE 36th International Symposium on Computer-Based Medical Systems (CBMS) 2023-June (2023) 123–128. doi:10.1109/CBMS58004.2023.00203.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10178743

[104] A. Vahadane, A. Joshi, K. Madan, T. R. Dastidar, A. Vahadane, A. Joshi, K. Madan, T. R. Dastidar, V. A., J. A., M. K., D. T.R., Detection of diabetic macular edema in optical coherence tomography scans using patch based deep learning, 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018) 2018-April (2018) 1427–1430. doi:10.1109/ISBI.2018.8363840.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8363840

[105] A. A. Movassagh, M. Jajroudi, A. Homayoun Jafari, E. Khalili Pour, H. Farrokhpour, H. Faghihi, H. Riazi, H. ArabAlibeik, M. AA, J. M, H. J. A, K. P. E, F. H, F. H, R. H, A. H, Quantifying the characteristics of diabetic retinopathy in macular optical coherence tomography angiography images: A few-shot learning and explainable artificial intelligence approach, Cureus 17 (1) (2025) e76746. doi:10.7759/cureus.76746.
URL https://pubmed.ncbi.nlm.nih.gov/39897224/

[106] D. N. Phu, A. Q. Tran, N. Q. Tran, T. H. Dang, D. N. Phu, A. Q. Tran, N. Q. Tran, T. H. Dang, P. D.N., T. A.Q., T. N.Q., D. T.H., The classification of optical coherence tomography images using machine learning for macular hole and diabetic macular edema diagnoses, Proceedings - 12th IEEE International Conference on Control, Automation and Information Sciences, ICCAIS 2023 (2023) 634–638doi:10.1109/ICCAIS59597.2023.10382327.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10382327

[107] K. Pavithra, P. Kumar, M. Geetha, S. Bhandary, K. Pavithra, P. Kumar, M. Geetha, S. Bhandary, P. K., K. P., B. S., Statistical analysis of deep learning models for diabetic macular edema classification using oct images, 2022 IEEE International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics, DISCOVER 2022 - Proceedings (2022) 305–310doi:10.1109/DISCOVER55800.2022.9974917.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9974917

[108] M. Mirsalehi, M. Schwemm, E. Flockerzi, N. Szentmáry, A. D. Abdin, B. Seitz, A. Langenbucher, M. M, S. M, F. E, S. N, A. AD, S. B, L. A, A. A.D., Deep learning-based diagnosis of corneal condition by using raw optical coherence tomography data, Diagnostics (Basel, Switzerland) 15 (24) (2025). doi:10.3390/diagnostics15243115.
URL https://pubmed.ncbi.nlm.nih.gov/41464119/

[109] S. Baba, P. Kumari, P. Saxena, B. S., K. P., S. P., Retinal disease classification using custom cnn model from oct images, Procedia Computer Science 235 (2024) 3142–3152. doi:10.1016/j.procs.2024.04.297.
URL https://www.scopus.com/pages/publications/85196390950?origin=resultslist

[110] C. Lam, Y. L. Wong, Z. Tang, X. Hu, T. X. Nguyen, D. Yang, S. Zhang, J. Ding, S. K. H. Szeto, A. R. Ran, C. Y. Cheung, L. C, W. YL, T. Z, H. X, N. TX, Y. D, Z. S, D. J, S. SKH, R. AR, C. CY, W. Y.L., N. T.X., S. S.K.H., R. A.R., C. C.Y., Performance of artificial intelligence in detecting diabetic macular edema from fundus photography and optical coherence tomography images: A systematic review and meta-analysis, Diabetes care 47 (2) (2024) 304–319. doi:10.2337/dc23-0993.
URL https://pubmed.ncbi.nlm.nih.gov/38241500/

[111] J. Smida, A. S. C. Bose, N. A. Jarallah, A. Smida, G. Koushaji, J. Smida, A. S. C. Bose, N. Al Jarallah, A. Smida, G. Koushaji, S. J., B. A.S.C., A. J. N., S. A., K. G., Classification of fluid filled abnormalities in retinal optical coherence tomography images using chicken swarm optimized deep learning classifier, International Conference on Electrical, Computer, Communications and Mechatronics Engineering, ICECCME 2023 (2023) 1–8doi:10.1109/ICECCME57830.2023.10252324.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10252324

[112] C. Xu, H. Zheng, K. Liu, Y. Chen, C. Ye, C. Niu, S. Jin, Y. Li, H. Gao, J. Hu, Y. Zou, X. He, X. C., Z. H., L. K., C. Y., Y. C., N. C., J. S., L. Y., G. H., H. J., Z. Y., H. X., Deep learning for retina structural biomarker classification using oct images, Proceedings of SPIE - The International Society for Optical Engineering 13164 (2024). doi:10.1117/12.3026739.
URL https://www.scopus.com/pages/publications/85192828253?origin=resultslist

[113] Y. Wang, C. Galang, W. R. Freeman, A. Warter, A. Heinke, D. U. G. Bartsch, T. Q. Nguyen, C. An, Y. Wang, C. Galang, W. R. Freeman, A. Warter, A. Heinke, D.-U. G. Bartsch, T. Q. Nguyen, C. An, W. Y, G. C, F. WR, W. A, H. A, B. DG, N. TQ, A. C, F. W.R., B. D.-U.G., N. T.Q., Retinal oct layer segmentation via joint motion correction and graph-assisted 3d neural network, IEEE Access 11 (2023) 103319–103332. doi:10.1109/ACCESS.2023.3317011.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10254538

[114] P. Seeböck, W.-D. Vogl, S. M. Waldstein, J. I. Orlando, M. Baratsits, T. Alten, M. Arikan, G. Mylonas, H. Bogunović, U. Schmidt-Erfurth, S. P, V. WD, W. SM, O. JI, B. M, A. T, A. M, M. G, B. H, S.-E. U, V. W.-D., W. S.M., O. J.I., Linking function and structure with resensnet: Predicting retinal sensitivity from oct using deep learning, Ophthalmology. Retina 6 (6) (2022) 501–511. doi:10.1016/j.oret.2022.01.021.
URL https://pubmed.ncbi.nlm.nih.gov/35134543/

[115] J. Bridge, S. P. Harding, Y. Zheng, B. J., H. S.P., Z. Y., Daisy descriptors combined with deep learning to diagnose retinal disease from high resolution 2d oct images, Communications in Computer and Information Science 1065 CCIS (2020) 489–496. doi:10.1007/978-3-030-39343-4_42.
URL https://www.scopus.com/pages/publications/85079091090?origin=resultslist

[116] A. Alfahaid, T. Morris, T. Cootes, P. A. Keane, H. Khalid, N. Pontikos, P. Sergouniotis, K. Balaskas, A. A., M. T., C. T., K. P.A., K. H., P. N., S. P., B. K., A hybrid machine learning approach using lbp descriptor and pca for age-related macular degeneration classification in octa images, Communications in Computer and Information Science 1065 CCIS (2020) 231–241. doi:10.1007/978-3-030-39343-4_20.
URL https://www.scopus.com/pages/publications/85079092660?origin=resultslist

[117] B. Lv, S. Li, Y. Liu, W. Wang, H. Li, X. Zhang, Y. Sha, X. Yang, Y. Yang, Y. Wang, C. Zhang, Y. Wang, C. Lv, G. Xie, K. Wang, L. B, L. S, L. Y, W. W, L. H, Z. X, S. Y, Y. X, Y. Y, W. Y, Z. C, L. C, X. G, W. K, Development and validation of an explainable artificial intelligence framework for macular disease diagnosis based on optical coherence tomography images, Retina (Philadelphia, Pa.) 42 (3) (2022) 456–464. doi:10.1097/IAE.0000000000003325.
URL https://pubmed.ncbi.nlm.nih.gov/34723902/

[118] D. J. B. Saini, R. Sivakami, R. Venkatesh, C. Raghava, P. Sandeep Dwarkanath, T. M. K. Anwer, L. K. Smirani, S. H. Ahammad, U. Pamula, M. Amzad Hossain, A. N. Z. Rashed, S. D.J.B., A. T.M.K., S. L.K., A. S.H., A. H. M., R. A.N.Z., Convolution neural network model for predicting various lesion-based diseases in diabetic macula edema in optical coherence tomography images, Biomedical Signal Processing and Control 86 (2023). doi:10.1016/j.bspc.2023.105180.
URL https://www.scopus.com/pages/publications/85165713022?origin=resultslist

[119] H.-Y. Hsu, Y.-B. Chou, Y.-C. Jheng, Z.-K. Kao, H.-Y. Huang, H.-R. Chen, D.-K. Hwang, S.-J. Chen, S.-H. Chiou, Y.-T. Wu, H. HY, C. YB, J. YC, K. ZK, H. HY, C. HR, H. DK, C. SJ, C. SH, W. YT, H. H.-Y., C. Y.-B., J. Y.-C., K. Z.-K., H. H.-Y., C. H.-R., H. D.-K., C. S.-J., C. S.-H., W. Y.-T., Automatic segmentation of retinal fluid and photoreceptor layer from optical coherence tomography images of diabetic macular edema patients using deep learning and associations with visual acuity, Biomedicines 10 (6) (2022). doi:10.3390/biomedicines10061269.
URL https://pubmed.ncbi.nlm.nih.gov/35740291/

[120] M. T. Reza, F. Ahmed, S. Sharar, A. A. Rasel, M. T. Reza, F. Ahmed, S. Sharar, A. A. Rasel, R. M.T., A. F., S. S., R. A.A., Interpretable retinal disease classification from oct images using deep neural network and explainable ai, Proceedings of International Conference on Electronics, Communications and Information Technology, ICECIT 2021 (2021) 1–4doi:10.1109/ICECIT54077.2021.9641066.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9641066

[121] S. Manikandan, R. Raman, R. Rajalakshmi, S. Tamilselvi, R. J. Surya, M. S, R. R, R. R, S. RJ, R. Janani Surya, Deep learning-based detection of diabetic macular edema using optical coherence tomography and fundus images: A meta-analysis, Indian journal of ophthalmology 71 (5) (2023) 1783–1796. doi:10.4103/IJO.IJO_2614_22.
URL https://pubmed.ncbi.nlm.nih.gov/37203031/

[122] V. A. Dos Santos, L. Schmetterer, H. Stegmann, M. Pfister, A. Messner, G. Schmidinger, G. Garhofer, R. M. Werkmeister, D. S. VA, S. L, S. H, P. M, M. A, S. G, G. G, W. RM, D. S. V.A., W. R.M., Corneanet: fast segmentation of cornea oct scans of healthy and keratoconic eyes using deep learning, Biomedical optics express 10 (2) (2019) 622–641. doi:10.1364/BOE.10.000622.
URL https://pubmed.ncbi.nlm.nih.gov/30800504/

[123] F. Ricardi, J. Oakley, D. Russakoff, G. Boscia, P. Caselgrandi, F. Gelormini, A. Ghilardi, G. Pintore, T. Tibaldi, P. Marolo, F. Bandello, M. Reibaldi, E. Borrelli, R. F, O. J, R. D, B. G, C. P, G. F, G. A, P. G, T. T, M. P, B. F, R. M, B. E, Validation of a deep learning model for automatic detection and quantification of five oct critical retinal features associated with neovascular age-related macular degeneration, British Journal of Ophthalmology 108 (10) (2024) 1436–1442. doi:10.1136/bjo-2023-324647.
URL https://pubmed.ncbi.nlm.nih.gov/38485214/

[124] S. Gurumoorthy, N. Meena Lochana Nayaki, K. E. Vendhan, G. S., V. K.E., The role of artificial intelligence in monitoring glaucoma progression using optical coherence tomography, Indian Journal of Clinical and Experimental Ophthalmology 11 (4) (2025) 631–640. doi:10.18231/j.ijceo.11108.1761806059.
URL https://www.scopus.com/pages/publications/105026345646?origin=resultslist

[125] K. Alsaih, M. Z. Yusoff, T. B. Tang, I. Faye, F. Mériaudeau, K. Alsaih, M. Yusoff, T. Tang, I. Faye, F. Meriaudeau, Retinal fluids segmentation using volumetric deep neural networks on optical coherence tomography scans, Proceedings - 10th IEEE International Conference on Control System, Computing and Engineering, ICCSCE 2020 (2020) 68–72doi:10.1109/ICCSCE50387.2020.9204951.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9204951

[126] Z. Meng, Y. Chen, H. Li, Y. Zhang, X. Yao, Y. Meng, W. Shi, Y. Liang, Y. Hu, D. Liu, M. Xie, B. Yan, J. Luo, M. Z, C. Y, L. H, Z. Y, Y. X, M. Y, S. W, L. Y, H. Y, L. D, X. M, Y. B, L. J, Machine learning and optical coherence tomography-derived radiomics analysis to predict persistent diabetic macular edema in patients undergoing anti-vegf intravitreal therapy, Journal of translational medicine 22 (1) (2024) 358. doi:10.1186/s12967-024-05141-7.
URL https://pubmed.ncbi.nlm.nih.gov/38627718/

[127] H. Xie, J.-K. Wang, R. H. Kardon, M. K. Garvin, X. Wu, X. H., W. J.-K., K. R.H., G. M.K., W. X., Automated macular oct retinal surface segmentation in cases of severe glaucoma using deep learning, Progress in Biomedical Optics and Imaging - Proceedings of SPIE 12032 (2022). doi:10.1117/12.2611859.
URL https://www.scopus.com/pages/publications/85131935220?origin=resultslist

[128] R. Vasanthi, S. S. S. Sowmya, S. S. Madhura, D. Menaga, R. Vasanthi, S. Salai Swarna Sowmya, S. Sri Madhura, D. Menaga, Macula oct disease images classification using deep learning techniques, 2024 5th International Conference on Smart Electronics and Communication (ICOSEC) (2024) 1142–1149doi:10.1109/ICOSEC61587.2024.10722416.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10722416

[129] G. Muni Nagamani, E. Rayachoti, R. E., Deep learning network (dl-net) based classification and segmentation of multi-class retinal diseases using oct scans, Biomedical Signal Processing and Control 88 (2024). doi:10.1016/j.bspc.2023.105619.
URL https://www.scopus.com/pages/publications/85175456715?origin=resultslist

[130] Q. Li, S. Li, Z. He, H. Guan, R. Chen, Y. Xu, T. Wang, S. Qi, J. Mei, W. Wang, L. Q, L. S, H. Z, G. H, C. R, X. Y, W. T, Q. S, M. J, W. W, Deepretina: Layer segmentation of retina in oct images using deep learning, Translational vision science & technology 9 (2) (2020) 61. doi:10.1167/tvst.9.2.61.
URL https://pubmed.ncbi.nlm.nih.gov/33329940/

[131] H. K. Sharma, R. Choudhary, S. Kumar, T. Choudhury, S. H.K., C. R., K. S., C. T., Retinal optical coherence tomography classification using deep learning, Lecture Notes in Networks and Systems 490 (2023) 225–235. doi:10.1007/978-981-19-4052-1_24.
URL https://www.scopus.com/pages/publications/85138815782?origin=resultslist

[132] Q. Wu, B. Zhang, Y. Hu, B. Liu, D. Cao, D. Yang, Q. Peng, P. Zhong, X. Zeng, Y. Xiao, C. Li, Y. Fang, S. Feng, M. Huang, H. Cai, X. Yang, H. Yu, W. Q, Z. B, H. Y, L. B, C. D, Y. D, P. Q, Z. P, Z. X, X. Y, L. C, F. Y, F. S, H. M, C. H, Y. X, Y. H, Detection of morphologic patterns of diabetic macular edema using a deep learning approach based on optical coherence tomography images, Retina (Philadelphia, Pa.) 41 (5) (2021) 1110–1117. doi:10.1097/IAE.0000000000002992.
URL https://pubmed.ncbi.nlm.nih.gov/33031250/

[133] K. Karthik, M. Mahadevappa, K. K., M. M., Deep learning with adaptive convolutions for classification of retinal diseases via optical coherence tomography, Image and Vision Computing 146 (2024). doi:10.1016/j.imavis.2024.105044.
URL https://www.scopus.com/pages/publications/85192063542?origin=resultslist

[134] J.-A. Kim, H. Yoon, D. Lee, M. Kim, J. Choi, E. J. Lee, T.-W. Kim, K. JA, Y. H, L. D, K. M, C. J, L. EJ, K. TW, Development of a deep learning system to detect glaucoma using macular vertical optical coherence tomography scans of myopic eyes, Scientific reports 13 (1) (2023) 8040. doi:10.1038/s41598-023-34794-5.
URL https://pubmed.ncbi.nlm.nih.gov/37198215/

[135] E. Hassan, S. Elmougy, M. R. Ibraheem, M. S. Hossain, K. AlMutib, A. Ghoneim, S. A. AlQahtani, F. M. Talaat, H. E, E. S, I. MR, H. MS, A. K, G. A, A. SA, T. FM, I. M.R., H. M.S., A. S.A., T. F.M., Enhanced deep learning model for classification of retinal optical coherence tomography images, Sensors (Basel, Switzerland) 23 (12) (2023). doi:10.3390/s23125393.
URL https://pubmed.ncbi.nlm.nih.gov/37420558/

[136] J. Cao, K. You, K. Jin, L. Lou, Y. Wang, M. Chen, X. Pan, J. Shao, Z. Su, J. Wu, J. Ye, C. J, Y. K, J. K, L. L, W. Y, C. M, P. X, S. J, S. Z, W. J, Y. J, Prediction of response to anti-vascular endothelial growth factor treatment in diabetic macular oedema using an optical coherence tomography-based machine learning method, Acta ophthalmologica 99 (1) (2021) e19–e27. doi:10.1111/aos.14514.
URL https://pubmed.ncbi.nlm.nih.gov/32573116/

[137] Y. Sun, H. Zhang, X. Yao, S. Y, Z. H, Y. X, Automatic diagnosis of macular diseases from oct volume based on its two-dimensional feature map and convolutional neural network with attention mechanism, Journal of biomedical optics 25 (9) (2020). doi:10.1117/1.JBO.25.9.096004.
URL https://pubmed.ncbi.nlm.nih.gov/32940026/

[138] Z. Zhou, C. Niu, H. Yu, J. Zhao, Y. Wang, C. Dai, Z. Z., N. C., Y. H., Z. J., W. Y., D. C., Diagnosis of retinal diseases using the vision transformer model based on optical coherence tomography images, Proceedings of SPIE - The International Society for Optical Engineering 12601 (2023). doi:10.1117/12.2665918.
URL https://www.scopus.com/pages/publications/85159652005?origin=resultslist

[139] L. Mou, L. Liang, Z. Gao, X. Wang, M. L., L. L., G. Z., W. X., A multi-scale anomaly detection framework for retinal oct images based on the bayesian neural network, Biomedical Signal Processing and Control 75 (2022). doi:10.1016/j.bspc.2022.103619.
URL https://www.scopus.com/pages/publications/85125700276?origin=resultslist

[140] K. Hou, C. Bradley, P. Herbert, C. Johnson, M. Wall, P. Y. Ramulu, M. Unberath, J. Yohannan, H. K, B. C, H. P, J. C, W. M, R. PY, U. M, Y. J, Predicting visual field worsening with longitudinal oct data using a gated transformer network, Ophthalmology 130 (8) (2023) 854–862. doi:10.1016/j.ophtha.2023.03.019.
URL https://pubmed.ncbi.nlm.nih.gov/37003520/

[141] S. Wang, S. Feng, Z. Wang, Z. Ji, J. Liu, W. Chen, B. Fu, R. Liu, W. Chen, Y. Dai, C. Li, W. S, F. S, W. Z, J. Z, L. J, C. W, F. B, L. R, D. Y, L. C, Structural-prior guided and feature-enhanced transformer with masked image modeling pretraining for retinal layers and fluid segmentation in macular edema oct images, Biomedical optics express 16 (12) (2025) 5096–5117. doi:10.1364/BOE.574769.
URL https://pubmed.ncbi.nlm.nih.gov/41394497/

[142] N. Krishna, N. Kenchappa, K. N., K. N., Optimizing glaucoma diagnosis using fundus and optical coherence tomography image fusion based on multi-modal convolutional neural network approach, International Journal of Electrical and Computer Engineering 14 (4) (2024) 4005–4017. doi:10.11591/ijece.v14i4.pp4005-4017.
URL https://www.scopus.com/pages/publications/85195097843?origin=resultslist

[143] S. Sotoudeh-Paima, A. Jodeiri, F. Hajizadeh, H. Soltanian-Zadeh, S.-P. S, J. A, H. F, S.-Z. H, Multi-scale convolutional neural network for automated amd classification using retinal oct images, Computers in biology and medicine 144 (2022) 105368. doi:10.1016/j.compbiomed.2022.105368.
URL https://pubmed.ncbi.nlm.nih.gov/35259614/

[144] M. M. Hasan, J. Phu, H. Wang, A. Sowmya, E. Meijering, M. Kalloniatis, H. MM, P. J, W. H, S. A, M. E, K. M, H. M.M., Predicting visual field global and local parameters from oct measurements using explainable machine learning, Scientific reports 15 (1) (2025) 5685. doi:10.1038/s41598-025-89557-1.
URL https://pubmed.ncbi.nlm.nih.gov/39956834/

[145] M. Ratanapunperm, T. Khemniwat, S. Deepaisarn, P. Aimmanee, R. M., K. T., D. S., A. P., Automated detection of unhealthy retinal optical coherence tomography images using geometrical features for machine learning modeling, Lecture Notes in Networks and Systems 1510 LNNS (2026) 369–380. doi:10.1007/978-981-96-9275-0_32.
URL https://www.scopus.com/pages/publications/105028303707?origin=resultslist

[146] A. Rodríguez-Miguel, C. Arruabarrena, G. Allendes, M. Olivera, J. Zarranz-Ventura, M. A. Teus, R.-M. A, A. C, A. G, O. M, Z.-V. J, T. MA, T. M.A., Hybrid deep learning models for the screening of diabetic macular edema in optical coherence tomography volumes, Scientific reports 14 (1) (2024) 17633. doi:10.1038/s41598-024-68489-2.
URL https://pubmed.ncbi.nlm.nih.gov/39085461/

[147] D. Nagasato, H. Tabuchi, H. Masumoto, H. Enno, N. Ishitobi, M. Kameoka, M. Niki, Y. Mitamura, N. D, T. H, M. H, E. H, I. N, K. M, N. M, M. Y, Automated detection of a nonperfusion area caused by retinal vein occlusion in optical coherence tomography angiography images using deep learning, PloS one 14 (11) (2019) e0223965. doi:10.1371/journal.pone.0223965.
URL https://pubmed.ncbi.nlm.nih.gov/31697697/

[148] A. Kumar, M. V. Ajit, A. Kumar, M. V. Ajit, K. A., A. M.V., Diagnosis of retinal disorders from optical coherence tomography images using cnn, 2025 3rd International Conference on Data Science and Information System (ICDSIS) (2025) 1–8doi:10.1109/ICDSIS65355.2025.11071143.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11071143

[149] H. J. Kwon, J. Heo, S. H. Park, S. W. Park, I. Byon, K. HJ, H. J, P. SH, P. SW, B. I, K. H.J., P. S.H., P. S.W., Accuracy of generative deep learning model for macular anatomy prediction from optical coherence tomography images in macular hole surgery, Scientific reports 14 (1) (2024) 6913. doi:10.1038/s41598-024-57562-5.
URL https://pubmed.ncbi.nlm.nih.gov/38519532/

[150] L. Jiang, J. Hu, Z. Wang, G. Yuan, C. Huang, Z. Xiong, M. Zhang, W. Yang, Y. Tong, J. L, H. J, W. Z, Y. G, H. C, X. Z, Z. M, Y. W, T. Y, Transdualsegnet: transformer dual segment network for retinal vasculature segmentation in oct, Quantitative imaging in medicine and surgery 15 (10) (2025) 9338–9350. doi:10.21037/qims-2024-2842.
URL https://pubmed.ncbi.nlm.nih.gov/41081166/

[151] G. Pinedo-Diaz, B. Lorenz, S. H. Künzel, S. Thiele, S. Ortega-Cisneros, E. B. Corrochano, F. G. Holz, A. Effland, P.-D. G, L. B, K. SH, T. S, O.-C. S, C. EB, H. FG, E. A, K. S.H., C. E.B., H. F.G., Deep learning-based sd-oct layer segmentation quantifies outer retina changes in patients with biallelic rpe65 mutations undergoing gene therapy, Investigative ophthalmology & visual science 66 (1) (2025) 5. doi:10.1167/iovs.66.1.5.
URL https://pubmed.ncbi.nlm.nih.gov/39745677/

[152] A. Szeskin, R. Yehuda, O. Shmueli, J. Levy, L. Joskowicz, S. A, Y. R, S. O, L. J, J. L, A column-based deep learning method for the detection and quantification of atrophy associated with amd in oct scans, Medical image analysis 72 (2021) 102130. doi:10.1016/j.media.2021.102130.
URL https://pubmed.ncbi.nlm.nih.gov/34198041/

[153] Puneet, R. Kumar, M. Gupta, K. R, G. M, Optical coherence tomography image based eye disease detection using deep convolutional neural network, Health information science and systems 10 (1) (2022) 13. doi:10.1007/s13755-022-00182-y.
URL https://pubmed.ncbi.nlm.nih.gov/35756852/

[154] Z. Xu, Y. Yang, H. Chen, R. Han, X. Han, J. Zhao, W. Yu, Z. Yang, Y. Chen, X. Z, Y. Y, C. H, H. R, H. X, Z. J, Y. W, Y. Z, C. Y, Enhancing pathological myopia diagnosis: a bimodal artificial intelligence approach integrating fundus and optical coherence tomography imaging for precise atrophy, traction and neovascularisation grading, British Journal of Ophthalmology 109 (10) (2025) 1179–1186. doi:10.1136/bjo-2024-326252.
URL https://pubmed.ncbi.nlm.nih.gov/40393796/

[155] D. Liu, X. Xu, J. Zhao, N. Tian, Y. An, X. Liu, J. Yu, X. Yang, T. Niu, L. D., X. X., Z. J., T. N., A. Y., L. X., Y. J., Y. X., N. T., Analysis of early screening results for high-risk populations of retinal diseases based on artificial intelligence optical coherence tomography technology, Chinese Journal of Ocular Fundus Diseases 42 (2) (2026) 127–133. doi:10.3760/cma.j.cn511434-20251009-00423.
URL https://www.scopus.com/pages/publications/105031493897?origin=resultslist

[156] A. A. Slamaa, S. A.A., A deep learning-based eye disease diagnosis using oct imaging and se-enhanced cnns, Applied Mathematics and Information Sciences 19 (5) (2025) 1153–1165. doi:10.18576/amis/190515.
URL https://www.scopus.com/pages/publications/105022130649?origin=resultslist

[157] Y. Z. Tun, P. Aimmanee, Y. Z. Tun, P. Aimmanee, T. Y.Z., A. P., Studies of plausibility using deep learning techniques to automatically detect retinal neovascularization in enface optical coherence tomography angiography images, 2025 13th International Conference on Bioinformatics and Computational Biology, ICBCB 2025 (2025) 35–41doi:10.1109/ICBCB64873.2025.11198054.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11198054

[158] S. Borkovkina, A. Camino, W. Janpongsri, M. V. Sarunic, Y. Jian, B. S, C. A, J. W, S. MV, J. Y, S. M.V., Real-time retinal layer segmentation of oct volumes with gpu accelerated inferencing using a compressed, low-latency neural network, Biomedical optics express 11 (7) (2020) 3968–3984. doi:10.1364/BOE.395279.
URL https://pubmed.ncbi.nlm.nih.gov/33014579/

[159] H. Xu, R. Lv, X. H., L. R., Adcformer: A hybrid model of adaptive depth-wise convolution and transformer for retinal macular edema segmentation in oct images, Biomedical Signal Processing and Control 108 (2025). doi:10.1016/j.bspc.2025.107949.
URL https://www.scopus.com/pages/publications/105003820322?origin=resultslist

[160] K. A. Thakoor, X. Li, E. Tsamis, Z. Z. Zemborain, C. G. De Moraes, P. Sajda, D. C. Hood, T. KA, L. X, T. E, Z. ZZ, D. M. CG, S. P, H. DC, T. K.A., Z. Z.Z., D. M. C.G., H. D.C., Strategies to improve convolutional neural network generalizability and reference standards for glaucoma detection from oct scans, Translational vision science & technology 10 (4) (2021) 16. doi:10.1167/tvst.10.4.16.
URL https://pubmed.ncbi.nlm.nih.gov/34003990/

[161] B. S. Brasil, A. R. de Alexandria, G. de Freitas Guimarães, B. S. Brasil, A. R. De Alexandria, G. De Freitas Guimaraes, B. B.S., D. A. A.R., D. F. G. G., Artificial intelligence applied to the classification of retinal diseases in optical coherence tomography images, 2022 5th International Conference on Vocational Education and Electrical Engineering: The Future of Electrical Engineering, Informatics, and Educational Technology Through the Freedom of Study in the Post-Pandemic Era, ICVEE 2022 - Proceeding (2022) 78–83doi:10.1109/ICVEE57061.2022.9930121.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9930121

[162] S. R, A. Sharma, R. Srivats, K. V, S. K. Shanmugam, R. Suganya, A. Sharma, R. Srivats, V. Kalyanasundaram, S. K. Shanmugam, S. A., S. S.K., Interpretable deep learning for automated retinal damage detection in optical coherence tomography, Proceedings - 2024 IEEE International Conference on Big Data, BigData 2024 (2024) 6101–6107doi:10.1109/BigData62323.2024.10825636.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10825636

[163] A. Alqudah, A. M. Alqudah, M. Altantawi, A. A., A. A.M., A. M., Artificial intelligence hybrid system for enhancing retinal diseases classification using automated deep features extracted from oct images, International Journal of Intelligent Systems and Applications in Engineering 9 (3) (2021) 91–100. doi:10.18201/IJISAE.2021.236.
URL https://www.scopus.com/pages/publications/85118194383?origin=resultslist

[164] D. Rastogi, R. P. Padhy, P. K. Sa, D. Rastogi, R. P. Padhy, P. K. Sa, R. D., P. R.P., S. P.K., Detection of retinal disorders in optical coherence tomography using deep learning, 2019 10th International Conference on Computing, Communication and Networking Technologies, ICCCNT 2019 (2019) 1–7doi:10.1109/ICCCNT45670.2019.8944406.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8944406

[165] A. A. Jammal, A. C. Thompson, N. G. Ogata, E. B. Mariottoni, C. N. Urata, V. P. Costa, F. A. Medeiros, J. AA, T. AC, O. NG, M. EB, U. CN, C. VP, M. FA, J. A.A., T. A.C., O. N.G., M. E.B., U. C.N., C. V.P., M. F.A., Detecting retinal nerve fibre layer segmentation errors on spectral domain-optical coherence tomography with a deep learning algorithm, Scientific reports 9 (1) (2019) 9836. doi:10.1038/s41598-019-46294-6.
URL https://pubmed.ncbi.nlm.nih.gov/31285505/

[166] S. F. Rabbi, M. A. Mamun, M. F. Faruk, S. M. M. Hasan, A. Y. Srizon, S. F. Rabbi, M. A. Mamun, M. F. Faruk, S. Mahedy Hasan, A. Y. Srizon, R. S.F., M. M.A., F. M.F., S. A.Y., A multi-branch and attention based cnn architecture for the classification of retinal diseases from oct images, 2023 International Conference on Information and Communication Technology for Sustainable Development, ICICT4SD 2023 - Proceedings (2023) 36–40doi:10.1109/ICICT4SD59951.2023.10303384.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10303384

[167] Y. George, B. Antony, H. Ishikawa, G. Wollstein, J. Schuman, R. Garnavi, G. Y., A. B., I. H., W. G., S. J., G. R., 3d-cnn for glaucoma detection using optical coherence tomography, Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 11855 LNCS (2019) 52–59. doi:10.1007/978-3-030-32956-3_7.
URL https://www.scopus.com/pages/publications/85075660036?origin=resultslist

[168] S. Sun, A. Ha, Y. K. Kim, B. W. Yoo, H. C. Kim, K. H. Park, S. S, H. A, K. YK, Y. BW, K. HC, P. KH, Dual-input convolutional neural network for glaucoma diagnosis using spectral-domain optical coherence tomography, The British journal of ophthalmology 105 (11) (2021) 1555–1560. doi:10.1136/bjophthalmol-2020-316274.
URL https://pubmed.ncbi.nlm.nih.gov/32920530/

[169] Z. Li, X. Zhao, A. Yin, C. Guo, L. Chen, L. Z., Z. X., Y. A., G. C., C. L., Research on deep learning-based intelligent diagnosis algorithms for oct medical images of macular edema, Communications in Computer and Information Science 1142 CCIS (2019) 615–623. doi:10.1007/978-3-030-36808-1_67.
URL https://www.scopus.com/pages/publications/85089616797?origin=resultslist

[170] Y. Wang, C. Galang, W. R. Freeman, T. Q. Nguyen, C. An, Y. Wang, C. Galang, W. R. Freeman, T. Q. Nguyen, C. An, W. Y, G. C, F. WR, N. TQ, A. C, F. W.R., N. T.Q., Joint motion correction and 3d segmentation with graph-assisted neural networks for retinal oct, 2022 IEEE International Conference on Image Processing (ICIP) 2022 (2022) 766–770. doi:10.1109/ICIP46576.2022.9898072.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9898072

[171] E. B. Mariottoni, S. Datta, L. S. Shigueoka, A. A. Jammal, I. M. Tavares, R. Henao, L. Carin, F. A. Medeiros, M. EB, D. S, S. LS, J. AA, T. IM, H. R, C. L, M. FA, M. E.B., S. L.S., J. A.A., T. I.M., M. F.A., Deep learning-assisted detection of glaucoma progression in spectral-domain oct, Ophthalmology. Glaucoma 6 (3) (2023) 228–238. doi:10.1016/j.ogla.2022.11.004.
URL https://pubmed.ncbi.nlm.nih.gov/36410708/

[172] G. C. Y. Chan, R. Kamble, H. Müller, S. A. A. Shah, T. B. Tang, F. Mériaudeau, G. C. Y. Chan, R. Kamble, H. Muller, S. A. A. Shah, T. B. Tang, F. Meriaudeau, C. GCY, K. R, M. H, S. SAA, T. TB, M. F, C. G.C.Y., S. S.A.A., Fusing results of several deep learning architectures for automatic classification of normal and diabetic macular edema in optical coherence tomography, Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference 2018 (2018) 670–673. doi:10.1109/EMBC.2018.8512371.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8512371

[173] K. Bitton, P. Zéboulon, W. Ghazal, M. Rizk, S. Elahi, D. Gatinel, B. K, Z. P, G. W, R. M, E. S, G. D, Deep learning model for the detection of corneal edema before descemet membrane endothelial keratoplasty on optical coherence tomography images, Translational vision science & technology 11 (12) (2022) 19. doi:10.1167/tvst.11.12.19.
URL https://pubmed.ncbi.nlm.nih.gov/36583911/

[174] E. K. Akpek, G. Li, A. J. Aldave, G. Amescua, K. A. Colby, M. S. Cortina, J. de la Cruz, J.-M. A. Parel, T. Schmiedel, A. EK, L. G, A. AJ, A. G, C. KA, C. MS, de la Cruz J, P. JA, S. T, A. E.K., A. A.J., C. K.A., C. M.S., P. J.-M.A., Artificial intelligence aided analysis of anterior segment optical coherence tomography imaging to monitor the device-cornea joint after synthetic cornea implantation, Translational vision science & technology 14 (6) (2025) 1. doi:10.1167/tvst.14.6.1.
URL https://pubmed.ncbi.nlm.nih.gov/40455039/

[175] H. Kumar, Y. Bagdasarova, S. Song, D. G. Hickey, A. C. Cohn, M. Okada, R. P. Finger, J. H. Terheyden, R. E. Hogg, P.-H. Gabrielle, L. Arnould, M. Jannaud, X. Hadoux, P. van Wijngaarden, C. J. Abbott, L. A. B. Hodgson, R. Schwartz, A. Tufail, E. Y. Chew, C. S. Lee, E. L. Fletcher, M. Bahlo, B. R. E. Ansell, A. Pébay, R. H. Guymer, A. Y. Lee, Z. Wu, K. H, B. Y, S. S, H. DG, C. AC, O. M, F. RP, T. JH, H. RE, G. PH, A. L, J. M, H. X, van Wijngaarden P, A. CJ, H. LAB, S. R, T. A, C. EY, L. CS, F. EL, B. M, A. BRE, P. A, G. RH, L. AY, W. Z, Deep learning-based detection of reticular pseudodrusen in age-related macular degeneration on optical coherence tomography, medRxiv : the preprint server for health sciences (2024). doi:10.1101/2024.09.11.24312817.
URL https://pubmed.ncbi.nlm.nih.gov/39314940/

[176] Z. Chen, H. Zhang, E. F. Linton, B. A. Johnson, Y. J. Choi, M. J. Kupersmith, M. Sonka, M. K. Garvin, R. H. Kardon, J.-K. Wang, C. Z, Z. H, L. EF, J. BA, C. YJ, K. MJ, S. M, G. MK, K. RH, W. JK, Y. J. Choi, L. E.F., J. B.A., C. Y.J.A.E., K. M.J., G. M.K., K. R.H., W. J.-K., Hybrid deep learning and optimal graph search method for optical coherence tomography layer segmentation in diseases affecting the optic nerve, Biomedical optics express 15 (6) (2024) 3681–3698. doi:10.1364/BOE.516045.
URL https://pubmed.ncbi.nlm.nih.gov/38867777/

[177] S. Touma, B. A. Hammou, F. Antaki, M. C. Boucher, R. Duval, T. S, H. BA, A. F, B. MC, D. R, Comparing code-free deep learning models to expert-designed models for detecting retinal diseases from optical coherence tomography, International journal of retina and vitreous 10 (1) (2024) 37. doi:10.1186/s40942-024-00555-3.
URL https://pubmed.ncbi.nlm.nih.gov/38671486/

[178] C. Zheng, X. Xie, L. Huang, B. Chen, J. Yang, J. Lu, T. Qiao, Z. Fan, M. Zhang, Z. C, X. X, H. L, C. B, Y. J, L. J, Q. T, F. Z, Z. M, Detecting glaucoma based on spectral domain optical coherence tomography imaging of peripapillary retinal nerve fiber layer: a comparison study between hand-crafted features and deep learning model, Graefe's Archive for Clinical and Experimental Ophthalmology 258 (3) (2020) 577–585. doi:10.1007/s00417-019-04543-4.
URL https://pubmed.ncbi.nlm.nih.gov/31811363/

[179] S. Moon, J. H. Lee, H. Choi, S. Y. Lee, J. Lee, M. S, L. JH, C. H, L. SY, L. J, L. J.H., L. S.Y., Deep learning approaches to predict 10-2 visual field from wide-field swept-source optical coherence tomography en face images in glaucoma, Scientific reports 12 (1) (2022) 21041. doi:10.1038/s41598-022-25660-x.
URL https://pubmed.ncbi.nlm.nih.gov/36471039/

[180] S. A. Kamran, S. Saha, A. S. Sabbir, A. Tavakkoli, K. S.A., S. S., S. A.S., T. A., A comprehensive set of novel residual blocks for deep learning architectures for diagnosis of retinal diseases from optical coherence tomography images, Advances in Intelligent Systems and Computing 1232 (2021) 25–48. doi:10.1007/978-981-15-6759-9_2.
URL https://www.scopus.com/pages/publications/85091580584?origin=resultslist

[181] F. A. Medeiros, A. A. Jammal, A. C. Thompson, M. FA, J. AA, T. AC, M. F.A., J. A.A., T. A.C., From machine to machine: An oct-trained deep learning algorithm for objective quantification of glaucomatous damage in fundus photographs, Ophthalmology 126 (4) (2019) 513–521. doi:10.1016/j.ophtha.2018.12.033.
URL https://pubmed.ncbi.nlm.nih.gov/30578810/

[182] P. Riazi Esfahani, A. J. Reddy, N. Nawathey, M. S. Ghauri, M. Min, H. Wagh, N. Tak, R. Patel, R. E. P, R. AJ, N. N, G. MS, M. M, W. H, T. N, P. R, Deep learning classification of drusen, choroidal neovascularization, and diabetic macular edema in optical coherence tomography (oct) images, Cureus 15 (7) (2023) e41615. doi:10.7759/cureus.41615.
URL https://pubmed.ncbi.nlm.nih.gov/37565126/

[183] O. Leingang, S. Riedl, J. Mai, G. S. Reiter, G. Faustmann, P. Fuchs, H. P. N. Scholl, S. Sivaprasad, D. Rueckert, A. Lotery, U. Schmidt-Erfurth, H. Bogunović, L. O, R. S, M. J, R. GS, F. G, F. P, S. HPN, S. S, R. D, L. A, S.-E. U, B. H, R. G.S., S. H.P.N., Automated deep learning-based amd detection and staging in real-world oct datasets (pinnacle study report 5), Scientific reports 13 (1) (2023) 19545. doi:10.1038/s41598-023-46626-7.
URL https://pubmed.ncbi.nlm.nih.gov/37945665/

[184] C. L. Hung, K. H. Lin, Y. K. Lee, C. Lin, Y. T. Tsai, C.-L. Hung, K.-H. Lin, Y.-K. Lee, C. Lin, Y.-T. Tsai, H. C.-L., L. K.-H., L. Y.-K., L. C., T. Y.-T., Macular holes detection using deep learning on optical coherence tomography images, Proceedings - 2023 2023 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2023 (2023) 3376–3381doi:10.1109/BIBM58861.2023.10385298.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10385298

[185] G. Kalra, H. Cetin, J. Whitney, S. Yordi, Y. Cakir, C. McConville, V. Whitmore, M. Bonnay, J. L. Reese, S. K. Srivastava, J. P. Ehlers, K. G, C. H, W. J, Y. S, C. Y, M. C, W. V, B. M, R. JL, S. SK, E. JP, R. J.L., S. S.K., E. J.P., Automated identification and segmentation of ellipsoid zone at-risk using deep learning on sd-oct for predicting progression in dry amd, Diagnostics (Basel, Switzerland) 13 (6) (2023). doi:10.3390/diagnostics13061178.
URL https://pubmed.ncbi.nlm.nih.gov/36980486/

[186] P. Kaushik, S. Choudhary, P. Kaushik, S. Choudhary, K. P., C. S., Automated retinal disease detection: Vgg16-based deep learning approach for oct image classification, 2025 Fourth International Conference on Power, Control and Computing Technologies (ICPC2T) (2025) 1–5doi:10.1109/ICPC2T63847.2025.10958598.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10958598

[187] A. Pereira, J. D. Oakley, S. K. Sodhi, D. B. Russakoff, N. Choudhry, P. A, O. JD, S. SK, R. DB, C. N, O. J.D., S. S.K., R. D.B., Proof-of-concept analysis of a deep learning model to conduct automated segmentation of oct images for macular hole volume, Ophthalmic surgery, lasers & imaging retina 53 (4) (2022) 208–214. doi:10.3928/23258160-20220315-02.
URL https://pubmed.ncbi.nlm.nih.gov/35417293/

[188] X. Liang, S. Luo, Z. Liu, Y. Liu, S. Luo, K. Zhang, L. Li, L. X, L. S, L. Z, L. Y, Z. K, L. L, Unsupervised machine learning analysis of optical coherence tomography radiomics features for predicting treatment outcomes in diabetic macular edema, Scientific reports 15 (1) (2025) 13389. doi:10.1038/s41598-025-96988-3.
URL https://pubmed.ncbi.nlm.nih.gov/40251316/

[189] X. Ye, K. Gao, S. He, X. Zhong, Y. Shen, Y. Wang, H. Shao, L. Shen, Y. X, G. K, H. S, Z. X, S. Y, W. Y, S. H, S. L, Artificial intelligence-based quantification of central macular fluid volume and va prediction for diabetic macular edema using oct images, Ophthalmology and therapy 12 (5) (2023) 2441–2452. doi:10.1007/s40123-023-00746-5.
URL https://pubmed.ncbi.nlm.nih.gov/37318706/

[190] ç. Aykat, S. Senan, A. Ş., S. S., Using machine learning to detect different eye diseases from oct images, International Journal of Computational and Experimental Science and Engineering 9 (2) (2023) 62–67. doi:10.22399/ijcesen.1297655.
URL https://www.scopus.com/pages/publications/85187622127?origin=resultslist

[191] G. Atteia, N. Abdel Samee, E.-S. M. El-Kenawy, A. Ibrahim, A. G., A. S. N., E.-K. E.-S.M., I. A., Cnn-hyperparameter optimization for diabetic maculopathy diagnosis in optical coherence tomography and fundus retinography, Mathematics 10 (18) (2022). doi:10.3390/math10183274.
URL https://www.scopus.com/pages/publications/85138632689?origin=resultslist

[192] Y. Huang, J. Hu, Y. Huang, J. Hu, H. Y., H. J., Residual neural network based classification of macular edema in oct, 2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI) 2019-November (2019) 736–743. doi:10.1109/ICTAI.2019.00107.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8995439

[193] C.-W. Wu, H.-L. Shen, C.-J. Lu, S.-H. Chen, H.-Y. Chen, W. CW, S. HL, L. CJ, C. SH, C. HY, W. C.-W., S. H.-L., L. C.-J., C. S.-H., C. H.-Y., Comparison of different machine learning classifiers for glaucoma diagnosis based on spectralis oct, Diagnostics (Basel, Switzerland) 11 (9) (2021). doi:10.3390/diagnostics11091718.
URL https://pubmed.ncbi.nlm.nih.gov/34574059/

[194] M. Al-Zubaidy, A. Stankiewicz, M. Anderson, J. Reed, V. Corona, R. Pope, B. Obara, M. S. Habib, D. H. Steel, A.-Z. M., S. A., A. M., R. J., C. V., P. R., O. B., H. M.S., S. D.H., A scoping review of the use of artificial intelligence models in automated oct analysis and prediction of treatment outcomes in diabetic macular oedema, Informatics in Medicine Unlocked 57 (2025). doi:10.1016/j.imu.2025.101676.
URL https://www.scopus.com/pages/publications/105011972572?origin=resultslist

[195] M. R. Ahmed, M. S. Shehata, P. Lasserre, A. MR, S. MS, L. P, A. M.R., S. M.S., Integrating lightweight convolutional neural network with entropy-informed channel attention and adaptive spatial attention for oct-based retinal disease classification, Computers in biology and medicine 190 (2025) 110071. doi:10.1016/j.compbiomed.2025.110071.
URL https://pubmed.ncbi.nlm.nih.gov/40174502/

[196] A. Camino, Z. Wang, J. Wang, M. E. Pennesi, P. Yang, D. Huang, D. Li, Y. Jia, C. A, W. Z, W. J, P. ME, Y. P, H. D, L. D, J. Y, P. M.E., Deep learning for the segmentation of preserved photoreceptors on en face optical coherence tomography in two inherited retinal diseases, Biomedical optics express 9 (7) (2018) 3092–3105. doi:10.1364/BOE.9.003092.
URL https://pubmed.ncbi.nlm.nih.gov/29984085/

[197] S. Vahdati, D. B. Nagaraj, M. Bryan, S. Moazemi, S. Gründer-Fahrer, M. Martin, V. S., N. D.B., B. M., M. S., G.-F. S., M. M., Utilizing transformers on oct imagery and metadata for treatment response prediction in macular edema patients, Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 13886 LNCS (2023) 3–15. doi:10.1007/978-3-031-31438-4_1.
URL https://www.scopus.com/pages/publications/85161389543?origin=resultslist

[198] K. Thakoor, D. Bordbar, J. Yao, O. Moussa, R. Chen, P. Sajda, K. Thakoor, D. Bordbar, J. Yao, O. Moussa, R. Chen, P. Sajda, T. K., B. D., Y. J., M. O., C. R., S. P., Hybrid 3d-2d deep learning for detection of neovascularage-related macular degeneration using optical coherence tomography b-scans and angiography volumes, 2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI) 2021-April (2021) 1600–1604. doi:10.1109/ISBI48211.2021.9434111.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9434111

[199] Z. Wu, L. Ji, W. Tang, S. Zhou, Q. Yan, X. Chen, J. Zhu, J. Ma, Y. Zhao, W. He, W. Z, J. L, T. W, Z. S, Y. Q, C. X, Z. J, M. J, Z. Y, H. W, Prediction of coronary artery disease using retinal optical coherence tomography angiography imaging and electronic health records: a multimodal machine learning approach, BMC cardiovascular disorders 26 (1) (2026). doi:10.1186/s12872-026-05660-9.
URL https://pubmed.ncbi.nlm.nih.gov/41857509/

[200] M. Christopher, C. Bowd, A. Belghith, M. H. Goldbaum, R. N. Weinreb, M. A. Fazio, C. A. Girkin, J. M. Liebmann, L. M. Zangwill, C. M, B. C, B. A, G. MH, W. RN, F. MA, G. CA, L. JM, Z. LM, Deep learning approaches predict glaucomatous visual field damage from oct optic nerve head en face images and retinal nerve fiber layer thickness maps, Ophthalmology 127 (3) (2020) 346–356. doi:10.1016/j.ophtha.2019.09.036.
URL https://pubmed.ncbi.nlm.nih.gov/31718841/

[201] A. Beuse, D. A. Wenzel, M. S. Spitzer, K. U. Bartz-Schmidt, M. Schultheiss, S. Poli, C. Grohmann, B. A, W. DA, S. MS, B.-S. KU, S. M, P. S, G. C, W. D.A., S. M.S., B.-S. K.U., Automated detection of central retinal artery occlusion using oct imaging via explainable deep learning, Ophthalmology science 5 (2) (2025) 100630. doi:10.1016/j.xops.2024.100630.
URL https://pubmed.ncbi.nlm.nih.gov/39669299/

[202] F. Hosseini, F. Asadi, R. Rabiei, F. Kiani, R. E. Harari, H. F, A. F, R. R, K. F, H. RE, H. R.E., Applications of artificial intelligence in diagnosis of uncommon cystoid macular edema using optical coherence tomography imaging: A systematic review, Survey of ophthalmology 69 (6) (2024) 937–944. doi:10.1016/j.survophthal.2024.06.005.
URL https://pubmed.ncbi.nlm.nih.gov/38942125/

[203] M. M. Hasan, J. Phu, H. Wang, A. Sowmya, M. Kalloniatis, E. Meijering, H. MM, P. J, W. H, S. A, K. M, M. E, Oct-based diagnosis of glaucoma and glaucoma stages using explainable machine learning, Scientific reports 15 (1) (2025) 3592. doi:10.1038/s41598-025-87219-w.
URL https://pubmed.ncbi.nlm.nih.gov/39875492/

[204] K. Alsaih, M. Z. Yusoff, T. B. Tang, I. Faye, F. Mériaudeau, Y. MZ, T. TB, Deep learning architectures analysis for age-related macular degeneration segmentation on optical coherence tomography scans, Computer methods and programs in biomedicine 195 (2020) 105566. doi:10.1016/j.cmpb.2020.105566.
URL https://pubmed.ncbi.nlm.nih.gov/32504911/

[205] D. Bizios, A. Heijl, J. L. Hougaard, B. Bengtsson, B. D, H. A, H. JL, B. B, H. J.L., Machine learning classifiers for glaucoma diagnosis based on classification of retinal nerve fibre layer thickness parameters measured by stratus oct, Acta ophthalmologica 88 (1) (2010) 44–52. doi:10.1111/j.1755-3768.2009.01784.x.
URL https://pubmed.ncbi.nlm.nih.gov/20064122/

[206] X. Tan, X. Chen, Q. Meng, F. Shi, D. Xiang, Z. Chen, L. Pan, W. Zhu, T. X, C. X, M. Q, S. F, X. D, C. Z, P. L, Z. W, Oct(2)former: A retinal oct-angiography vessel segmentation transformer, Computer methods and programs in biomedicine 233 (2023) 107454. doi:10.1016/j.cmpb.2023.107454.
URL https://pubmed.ncbi.nlm.nih.gov/36921468/

[207] V. K. Chintamaneni, S. K. Debbadi, S. S. Kandula, R. S. Gargees, V. K. Chintamaneni, S. K. Debbadi, S. S. Kandula, R. S. Gargees, C. V.K., D. S.K., K. S.S., G. R.S., Hybrid deep learning/machine learning model for retinal diseases classifications using oct images, 2024 2nd International Conference on Artificial Intelligence, Blockchain, and Internet of Things, AIBThings 2024 - Proceedings (2024) 1–5doi:10.1109/AIBThings63359.2024.10863314.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10863314

[208] Z. Tang, X. Wang, A. R. Ran, D. Yang, A. Ling, J. C. Yam, X. Zhang, S. K. H. Szeto, J. Chan, C. Y. K. Wong, V. W. K. Hui, C. K. M. Chan, T. Y. Wong, C.-Y. Cheng, C. Sabanayagam, Y. C. Tham, G. Liew, G. Anantharaman, R. Raman, Y. Cai, H. Che, L. Luo, Q. Liu, Y. L. Wong, A. K. Y. Ngai, V. L. Yuen, N. Kei, T. Y. Y. Lai, H. Chen, C. C. Tham, P.-A. Heng, C. Y. Cheung, T. Z, W. X, R. AR, Y. D, L. A, Y. JC, Z. X, S. SKH, C. J, W. CYK, H. VWK, C. CKM, W. TY, C. CY, S. C, T. YC, L. G, A. G, R. R, C. Y, C. H, L. L, L. Q, W. YL, N. AKY, Y. VL, K. N, L. TYY, C. H, T. CC, H. PA, C. CY, R. A.R., Y. J.C., S. S.K.H., W. C.Y.K., H. V.W.K., C. C.K.M., W. T.Y., C. C.-Y., T. Y.C., W. Y.L., N. A.K.Y., Y. V.L., L. T.Y.Y., T. C.C., H. P.-A., C. C.Y., Deep learning-based image quality assessment for optical coherence tomography macular scans: a multicentre study, British Journal of Ophthalmology 108 (11) (2024) 1555–1563. doi:10.1136/bjo-2023-323871.
URL https://pubmed.ncbi.nlm.nih.gov/39033014/

[209] G. An, K. Omodaka, K. Hashimoto, S. Tsuda, Y. Shiga, N. Takada, T. Kikawa, H. Yokota, M. Akiba, T. Nakazawa, A. G, O. K, H. K, T. S, S. Y, T. N, K. T, Y. H, A. M, N. T, Glaucoma diagnosis with machine learning based on optical coherence tomography and color fundus images, Journal of healthcare engineering 2019 (2019) 4061313. doi:10.1155/2019/4061313.
URL https://pubmed.ncbi.nlm.nih.gov/30911364/

[210] H.-Y. Li, D.-X. Wang, L. Dong, W.-B. Wei, L. HY, W. DX, D. L, W. WB, L. H.-Y., W. D.-X., W. W.-B., Deep learning algorithms for detection of diabetic macular edema in oct images: A systematic review and meta-analysis, European journal of ophthalmology 33 (1) (2023) 278–290. doi:10.1177/11206721221094786.
URL https://pubmed.ncbi.nlm.nih.gov/35473414/

[211] D. Wang, K. Huang, Q. Chen, D. Wang, K. Huang, Q. Chen, W. D., H. K., C. Q., Progressive neural networks for continuous classification of retinal optical coherence tomography images, 2023 Eleventh International Conference on Advanced Cloud and Big Data (CBD) (2023) 156–161doi:10.1109/CBD63341.2023.00036.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10516494

[212] B. Hassan, S. Qin, R. Ahmed, T. Hassan, A. H. Taguri, S. Hashmi, N. Werghi, H. B, Q. S, A. R, H. T, T. AH, H. S, W. N, T. A.H., Deep learning based joint segmentation and characterization of multi-class retinal fluid lesions on oct scans for clinical use in anti-vegf therapy, Computers in biology and medicine 136 (2021) 104727. doi:10.1016/j.compbiomed.2021.104727.
URL https://pubmed.ncbi.nlm.nih.gov/34385089/

[213] M. Rithani, R. P. Kumar, V. Chatiyode, J. S. Anusha, M. Rithani, R. P. Kumar, V. Chatiyode, J. S. Anusha, K. R.P., C. V., A. J.S., Advancing ophthalmic diagnostics: Employing deep learning for precision detection of retinal damage in oct images, 2nd IEEE International Conference on Data Science and Network Security, ICDSNS 2024 (2024) 1–7doi:10.1109/ICDSNS62112.2024.10690853.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10690853

[214] R. Arefin, M. D. Samad, F. A. Akyelken, A. Davanian, R. Arefin, M. D. Samad, F. A. Akyelken, A. Davanian, A. R, S. MD, A. FA, D. A, S. M.D., A. F.A., Non-transfer deep learning of optical coherence tomography for post-hoc explanation of macular disease classification, Proceedings - 2021 IEEE 9th International Conference on Healthcare Informatics, ISCHI 2021 2021 (2021) 48–52. doi:10.1109/ICHI52183.2021.00020.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9565709

[215] A. Mlaouhi, Z. Haddad, H. Zgolli, H. Tabia, D. Sidibé, N. Khlifa, A. Mlaouhi, Z. Haddad, H. Zgolli, H. Tabia, D. Sidibe, N. Khlifa, M. A., H. Z., Z. H., T. H., S. D., K. N., Advanced deep learning techniques for evaluating oct image quality and detecting retinal pathologies, Proceedings of IEEE/ACS International Conference on Computer Systems and Applications, AICCSA (2025) 1–2doi:10.1109/AICCSA66935.2025.11315355.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11315355

[216] P. B, S. S, D. M. R, D. K. S, B. Padmanaban, S. Subiksha, R. Daniel Mathews, S. Dilip Kumar, Dual-stream cnn-vit framework for interpretable glaucoma detection from oct and fundus images, 2025 9th International Conference on Inventive Systems and Control (ICISC) (2025) 40–45doi:10.1109/ICISC65841.2025.11188172.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11188172

[217] T. K. Yoo, J. Y. Choi, J. G. Seo, B. Ramasubramanian, S. Selvaperumal, D. W. Kim, Y. TK, C. JY, S. JG, R. B, S. S, K. DW, The possibility of the combination of oct and fundus images for improving the diagnostic accuracy of deep learning for age-related macular degeneration: a preliminary experiment, Medical & biological engineering & computing 57 (3) (2019) 677–687. doi:10.1007/s11517-018-1915-z.
URL https://pubmed.ncbi.nlm.nih.gov/30349958/

[218] X.-N. Wang, Z. Guan, B. Qian, T. Chen, Q. Wu, W. X.-N., G. Z., Q. B., C. T., W. Q., A deep learning system for the detection of optic disc neovascularization in diabetic retinopathy using optical coherence tomography angiography images, Visual Computer 41 (2) (2025) 1293–1302. doi:10.1007/s00371-024-03418-y.
URL https://www.scopus.com/pages/publications/85192752693?origin=resultslist

[219] A. Sharma, A. V. Khanna, M. Bhargava, A. Sharma, A. V. Khanna, M. Bhargava, S. A., K. A.V., B. M., Multi-label classification of retinal disorders in optical coherence tomography using deep learning, 2021 Second International Conference on Electronics and Sustainable Communication Systems (ICESC) (2021) 1750–1757doi:10.1109/ICESC51422.2021.9532711.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9532711

[220] C. Bowd, A. Belghith, L. M. Zangwill, M. Christopher, M. H. Goldbaum, R. Fan, J. Rezapour, S. Moghimi, A. Kamalipour, H. Hou, R. N. Weinreb, B. C, B. A, Z. LM, C. M, G. MH, F. R, R. J, M. S, K. A, H. H, W. RN, Z. L.M., G. M.H., W. R.N., Deep learning image analysis of optical coherence tomography angiography measured vessel density improves classification of healthy and glaucoma eyes, American journal of ophthalmology 236 (2022) 298–308. doi:10.1016/j.ajo.2021.11.008.
URL https://pubmed.ncbi.nlm.nih.gov/34780803/

[221] Y. Gueddena, N. Aboudi, H. Zgolli, S. Mabrouk, D. Sidibe, H. Tabia, N. Khlifa, G. Y, A. N, Z. H, M. S, S. D, T. H, K. N, A new intelligent system based deep learning to detect dme and amd in oct images, International ophthalmology 44 (1) (2024) 191. doi:10.1007/s10792-024-03115-8.
URL https://pubmed.ncbi.nlm.nih.gov/38653842/

[222] K. Syed, S. Kareemulla, U. S. Rani, S. Maity, K. Syed, S. Kareemulla, U. S. Rani, S. Maity, S. K., K. S., R. U.S., M. S., Automated detection of glaucoma using oct rnflt maps and supervised deep learning models: A comparative study of cnn, mobilenet, resnet50, and vgg16 architectures, 2025 International Conference on

Intelligent Communication Networks and Computational Techniques (ICICNCT) (2025) 1–8doi:10.1109/ICICNCT66124.2025.11232589.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11232589

[223] T. S. Apon, M. M. Hasan, A. Islam, M. G. R. Alam, T. S. Apon, M. M. Hasan, A. Islam, M. G. R. Alam, A. T.S., H. M.M., I. A., A. M.G.R., Demystifying deep learning models for retinal oct disease classification using explainable ai, 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering, CSDE 2021 (2021) 1–6doi:10.1109/CSDE53843.2021.9718400.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9718400

[224] T. K. Shivaprasad, P. Kumar, V. Mayya, S. V. Bhandary, T. Shreekumar, T. Shivaprasad, P. Kumar, V. Mayya, S. V. Bhandary, T. Shreekumar, K. P., M. V., B. S.V., A comprehensive review of deep learning-based retinal layer integrity assessment in optical coherence tomography (oct), IEEE Access 13 (2025) 194421–194434. doi:10.1109/ACCESS.2025.3632550.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11244904

[225] S. Choi, B. Jeoun, J. Anh, J. Jeong, Y. Choi, D. Kwon, U. Kim, S. Shin, C. S., J. B., A. J., J. J., C. Y., K. D., K. U., S. S., A vision transformer based deep learning architecture for automatic diagnosis of diabetic retinopathy in optical coherence tomography angiography, Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 13597 LNCS (2023) 135–145. doi:10.1007/978-3-031-33658-4_13.
URL https://www.scopus.com/pages/publications/85173565088?origin=resultslist

[226] S. G, N. Yaligar, K. R. Kumar, A. Padaki, R. Dawe, S. Katagi, G. Sharon, N. Yaligar, K. R. Kumar, A. Padaki, R. Dawe, S. Katagi, Y. N., K. K.R., P. A., D. R., K. S., State-of-the-art deep learning strategies for multi-class classification of retinal oct images, 2nd IEEE International Conference on Advances in Information Technology, ICAIT 2024 - Proceedings 1 (2024) 1–7. doi:10.1109/ICAIT61638.2024.10690676.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10690676

[227] H. Xie, W. Xu, Y. X. Wang, X. Wu, X. H, X. W, W. YX, W. X, W. Y.X., Deep learning network with differentiable dynamic programming for retina oct surface segmentation, Biomedical optics express 14 (7) (2023) 3190–3202. doi:10.1364/BOE.492670.
URL https://pubmed.ncbi.nlm.nih.gov/37497505/

[228] D. Abhiram, R. A. Flarence, K. Anuradha, V. Srilakshmi, A. D., F. R.A., Deep learning and oct imaging: A novel ensemble approach for eye disease diagnosis, Cognitive Science and Technology Part F1413 (2026) 57–63. doi:10.1007/978-981-95-0144-1_7.
URL https://www.scopus.com/pages/publications/105029427342?origin=resultslist

[229] Y. D. Sanchez, B. Quijano, F. D. Padilla, O. Perdomo, F. A. González, S. Y.D., Q. B., P. F.D., P. O., G. F.A., Segmentation of retinal fluids and hyperreflective foci using deep learning approach in optical coherence tomography scans, Proceedings of SPIE - The International Society for Optical Engineering 11583 (2020). doi:10.1117/12.2579934.
URL https://www.scopus.com/pages/publications/85096838966?origin=resultslist

[230] M. Abini, S. S. S. Priya, P. S.S.S., Automatic detection and classification of diabetic retinopathy from optical coherence tomography angiography images using deep learning-a review, AIUB Journal of Science and Engineering 23 (3) (2025) 277–297. doi:10.53799/ajse.v23i3.1361.
URL https://www.scopus.com/pages/publications/85217138786?origin=resultslist

[231] N.-N. Shi, J. Li, G.-H. Liu, M.-F. Cao, S. NN, L. J, L. GH, C. MF, S. N.-N., L. G.-H., C. M.-F., Artificial intelligence for the detection of glaucoma with sd-oct images: a systematic review and meta-analysis, International journal of ophthalmology 17 (3) (2024) 408–419. doi:10.18240/ijo.2024.03.02.
URL https://pubmed.ncbi.nlm.nih.gov/38721504/

[232] S. A P, S. Kar, G. S, V. P. Gopi, P. Palanisamy, A. P. S, K. S, S. G, G. VP, G. V.P., Octnet: A lightweight cnn for retinal disease classification from optical coherence tomography images, Computer methods and programs in biomedicine 200 (2021) 105877. doi:10.1016/j.cmpb.2020.105877.
URL https://pubmed.ncbi.nlm.nih.gov/33339630/

[233] M. G. İnam, O. İnam, X. Yang, Q. Zeng, G. Tezel, İnam MG, İnam O, Y. X, Z. Q, T. G, İnam M.G., Integrating retinal segmentation metrics with machine learning for predictions from mouse sd-oct scans, Current eye research 50 (5) (2025) 502–511. doi:10.1080/02713683.2025.2456783.
URL https://pubmed.ncbi.nlm.nih.gov/39849306/

[234] J. Loo, G. J. Jaffe, J. L. Duncan, D. G. Birch, S. Farsiu, L. J, J. GJ, D. JL, B. DG, F. S, J. G.J., D. J.L., B. D.G., Validation of a deep learning-based algorithm for segmentation of the ellipsoid zone on optical coherence tomography images of an ush2a-related retinal degeneration clinical trial, Retina (Philadelphia, Pa.) 42 (7) (2022) 1347–1355. doi:10.1097/IAE.0000000000003448.
URL https://pubmed.ncbi.nlm.nih.gov/35174801/

[235] M. Subramanian, K. Shanmugavadivel, O. S. Naren, K. Premkumar, K. Rankish, M. Subramanian, K. Shanmugavadivel, O. S. Naren, K. Premkumar, K. Rankish, S. M., S. K., N. O.S., Classification of retinal oct images using deep learning, 2022 International Conference on Computer Communication and Informatics, ICCCI 2022 (2022) 1–7doi:10.1109/ICCCI54379.2022.9740985.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9740985

[236] J. M. Han, J. Han, J. Ko, J. Jung, J. I. Park, J. S. Hwang, J. Yoon, J. H. Jung, D. D.-J. Hwang, H. JM, H. J, K. J, J. J, P. JI, H. JS, Y. J, J. JH, H. DD, H. J.M., P. J.I., H. J.S., J. J.H., H. D.D.-J., Anti-vegf treatment outcome prediction based on optical coherence tomography images in neovascular age-related macular degeneration using a deep neural network, Scientific reports 14 (1) (2024) 28253. doi:10.1038/s41598-024-79034-6.
URL https://pubmed.ncbi.nlm.nih.gov/39548212/

[237] N. Wongchaisuwat, J. Wang, E. S. White, T. S. Hwang, Y. Jia, S. T. Bailey, W. N, W. J, W. ES, H. TS, J. Y, B. ST, W. E.S., H. T.S., B. S.T., Detection of macular neovascularization in eyes presenting with macular edema using oct angiography and a deep learning model, Ophthalmology. Retina 9 (4) (2025) 378–385. doi:10.1016/j.oret.2024.10.017.
URL https://pubmed.ncbi.nlm.nih.gov/39461425/

[238] Y.-T. Lin, X. Xiong, Y.-P. Zheng, Q. Zhou, L. YT, X. X, Z. YP, Z. Q, L. Y.-T., Z. Y.-P., Transfer learning and multi-feature fusion-based deep learning model for idiopathic macular hole diagnosis and grading from optical coherence tomography images, Clinical ophthalmology (Auckland, N.Z.) 19 (2025) 1593–1607. doi:10.2147/OPTH.S521558.
URL https://pubmed.ncbi.nlm.nih.gov/40396157/

[239] Z. Li, K. Cheng, P. Qin, Y. Dong, C. Yang, X. Jiang, Z. Li, K. Cheng, P. Qin, Y. Dong, C. Yang, X. Jiang, L. Z., C. K., Q. P., D. Y., Y. C., J. X., Retinal oct image classification based on domain adaptation convolutional neural networks, Proceedings - 2021 14th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics, CISP-BMEI 2021 (2021) 1–5doi:10.1109/CISP-BMEI53629.2021.9624429.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9624429

[240] S. Patnaik, A. Subasi, P. S., S. A., Artificial intelligence-based retinal disease classification using optical coherence tomography images, Applications of Artificial Intelligence in Medical Imaging (2022) 305–319doi:10.1016/B978-0-443-18450-5.00009-8.
URL https://www.scopus.com/pages/publications/85150137095?origin=resultslist

[241] E. Khalili Pour, K. Rezaee, H. Azimi, S. M. Mirshahvalad, B. Jafari, K. Fadakar, H. Faghihi, A. Mirshahi, F. Ghassemi, N. Ebrahimiadib, M. Mirghorbani, F. Bazvand, H. Riazi-Esfahani, M. Riazi Esfahani, K. P. E, R. K, A. H, M. SM, J. B, F. K, F. H, M. A, G. F, E. N, M. M, B. F, R.-E. H, R. E. M, M. S.M., Automated machine learning-based classification of proliferative and non-proliferative diabetic retinopathy using optical

coherence tomography angiography vascular density maps, Graefe's Archive for Clinical and Experimental Ophthalmology 261 (2) (2023) 391–399. doi:10.1007/s00417-022-05818-z.
URL https://pubmed.ncbi.nlm.nih.gov/36050474/

[242] S. Chhabria, S. Khedkar, P. Dongre, S. Jangde, S. Rangari, S. Selokar, S. Potle, S. Chhabria, S. Khedkar, P. Dongre, S. Jangde, S. Rangari, S. Selokar, S. Potle, C. S., K. S., D. P., J. S., R. S., S. S., P. S., Oct vision: Retinal disease detection using deep learning, 2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Applications, ICAIQSA 2024 - Proceedings (2024) 1–6doi:10.1109/ICAIQSA64000.2024.10882359.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10882359

[243] K. Pin, Y. Nam, S. Ha, J. W. Han, K. Pin, Y. Nam, S. Ha, J. W. Han, P. K., N. Y., H. S., H. J.W., Deep learning based on ensemble to diagnose of retinal disease using optical coherence tomography, Proceedings - 2021 International Conference on Computational Science and Computational Intelligence, CSCI 2021 (2021) 661–664doi:10.1109/CSCI54926.2021.00174.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9799021

[244] J. Saeidian, H. Azimi, Z. Azimi, P. Pouya, H. Asadigandomani, H. Riazi-Esfahani, A. Hayati, K. Daneshvar, E. Khalili Pour, S. J, A. H, A. Z, P. P, A. H, R.-E. H, H. A, D. K, K. P. E, Segmentation of choroidal area in optical coherence tomography images using a transfer learning-based conventional neural network: a focus on diabetic retinopathy and a literature review, BMC medical imaging 24 (1) (2024) 281. doi:10.1186/s12880-024-01459-2.
URL https://pubmed.ncbi.nlm.nih.gov/39425019/

[245] M. Hu, B. Wu, D. Lu, J. Xie, Y. Chen, Z. Yang, W. Dai, H. M, W. B, L. D, X. J, C. Y, Y. Z, D. W, Two-step hierarchical neural network for classification of dry age-related macular degeneration using optical coherence tomography images, Frontiers in medicine 10 (2023) 1221453. doi:10.3389/fmed.2023.1221453.
URL https://pubmed.ncbi.nlm.nih.gov/37547613/

[246] Y. Hashimoto, R. Asaoka, T. Kiwaki, H. Sugiura, S. Asano, H. Murata, Y. Fujino, M. Matsuura, A. Miki, K. Mori, Y. Ikeda, T. Kanamoto, J. Yamagami, K. Inoue, M. Tanito, K. Yamanishi, H. Y, A. R, K. T, S. H, A. S, M. H, F. Y, M. M, M. A, M. K, I. Y, K. T, Y. J, I. K, T. M, Y. K, Deep learning model to predict visual field in central 10° from optical coherence tomography measurement in glaucoma, British Journal of Ophthalmology 105 (4) (2021) 507–513. doi:10.1136/bjophthalmol-2019-315600.
URL https://pubmed.ncbi.nlm.nih.gov/32593978/

[247] M. Ashtari-Majlan, D. Masip, M. Ashtari-Majlan, D. Masip, A.-M. M, M. D, Spatial-aware transformer-gru framework for enhanced glaucoma diagnosis from 3d oct imaging, IEEE Journal of Biomedical and Health Informatics 29 (9) (2025) 6167–6175. doi:10.1109/JBHI.2025.3550394.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10921670

[248] B. Ait Hammou, F. Antaki, M.-C. Boucher, R. Duval, A. H. B, A. F, B. MC, D. R, B. M.-C., Mbt: Model-based transformer for retinal optical coherence tomography image and video multi-classification, International journal of medical informatics 178 (2023) 105178. doi:10.1016/j.ijmedinf.2023.105178.
URL https://pubmed.ncbi.nlm.nih.gov/37657204/

[249] A. Mousavian, J. Jarkaneh, M. Pourhomayoun, M. Kaur, R. Chatterjee, S. Besharati, K. Nouri-Mahdavi, N. Amini, A. Mousavian, R. Chatterjee, J. Jarkaneh, S. Besharati, M. Pourhomayoun, K. Nouri-Mahdavi, M. Kaur, N. Amini, M. A., C. R., J. J., B. S., P. M., N.-M. K., K. M., A. N., Deep learning image analysis of macular optical coherence tomography angiography images for detection of progression in glaucoma, 2024 IEEE First International Conference on Artificial Intelligence for Medicine, Health and Care (AIMHC) (2024) 130–136doi:10.1109/AIMHC59811.2024.00031.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10504332

[250] S. Islam, R. C. Deo, P. D. Barua, J. Soar, U. R. Acharya, I. S, D. RC, B. PD, S. J, A. UR, D. R.C., B. P.D., A. U.R., Novel deep learning model for glaucoma detection using fusion of fundus and optical coherence tomography images, Sensors (Basel, Switzerland) 25 (14) (2025). doi:10.3390/s25144337.
URL https://pubmed.ncbi.nlm.nih.gov/40732464/

[251] M. Stanojević, D. Drašković, B. Nikolić, S. M., D. D., N. B., Retinal disease classification based on optical coherence tomography images using convolutional neural networks, Journal of Electronic Imaging 32 (3) (2023). doi:10.1117/1.JEI.32.3.032004.
URL https://www.scopus.com/pages/publications/85164218224?origin=resultslist

[252] H. Lee, N. Kim, N. H. Kim, H. Chung, H. C. Kim, L. H, K. N, K. NH, C. H, K. HC, K. N.H., K. H.C., Generative deep learning approach to predict posttreatment optical coherence tomography images of age-related macular degeneration after 12 months, Retina (Philadelphia, Pa.) 45 (6) (2025) 1184–1191. doi:10.1097/IAE.0000000000004409.
URL https://pubmed.ncbi.nlm.nih.gov/39841905/

[253] E. R. Vázquez, M. N. B. Rodríguez, E. López-Varela, M. G. Penedo, V. E.R., R. M.N.B., L.-V. E., P. M.G., Deep learning for segmentation of optic disc and retinal layers in peripapillary optical coherence tomography images, Proceedings of SPIE - The International Society for Optical Engineering 12701 (2023). doi:10.1117/12.2680545.
URL https://www.scopus.com/pages/publications/85172915489?origin=resultslist

[254] L. V, A. L. R, S. K. G, V. Latha, L. Ashok, K. Sreeni, Automated macular disease detection using retinal optical coherence tomography images by fusion of deep learning networks, 2021 National Conference on Communications, NCC 2021 (2021) 1–6doi:10.1109/NCC52529.2021.9530171.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9530171

[255] Z. Fan, X. Liu, Y. Zhang, J. Zhang, F. Z., L. X., Z. Y., Z. J., Multi-scale cnn-swin transformer network with boundary supervision for multiclass biomarker segmentation in retinal oct images, Biocybernetics and Biomedical Engineering 45 (3) (2025) 340–356. doi:10.1016/j.bbe.2025.05.002.
URL https://www.scopus.com/pages/publications/105005371320?origin=resultslist

[256] M. Lin, G. Bao, X. Sang, Y. Wu, L. M, B. G, S. X, W. Y, Recent advanced deep learning architectures for retinal fluid segmentation on optical coherence tomography images, Sensors (Basel, Switzerland) 22 (8) (2022). doi:10.3390/s22083055.
URL https://pubmed.ncbi.nlm.nih.gov/35459040/

[257] A. K. Kakumani, K. Tanuja, J. S. Yetukuri, A. K. Kakumani, K. Tanuja, J. S. Yetukuri, K. A.K., T. K., Y. J.S., Vision transformers for retinal disease classification using optical coherence tomography images, 2025 International Conference on Computer, Electrical & Communication Engineering (ICCECE) (2025) 1–5doi:10.1109/ICCECE61355.2025.10941428.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10941428

[258] Z. Bi, J. Li, Q. Liu, Z. Fang, B. Z, L. J, L. Q, F. Z, Deep learning-based optical coherence tomography and retinal images for detection of diabetic retinopathy: a systematic and meta analysis, Frontiers in endocrinology 16 (2025) 1485311. doi:10.3389/fendo.2025.1485311.
URL https://pubmed.ncbi.nlm.nih.gov/40171193/

[259] F. V. Jacob, G. P, F. V. Jacob, P. Gethsia, J. F.V., Adaptive and explainable deep learning for automated retinal disease diagnosis using oct images, 2025 9th International Conference on Electronics, Communication and Aerospace Technology (ICECA) (2025) 1677–1683doi:10.1109/ICECA66444.2025.11382772.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11382772

[260] O. Perdomo, H. Rios, F. J. Rodríguez, S. Otálora, F. Meriaudeau, H. Müller, F. A. González, P. O, R. H, R. FJ, O. S, M. F, M. H, G. FA, R. F.J., G. F.A., Classification of diabetes-related retinal diseases using a deep learning approach in optical coherence tomography, Computer methods and programs in biomedicine 178 (2019) 181–189. doi:10.1016/j.cmpb.2019.06.016.
URL https://pubmed.ncbi.nlm.nih.gov/31416547/

[261] E. Midena, L. Toto, L. Frizziero, G. Covello, T. Torresin, G. Midena, L. Danieli, E. Pilotto, M. Figus, C. Mariotti, M. Lupidi, M. E, T. L, F. L, C. G, T. T, M. G, D. L, P. E, F. M, M. C, L. M, Validation of an automated artificial intelligence algorithm for the quantification of major oct parameters in diabetic macular edema, Journal of clinical medicine 12 (6) (2023). doi:10.3390/jcm12062134.
URL https://pubmed.ncbi.nlm.nih.gov/36983137/

[262] M. K. Nigjeh, M. K. Nigjeh, S. E. Umbaugh, N. M.K., U. S.E., Automated retinal disorders classification: leveraging digital image enhancement techniques and deep learning on oct images, Proceedings of SPIE - The International Society for Optical Engineering 13137 (2024). doi:10.1117/12.3028346.
URL https://www.scopus.com/pages/publications/85207234429?origin=resultslist

[263] R. Liu, Q. Li, F. Xu, S. Wang, J. He, Y. Cao, F. Shi, X. Chen, J. Chen, L. R, L. Q, X. F, W. S, H. J, C. Y, S. F, C. X, C. J, Application of artificial intelligence-based dual-modality analysis combining fundus photography and optical coherence tomography in diabetic retinopathy screening in a community hospital, Biomedical engineering online 21 (1) (2022) 47. doi:10.1186/s12938-022-01018-2.
URL https://pubmed.ncbi.nlm.nih.gov/35859144/

[264] M.-X. Li, S.-Q. Yu, W. Zhang, H. Zhou, X. Xu, T.-W. Qian, Y.-J. Wan, L. MX, Y. SQ, Z. W, Z. H, X. X, Q. TW, W. YJ, L. M.-X., Y. S.-Q., Q. T.-W., W. Y.-J., Segmentation of retinal fluid based on deep learning: application of three-dimensional fully convolutional neural networks in optical coherence tomography images, International journal of ophthalmology 12 (6) (2019) 1012–1020. doi:10.18240/ijo.2019.06.22.
URL https://pubmed.ncbi.nlm.nih.gov/31236362/

[265] M. M. Abdelsalam, A. M.M., Effective blood vessels reconstruction methodology for early detection and classification of diabetic retinopathy using octa images by artificial neural network, Informatics in Medicine Unlocked 20 (2020). doi:10.1016/j.imu.2020.100390.
URL https://www.scopus.com/pages/publications/85088016584?origin=resultslist

[266] V. Sudha, S. P. Shanmugam, D. Anitha, R. Raja, S. S.P., Retraction:mac-leonets-a novel optimized hybrid convolutional neural networks for the segmentation and diagnosis of edema diseases using retinal oct images, Journal of Intelligent and Fuzzy Systems 44 (6) (2023) 10605–10620. doi:10.3233/JIFS-230128.
URL https://www.scopus.com/pages/publications/85166395209?origin=resultslist

[267] M. Rahil, B. N. Anoop, G. N. Girish, A. R. Kothari, S. G. Koolagudi, J. Rajan, M. Rahil, B. Anoop, G. Girish, A. R. Kothari, S. G. Koolagudi, J. Rajan, R. M., K. A.R., K. S.G., R. J., A deep ensemble learning-based cnn architecture for multiclass retinal fluid segmentation in oct images, IEEE Access 11 (2023) 17241–17251. doi:10.1109/ACCESS.2023.3244922.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10043848

[268] D. Alonso-Caneiro, J. Kugelman, J. Hamwood, S. A. Read, S. J. Vincent, F. K. Chen, M. J. Collins, A.-C. D., K. J., H. J., R. S.A., V. S.J., C. F.K., C. M.J., Automatic retinal and choroidal boundary segmentation in oct images using patch-based supervised machine learning methods, Lecture Notes in Computer Science 11367 LNCS (2019) 215–228. doi:10.1007/978-3-030-21074-8_17.
URL https://www.scopus.com/pages/publications/85068458758?origin=resultslist

[269] A. Sampath Kumar, T. Schlosser, H. Langner, M. Ritter, D. Kowerko, S. K. A, S. T, L. H, R. M, K. D, Improving oct image segmentation of retinal layers by utilizing a machine learning based multistage system of stacked multiscale encoders and decoders, Bioengineering (Basel, Switzerland) 10 (10) (2023). doi:10.3390/bioengineering10101177.
URL https://pubmed.ncbi.nlm.nih.gov/37892907/

[270] R. Huang, R. Huang, H. R., Automated deep learning framework for retinal disease classification from oct images: A large-scale study across four categories, 2024 International Conference on Image Processing, Computer Vision and Machine Learning, ICICML 2024 (2024) 266–270doi:10.1109/ICICML63543.2024.10958147.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10958147

[271] T. Murata, T. Hirano, H. Mizobe, S. Toba, M. T, H. T, M. H, T. S, Oct-angiography based artificial intelligence-inferred fluorescein angiography for leakage detection in retina [invited], Biomedical optics express 14 (11) (2023) 5851–5860. doi:10.1364/BOE.506467.
URL https://pubmed.ncbi.nlm.nih.gov/38021144/

[272] N. Yamsani, P. Nasra, R. Badhani, A. Kharola, N. Yamsani, P. Nasra, R. Badhani, A. Kharola, Y. N., N. P., B. R., K. A., Deep learning for retinal oct disease classification using efficientnetb0: A transfer learning approach, Proceedings of 6th International Conference on IoT Based Control Networks and Intelligent Systems, ICICNIS 2025 (2025) 1846–1851doi:10.1109/ICICNIS66685.2025.11315745.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11315745

[273] B. Kucukgoz, M. M. Yapici, D. C. Murphy, E. Spowart, D. H. Steel, B. Obara, B. Kucukgoz, M. M. Yapici, D. C. Murphy, E. Spowart, D. H. Steel, B. Obara, K. B., Y. M.M., M. D.C., S. E., S. D.H., O. B., Deep learning using preoperative optical coherence tomography images to predict visual acuity following surgery for idiopathic full-thickness macular holes, IEEE Access 12 (2024) 32911–32926. doi:10.1109/ACCESS.2024.3369676.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10445126

[274] D. Mahalakshmi, S. J. J. Thangaraj, D. Mahalakshmi, S. J. J. Thangaraj, T. S.J.J., A deep learning model using vgg19 and random forest for amd stage classification on retinal oct images, 2025 2nd International Conference on Computing and Data Science, ICCDS 2025 (2025) 1–6doi:10.1109/ICCDS64403.2025.11209005.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11209005

[275] H. Liu, W. Gao, L. Yang, D. Wu, D. Zhao, K. Chen, J. Liu, Y. Ye, R. X. Xu, M. Sun, L. H, G. W, Y. L, W. D, Z. D, C. K, L. J, Y. Y, X. RX, S. M, X. R.X., Semantic uncertainty guided cross-transformer for enhanced macular edema segmentation in oct images, Computers in biology and medicine 174 (2024) 108458. doi:10.1016/j.compbiomed.2024.108458.
URL https://pubmed.ncbi.nlm.nih.gov/38631114/

[276] J. R. Mapanao, P. L. Lozano, J. R. Mapanao, P. L. Lozano, M. J.R., L. P.L., Utilizing swin transformer for the classification of ophthalmic diseases in optical coherence tomography (oct) images: A novel approach, Proceedings of the IEEE International Conference on Computer Communication and the Internet, ICCCI (2024) (2024) 94–100. doi:10.1109/ICCCI62159.2024.10674599.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10674599

[277] D. Kim, S. B. Seo, S. J. Park, H.-K. Cho, K. D, S. SB, P. SJ, C. HK, S. S.B., P. S.J., C. H.-K., Deep learning visual field global index prediction with optical coherence tomography parameters in glaucoma patients, Scientific reports 13 (1) (2023) 18304. doi:10.1038/s41598-023-43104-y.
URL https://pubmed.ncbi.nlm.nih.gov/37880259/

[278] D. Rivas-Villar, A. R. Motschi, M. Pircher, C. K. Hitzenberger, M. Schranz, P. K. Roberts, U. Schmidt-Erfurth, H. Bogunović, R.-V. D, M. AR, P. M, H. CK, S. M, R. PK, S.-E. U, B. H, M. A.R., H. C.K., R. P.K., Automated inter-device 3d oct image registration using deep learning and retinal layer segmentation, Biomedical optics express 14 (7) (2023) 3726–3747. doi:10.1364/BOE.493047.
URL https://pubmed.ncbi.nlm.nih.gov/37497506/

[279] X. Liu, D. Liu, B. Li, S. Wang, L. X., L. D., L. B., W. S., Deep learning based fluid segmentation in retinal optical coherence tomography images, Lecture Notes in Computer Science 11643 LNCS (2019) 337–345. doi:10.1007/978-3-030-26763-6_33.
URL https://www.scopus.com/pages/publications/85070711243?origin=resultslist

[280] G. Sanchez-Rodriguez, L. Lou, M. T. Pardue, A. J. Feola, S.-R. G, L. L, P. MT, F. AJ, P. M.T., F. A.J., Retoctnet: Deep learning-based segmentation of oct images following retinal ganglion cell injury, Translational vision science & technology 14 (2) (2025) 4. doi:10.1167/tvst.14.2.4.
URL https://pubmed.ncbi.nlm.nih.gov/39903165/

[281] S. Shroff, D. P. Rao, F. M. Savoy, S. Shruthi, C.-K. Hsu, Z. S. Pradhan, P. V. Jayasree, A. Sivaraman, S. Sengupta, R. Shetty, H. L. Rao, S. S, R. DP, S. FM, H. CK, P. ZS, J. PV, S. A, S. S, S. R, R. HL, R. D.P., S. F.M., H. C.-K., P. Z.S., R. H.L., Agreement of a novel artificial intelligence software with optical coherence tomography and manual grading of the optic disc in glaucoma, Journal of glaucoma 32 (4) (2023) 280–286. doi:10.1097/IJG.0000000000002147.
URL https://pubmed.ncbi.nlm.nih.gov/36730188/

[282] Z. Wu, W. Cai, H. Xie, S. Chen, Y. Wang, B. Lei, Y. Zheng, L. Lu, W. Z, C. W, X. H, C. S, W. Y, L. B, Z. Y, L. L, Predicting optical coherence tomography-derived high myopia grades from fundus photographs using deep learning, Frontiers in medicine 9 (2022) 842680. doi:10.3389/fmed.2022.842680.
URL https://pubmed.ncbi.nlm.nih.gov/35308524/

[283] K. Ma, J. Zhang, X. Hu, Y. Chen, G. Wang, Y. Huang, J. Yuan, P. Xiao, M. K., Z. J., H. X., C. Y., W. G., H. Y., Y. J., X. P., Deep learning-based segmentation and quantitative analysis of retinal microstructures in optical coherence tomography angiography images using rs_unet3+, Intelligent Medicine 6 (1) (2026) 12–23. doi:10.1016/j.imed.2025.05.008.
URL https://www.scopus.com/pages/publications/105029265265?origin=resultslist

[284] M. M. Noori Al-Tekreeti, M. Y. Agici, N. A.-T. M.M., A. M.Y., Classification and prediction retinal oct images by cnn algorithm, ISMSIT 2022 - 6th International Symposium on Multidisciplinary Studies and Innovative Technologies, Proceedings (2022) 424–428doi:10.1109/ISMSIT56059.2022.9932698.
URL https://www.scopus.com/pages/publications/85142845454?origin=resultslist

[285] Y. Kihara, M. Shen, Y. Shi, X. Jiang, L. Wang, R. Laiginhas, C. Lyu, J. Yang, J. Liu, R. Morin, R. Lu, H. Fujiyoshi, W. J. Feuer, G. Gregori, P. J. Rosenfeld, A. Y. Lee, K. Y, S. M, S. Y, J. X, W. L, L. R, L. C, Y. J, L. J, M. R, L. R, F. H, F. WJ, G. G, R. PJ, L. AY, F. W.J., R. P.J., L. A.Y., Detection of nonexudative macular neovascularization on structural oct images using vision transformers, Ophthalmology science 2 (4) (2022) 100197. doi:10.1016/j.xops.2022.100197.
URL https://pubmed.ncbi.nlm.nih.gov/36531577/

[286] M. Pekala, N. Joshi, T. Y. A. Liu, N. M. Bressler, D. C. DeBuc, P. Burlina, L. TYA, B. NM, D. DC, L. T.Y.A., D. D.C., Deep learning based retinal oct segmentation, Computers in biology and medicine 114 (2019) 103445. doi:10.1016/j.compbiomed.2019.103445.
URL https://pubmed.ncbi.nlm.nih.gov/31561100/

[287] M. Treder, J. L. Lauermann, N. Eter, T. M, L. JL, E. N, L. J.L., Automated detection of exudative age-related macular degeneration in spectral domain optical coherence tomography using deep learning, Graefe's Archive for Clinical and Experimental Ophthalmology 256 (2) (2018) 259–265. doi:10.1007/s00417-017-3850-3.
URL https://pubmed.ncbi.nlm.nih.gov/29159541/

[288] R. G. Mundada, D. D. Nawgaje, R. G. Mundada, D. D. Nawgaje, M. R.G., N. D.D., Mdlgo: Integrated multimodal deep learning framework for enhanced precision in glaucoma diagnosis using oct a-scans, 2024 International Conference on Emerging Smart Computing and Informatics, ESCI 2024 (2024) 1–7doi:10.1109/ESCI59607.2024.10497193.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10497193

[289] D. Lu, M. Heisler, S. Lee, G. W. Ding, E. Navajas, M. V. Sarunic, M. F. Beg, L. D, H. M, L. S, D. GW, N. E, S. MV, B. MF, D. G.W., S. M.V., B. M.F., Deep-learning based multiclass retinal fluid segmentation and detection in optical coherence tomography images using a fully convolutional neural network, Medical image analysis 54 (2019) 100–110. doi:10.1016/j.media.2019.02.011.
URL https://pubmed.ncbi.nlm.nih.gov/30856455/

[290] J. Wang, J. Wang, D. Chen, X. Wu, Z. Xu, X. Yu, S. Sheng, X. Lin, X. Chen, J. Wu, H. Ying, W. Xu, W. J, C. D, W. X, X. Z, Y. X, S. S, L. X, C. X, W. J, Y. H, X. W, Prediction of postoperative visual acuity in patients with age-related cataracts using macular optical coherence tomography-based deep learning method, Frontiers in medicine 10 (2023) 1165135. doi:10.3389/fmed.2023.1165135.
URL https://pubmed.ncbi.nlm.nih.gov/37250634/

[291] C. Brandão Fantozzi, L. M. Peres, J. S. Neto, C. C. Brandão, R. C. Guido, R. C. Siqueira, B. ao Fantozzi C, P. LM, N. JS, B. ao CC, G. RC, S. RC, P. L.M., N. J.S., B. ao C.C., G. R.C., S. R.C., A comparative study between clinical optical coherence tomography (oct) analysis and artificial intelligence-based quantitative evaluation in the diagnosis of diabetic macular edema, Vision (Basel, Switzerland) 9 (3) (2025). doi:10.3390/vision9030075.
URL https://pubmed.ncbi.nlm.nih.gov/40981318/

[292] G. François, P. Sayeh, M. Benoit, G. François, P. Sayeh, M. Benoit, F. G., S. P., B. M., Deep convolutional neural network prediction for glaucoma detection using oct and oct-angiography disc- and macula-centered images and their combined power, 2024 IEEE International Conference on Image Processing (ICIP) (2024) 2088–2094doi:10.1109/ICIP51287.2024.10647463.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10647463

[293] F. Li, H. Chen, Z. Liu, X.-D. Zhang, M.-S. Jiang, Z.-Z. Wu, K.-Q. Zhou, L. F, C. H, L. Z, Z. XD, J. MS, W. ZZ, Z. KQ, Z. X.-D., J. M.-S., W. Z.-Z., Z. K.-Q., Deep learning-based automated detection of retinal diseases using optical coherence tomography images, Biomedical optics express 10 (12) (2019) 6204–6226. doi:10.1364/BOE.10.006204.
URL https://pubmed.ncbi.nlm.nih.gov/31853395/

[294] M. Hasan, A. M. Pathan, Z. Rafa, M. R. M. M. Tahsin, A. Hossain, R. U. Islam, M. R. A. Rashid, A. W. Reza, S. H. Ripon, M. Hasan, A. M. Pathan, Z. Rafa, M. R. M. M. Tahsin, A. Hossain, R. Ul Islam, M. R. A. R. Ahmed, W. Reza, S. H. Ripon, H. M., P. A.M., R. Z., T. M.R.M.M., H. A., U. I. R., A. M.R.A.R., R. W., R. S.H., Attention-guided neural networks for explainable retinal disease detection in oct images, 2025 IEEE 2nd International Conference on Computing, Applications and Systems, COMPAS 2025 (2025) 1–6doi:10.1109/COMPAS67506.2025.11381821.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11381821

[295] J. Kugelman, D. Alonso-Caneiro, S. A. Read, S. J. Vincent, M. J. Collins, K. J, A.-C. D, R. SA, V. SJ, C. MJ, R. S.A., V. S.J., C. M.J., Automatic segmentation of oct retinal boundaries using recurrent neural networks and graph search, Biomedical optics express 9 (11) (2018) 5759–5777. doi:10.1364/BOE.9.005759.
URL https://pubmed.ncbi.nlm.nih.gov/30460160/

[296] Z. Li, Y. Han, X. Yang, L. Z, H. Y, Y. X, Multi-fundus diseases classification using retinal optical coherence tomography images with swin transformer v2, Journal of imaging 9 (10) (2023). doi:10.3390/jimaging9100203.
URL https://pubmed.ncbi.nlm.nih.gov/37888310/

[297] G. Aresta, T. Araújo, U. Schmidt-Erfurth, H. Bogunovic, A. G, A. T, S.-E. U, B. H, Anomaly detection in retinal oct images with deep learning-based knowledge distillation, Translational vision science & technology 14 (3) (2025) 26. doi:10.1167/tvst.14.3.26.
URL https://pubmed.ncbi.nlm.nih.gov/40146150/

[298] R. Almasi, A. Vafaei, E. Kazeminasab, H. Rabbani, A. R, V. A, K. E, R. H, Automatic detection of microaneurysms in optical coherence tomography images of retina using convolutional neural networks and transfer learning, Scientific reports 12 (1) (2022) 13975. doi:10.1038/s41598-022-18206-8.
URL https://pubmed.ncbi.nlm.nih.gov/35978087/

[299] A. Manassakorn, S. Auethavekiat, V. Sa-Ing, S. Chansangpetch, K. Ratanawongphaibul, N. Uramphorn, V. Tantisevi, A. Manassakorn, S. Auethavekiat, V. Sa-Ing, S. Chansangpetch, K. Ratanawongphaibul, N. Uramphorn, V. Tantisevi, M. A., A. S., S.-I. V., C. S., R. K., U. N., T. V., Glaunet: Glaucoma diagnosis for octa imaging using a new cnn architecture, IEEE Access 10 (2022) 95613–95622. doi:10.1109/ACCESS.2022.3204029.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9875271

[300] N. Kaothanthong, J. Limwattanayingyong, S. Silpa-Archa, M. Tadarati, A. Amphornphruet, P. Singhanetr, P. Lalitwongsa, P. Chantangphol, A. Amornpetchsathaporn, M. Chainakul, P. Ruamviboonsuk, K. N, L. J, S.-A. S, T. M, A. A, S. P, L. P, C. P, A. A, C. M, R. P, The classification of common macular diseases using deep learning on optical coherence tomography images with and without prior automated segmentation, Diagnostics (Basel, Switzerland) 13 (2) (2023). doi:10.3390/diagnostics13020189.
URL https://pubmed.ncbi.nlm.nih.gov/36672999/

[301] J. Li, Y. Ling, J. He, P. Jin, J. Zhu, H. Zou, X. Xu, S. Shao, Y. Gan, Y. Su, L. J., L. Y., H. J., J. P., Z. J., Z. H., X. X., S. S., G. Y., S. Y., A gcn-assisted deep learning method for peripapillary retinal layer segmentation in oct images, Progress in Biomedical Optics and Imaging - Proceedings of SPIE 11630 (2021). doi:10.1117/12.2582905.
URL https://www.scopus.com/pages/publications/85108697358?origin=resultslist

[302] P. Bidwai, S. Gite, B. Pradhan, H. Gupta, A. Alamri, B. P, G. S, P. B, G. H, A. A, Harnessing deep learning for detection of diabetic retinopathy in geriatric group using optical coherence tomography angiography-octa: A promising approach, MethodsX 13 (2024) 102910. doi:10.1016/j.mex.2024.102910.
URL https://pubmed.ncbi.nlm.nih.gov/39280760/

[303] M. S. Islam, S. S. Bristy, T. Azam, M. H. Rahman, S. S. Jennifer, A. W. Reza, I. M.S., B. S.S., A. T., R. M.H., J. S.S., R. A.W., Detection of retinal disease from optical coherence tomography images using cnn models, Lecture Notes in Networks and Systems 1167 LNNS (2024) 14–21. doi:10.1007/978-3-031-73318-5_2.
URL https://www.scopus.com/pages/publications/85215699235?origin=resultslist

[304] N. Takahashi, J.-K. Wang, M. R. Tamplin, E. M. Binkley, M. K. Garvin, I. M. Grumbach, R. H. Kardon, T. N., W. J.-K., T. M.R., B. E.M., G. M.K., G. I.M., K. R.H., Classification of range of oct-angiography capillary density using multichannel deep learning models in diabetic retinopathy, aging macular degeneration, and radiation retinopathy, Progress in Biomedical Optics and Imaging - Proceedings of SPIE 13407 (2025). doi:10.1117/12.3046775.
URL https://www.scopus.com/pages/publications/105004411637?origin=resultslist

[305] F. R. Silva, V. G. Vidotti, F. Cremasco, M. Dias, E. S. Gomi, V. P. Costa, S. FR, V. VG, C. F, D. M, G. ES, C. VP, S. F.R., V. V.G., G. E.S., C. V.P., Sensitivity and specificity of machine learning classifiers for glaucoma diagnosis using spectral domain oct and standard automated perimetry, Arquivos brasileiros de oftalmologia 76 (3) (2013) 170–4. doi:10.1590/s0004-27492013000300008.
URL https://pubmed.ncbi.nlm.nih.gov/23929078/

[306] E. Haihong, J. Ding, L. Yuan, E. Haihong, J. Ding, L. Yuan, D. J., Y. L., Sae-wamd: A self-attention enhanced convolution neural network for fine-grained classification of wet age-related macular degeneration using oct images, 2022 International Conference on Image Processing, Computer Vision and Machine Learning, ICICML 2022 (2022) 619–627doi:10.1109/ICICML57342.2022.10009714.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10009714

[307] K. Madan, A. Wagh, P. Pranjal, K. Madan, A. Wagh, P. Pranjal, M. K., W. A., P. P., Deep learning-based retinal oct segmentation: A comparative study, 2025 17th IEEE International Conference on Computational Intelligence and Communication Networks, CICN 2025 (2025) 412–418doi:10.1109/CICN67655.2025.11368110.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11368110

[308] J.-H. Wu, T. Nishida, R. N. Weinreb, J.-W. Lin, W. JH, N. T, W. RN, L. JW, Performances of machine learning in detecting glaucoma using fundus and retinal optical coherence tomography images: A meta-analysis, American journal of ophthalmology 237 (2022) 1–12. doi:10.1016/j.ajo.2021.12.008.
URL https://pubmed.ncbi.nlm.nih.gov/34942113/

[309] S. K. Devalla, T. H. Pham, S. K. Panda, L. Zhang, G. Subramanian, A. Swaminathan, C. Z. Yun, M. Rajan, S. Mohan, R. Krishnadas, V. Senthil, J. M. S. De Leon, T. A. Tun, C.-Y. Cheng, L. Schmetterer, S. Perera, T. Aung, A. H. Thiéry, M. J. A. Girard, D. SK, P. TH, P. SK, Z. L, S. G, S. A, Y. CZ, R. M, M. S, K. R, S. V, D. L. JMS, T. TA, C. CY, S. L, P. S, A. T, T. AH, G. MJA, C. Z. Yun, D. S.K., P. T.H., P. S.K., Y. C.Z.H.I., de Leon J.M.S., T. T.A., C. C.-Y., T. A.H., G. M.J.A., Towards label-free 3d segmentation of optical coherence tomography images of the optic nerve head using deep learning, Biomedical optics express 11 (11) (2020) 6356–6378. doi:10.1364/BOE.395934.
URL https://pubmed.ncbi.nlm.nih.gov/33282495/

[310] X. Liu, T. K. Ali, P. Singh, A. Shah, S. M. McKinney, P. Ruamviboonsuk, A. W. Turner, P. A. Keane, P. Chotcomwongse, V. Nganthavee, M. Chia, J. Huemer, J. Cuadros, R. Raman, G. S. Corrado, L. Peng, D. R. Webster, N. Hammel, A. V. Varadarajan, Y. Liu, R. Chopra, P. Bavishi, L. X, A. TK, S. P, S. A, M. SM, R. P, T. AW, K. PA, C. P, N. V, C. M, H. J, C. J, R. R, C. GS, P. L, W. DR, H. N, V. AV, L. Y, C. R, B. P, A. T.K., M. S.M., T. A.W., K. P.A., C. G.S., W. D.R., V. A.V., Deep learning to detect oct-derived diabetic macular edema from color retinal photographs: A multicenter validation study, Ophthalmology. Retina 6 (5) (2022) 398–410. doi:10.1016/j.oret.2021.12.021.
URL https://pubmed.ncbi.nlm.nih.gov/34999015/

[311] S. Mukherjee, T. De Silva, C. Duic, G. Jayakar, T. D. L. Keenan, A. T. Thavikulwat, E. Chew, C. Cukras, M. S, D. S. T, D. C, J. G, K. TDL, T. AT, C. E, C. C, K. T.D.L., T. A.T., Validation of deep learning-based automatic retinal layer segmentation algorithms for age-related macular degeneration with 2 spectral-domain oct devices, Ophthalmology science 5 (3) (2025) 100670. doi:10.1016/j.xops.2024.100670.
URL https://pubmed.ncbi.nlm.nih.gov/40091912/

[312] N. A. Mumtaz, A. Luthfiarta, A. Nugraha, I. N. Dewi, U. M. Syarifah, D. Uqul, N. A. Mumtaz, A. Luthfiarta, A. Nugraha, I. N. Dewi, U. M. Syarifah, D. Uqul, M. N.A., L. A., N. A., D. I.N., S. U.M., U. D., A hybrid cbam-vision transformer for enhanced classification of retinal diseases in optical coherence tomography imagery, Proceedings - 2025 International Seminar on Application for Technology of Information and Communication: Advancing Electrical Engineering and Informatics Through Quantum Computing and Intelligence Technologies, iSemantic 2025 (2025) 345–350doi:10.1109/ISemantic67418.2025.11292304.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11292304

[313] M. Ghazal, S. S. Ali, A. H. Mahmoud, A. M. Shalaby, A. El-Baz, M. Ghazal, S. S. Ali, A. H. Mahmoud, A. M. Shalaby, A. El-Baz, G. M., A. S.S., M. A.H., S. A.M., E.-B. A., Accurate detection of non-proliferative diabetic retinopathy in optical coherence tomography images using convolutional neural networks, IEEE Access 8 (2020) 34387–34397. doi:10.1109/ACCESS.2020.2974158.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9011786

[314] T. Schlegl, S. M. Waldstein, H. Bogunovic, F. Endstraßer, A. Sadeghipour, A.-M. Philip, D. Podkowinski, B. S. Gerendas, G. Langs, U. Schmidt-Erfurth, S. T, W. SM, B. H, E. F, S. A, P. AM, P. D, G. BS, L. G, S.-E. U, W. S.M., P. A.-M., G. B.S., Fully automated detection and quantification of macular fluid in oct using deep learning, Ophthalmology 125 (4) (2018) 549–558. doi:10.1016/j.ophtha.2017.10.031.
URL https://pubmed.ncbi.nlm.nih.gov/29224926/

[315] M. B. Özdaş, F. Uysal, F. Hardalaç, Özdaş MB, U. F, H. F, Özdaş M.B., Classification of retinal diseases in optical coherence tomography images using artificial intelligence and firefly algorithm, Diagnostics (Basel, Switzerland) 13 (3) (2023). doi:10.3390/diagnostics13030433.
URL https://pubmed.ncbi.nlm.nih.gov/36766537/

[316] R. Alizadeh Eghtedar, A. Vard, M. Malekahmadi, A. Peyman, A. E. R, V. A, M. M, P. A, A new computer-aided diagnosis tool based on deep learning methods for automatic detection of retinal disorders from oct images, International ophthalmology 44 (1) (2024) 110. doi:10.1007/s10792-024-03033-9.
URL https://pubmed.ncbi.nlm.nih.gov/38396074/

[317] G. Kaur, R. Sridhar, R. N. K. Dhenia, I. J. Kanani, G. Kaur, R. Sridhar, R. N. K. Dhenia, I. J. Kanani, K. G., S. R., D. R.N.K., K. I.J., Explainable deep learning approach for automated classification of retinal diseases using oct images and grad-cam, 2025 IEEE 4th International Conference for Advancement in Technology, ICONAT 2025 (2025) 1–6doi:10.1109/ICONAT66879.2025.11362792.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11362792

[318] N. Rajagopalan, V. Narasimhan, S. Kunnavakkam Vinjimoor, J. Aiyer, R. N., N. V., K. V. S., A. J., Deep cnn framework for retinal disease diagnosis using optical coherence tomography images, Journal of Ambient Intelligence and Humanized Computing 12 (7) (2021) 7569–7580. doi:10.1007/s12652-020-02460-7.
URL https://www.scopus.com/pages/publications/85089663468?origin=resultslist

[319] I. Kayadibi, G. E. Güraksın, K. I., G. G.E., An early retinal disease diagnosis system using oct images via cnn-based stacking ensemble learning, International Journal for Multiscale Computational Engineering 21 (1) (2023) 1–25. doi:10.1615/IntJMultCompEng.2022043544.
URL https://www.scopus.com/pages/publications/85142335400?origin=resultslist

[320] E. B. Mariottoni, A. A. Jammal, C. N. Urata, S. I. Berchuck, A. C. Thompson, T. Estrela, F. A. Medeiros, M. EB, J. AA, U. CN, B. SI, T. AC, E. T, M. FA, M. E.B., J. A.A., U. C.N., B. S.I., T. A.C., M. F.A., Quantification of retinal nerve fibre layer thickness on optical coherence tomography with a deep learning segmentation-free approach, Scientific reports 10 (1) (2020) 402. doi:10.1038/s41598-019-57196-y.
URL https://pubmed.ncbi.nlm.nih.gov/31941958/

[321] A. A. Rani, C. Karthikeyini, C. R. Ravi, R. A.A., R. C.R., Eye disease prediction using deep learning and attention on oct scans, SN Computer Science 5 (8) (2024). doi:10.1007/s42979-024-03451-7.
URL https://www.scopus.com/pages/publications/85209713496?origin=resultslist

[322] K. A. Thakoor, S. C. Koorathota, D. C. Hood, P. Sajda, K. A. Thakoor, S. C. Koorathota, D. C. Hood, P. Sajda, T. KA, K. SC, H. DC, S. P, T. K.A., K. S.C., H. D.C., Robust and interpretable convolutional neural networks to detect glaucoma in optical coherence tomography images, IEEE Transactions on Biomedical Engineering 68 (8) (2021) 2456–2466. doi:10.1109/TBME.2020.3043215.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9286420

[323] A. Hayati, M. R. Abdol Homayuni, R. Sadeghi, H. Asadigandomani, M. Dashtkoohi, S. Eslami, M. Soleimani, H. A, A. H. MR, S. R, A. H, D. M, E. S, S. M, A. H. M.R., Advancing diabetic retinopathy screening: A systematic review of artificial intelligence and optical coherence tomography angiography innovations, Diagnostics (Basel, Switzerland) 15 (6) (2025). doi:10.3390/diagnostics15060737.
URL https://pubmed.ncbi.nlm.nih.gov/40150080/

[324] S. Schürer-Waldheim, P. Seeböck, H. Bogunović, B. S. Gerendas, U. Schmidt-Erfurth, S. Schurer-Waldheim, P. Seebock, H. Bogunovic, B. S. Gerendas, U. Schmidt-Erfurth, S.-W. S, S. P, B. H, G. BS, S.-E. U, G. B.S., Robust fovea detection in retinal oct imaging using deep learning, IEEE Journal of Biomedical and Health Informatics 26 (8) (2022) 3927–3937. doi:10.1109/JBHI.2022.3166068.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9754240

[325] Y. Hashimoto, T. Kiwaki, H. Sugiura, S. Asano, H. Murata, Y. Fujino, M. Matsuura, A. Miki, K. Mori, Y. Ikeda, T. Kanamoto, J. Yamagami, K. Inoue, M. Tanito, K. Yamanishi, R. Asaoka, H. Y, K. T, S. H, A. S, M. H, F. Y, M. M, M. A, M. K, I. Y, K. T, Y. J, I. K, T. M, Y. K, A. R, Predicting 10-2 visual field from optical coherence tomography in glaucoma using deep learning corrected with 24-2/30-2 visual field, Translational vision science & technology 10 (13) (2021) 28. doi:10.1167/tvst.10.13.28.
URL https://pubmed.ncbi.nlm.nih.gov/34812893/

[326] A. C. Mandal, A. Phatak, M. A.C., P. A., Optimizing deep learning based retinal diseases classification on optical coherence tomography scans, Proceedings of SPIE - The International Society for Optical Engineering 12632 (2023). doi:10.1117/12.2672249.
URL https://www.scopus.com/pages/publications/85184293161?origin=resultslist

[327] E. Parra-Mora, A. C. nas Gordon, R. Proença, L. A. da Silva Cruz, E. Parra-Mora, A. Cazanas-Gordon, R. Proenca, L. A. Da Silva Cruz, P.-M. E., C.-G. A., P. R., D. S. C. L.A., Epiretinal membrane detection in optical coherence tomography retinal images using deep learning, IEEE Access 9 (2021) 99201–99219. doi:10.1109/ACCESS.2021.3095655.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9477630

[328] E. Mourgues, V. Saunier, D. Smadja, D. Touboul, V. Saunier, M. E, S. V, S. D, T. D, Forme fruste keratoconus detection with oct corneal topography using artificial intelligence algorithms, Journal of cataract and refractive surgery 50 (12) (2024) 1247–1253. doi:10.1097/j.jcrs.0000000000001542.
URL https://pubmed.ncbi.nlm.nih.gov/39223730/

[329] R. Bunod, M. Lubrano, A. Pirovano, G. Chotard, E. Brasnu, S. Berlemont, A. Labbé, E. Augstburger, C. Baudouin, B. R, L. M, P. A, C. G, B. E, B. S, L. A, A. E, B. C, A deep learning system using optical coherence tomography angiography to detect glaucoma and anterior ischemic optic neuropathy, Journal of clinical medicine 12 (2) (2023). doi:10.3390/jcm12020507.
URL https://pubmed.ncbi.nlm.nih.gov/36675435/

[330] D. A. Pon Daniel, M. A. Santhana Sagaya, S. Chidambaranathan, P. D. D.A., S. S. M.A., Deep learning-based neuro computing to classify and diagnosis of ophthalmology by oct, Integrating Neurocomputing with Artificial Intelligence (2025) 159–187doi:10.1002/9781394335718.ch10.
URL https://www.scopus.com/pages/publications/105014660052?origin=resultslist

[331] T. Khatibi, K. T., Context-aware diagnosis of age-related macular degeneration (amd) stages: a unified approach using self-supervised u-net and graph neural networks on oct imaging, Iran Journal of Computer Science 8 (4) (2025) 1835–1851. doi:10.1007/s42044-025-00290-4.
URL https://www.scopus.com/pages/publications/105007335650?origin=resultslist

[332] R. Arian, A. Vard, R. Kafieh, G. Plonka, H. Rabbani, A. R, V. A, K. R, P. G, R. H, A new convolutional neural network based on combination of circlets and wavelets for macular oct classification, Scientific reports 13 (1) (2023) 22582. doi:10.1038/s41598-023-50164-7.
URL https://pubmed.ncbi.nlm.nih.gov/38114582/

[333] X. Liu, Y. Ding, Y. Zhang, J. Tang, L. X., D. Y., Z. Y., T. J., Multi-scale local-global transformer with contrastive learning for biomarkers segmentation in retinal oct images, Biocybernetics and Biomedical Engineering 44 (1) (2024) 231–246. doi:10.1016/j.bbe.2024.02.001.
URL https://www.scopus.com/pages/publications/85185293454?origin=resultslist

[334] S. Vishnu Priyan, R. Vinod Kumar, C. Moorthy, V. S. Nishok, N. VS, S. V. Priyan, R. V. Kumar, P. S.V., K. R.V., A fusion of deep neural networks and game theory for retinal disease diagnosis with oct images, Journal of X-ray science and technology 32 (4) (2024) 1011–1039. doi:10.3233/XST-240027.
URL https://pubmed.ncbi.nlm.nih.gov/38759091/

[335] A. C. Thompson, A. A. Jammal, S. I. Berchuck, E. B. Mariottoni, F. A. Medeiros, T. AC, J. AA, B. SI, M. EB, M. FA, T. A.C., J. A.A., B. S.I., M. E.B., M. F.A., Assessment of a segmentation-free deep learning algorithm for diagnosing glaucoma from optical coherence tomography scans, JAMA ophthalmology 138 (4) (2020) 333–339. doi:10.1001/jamaophthalmol.2019.5983.
URL https://pubmed.ncbi.nlm.nih.gov/32053142/

[336] J. Wang, G. Deng, W. Li, Y. Chen, F. Gao, H. Liu, Y. He, G. Shi, W. J, D. G, L. W, C. Y, G. F, L. H, H. Y, S. G, Deep learning for quality assessment of retinal oct images, Biomedical optics express 10 (12) (2019) 6057–6072. doi:10.1364/BOE.10.006057.
URL https://pubmed.ncbi.nlm.nih.gov/31853385/

[337] K. Garraoui, I. Rahmany, S. Dhahri, H. Tabia, D. Sidibé, H. Zgolli, N. Khlifa, G. K., R. I., D. S., T. H., S. D., Z. H., K. N., A deep learning approach for predicting the response to anti-vegf treatment in diabetic macular edema patients using optical coherence tomography images, International Conference on Agents and Artificial Intelligence 2 (2025) 453–462. doi:10.5220/0013181700003890.
URL https://www.scopus.com/pages/publications/105001706165?origin=resultslist

[338] A. Laouarem, C. Kara-Mohamed, E.-B. Bourennane, A. Hamdi-Cherif, L. A, K.-M. C, B. EB, H.-C. A, B. E.-B., Htc-retina: A hybrid retinal diseases classification model using transformer-convolutional neural network from optical coherence tomography images, Computers in biology and medicine 178 (2024) 108726. doi:10.1016/j.compbiomed.2024.108726.
URL https://pubmed.ncbi.nlm.nih.gov/38878400/

[339] R. Tennakoon, A. K. Gostar, R. Hoseinnezhad, A. Bab-Hadiashar, R. Tennakoon, A. K. Gostar, R. Hoseinnezhad, A. Bab-Hadiashar, T. R., G. A.K., H. R., B.-H. A., Retinal fluid segmentation in oct images using adversarial loss based convolutional neural networks, 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018) 2018-April (2018) 1436–1440. doi:10.1109/ISBI.2018.8363842.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8363842

[340] M. Zhao, S. Zhu, H. Shan, J. Feng, Z. M., Z. S., S. H., F. J., Automatic classification of retinal oct images based on convolutional neural network, Proceedings of SPIE - The International Society for Optical Engineering 11511 (2020). doi:10.1117/12.2567986.
URL https://www.scopus.com/pages/publications/85092576295?origin=resultslist

[341] G. Aresta, T. Araújo, B. Fazekas, J. Mai, U. Schmidt-Erfurth, H. Bogunović, G. Aresta, T. Araujo, B. Fazekas, J. Mai, U. Schmidt-Erfurth, H. Bogunovic, A. G., A. T., F. B., M. J., S.-E. U., B. H., Interactive deep learning-based retinal oct layer segmentation refinement by regressing translation maps, IEEE Access 12 (2024) 47009–47023. doi:10.1109/ACCESS.2024.3379015.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10474349

[342] Y. Jian, S. Borkovkina, W. Japongsori, A. Camino, M. V. Sarunic, Y. Jian, S. Borkovkina, W. Japongsori, A. Camino, M. V. Sarunic, J. Y., B. S., J. W., C. A., S. M.V., Real-time retinal layer segmentation of adaptive optics optical coherence tomography angiography with deep learning, 2020 IEEE Photonics Conference, IPC 2020 - Proceedings (2020) 1–2doi:10.1109/IPC47351.2020.9252343.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9252343

[343] S. Kumar, V. Dixit, M. Gupta, K. S., D. V., G. M., Deep learning approaches for retinal disease diagnosis: Insights from fundus and oct analysis, EPJ Web of Conferences 328 (2025). doi:10.1051/epjconf/202532801041.
URL https://www.scopus.com/pages/publications/105011084057?origin=resultslist

[344] F. Y. Shih, H. Patel, S. F.Y., P. H., Deep learning classification on optical coherence tomography retina images, International Journal of Pattern Recognition and Artificial Intelligence 34 (8) (2020). doi:10.1142/S0218001420520023.
URL https://www.scopus.com/pages/publications/85074088077?origin=resultslist

[345] J. Hong, A. A. Nanji, R. D. Stutzman, W. D. Chamberlain, X. Song, D. Huang, Y. Li, H. J, N. AA, S. RD, C. WD, S. X, H. D, L. Y, Artificial intelligence-driven detection of lasik using corneal optical coherence tomography maps, Translational vision science & technology 14 (3) (2025) 17. doi:10.1167/tvst.14.3.17.
URL https://pubmed.ncbi.nlm.nih.gov/40094700/

[346] X. Liu, C. Xue, M. Li, Y. Guo, W. Zhang, L. X, X. C, L. M, G. Y, Z. W, Retinal and choroidal microvasculature and structural analysis in octa for refractive amblyopia diagnosis using machine learning, Journal of optometry 18 (3) (2025) 100555. doi:10.1016/j.optom.2025.100555.
URL https://pubmed.ncbi.nlm.nih.gov/40334350/

[347] P. Powroznik, M. Skublewska-Paszkowska, R. Rejdak, K. Nowomiejska, P. P., S.-P. M., R. R., N. K., Automatic method of macular diseases detection using deep cnn-gru network in oct images, Acta Mechanica et Automatica 18 (4) (2024) 197–206. doi:10.2478/ama-2024-0074.
URL https://www.scopus.com/pages/publications/85210276659?origin=resultslist

[348] K. Gopinath, S. B. Rangrej, J. Sivaswamy, K. Gopinath, S. B. Rangrej, J. Sivaswamy, G. K., R. S.B., S. J., A deep learning framework for segmentation of retinal layers from oct images, 2017 4th IAPR Asian Conference on Pattern Recognition (ACPR) (2017) 888–893doi:10.1109/ACPR.2017.121.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8575940

[349] H. S. Ahmed, C. J. Thrishulamurthy, A. HS, T. CJ, Advancing diabetic retinopathy diagnosis: Leveraging optical coherence tomography imaging with convolutional neural networks, Romanian journal of ophthalmology 67 (4) (2023) 398–402. doi:10.22336/rjo.2023.63.
URL https://pubmed.ncbi.nlm.nih.gov/38239418/

[350] M. Mirsalehi, A. Langenbucher, M. M, L. A, A deep learning-based graphical user interface for predicting corneal ectasia scores from raw optical coherence tomography data, Diagnostics (Basel, Switzerland) 16 (2) (2026). doi:10.3390/diagnostics16020310.
URL https://pubmed.ncbi.nlm.nih.gov/41594286/

[351] V. Mohammadzadeh, Y. Liang, S. Moghimi, P. Xie, T. Nishida, G. Mahmoudinezhad, M. Eslani, E. Walker, A. Kamalipour, E. Micheletti, J.-H. Wu, M. Christopher, L. M. Zangwill, T. Javidi, R. N. Weinreb, M. V, L. Y, M. S, X. P, N. T, M. G, E. M, W. E, K. A, M. E, W. JH, C. M, Z. LM, J. T, W. RN, W. J.-H., Z. L.M., W. R.N., Detection of glaucoma progression on longitudinal series of en-face macular optical coherence tomography angiography images with a deep learning model, British Journal of Ophthalmology 108 (12) (2024) 1688–1693. doi:10.1136/bjo-2023-324528.
URL https://pubmed.ncbi.nlm.nih.gov/39117359/

[352] F. Li, W. Pan, W. Xiang, H. Zou, L. F, P. W, X. W, Z. H, Automatic segmentation of multitype retinal fluid from optical coherence tomography images using semisupervised deep learning network, British Journal of Ophthalmology 107 (9) (2023) 1350–1355. doi:10.1136/bjophthalmol-2022-321348.
URL https://pubmed.ncbi.nlm.nih.gov/35697498/

[353] R. V. Teja, S. R. Manne, A. Goud, M. A. Rasheed, K. K. Dansingani, J. Chhablani, K. K. Vupparaboina, S. Jana, R. V. Teja, S. Reddy Manne, A. Goud, M. A. Rasheed, K. K. Dansingani, J. Chhablani, K. K. Vupparaboina, S. Jana, T. RV, R. M. S, G. A, R. MA, D. KK, C. J, V. KK, J. S, T. R.V., R. M.A., D. K.K., V. K.K., Classification and quantification of retinal cysts in oct b-scans: Efficacy of machine learning methods, Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference 2019 (2019) 48–51. doi:10.1109/EMBC.2019.8857075.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8857075

[354] H. Xiong, Q. S. You, Y. Guo, J. Wang, B. Wang, L. Gao, C. J. Flaxel, S. T. Bailey, T. S. Hwang, Y. Jia, X. H, Y. QS, G. Y, W. J, W. B, G. L, F. CJ, B. ST, H. TS, J. Y, Y. Q.S., F. C.J., B. S.T., H. T.S., Deep learning-based signal-independent assessment of macular avascular area on 6×6 mm optical coherence tomography angiogram in diabetic retinopathy: a comparison to instrument-embedded software, British Journal of Ophthalmology 107 (1) (2023) 84–89. doi:10.1136/bjophthalmol-2020-318646.
URL https://pubmed.ncbi.nlm.nih.gov/34518161/

[355] A. Naik, B. Pavana, K. Sooda, N. A., S. K., Retinal disease classification from retinal-oct images using deep learning methods, Lecture Notes on Data Engineering and Communications Technologies 132 (2022) 95–104. doi:10.1007/978-981-19-2347-0_8.
URL https://www.scopus.com/pages/publications/85135525398?origin=resultslist

[356] W. Wei, J. Southern, K. Zhu, Y. Li, M. F. Cordeiro, K. Veselkov, W. W, S. J, Z. K, L. Y, C. MF, V. K, C. M.F., Deep learning to detect macular atrophy in wet age-related macular degeneration using optical coherence tomography, Scientific reports 13 (1) (2023) 8296. doi:10.1038/s41598-023-35414-y.
URL https://pubmed.ncbi.nlm.nih.gov/37217770/

[357] R. Darooei, M. Nazari, R. Kafieh, H. Rabbani, D. R, N. M, K. R, R. H, Loss-modified transformer-based u-net for accurate segmentation of fluids in optical coherence tomography images of retinal diseases, Journal of medical signals and sensors 13 (4) (2023) 253–260. doi:10.4103/jmss.jmss_52_22.
URL https://pubmed.ncbi.nlm.nih.gov/37809015/

[358] M. R. Oishe, S. Mahedy Hasan, M. F. Zibran, O. M.R., Z. M.F., Breaking the mold: Vit-cnn fusion for enhanced glaucoma prediction in oct images, Lecture Notes in Computer Science 15166 LNAI (2025) 326–344. doi:10.1007/978-3-031-81596-6_29.
URL https://www.scopus.com/pages/publications/105010226646?origin=resultslist

[359] G. Gao, Y. Chen, X. Zhang, W. Sun, W. Shen, G. G, C. Y, Z. X, S. W, S. W, Machine learning-based prediction of persistent diabetic macular edema using oct-derived structural biomarkers and clinical features, European journal of medical research (2026). doi:10.1186/s40001-026-04064-x.
URL https://pubmed.ncbi.nlm.nih.gov/41723562/

[360] F. Gan, F.-P. Wu, Y.-L. Zhong, G. F, W. FP, Z. YL, W. F.-P., Z. Y.-L., Artificial intelligence method based on multi-feature fusion for automatic macular edema (me) classification on spectral-domain optical coherence tomography (sd-oct) images, Frontiers in neuroscience 17 (2023) 1097291. doi:10.3389/fnins.2023.1097291.
URL https://pubmed.ncbi.nlm.nih.gov/36793539/

[361] R. Rajan, S. Kumar, R. R., Deep learning architectures for oct images retinal disease classification, SN Computer Science 6 (2) (2025). doi:10.1007/s42979-025-03715-w.
URL https://www.scopus.com/pages/publications/85219714541?origin=resultslist

[362] D. Le, M. N. Alam, J. I. Lim, R. Chan, X. Yao, L. D., A. M.N., L. J.I., Y. X., Deep learning for objective octa detection of diabetic retinopathy, Progress in Biomedical Optics and Imaging - Proceedings of SPIE 11218 (2020). doi:10.1117/12.2546586.
URL https://www.scopus.com/pages/publications/85094193798?origin=resultslist

[363] T. Araújo, G. Aresta, U. Schmidt-Erfurth, H. Bogunović, A. T, A. G, S.-E. U, B. H, Few-shot out-of-distribution detection for automated screening in retinal oct images using deep learning, Scientific reports 13 (1) (2023) 16231. doi:10.1038/s41598-023-43018-9.
URL https://pubmed.ncbi.nlm.nih.gov/37758754/

[364] A. Jeya Prabha, M. Sameera Fathimal, G. Meghana, S. Angeline Kirubha, Application program interface for automatic segmentation of retinal layers and fluids in optical coherence tomography - neovascular age related macular degeneration retinal images using deep learning models, International Journal of Imaging Systems and Technology 34 (2) (2024). doi:10.1002/ima.23002.
URL https://www.scopus.com/pages/publications/85177735147?origin=resultslist

[365] H. Hsu, C. Lin, C. Lu, J. Wang, T. Huang, H.-C. Hsu, C.-H. Lin, C.-K. Lu, J.-K. Wang, T.-L. Huang, H. H.-C., L. C.-H., L. C.-K., W. J.-K., H. T.-L., A lightweight cnn net for amd detection using oct volumes, Digest of Technical Papers - IEEE International Conference on Consumer Electronics 2022-January (2022) 01–04. doi:10.1109/ICCE53296.2022.9730562.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9730562

[366] B. Y. Xu, M. Chiang, A. A. Pardeshi, S. Moghimi, R. Varma, X. BY, C. M, P. AA, M. S, V. R, Deep neural network for scleral spur detection in anterior segment oct images: The chinese american eye study, Translational vision science & technology 9 (2) (2020) 18. doi:10.1167/tvst.9.2.18.
URL https://pubmed.ncbi.nlm.nih.gov/32818079/

[367] Y. Shin, H. Cho, H. C. Jeong, M. Seong, J.-W. Choi, W. J. Lee, S. Y, C. H, J. HC, S. M, C. JW, L. WJ, J. H.C., C. J.-W., L. W.J., Deep learning-based diagnosis of glaucoma using wide-field optical coherence tomography images, Journal of glaucoma 30 (9) (2021) 803–812. doi:10.1097/IJG.0000000000001885.
URL https://pubmed.ncbi.nlm.nih.gov/33979115/

[368] J. Yang, B. Wu, J. Wang, Y. Lu, Z. Zhao, Y. Ding, K. Tang, F. Lu, L. Ma, Y. J, W. B, W. J, L. Y, Z. Z, D. Y, T. K, L. F, M. L, Dry age-related macular degeneration classification from optical coherence tomography images based on ensemble deep learning architecture, Frontiers in medicine 11 (2024) 1438768. doi:10.3389/fmed.2024.1438768.
URL https://pubmed.ncbi.nlm.nih.gov/39444813/

[369] O. Tan, D. S. Greenfield, B. A. Francis, R. Varma, J. S. Schuman, D. Huang, D. Choi, T. O, G. DS, F. BA, V. R, S. JS, H. D, C. D, A hybrid deep learning classification of perimetric glaucoma using peripapillary nerve fiber layer reflectance and other oct parameters from three anatomy regions, ArXiv (2024).
URL https://pubmed.ncbi.nlm.nih.gov/38883241/

[370] L. Xu, R. Asaoka, T. Kiwaki, H. Murata, Y. Fujino, M. Matsuura, Y. Hashimoto, S. Asano, A. Miki, K. Mori, Y. Ikeda, T. Kanamoto, J. Yamagami, K. Inoue, M. Tanito, K. Yamanishi, X. L, A. R, K. T, M. H, F. Y, M. M, H. Y, A. S, M. A, M. K, I. Y, K. T, Y. J, I. K, T. M, Y. K, Predicting the glaucomatous central 10-degree visual field from optical coherence tomography using deep learning and tensor regression, American journal of ophthalmology 218 (2020) 304–313. doi:10.1016/j.ajo.2020.04.037.
URL https://pubmed.ncbi.nlm.nih.gov/32387432/

[371] B. Ait Hammou, R. Duval, M.-C. Boucher, F. Cheriet, A. H. B, D. R, B. MC, C. F, B. M.-C., Dtg: Dual transformers-based generative adversarial networks for retinal 2d/3d oct image classification, Medical image analysis 109 (2026) 103915. doi:10.1016/j.media.2025.103915.
URL https://pubmed.ncbi.nlm.nih.gov/41518764/

[372] M. Heisler, S. Karst, J. Lo, Z. Mammo, T. Yu, S. Warner, D. Maberley, M. F. Beg, E. V. Navajas, M. V. Sarunic, H. M, K. S, L. J, M. Z, Y. T, W. S, M. D, B. MF, N. EV, S. MV, B. M.F., N. E.V., S. M.V., Ensemble deep learning for diabetic retinopathy detection using optical coherence tomography angiography, Translational vision science & technology 9 (2) (2020) 20. doi:10.1167/tvst.9.2.20.
URL https://pubmed.ncbi.nlm.nih.gov/32818081/

[373] T. D. L. Keenan, T. E. Clemons, A. Domalpally, M. J. Elman, M. Havilio, E. Agrón, G. Benyamini, E. Y. Chew, K. TDL, C. TE, D. A, E. MJ, H. M, A. E, B. G, C. EY, K. T.D.L., C. T.E., E. M.J., C. E.Y., Retinal specialist versus artificial intelligence detection of retinal fluid from oct: Age-related eye disease study 2: 10-year follow-on study, Ophthalmology 128 (1) (2021) 100–109. doi:10.1016/j.ophtha.2020.06.038.
URL https://pubmed.ncbi.nlm.nih.gov/32598950/

[374] M. Elkholy, M. A. Marzouk, E. M., M. M.A., Deep learning-based classification of eye diseases using convolutional neural network for oct images, Frontiers in Computer Science 5 (2023). doi:10.3389/fcomp.2023.1252295.
URL https://www.scopus.com/pages/publications/85183857727?origin=resultslist

[375] C. Panthier, P. Zeboulon, H. Rouger, J. Bijon, D. Gatinel, P. C, Z. P, R. H, B. J, G. D, Catalyze: a deep learning approach for cataract assessment and grading on ss-oct images, Journal of cataract and refractive surgery 51 (3) (2025) 222–228. doi:10.1097/j.jcrs.0000000000001598.
URL https://pubmed.ncbi.nlm.nih.gov/39680679/

[376] C. Yu, S. Xie, S. Niu, Z. Ji, W. Fan, S. Yuan, Q. Liu, Q. Chen, Y. C, X. S, N. S, J. Z, F. W, Y. S, L. Q, C. Q, Hyper-reflective foci segmentation in sd-oct retinal images with diabetic retinopathy using deep convolutional neural networks, Medical physics 46 (10) (2019) 4502–4519. doi:10.1002/mp.13728.
URL https://pubmed.ncbi.nlm.nih.gov/31315159/

[377] Y. Guo, T. T. Hormel, H. Xiong, J. Wang, T. S. Hwang, Y. Jia, G. Y, H. TT, X. H, W. J, H. TS, J. Y, H. T.T., H. T.S., Automated segmentation of retinal fluid volumes from structural and angiographic optical coherence tomography using deep learning, Translational vision science & technology 9 (2) (2020) 54. doi:10.1167/tvst.9.2.54.
URL https://pubmed.ncbi.nlm.nih.gov/33110708/

[378] F. Eckardt, R. Mittas, N. Horlava, J. Schiefelbein, B. Asani, S. Michalakis, M. Gerhardt, C. Priglinger, D. Keeser, N. Koutsouleris, S. Priglinger, F. Theis, T. Peng, B. Schworm, E. F, M. R, H. N, S. J, A. B, M. S, G. M, P. C, K. D, K. N, P. S, T. F, P. T, S. B, Deep learning-based retinal layer segmentation in optical coherence tomography scans of patients with inherited retinal diseases, Klinische Monatsblatter fur Augenheilkunde 242 (9) (2025) 903–911. doi:10.1055/a-2227-3742.
URL https://pubmed.ncbi.nlm.nih.gov/38086412/

[379] T. Schlosser, F. Beuth, T. Meyer, A. Sampath Kumar, G. Stolze, O. Furashova, K. Engelmann, D. Kowerko, Visual acuity prediction on real-life patient data using a machine learning based multistage system, Scientific Reports 14 (1) (2024) 5532. doi:10.1038/s41598-024-54482-2.
URL https://doi.org/10.1038/s41598-024-54482-2

[380] I. T. Fırat, M. Fırat, H. Erbali, T. Tuncer, F. İT, F. M, E. H, T. T, F. İ.T., Predicting hfa 30-2 visual fields with deep learning from multimodal oct-fundus feature fusion and structure-function discordance analysis, Journal of imaging informatics in medicine (2026). doi:10.1007/s10278-025-01798-8.
URL https://pubmed.ncbi.nlm.nih.gov/41559508/

[381] X. Wei, R. Sui, W. X, S. R, A review of machine learning algorithms for retinal cyst segmentation on optical coherence tomography, Sensors (Basel, Switzerland) 23 (6) (2023). doi:10.3390/s23063144.
URL https://pubmed.ncbi.nlm.nih.gov/36991857/

[382] H. Zhu, J. Ji, J.-W. Lin, J. Wang, Y. Zheng, P. Xie, C. Liu, T. K. Ng, J. Huang, Y. Xiong, H. Wu, L. Lin, M. Zhang, G. Zhang, Z. H, J. J, L. JW, W. J, Z. Y, X. P, L. C, N. TK, H. J, X. Y, W. H, L. L, Z. M, Z. G, L. J.-W., N. T.K., Development and validation of a 3-d deep learning system for diabetic macular oedema classification on optical coherence tomography images, BMJ open 15 (5) (2025) e099167. doi:10.1136/bmjopen-2025-099167.
URL https://pubmed.ncbi.nlm.nih.gov/40449950/

[383] T. T. Hormel, W. T. Beaulieu, J. Wang, J. K. Sun, Y. Jia, H. TT, B. WT, W. J, S. JK, J. Y, H. T.T., B. W.T., S. J.K., Artificial intelligence versus rules-based approach for segmenting nonperfusion area in a drcr retina network optical coherence tomography angiography dataset, Investigative ophthalmology & visual science 66 (3) (2025) 22. doi:10.1167/iovs.66.3.22.
URL https://pubmed.ncbi.nlm.nih.gov/40062815/

[384] H. Mekonen, T. Tadesse, O. Krejcar, K. Abdella, D. Assefa, M. H., T. T., K. O., A. K., A. D., Gcc aware glaucoma detection using macula oct image analysis based on deep cnn, Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 14811 LNAI (2024) 319–333. doi:10.1007/978-3-031-70819-0_25.
URL https://www.scopus.com/pages/publications/85204537465?origin=resultslist

[385] H. Bogunović, V. Mares, G. S. Reiter, U. Schmidt-Erfurth, B. H, M. V, R. GS, S.-E. U, R. G.S., Predicting treat-and-extend outcomes and treatment intervals in neovascular age-related macular degeneration from retinal optical coherence tomography using artificial intelligence, Frontiers in medicine 9 (2022) 958469. doi:10.3389/fmed.2022.958469.
URL https://pubmed.ncbi.nlm.nih.gov/36017006/

[386] C. Germanese, A. Anwer, P. Eid, L.-A. Steinberg, C. Guenancia, P.-H. Gabrielle, C. Creuzot-Garcher, F. Meriaudeau, L. Arnould, S. LA, G. PH, Artificial intelligence-based prediction of neurocardiovascular risk score from retinal swept-source optical coherence tomography-angiography, Scientific reports 14 (1) (2024) 27089. doi:10.1038/s41598-024-78587-w.
URL https://pubmed.ncbi.nlm.nih.gov/39511360/

[387] L.-C. Sun, S.-I. Pao, K.-H. Huang, C.-Y. Wei, K.-F. Lin, P.-N. Chen, S. LC, P. SI, H. KH, W. CY, L. KF, C. PN, S. L.-C., P. S.-I., H. K.-H., W. C.-Y., L. K.-F., C. P.-N., Generative adversarial network-based deep learning approach in classification of retinal conditions with optical coherence tomography images, Graefe's Archive for Clinical and Experimental Ophthalmology 261 (5) (2023) 1399–1412. doi:10.1007/s00417-022-05919-9.
URL https://pubmed.ncbi.nlm.nih.gov/36441228/

[388] B. Braaf, S. Donner, N. Uribe-Patarroyo, B. E. Bouma, B. J. Vakoc, B. B, D. S, U.-P. N, B. BE, V. BJ, A neural network approach to quantify blood flow from retinal oct intensity time-series measurements, Scientific reports 10 (1) (2020) 9611. doi:10.1038/s41598-020-66158-8.
URL https://pubmed.ncbi.nlm.nih.gov/32541887/

[389] D. B. Russakoff, S. S. Mannil, J. D. Oakley, A. R. Ran, C. Y. Cheung, S. Dasari, M. Riyazzuddin, S. Nagaraj, H. L. Rao, D. Chang, R. T. Chang, R. DB, M. SS, O. JD, R. AR, C. CY, D. S, R. M, N. S, R. HL, C. D, C. RT, R. D.B., M. S.S., O. J.D., R. A.R., C. C.Y., R. H.L., C. R.T., A 3d deep learning system for detecting referable glaucoma using full oct macular cube scans, Translational vision science & technology 9 (2) (2020) 12. doi:10.1167/tvst.9.2.12.
URL https://pubmed.ncbi.nlm.nih.gov/32704418/

[390] P. Kaushik, S. Choudhary, P. Kaushik, S. Choudhary, K. P., C. S., Retinal disease detection: Using deep learning for accurate diagnosis with optical coherence tomography scans, 2024 International Conference on Advances in Computing, Communication and Materials (ICACCM) (2024) 1–6doi:10.1109/ICACCM61117.2024.11059086.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11059086

[391] K. Park, J. Kim, S. Kim, J. Shin, P. K, K. J, K. S, S. J, Prediction of visual field from swept-source optical coherence tomography using deep learning algorithms, Graefe's Archive for Clinical and Experimental Ophthalmology 258 (11) (2020) 2489–2499. doi:10.1007/s00417-020-04909-z.
URL https://pubmed.ncbi.nlm.nih.gov/32845372/

[392] C. Chase, A. Elsawy, T. Eleiwa, E. Ozcan, M. Tolba, M. Abou Shousha, C. C, E. A, E. T, O. E, T. M, A. S. M, Comparison of autonomous as-oct deep learning algorithm and clinical dry eye tests in diagnosis of dry eye disease, Clinical ophthalmology (Auckland, N.Z.) 15 (2021) 4281–4289. doi:10.2147/OPTH.S321764.
URL https://pubmed.ncbi.nlm.nih.gov/34707347/

[393] C. K. Yoon, H. W. Lee, H. W. Kim, J. L. Kim, Y. CK, L. HW, K. HW, K. JL, Y. C.K., L. H.W., K. H.W., K. J.L., Deep learning based retinal hard exudates quantification of optical coherence tomography, International journal of retina and vitreous 11 (1) (2025) 114. doi:10.1186/s40942-025-00715-z.
URL https://pubmed.ncbi.nlm.nih.gov/41107968/

[394] P. Zéboulon, C. Panthier, H. Rouger, J. Bijon, W. Ghazal, D. Gatinel, Z. P, P. C, R. H, B. J, G. W, G. D, Development and validation of a pixel wise deep learning model to detect cataract on swept-source optical coherence tomography images, Journal of optometry 15 Suppl 1 (Suppl 1) (2022) S43–S49. doi:10.1016/j.optom.2022.08.003.
URL https://pubmed.ncbi.nlm.nih.gov/36229338/

[395] S. Takinami, Y. Ueno, H. Mori, Y. Morita, J. Seita, T. Oshika, T. S, U. Y, M. H, M. Y, S. J, O. T, Advancement of cataract classification with artificial intelligence using anterior segment optical coherence tomography images with self-supervised vision transformer, Japanese journal of ophthalmology (2026). doi:10.1007/s10384-025-01323-y.
URL https://pubmed.ncbi.nlm.nih.gov/41593412/

[396] I. Banerjee, L. de Sisternes, J. A. Hallak, T. Leng, A. Osborne, P. J. Rosenfeld, G. Gregori, M. Durbin, D. Rubin, B. I, de Sisternes L, H. JA, L. T, O. A, R. PJ, G. G, D. M, R. D, H. J.A., R. P.J., Prediction of age-related macular degeneration disease using a sequential deep learning approach on longitudinal sd-oct imaging biomarkers, Scientific reports 10 (1) (2020) 15434. doi:10.1038/s41598-020-72359-y.
URL https://pubmed.ncbi.nlm.nih.gov/32963300/

[397] G. A. Muntean, A. Marginean, A. Groza, I. Damian, S. A. Roman, M. C. Hapca, M. V. Muntean, S. D. Nicoară, M. GA, M. A, G. A, D. I, R. SA, H. MC, M. MV, N. SD, M. G.A., R. S.A., H. M.C., M. M.V., N. S.D., The predictive capabilities of artificial intelligence-based oct analysis for age-related macular degeneration progression-a systematic review, Diagnostics (Basel, Switzerland) 13 (14) (2023). doi:10.3390/diagnostics13142464.
URL https://pubmed.ncbi.nlm.nih.gov/37510207/

[398] X. Liu, D. Zhang, X. Zhu, J. Tang, X. Liu, D. Zhang, X. Zhu, J. Tang, L. X., Z. D., Z. X., T. J., Vct-net: An octa retinal vessel segmentation network based on convolution and transformer, 2022 IEEE International Conference on Image Processing (ICIP) (2022) 2656–2660doi:10.1109/ICIP46576.2022.9898038.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9898038

[399] B. Awan, M. Elsaigh, M. Hesham Gamal, S. E. Elbahnasawy, M. Badee, A. B, E. M, H. G. M, E. SE, B. M, Integrating human expertise with artificial intelligence (ai) models for optical coherence tomography (oct) retinal fluid and pathology quantification: A systematic review, Cureus 17 (12) (2025) e100223. doi:10.7759/cureus.100223.
URL https://pubmed.ncbi.nlm.nih.gov/41466814/

[400] D. Allegrini, R. Raimondi, T. Sorrentino, D. Tripepi, E. Stradiotto, M. Caruso, F. P. De Rosa, M. R. Romano, A. D, R. R, S. T, T. D, S. E, C. M, D. R. FP, R. MR, D. R. F.P., R. M.R., The effect of optical degradation from cataract using a new deep learning optical coherence tomography segmentation algorithm, Graefe's Archive for Clinical and Experimental Ophthalmology 262 (2) (2024) 431–440. doi:10.1007/s00417-023-06261-4.
URL https://pubmed.ncbi.nlm.nih.gov/37843567/

[401] A. Heinke, H. Zhang, D. Deussen, C. M. B. Galang, A. Warter, F. G. P. Kalaw, D.-U. G. Bartsch, L. Cheng, C. An, T. Nguyen, W. R. Freeman, H. A, Z. H, D. D, G. CMB, W. A, K. FGP, B. DG, C. L, A. C, N. T, F. WR, G. C.M.B., K. F.G.P., B. D.-U.G., F. W.R., Artificial intelligence for optical coherence tomography angiography-based disease activity prediction in age-related macular degeneration, Retina (Philadelphia, Pa.) 44 (3) (2024) 465–474. doi:10.1097/IAE.0000000000003977.
URL https://pubmed.ncbi.nlm.nih.gov/37988102/

[402] J. Hamwood, D. Alonso-Caneiro, S. A. Read, S. J. Vincent, M. J. Collins, H. J, A.-C. D, R. SA, V. SJ, C. MJ, R. S.A., V. S.J., C. M.J., Effect of patch size and network architecture on a convolutional neural network approach for automatic segmentation of oct retinal layers, Biomedical optics express 9 (7) (2018) 3049–3066. doi:10.1364/BOE.9.003049.
URL https://pubmed.ncbi.nlm.nih.gov/29984082/

[403] R. Venkatesh, P. Gandhi, A. Choudhary, G. Sehgal, K. Godani, S. Darade, R. Kathare, P. Hande, V. Prabhu, J. Chhablani, V. R, G. P, C. A, S. G, G. K, D. S, K. R, H. P, P. V, C. J, Predicting macular hole surgery outcomes: Integrating preoperative oct features with supervised machine learning statistical models, Indian journal of ophthalmology 73 (Suppl 1) (2025) S66–S71. doi:10.4103/IJO.IJO_1895_24.
URL https://pubmed.ncbi.nlm.nih.gov/39723867/

[404] M. M. Ajmal, R. Mumtaz, S. Mumtaz, P. Temdee, M. D. A. Asif, M. A. Khan, S. Khan, A. MM, M. R, M. S, T. P, A. MDA, K. MA, K. S, Deep learning based eye disease classification using optical coherence tomography (oct) images, Experimental eye research 268 (2026) 111017. doi:10.1016/j.exer.2026.111017.
URL https://pubmed.ncbi.nlm.nih.gov/41967163/

[405] A. Serener, S. Serte, A. Serener, S. Serte, S. A., S. S., Dry and wet age-related macular degeneration classification using oct images and deep learning, 2019 Scientific Meeting on Electrical-Electronics & Biomedical Engineering and Computer Science (EBBT) (2019) 1–4doi:10.1109/EBBT.2019.8741768.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8741768

[406] T. Kepp, H. Sudkamp, C. Von Der Burchard, H. Schenke, P. Koch, G. Hüttmann, J. Roider, M. P. Heinrich, H. Handels, K. T., S. H., V. D. B. C., S. H., K. P., H. G., R. J., H. M.P., H. H., Segmentation of retinal low-cost optical coherence tomography images using deep learning, Progress in Biomedical Optics and Imaging - Proceedings of SPIE 11314 (2020). doi:10.1117/12.2551324.
URL https://www.scopus.com/pages/publications/85083073837?origin=resultslist

[407] A. V. Varadarajan, P. Bavishi, P. Ruamviboonsuk, P. Chotcomwongse, S. Venugopalan, A. Narayanaswamy, J. Cuadros, K. Kanai, G. Bresnick, M. Tadarati, S. Silpa-Archa, J. Limwattanayingyong, V. Nganthavee, J. R. Ledsam, P. A. Keane, G. S. Corrado, L. Peng, D. R. Webster, V. AV, B. P, R. P, C. P, V. S, N. A, C. J, K. K, B. G, T. M, S.-A. S, L. J, N. V, L. JR, K. PA, C. GS, P. L, W. DR, V. A.V., L. J.R., K. P.A., C. G.S., W. D.R., Predicting optical coherence tomography-derived diabetic macular edema grades from fundus photographs using deep learning, Nature communications 11 (1) (2020) 130. doi:10.1038/s41467-019-13922-8.
URL https://pubmed.ncbi.nlm.nih.gov/31913272/

[408] K. Park, J. Kim, J. Lee, P. K, K. J, L. J, A deep learning approach to predict visual field using optical coherence tomography, PloS one 15 (7) (2020) e0234902. doi:10.1371/journal.pone.0234902.
URL https://pubmed.ncbi.nlm.nih.gov/32628672/

[409] S. Kinder, S. McNamara, C. Clark, B. Bearce, U. Thakuria, Y. A. Veturi, G. Deitz, T. E. de Carlo Forest, N. Mandava, M. Y. Kahook, P. Singh, J. Kalpathy-Cramer, K. S, M. S, C. C, B. B, T. U, V. YA, D. G, de Carlo Forest TE, M. N, K. MY, S. P, K.-C. J, V. Y.A., de Carlo Forest T.E., K. M.Y., Optic cup and disc segmentation of fundus images using artificial intelligence externally validated with optical coherence tomography measurements, Translational vision science & technology 14 (6) (2025) 30. doi:10.1167/tvst.14.6.30.
URL https://pubmed.ncbi.nlm.nih.gov/40552928/

[410] S. Asano, R. Asaoka, H. Murata, Y. Hashimoto, A. Miki, K. Mori, Y. Ikeda, T. Kanamoto, J. Yamagami, K. Inoue, A. S, A. R, M. H, H. Y, M. A, M. K, I. Y, K. T, Y. J, I. K, Predicting the central 10 degrees visual field in glaucoma by applying a deep learning algorithm to optical coherence tomography images, Scientific reports 11 (1) (2021) 2214. doi:10.1038/s41598-020-79494-6.
URL https://pubmed.ncbi.nlm.nih.gov/33500462/

[411] A. Alzu'bi, S. Momany, A. Aleshawi, M. Tashtoush, R. Al-Dwairi, A. A, M. S, A. A, T. M, A.-D. R, The utility of artificial intelligence in characterization and detecting causes of macular edema: A spectral-domain oct-based algorithm study, Experimental eye research 260 (2025) 110619. doi:10.1016/j.exer.2025.110619.
URL https://pubmed.ncbi.nlm.nih.gov/40889611/

[412] A. Li, A. K. Tandon, G. Sun, M. J. Dinkin, C. Oliveira, L. A, T. AK, S. G, D. MJ, O. C, T. A.K., D. M.J., Early detection of optic nerve changes on optical coherence tomography using deep learning for risk-stratification of papilledema and glaucoma, Journal of Neuro-Ophthalmology 44 (1) (2024) 47–52. doi:10.1097/WNO.0000000000001945.
URL https://pubmed.ncbi.nlm.nih.gov/37494177/

[413] P. Zhang, G. Quellec, S. Matta, L. Borderie, A. Le Guilcher, T. Thiery, B. Cochener, M. Lamard, Z. P, Q. G, M. S, B. L, L. G. A, T. T, C. B, L. M, A robust deep learning classifier for screening multiple retinal diseases on optical coherence tomography, Scientific reports 15 (1) (2025) 35334. doi:10.1038/s41598-025-19286-y.
URL https://pubmed.ncbi.nlm.nih.gov/41068219/

[414] A. Heinke, H. Zhang, K. Broniarek, K. Michalska-Małecka, W. Elsner, C. M. B. Galang, D. N. Deussen, A. Warter, F. Kalaw, I. Nagel, A. Agnihotri, N. N. Mehta, J. E. Klaas, V. Schmelter, I. Kozak, S. L. Baxter, D.-U. Bartsch, L. Cheng, C. An, T. Nguyen, W. R. Freeman, H. A, Z. H, B. K, M.-M. K, E. W, G. CMB, D. DN,

W. A, K. F, N. I, A. A, M. NN, K. JE, S. V, K. I, B. SL, B. DU, C. L, A. C, N. T, F. WR, G. C.M.B., D. D.N., M. N.N., K. J.E., B. S.L., B. D.-U., F. W.R., Cross-instrument optical coherence tomography-angiography (octa)-based prediction of age-related macular degeneration (amd) disease activity using artificial intelligence, Scientific reports 14 (1) (2024) 27085. doi:10.1038/s41598-024-78327-0.
URL https://pubmed.ncbi.nlm.nih.gov/39511248/

[415] L. R, L. S, R. L, S. L, R. Loganathan, S. Latha, Enhanced amd detection in oct images using glcm texture features with machine learning and cnn methods, Biomedical physics & engineering express 11 (2) (2025). doi:10.1088/2057-1976/ada6bc.
URL https://pubmed.ncbi.nlm.nih.gov/39773983/

[416] A. Beuse, I. V. Lopes, M. S. Spitzer, V. Druchkiv, C. Grohmann, C. Skevas, B. A, L. IV, S. MS, D. V, G. C, S. C, L. I.V., S. M.S., Deep learning detection of retinal detachment: Optical coherence tomography staging and estimation of duration of macular detachment, PloS one 20 (9) (2025) e0329951. doi:10.1371/journal.pone.0329951.
URL https://pubmed.ncbi.nlm.nih.gov/40892865/

[417] M. Miranda, J. Santos-Oliveira, A. M. Mendonça, V. Sousa, T. Melo, A. Carneiro, M. M, S.-O. J, M. AM, S. V, M. T, C. Â, M. A.M., Human versus artificial intelligence: Validation of a deep learning model for retinal layer and fluid segmentation in optical coherence tomography images from patients with age-related macular degeneration, Diagnostics (Basel, Switzerland) 14 (10) (2024). doi:10.3390/diagnostics14100975.
URL https://pubmed.ncbi.nlm.nih.gov/38786273/

[418] A. C. Thompson, A. Falconi, R. M. Sappington, T. AC, F. A, S. RM, T. A.C., S. R.M., Deep learning and optical coherence tomography in glaucoma: Bridging the diagnostic gap on structural imaging, Frontiers in ophthalmology 2 (2022) 937205. doi:10.3389/fopht.2022.937205.
URL https://pubmed.ncbi.nlm.nih.gov/38983522/

[419] Y. F. Garcia-Marin, D. Alonso-Caneiro, D. Fisher, S. J. Vincent, M. J. Collins, G.-M. YF, A.-C. D, F. D, V. SJ, C. MJ, G.-M. Y.F., V. S.J., C. M.J., Patch-based cnn for corneal segmentation of as-oct images: Effect of the number of classes and image quality upon performance, Computers in biology and medicine 152 (2023) 106342. doi:10.1016/j.compbiomed.2022.106342.
URL https://pubmed.ncbi.nlm.nih.gov/36481759/

[420] R. Anandh, G. Indirani, Iot based automatic detection of glaucoma disease in oct and fundus images using deep learning techniques, Journal of Green Engineering 10 (12) (2020) 13621–13643.
URL https://www.scopus.com/pages/publications/85099164899?origin=resultslist

[421] H. Almushattat, J. Hensman, Y. El Allali, C. de Vente, C. I. Sánchez, C. J. F. Boon, A. H, H. J, E. A. Y, de Vente C, S. CI, B. CJF, S. C.I., B. C.J.F., Validation of a deep learning model for the automated detection and quantification of cystoid macular oedema on optical coherence tomography in patients with retinitis pigmentosa, Acta ophthalmologica 103 (7) (2025) e524–e531. doi:10.1111/aos.17518.
URL https://pubmed.ncbi.nlm.nih.gov/40396533/

[422] X. Chen, J. Wang, T. Qian, J. Wang, Y. Ding, S. Zhang, J. Liao, Q. Cheng, T. Yang, M. Mateen, Y. Fan, Z. Song, J. Chen, S. Li, J. Hu, W. Yan, H. Chen, W. Wu, J. Huang, T. Y. Wong, X. Xu, C. X, W. J, Q. T, D. Y, Z. S, L. J, C. Q, Y. T, M. M, F. Y, S. Z, C. J, L. S, H. J, Y. W, C. H, W. W, H. J, W. TY, X. X, W. T.Y., An artificial intelligence cloud platform for oct-based retinal anomalies screening system in real clinical environments, NPJ digital medicine 8 (1) (2025) 559. doi:10.1038/s41746-025-01959-7.
URL https://pubmed.ncbi.nlm.nih.gov/40883412/

[423] R. Oh, M. Kang, J. Ahn, E. K. Lee, K. Bae, U. C. Park, K. H. Park, C. K. Yoon, O. R, K. M, A. J, L. EK, B. K, P. UC, P. KH, Y. CK, L. E.K., P. U.C., P. K.H., Y. C.K., Prediction of axial length from macular optical coherence tomography using deep learning model, Translational vision science & technology 13 (9) (2024) 14. doi:10.1167/tvst.13.9.14.
URL https://pubmed.ncbi.nlm.nih.gov/39264604/

[424] J. Kim, L. Tran, J. Kim, L. Tran, K. J., T. L., Retinal disease classification from oct images using deep learning algorithms, 2021 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology, CIBCB 2021 (2021) 1–6doi:10.1109/CIBCB49929.2021.9562919.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9562919

[425] S. Yasuda, T. Hasegawa, S. Yoshimoto, K. Sugisawa, M. Ohno, Y. Shiotani, R. Teramatsu, H. Iemura, Y. Shibata, K. Ohno-Matsui, T. Yoshida, Y. S, H. T, Y. S, S. K, O. M, S. Y, T. R, I. H, S. Y, O.-M. K, Y. T, Deep learning-based prediction of visual field mean deviation from numeric oct data in glaucoma, Journal of imaging informatics in medicine (2026). doi:10.1007/s10278-026-01866-7.
URL https://pubmed.ncbi.nlm.nih.gov/41741850/

[426] J. Saeidian, T. Mahmoudi, H. Riazi-Esfahani, Z. Montazeriani, A. Khodabande, M. Zarei, N. Ebrahimiadib, B. Jafari, A. Afzal Aghaei, H. Azimi, E. Khalili Pour, S. J, M. T, R.-E. H, M. Z, K. A, Z. M, E. N, J. B, A. A. A, A. H, K. P. E, Automated assessment of the smoothness of retinal layers in optical coherence tomography images using a machine learning algorithm, BMC medical imaging 23 (1) (2023) 21. doi:10.1186/s12880-023-00976-w.
URL https://pubmed.ncbi.nlm.nih.gov/36732684/

[427] D. Romo-Bucheli, U. S. Erfurth, H. Bogunović, D. Romo-Bucheli, U. S. Erfurth, H. Bogunovic, R.-B. D, E. US, B. H, E. U.S., End-to-end deep learning model for predicting treatment requirements in neovascular amd from longitudinal retinal oct imaging, IEEE Journal of Biomedical and Health Informatics 24 (12) (2020) 3456–3465. doi:10.1109/JBHI.2020.3000136.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9108398

[428] M. Wilson, R. Chopra, M. Z. Wilson, C. Cooper, P. MacWilliams, Y. Liu, E. Wulczyn, D. Florea, C. O. Hughes, A. Karthikesalingam, H. Khalid, S. Vermeirsch, L. Nicholson, P. A. Keane, K. Balaskas, C. J. Kelly, W. M, C. R, W. MZ, C. C, M. P, L. Y, W. E, F. D, H. CO, K. A, K. H, V. S, N. L, K. PA, B. K, K. CJ, W. M.Z., H. C.O., K. P.A., K. C.J., Validation and clinical applicability of whole-volume automated segmentation of optical coherence tomography in retinal disease using deep learning, JAMA ophthalmology 139 (9) (2021) 964–973. doi:10.1001/jamaophthalmol.2021.2273.
URL https://pubmed.ncbi.nlm.nih.gov/34236406/

[429] H. M. S. S. Herath, S. L. P. Yasakethu, N. Madusanka, M. Yi, B.-I. Lee, H. HMSS, Y. SLP, M. N, Y. M, L. BI, L. B.-I., Comparative analysis of deep learning architectures for macular hole segmentation in oct images: A performance evaluation of u-net variants, Journal of imaging 11 (2) (2025). doi:10.3390/jimaging11020053.
URL https://pubmed.ncbi.nlm.nih.gov/39997555/

[430] L. Kandakji, S. Liu, S. Balal, I. Moghul, B. Allan, S. Tuft, D. Gore, N. Pontikos, K. L, L. S, B. S, M. I, A. B, T. S, G. D, P. N, An open-source deep learning framework for automated corneal segmentation in anterior segment optical coherence tomography with cross-device external validation, Cornea (2026). doi:10.1097/ICO.0000000000004130.
URL https://pubmed.ncbi.nlm.nih.gov/41962147/

[431] K. S. Kooner, A. Angirekula, A. H. Treacher, G. Al-Humimat, M. F. Marzban, A. Chen, R. Pradhan, N. Tunga, C. Wang, P. Ahuja, H. Zuberi, A. A. Montillo, K. KS, A. A, T. AH, A.-H. G, M. MF, C. A, P. R, T. N, W. C, A. P, Z. H, M. AA, K. K.S., T. A.H., M. M.F., M. A.A., Glaucoma diagnosis through the integration of optical coherence tomography/angiography and machine learning diagnostic models, Clinical ophthalmology (Auckland, N.Z.) 16 (2022) 2685–2697. doi:10.2147/OPTH.S367722.
URL https://pubmed.ncbi.nlm.nih.gov/36003072/

[432] L. Fang, D. Cunefare, C. Wang, R. H. Guymer, S. Li, S. Farsiu, F. L, C. D, W. C, G. RH, L. S, F. S, G. R.H., Automatic segmentation of nine retinal layer boundaries in oct images of non-exudative amd patients using deep learning and graph search, Biomedical optics express 8 (5) (2017) 2732–2744. doi:10.1364/BOE.8.002732.
URL https://pubmed.ncbi.nlm.nih.gov/28663902/

[433] C. C. S. Valentim, A. K. Wu, S. Yu, N. Manivannan, Q. Zhang, J. Cao, W. Song, V. Wang, H. Kang, A. Kalur, A. I. Iyer, T. Conti, R. P. Singh, K. E. Talcott, V. CCS, W. AK, Y. S, M. N, Z. Q, C. J, S. W, W. V, K. H, K. A, I. AI, C. T, S. RP, T. KE, V. C.C.S., W. A.K., I. A.I., S. R.P., T. K.E., Deep learning-based algorithm for the detection of idiopathic full thickness macular holes in spectral domain optical coherence tomography, International journal of retina and vitreous 10 (1) (2024) 9. doi:10.1186/s40942-024-00526-8.
URL https://pubmed.ncbi.nlm.nih.gov/38263402/

[434] M. Zhalechian, M. P. Van Oyen, M. S. Lavieri, C. G. De Moraes, C. A. Girkin, M. A. Fazio, R. N. Weinreb, C. Bowd, J. M. Liebmann, L. M. Zangwill, C. A. Andrews, J. D. Stein, Z. M, V. O. MP, L. MS, D. M. CG, G. CA, F. MA, W. RN, B. C, L. JM, Z. LM, A. CA, S. JD, V. O. M.P., L. M.S., D. M. C.G., G. C.A., F. M.A., W. R.N., L. J.M., Z. L.M., A. C.A., S. J.D., Augmenting kalman filter machine learning models with data from oct to predict future visual field loss: An analysis using data from the african descent and glaucoma evaluation study and the diagnostic innovation in glaucoma study, Ophthalmology science 2 (1) (2022) 100097. doi:10.1016/j.xops.2021.100097.
URL https://pubmed.ncbi.nlm.nih.gov/36246178/

[435] S. Fragiotta, F. Grassi, S. Abdolrahimzadeh, F. S., G. F., A. S., Implementing predictive models in artificial intelligence through oct biomarkers for age-related macular degeneration, Photonics 10 (2) (2023). doi:10.3390/photonics10020149.
URL https://www.scopus.com/pages/publications/85149068717?origin=resultslist

[436] M. Zang, P. Mukund, B. Forsyth, A. F. Laine, K. A. Thakoor, M. Zang, P. Mukund, B. Forsyth, A. F. Laine, K. A. Thakoor, Z. M, M. P, F. B, L. AF, T. KA, L. A.F., T. K.A., Predicting clinician fixations on glaucoma oct reports via cnn-based saliency prediction methods, IEEE Open Journal of Engineering in Medicine and Biology 5 (2024) 191–197. doi:10.1109/OJEMB.2024.3367492.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10440538

[437] Q. Jiang, Y. Fan, M. Li, S. Fang, W. Zhu, D. Xiang, T. Peng, X. Chen, X. Xu, F. Shi, J. Q., F. Y., L. M., F. S., Z. W., X. D., P. T., C. X., X. X., S. F., Hyformer: a hybrid transformer-cnn architecture for retinal oct image segmentation, Biomedical Optics Express 15 (11) (2024) 6156–6170. doi:10.1364/BOE.538959.
URL https://www.scopus.com/pages/publications/85209071652?origin=resultslist

[438] G. García, R. d. Amor, A. Colomer, V. Naranjo, G. Garcia, R. Del Amor, A. Colomer, V. Naranjo, G. G., D. A. R., C. A., N. V., Glaucoma detection from raw circumpapillary oct images using fully convolutional neural networks, 2020 IEEE International Conference on Image Processing (ICIP) 2020-October (2020) 2526–2530. doi:10.1109/ICIP40778.2020.9190916.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9190916

[439] Y. Guo, T. T. Hormel, L. Gao, Q. You, B. Wang, C. J. Flaxel, S. T. Bailey, D. Choi, D. Huang, T. S. Hwang, Y. Jia, G. Y, H. TT, G. L, Y. Q, W. B, F. CJ, B. ST, C. D, H. D, H. TS, J. Y, H. T.T., F. C.J., B. S.T., H. T.S., Quantification of nonperfusion area in montaged widefield oct angiography using deep learning in diabetic retinopathy, Ophthalmology science 1 (2) (2021) 100027. doi:10.1016/j.xops.2021.100027.
URL https://pubmed.ncbi.nlm.nih.gov/36249293/

[440] T. Sogawa, H. Tabuchi, D. Nagasato, H. Masumoto, Y. Ikuno, H. Ohsugi, N. Ishitobi, Y. Mitamura, S. T, T. H, N. D, M. H, I. Y, O. H, I. N, M. Y, Accuracy of a deep convolutional neural network in the detection of myopic macular diseases using swept-source optical coherence tomography, PloS one 15 (4) (2020) e0227240. doi:10.1371/journal.pone.0227240.
URL https://pubmed.ncbi.nlm.nih.gov/32298265/

[441] B. Chuter, V. Joshi, S. Hallaj, E. Walker, C. Bowd, A. Belghith, M. H. Goldbaum, A. Grzybowski, M. A. Fazio, C. A. Girkin, C. G. De Moraes, J. M. Liebmann, R. N. Weinreb, L. M. Zangwill, M. Christopher, C. B, J. V, H. S, W. E, B. C, B. A, G. MH, G. A, F. MA, G. CA, D. M. CG, L. JM, W. RN, Z. LM, C. M, G. M.H., F. M.A., G. C.A., D. M. C.G., L. J.M., W. R.N., Z. L.M., A novel multimodal implementation of a foundation

artificial intelligence model using optic nerve head fundus photographs and oct imaging for glaucoma detection, Ophthalmology science 6 (2) (2026) 101012. doi:10.1016/j.xops.2025.101012.
URL https://pubmed.ncbi.nlm.nih.gov/41631246/

[442] L. Wei, W. He, J. Wang, K. Zhang, Y. Du, J. Qi, J. Meng, X. Qiu, L. Cai, Q. Fan, Z. Zhao, Y. Tang, S. Ni, H. Guo, Y. Song, X. He, D. Ding, Y. Lu, X. Zhu, W. L, H. W, W. J, Z. K, D. Y, Q. J, M. J, Q. X, C. L, F. Q, Z. Z, T. Y, N. S, G. H, S. Y, H. X, D. D, L. Y, Z. X, An optical coherence tomography-based deep learning algorithm for visual acuity prediction of highly myopic eyes after cataract surgery, Frontiers in cell and developmental biology 9 (2021) 652848. doi:10.3389/fcell.2021.652848.
URL https://pubmed.ncbi.nlm.nih.gov/34124042/

[443] G. García, R. Del Amor, A. Colomer, R. Verdú-Monedero, J. Morales-Sánchez, V. Naranjo, G. G, D. A. R, C. A, V.-M. R, M.-S. J, N. V, Circumpapillary oct-focused hybrid learning for glaucoma grading using tailored prototypical neural networks, Artificial intelligence in medicine 118 (2021) 102132. doi:10.1016/j.artmed.2021.102132.
URL https://pubmed.ncbi.nlm.nih.gov/34412848/

[444] K. A. Barella, V. P. Costa, V. Gonçalves Vidotti, F. R. Silva, M. Dias, E. S. Gomi, B. KA, C. VP, G. V. V, S. FR, D. M, G. ES, Glaucoma diagnostic accuracy of machine learning classifiers using retinal nerve fiber layer and optic nerve data from sd-oct, Journal of ophthalmology 2013 (2013) 789129. doi:10.1155/2013/789129.
URL https://pubmed.ncbi.nlm.nih.gov/24369495/

[445] S. Kuwayama, Y. Ayatsuka, D. Yanagisono, T. Uta, H. Usui, A. Kato, N. Takase, Y. Ogura, T. Yasukawa, K. S, A. Y, Y. D, U. T, U. H, K. A, T. N, O. Y, Y. T, Automated detection of macular diseases by optical coherence tomography and artificial intelligence machine learning of optical coherence tomography images, Journal of ophthalmology 2019 (2019) 6319581. doi:10.1155/2019/6319581.
URL https://pubmed.ncbi.nlm.nih.gov/31093370/

[446] I. A. Viedma, D. Alonso-Caneiro, S. A. Read, M. J. Collins, V. IA, A.-C. D, R. SA, C. MJ, V. I.A., R. S.A., C. M.J., Oct retinal and choroidal layer instance segmentation using mask r-cnn, Sensors (Basel, Switzerland) 22 (5) (2022). doi:10.3390/s22052016.
URL https://pubmed.ncbi.nlm.nih.gov/35271165/

[447] I. A. Viedma, D. Alonso-Caneiro, S. A. Read, M. J. Collins, H. Hoseini-Yazdi, A deep learning multi-capture segmentation modality for retinal oct imaging, 2022 International Conference on Digital Image Computing: Techniques and Applications, DICTA 2022 (2022) 1–8doi:10.1109/DICTA56598.2022.10034625.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10034625

[448] A. Katuru, I. Y. Chung, I. Majid, L. Q. Shen, M. Wang, K. A, C. IY, M. I, S. LQ, W. M, C. I.Y., S. L.Q., Deep learning with disc photos or oct scans in glaucoma detection, Ophthalmology science 5 (6) (2025) 100877. doi:10.1016/j.xops.2025.100877.
URL https://pubmed.ncbi.nlm.nih.gov/40893625/

[449] Y. J. Lee, S. Sun, Y. K. Kim, J. W. Jeoung, K. H. Park, L. YJ, S. S, K. YK, J. JW, P. KH, Diagnostic ability of macular microvasculature with swept-source oct angiography for highly myopic glaucoma using deep learning, Scientific reports 13 (1) (2023) 5209. doi:10.1038/s41598-023-32164-9.
URL https://pubmed.ncbi.nlm.nih.gov/36997639/

[450] V. Mohammadzadeh, A. Vepa, C. Li, S. Wu, L. Chew, G. Mahmoudinezhad, E. Maltz, S. Sahin, A. Mylavarapu, K. Edalati, J. Martinyan, D. Yalzadeh, F. Scalzo, J. Caprioli, K. Nouri-Mahdavi, M. V, V. A, L. C, W. S, C. L, M. G, M. E, S. S, M. A, E. K, M. J, Y. D, S. F, C. J, N.-M. K, Prediction of central visual field measures from macular oct volume scans with deep learning, Translational vision science & technology 12 (11) (2023) 5. doi:10.1167/tvst.12.11.5.
URL https://pubmed.ncbi.nlm.nih.gov/37917086/

[451] Y. Fujinami-Yokokawa, N. Pontikos, L. Yang, K. Tsunoda, K. Yoshitake, T. Iwata, H. Miyata, K. Fujinami, O. B. O. Japan Eye Genetics Consortium, F.-Y. Y, P. N, Y. L, T. K, Y. K, I. T, M. H, F. K, J. E. G. C. OBO, Prediction of causative genes in inherited retinal disorders from spectral-domain optical coherence tomography utilizing deep learning techniques, Journal of ophthalmology 2019 (2019) 1691064. doi:10.1155/2019/1691064.
URL https://pubmed.ncbi.nlm.nih.gov/31093368/

[452] X. Zhang, Z. Xiao, R. Higashita, W. Chen, J. Yuan, J. Fang, Y. Hu, J. Liu, X. Zhang, Z. Xiao, R. Higashita, W. Chen, J. Yuan, J. Fang, Y. Hu, J. Liu, Z. X., X. Z., H. R., C. W., Y. J., F. J., H. Y., L. J., A novel deep learning method for nuclear cataract classification based on anterior segment optical coherence tomography images, Conference Proceedings - IEEE International Conference on Systems, Man and Cybernetics 2020-October (2020) 662–668. doi:10.1109/SMC42975.2020.9283218.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9283218

[453] X. Xiong, D. H. Steel, B. Obara, X. Xiong, D. H. Steel, B. Obara, X. X., S. D.H., O. B., Evaluation of 2d and 3d deep learning approaches for automatic segmentation of the retinal external limiting membrane in spectral domain optical coherence tomography images, ICIIBMS 2022 - 7th International Conference on Intelligent Informatics and Biomedical Sciences 7 (2022) 341–345. doi:10.1109/ICIIBMS55689.2022.9971669.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9971669

[454] L. Toto, A. Romano, M. Pavan, D. Degl'Innocenti, V. Olivotto, F. Formenti, P. Viggiano, E. Midena, R. Mastropasqua, T. L, R. A, P. M, D. D, O. V, F. F, V. P, M. E, M. R, A deep learning approach to hard exudates detection and disorganization of retinal inner layers identification on oct images, Scientific reports 14 (1) (2024) 16652. doi:10.1038/s41598-024-63844-9.
URL https://pubmed.ncbi.nlm.nih.gov/39030181/

[455] J. I. Orlando, B. S. Gerendas, S. Riedl, C. Grechenig, A. Breger, M. Ehler, S. M. Waldstein, H. Bogunović, U. Schmidt-Erfurth, O. JI, G. BS, R. S, G. C, B. A, E. M, W. SM, B. H, S.-E. U, Automated quantification of photoreceptor alteration in macular disease using optical coherence tomography and deep learning, Scientific reports 10 (1) (2020) 5619. doi:10.1038/s41598-020-62329-9.
URL https://pubmed.ncbi.nlm.nih.gov/32221349/

[456] Y. Ozturk, A. Ağın, B. Yelmi, N. Zorlutuna Kaymak, O. Y, A. A, Y. B, Z. K. N, Artificial intelligence in predicting macular hole surgery outcomes: a focus on optical coherence tomography parameters, BMC ophthalmology 25 (1) (2025) 419. doi:10.1186/s12886-025-04256-9.
URL https://pubmed.ncbi.nlm.nih.gov/40696275/

[457] N. Denk, C. Freichel, P. Valmaggia, N. Inglin, H. P. N. Scholl, P. Kaiser, S. Wise, M. Vezina, P. M. Maloca, D. N, F. C, V. P, I. N, S. HPN, K. P, W. S, V. M, M. PM, S. H.P.N., M. P.M., Cynomolgus monkey's retina volume reference database based on hybrid deep learning optical coherence tomography segmentation, Scientific reports 13 (1) (2023) 5797. doi:10.1038/s41598-023-32739-6.
URL https://pubmed.ncbi.nlm.nih.gov/37032376/

[458] M. H. Akpinar, A. Sengur, O. Faust, L. Tong, F. Molinari, U. R. Acharya, A. MH, S. A, F. O, T. L, M. F, A. UR, A. M.H., A. U.R., Artificial intelligence in retinal screening using oct images: A review of the last decade (2013-2023), Computer methods and programs in biomedicine 254 (2024) 108253. doi:10.1016/j.cmpb.2024.108253.
URL https://pubmed.ncbi.nlm.nih.gov/38861878/

[459] L. Buzura, M. L. Budileanu, R. Papara, H. Demea, R. Galatus, B. L., B. M.L., P. R., D. H., G. R., Macular edema degeneration classification on oct and fundus images with portable platform based on artificial intelligence methods, Proceedings of SPIE - The International Society for Optical Engineering 12144 (2022). doi:10.1117/12.2617520.
URL https://www.scopus.com/pages/publications/85133123514?origin=resultslist

[460] L. Siva Yamini, S. Shylu, G. Viveka, J. Sai Dheeraj, N. Srihari, An efficient retinal layer segmentation based on deep learning regression technique for early diagnosis of retinal diseases in oct and fundus images, Lecture Notes in Electrical Engineering 947 (2023) 887–891. doi:10.1007/978-981-19-5936-3_84.
URL https://www.scopus.com/pages/publications/85149990150?origin=resultslist

[461] P. L. Jancy, B. Latha, P. L. Jancy, B. Latha, J. P.L., Deep learning techniques for diabetic retinopathy diagnosis using optical coherence tomography: A review, 2022 International Conference on Advanced Computing Technologies and Applications, ICACTA 2022 (2022) 1–4doi:10.1109/ICACTA54488.2022.9753418.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9753418

[462] S. Liu, Y. Liu, X. Wu, H. Wang, Z. Jin, P. Wang, J. Feng, S. Chen, W. Zhou, L. S, L. Y, W. X, W. H, J. Z, W. P, F. J, C. S, Z. W, Efficacy and prognostic factors of anti-vegf treatment for neovascular age-related macular degeneration: An octa imaging-based deep learning analysis, Photodiagnosis and photodynamic therapy 55 (2025) 104701. doi:10.1016/j.pdpdt.2025.104701.
URL https://pubmed.ncbi.nlm.nih.gov/40683364/

[463] M. E. Subasi, S. Patnaik, A. Subasi, S. M.E., P. S., S. A., Optical coherence tomography image classification for retinal disease detection using artificial intelligence, Applications of Artificial Intelligence in Healthcare and Biomedicine (2024) 289–323doi:10.1016/B978-0-443-22308-2.00009-3.
URL https://www.scopus.com/pages/publications/85193408149?origin=resultslist

[464] L. M. Coulibaly, K. Birner, A. Zarghami, M. Gumpinger, S. Schürer-Waldheim, P. Fuchs, H. Bogunović, U. Schmidt-Erfurth, G. S. Reiter, C. LM, B. K, Z. A, G. M, S.-W. S, F. P, B. H, S.-E. U, R. GS, C. L.M., R. G.S., Repeatability of microperimetry in areas of retinal pigment epithelium and photoreceptor loss in geographic atrophy supported by artificial intelligence-based optical coherence tomography biomarker quantification, American journal of ophthalmology 271 (2025) 347–359. doi:10.1016/j.ajo.2024.11.005.
URL https://pubmed.ncbi.nlm.nih.gov/39547308/

[465] R. Krishna, R. Suganya, M. S. Yaganti, A. Asha, D. Jayanthi, N. K. Maurya, R. Krishna, R. Suganya, M. S. Yaganti, A. Asha, D. Jayanthi, N. K. Maurya, K. R., Y. M.S., M. N.K., Retinal disease classification in optical coherence tomography images using a convolutional neural network algorithm, 2024 IEEE 1st International Conference on Green Industrial Electronics and Sustainable Technologies, GIEST 2024 (2024) 1–6doi:10.1109/GIEST62955.2024.10959763.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10959763

[466] F. Dhaoui, A. Zrelli, F. Dhaoui, A. Zrelli, D. F., Z. A., Retinal diseases classification system using oct images combined with cnn models, 2023 International Symposium on Networks, Computers and Communications, ISNCC 2023 (2023) 1–6doi:10.1109/ISNCC58260.2023.10323745.
URL https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10323745

[467] H. Bogunović, W.-D. Vogl, P. Seeböck, A. Rivail, U. Schmidt-Erfurth, B. H., V. W.-D., S. P., R. A., S.-E. U., Retinal oct analysis and prediction with deep learning, Optics InfoBase Conference Papers Part F180-OCT-2020 (2020).
URL https://www.scopus.com/pages/publications/85091450828?origin=resultslist